



HCMC UNIVERSITY OF TECHNOLOGY AND EDUCATION
Faculty of Chemical and Food Technology



HCMUTE

ENVIRONMENTAL ENGINEERING CHEMISTRY

HÓA KỸ THUẬT MÔI TRƯỜNG

Chapter 1 INTRODUCTION



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❖ **Vị trí và mục tiêu môn học:**

- Tiếp nối và áp dụng các kiến thức của Hóa học vào Kỹ thuật môi trường.
- Môn học giới thiệu các kiến thức cốt lõi liên quan đến các quá trình hóa học được sử dụng trong kỹ thuật môi trường.

❖ **Ứng dụng môn học:**

Trong xử lý môi trường:

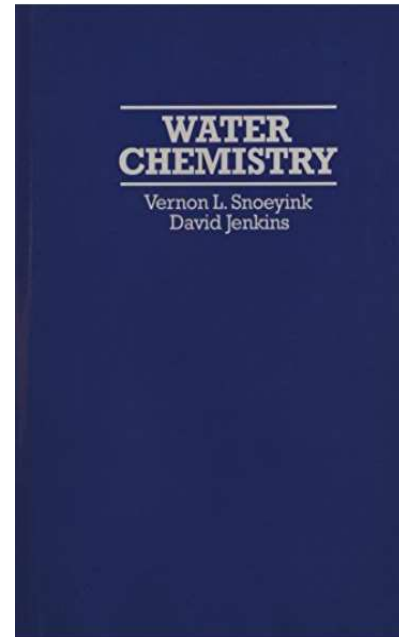
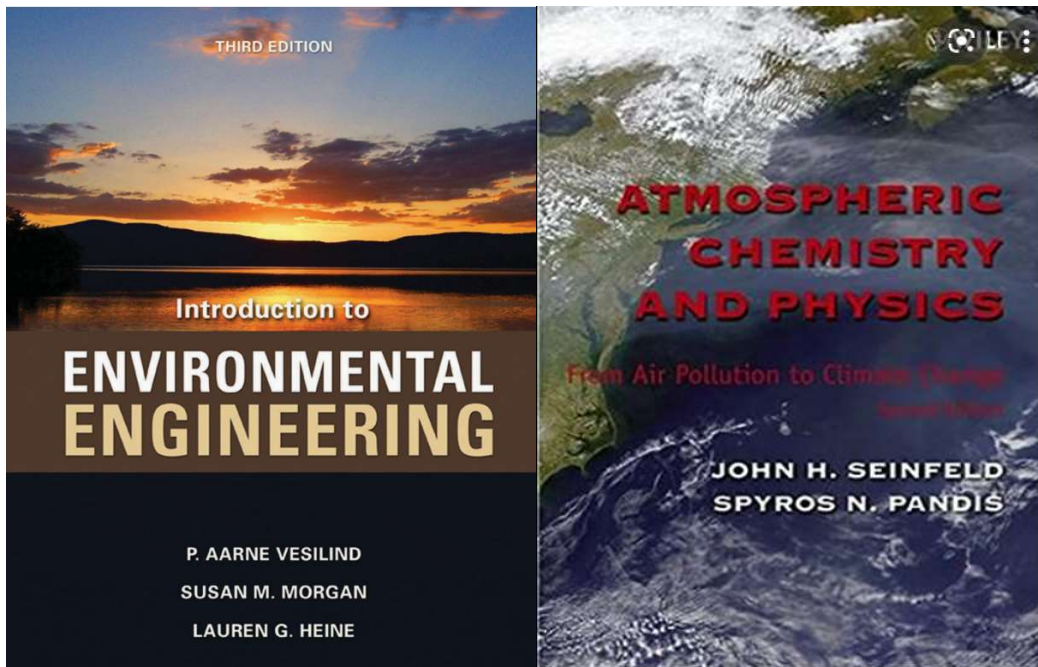
- Giải thích được các phản ứng và quá trình hóa học xảy ra trong môi trường và xử lý chất thải
- Hiểu được nguyên tắc chọn bề phản ứng trong xử lý môi trường

Trong nghiên cứu khoa học:

- Hỗ trợ thiết kế thí nghiệm.
- Thiết kế mô hình thực nghiệm.

❖ Tài liệu môn học:

1. Water chemistry



Water Chemistry (Spe Monographs)



Book Detail

Author : Vernon L. Snoeyink

Pages : 484 pages

Publisher : John Wiley & Sons 1980-04-03

Language : English

Book Synopsis

none

[.....more](#)

2. Introduction to Environmental Engineering
3. Atmospheric Chemistry and Physics

❖ Nội dung môn học:

Bài giảng được soạn xen
kẽ giữa tiếng anh và tiếng
việt

Chapter 1: Introduction (tuần 1)

Chapter 2: Chemical kinetics (tuần 2)

Chapter 3: Chemical equilibrium (tuần 3-4)

Chapter 4: Acid-base chemistry (tuần 5-6)

Chapter 5: Precipitation and dissolution (tuần 7)

Chapter 6: Adsorption and absorption (tuần 8-9)

Chapter 7: Advance oxidation processes (tuần 10)

Chapter 8: Reactor design (tuần 11-12)

Chapter 9: Atmospheric chemistry (tuần 12-13)

Chapter 10: Soil chemistry (tuần 14)

Chapter 11: Matter cycles in the environment – pollution and treatment (tuần 15)

❖ **Đánh giá môn học:**

- **Giữa kỳ (50%):**

- Bài tập về nhà: 30%
- Tham gia trên lớp (điểm danh, câu hỏi): 5%
- Kiểm tra giữa kỳ (trắc nghiệm hoặc tự luận): 15%

- **Cuối kỳ (50%):** Thi (trắc nghiệm + tự luận)



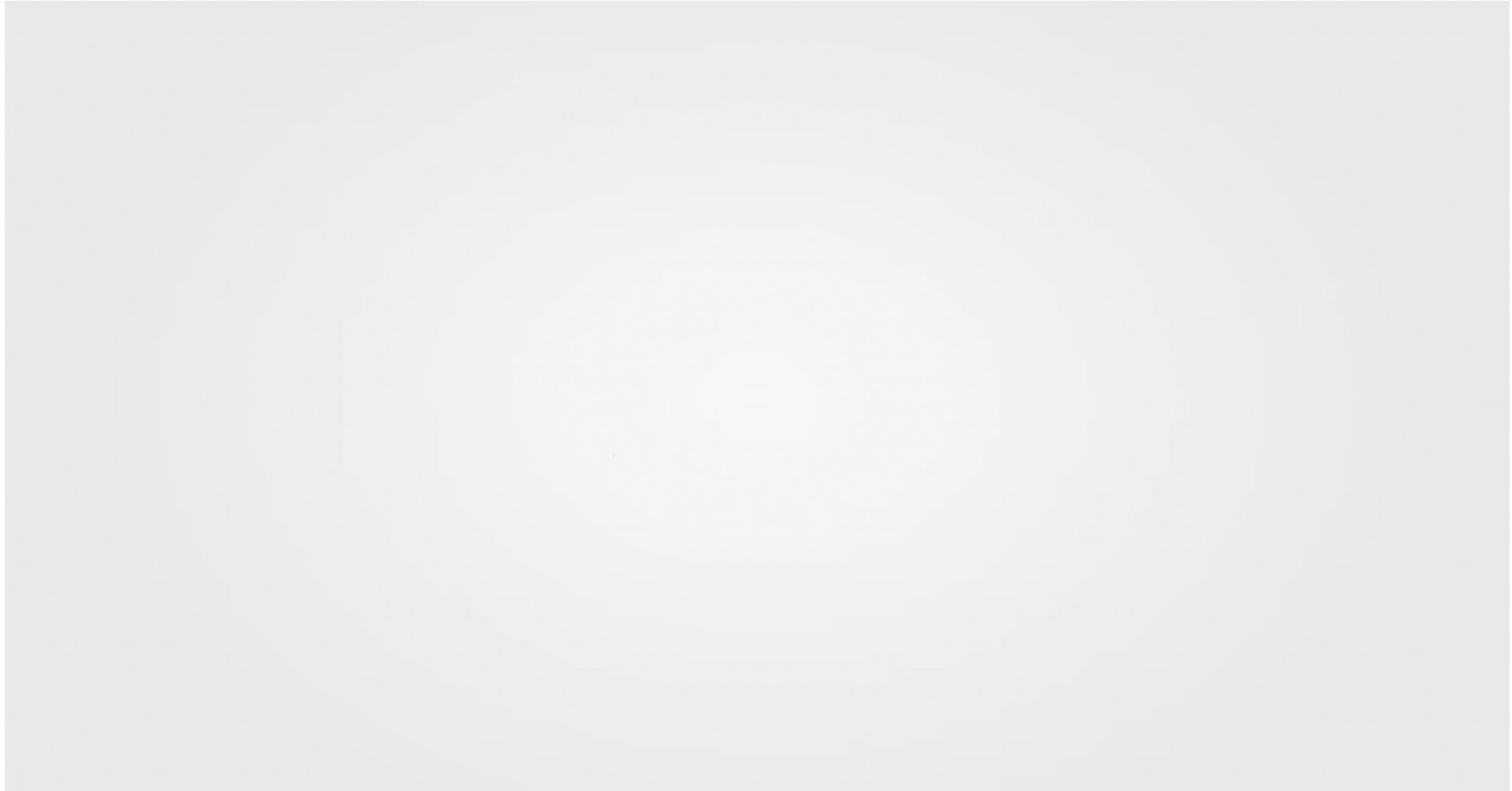
Lưu ý:

- SV tự nhớ số thứ tự của mình, GV hỏi bài mỗi ngày theo số thứ tự;
- Học online bắt buộc show webcam
- *Bài tập về nhà sẽ có điểm là -1 nếu phát hiện có sao chép*



Any questions or comments?



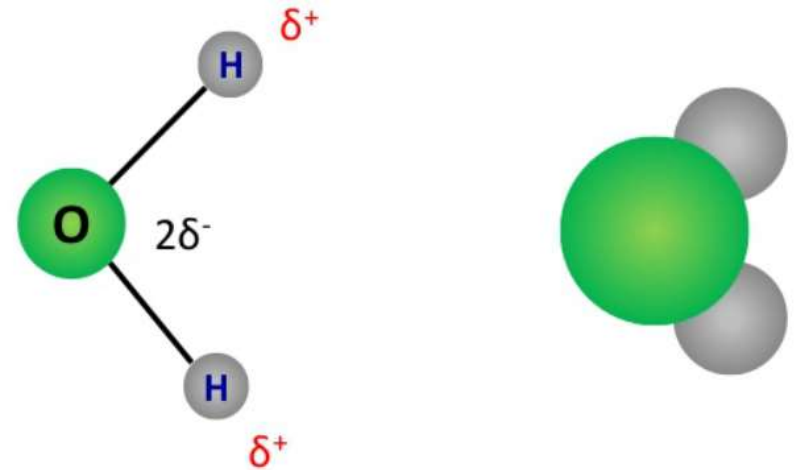


<https://www.youtube.com/watch?v=h0py6BFIFZw>

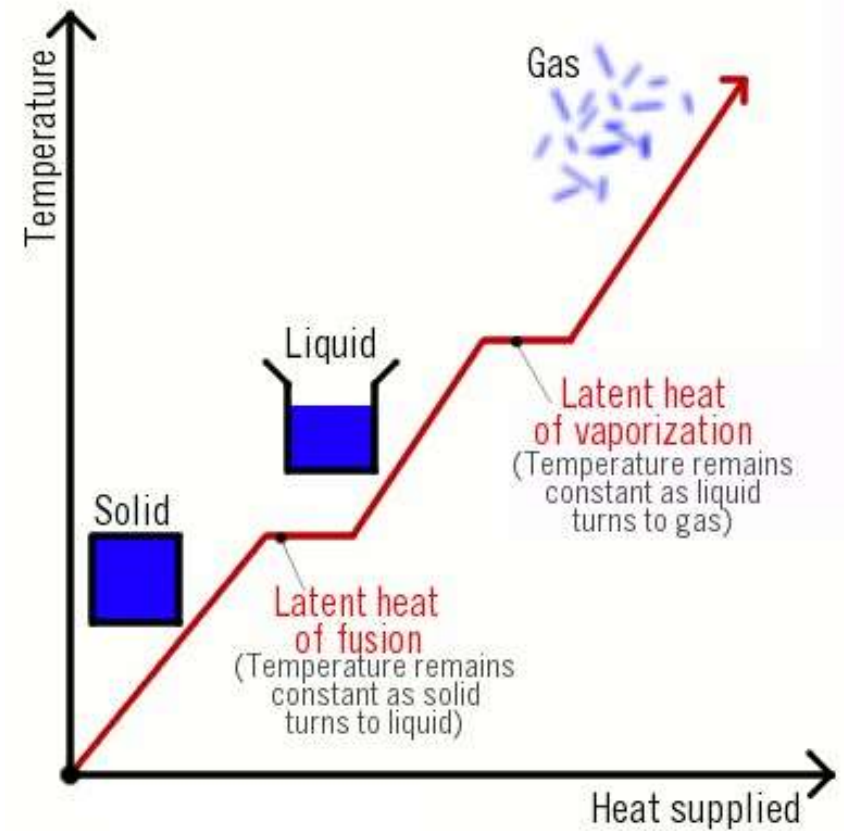
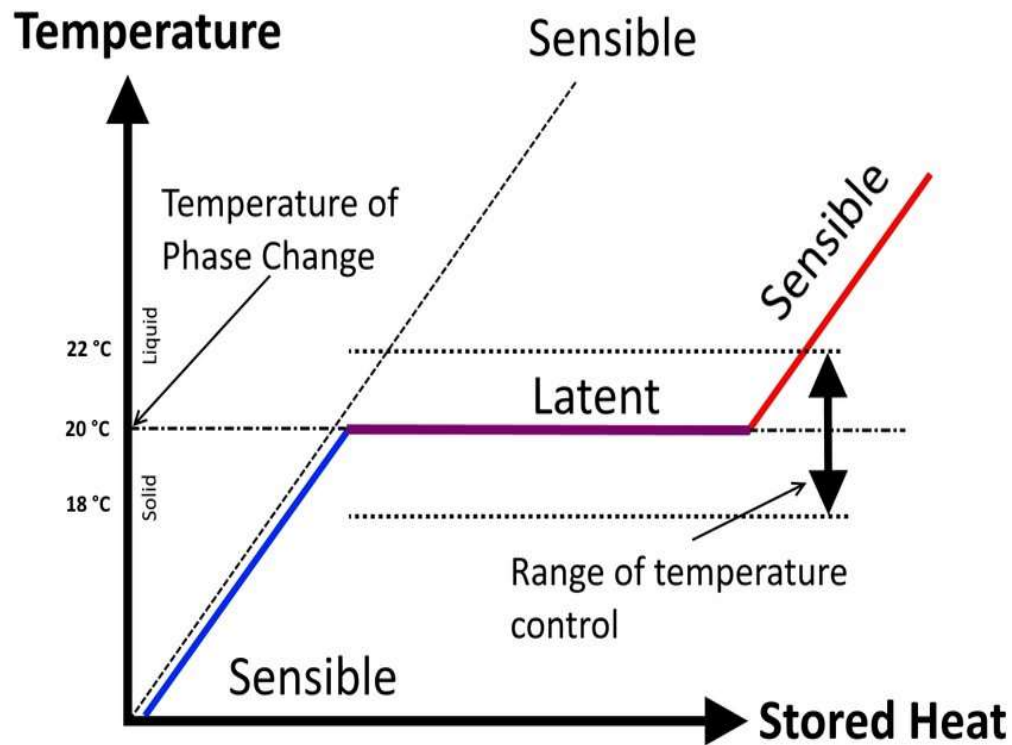


Water properties

- Strongly dipolar (tính lưỡng cực mạnh)
- Hydrogen bonding:
 - High latent heat of vaporation → high boiling point (nhiệt ẩn bay hơi cao → điểm sôi cao)
 - High latent heat of fusion → low freezing point (nhiệt ẩn tan chảy cao → điểm đông đặc thấp)
- Polar substance → universal solvent: excellent solvent for charged ions and polar molecules (phân cực → dung môi hòa tan tốt các ion và phân tử phân cực)
→ Transport (or transformation) media (môi trường thuận lợi cho các quá trình chuyển hóa và vận chuyển chất)



Water properties Sensible heat vs latent heat



Soluble ions

- IA - Alkali metals (Na and K ...)
 - Crystals containing these ions tend to be very soluble in water.
 - When dissolved in water, these ions tend to remain individual ions and not form complexes with other ions. → difficult to be removed from water
- IIA elements - Alkaline earth metals (Ca and Mg ...)
 - Many elements in this family are also called earth elements.
 - Ca and Mg easily form precipitates with SO_4^{2-} , OH^- , CO_3^{2-} , Cl^- , F^- , PO_4^{3-} , and etc.
- Halogen
 - Crystals containing these ions tend to be very soluble in water.
 - These anions do not form complexes with many other ions in aqueous solutions. → difficult to be removed from water

Soluble ions

- Elements from the upper right of the periodic table (CO_3^{2-} , NO_3^- , NO_2^- , PO_4^{3-} , SO_4^{2-} , OCl^- , S_2^- , HS^-)
 - Often these elements form multi-element compounds which can be considered as a reaction unit.
 - Some are not very reactive (NO_3^- , SO_4^{2-}); some react with water and form a variety of complex (CO_3^{2-} , PO_4^{3-} , HS^-)
- Transition elements (Fe, Co, Zn, Cd, Mn, ...)
 - Most of these elements will form complexes with a wide variety of inorganic and organic ions

Units of concentration

1. Mass concentration

w/v basis: mg/L

w/w basis: ppm(m) (mg/kg)

2. Molar concentration & activity

Molar concentration, $M = \frac{\text{moles of solute}}{\text{liter of solution}} = \frac{n \text{ (mole)}}{V \text{ (L)}}$

Activity coefficient, $\gamma = \frac{\text{apparent (active) conc.}}{\text{actual (analytical) conc.}} = \frac{\text{phân ly}}{\text{phân tích}}$

Activity, $a = \gamma M$

3. Equivalent and Normal concentration

Equivalent weight (D) = $\frac{\text{molecular weight}}{\text{Valence (ion charge)}} = \frac{M \text{ (g/mole)}}{Z \text{ (eq/mole)}}$

Normality = $\frac{\text{Mass of substance per liter}}{\text{Equivalent weight}}$ (eq/L) (nồng độ đương lượng)

Units of concentration

3. Equivalent and Normal concentration

$$\text{Equivalent weight (D)} = \frac{\text{molecular weight}}{\text{Valence (ion charge)}} = \frac{M \text{ (g/mole)}}{Z \text{ (eq/mole)}}$$

$$\text{Normality} = \frac{\text{Mass of substance per liter}}{\text{Equivalent weight}} \text{ (meq/L) (nồng độ đương lượng)}$$

Ion charge: số điện tích trao đổi, phụ thuộc vào phản ứng mà chất tham gia đã trao đổi bao nhiêu điện tích.

Units of concentration

4. Mass/volume concentrations as O₂: BOD, COD

5. Mass concentration as CaCO₃

For hardness, and alkalinity

$$\frac{\text{concentration in mg/L}}{\text{equivalent weight}} \times 50/\text{eq} = \frac{\text{mg}}{\text{L}} \text{ as CaCO}_3$$

Concentration in gas phase:

- ❖ Tỷ lệ hòa trộn (mixing ratio, C_x): là số mole của chất X trên số mole của không khí. Không đổi khi thay đổi áp suất, nhiệt độ và khối lượng riêng.

$$C_x = \text{mole}_x / \text{mole}_{\text{tổng}} = v_x / V_{\text{tổng}}$$

Đối với chất khí: Nồng độ phần mole = nồng độ phần thể tích

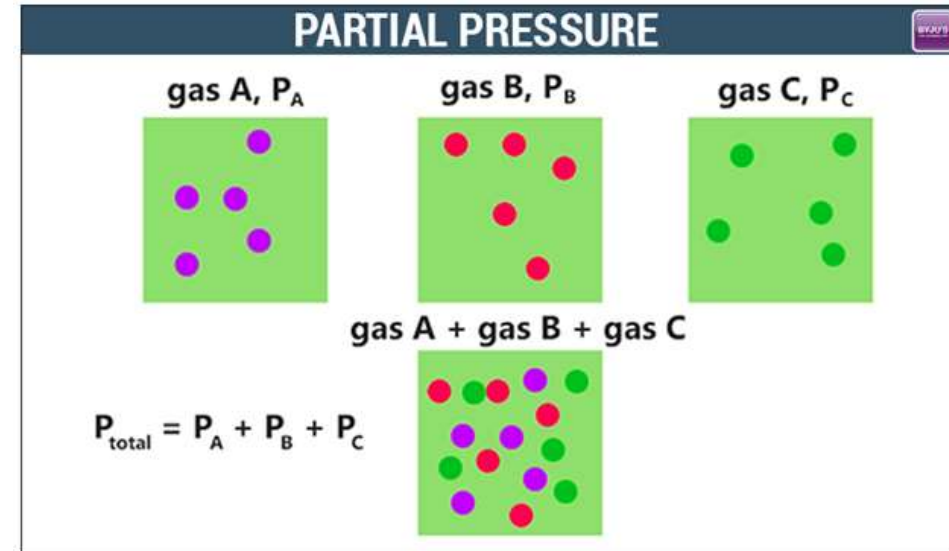
$$\text{ppm (v)} = C_x \times 10^6$$

- ppm (v) – parts per million
 - ppb (v) – parts per billion
 - ppt (v) – parts per trillion
- (v) means by volume

Concentration in gas phase:

- Áp suất riêng phần (partial pressure): là áp suất mà các phân tử X gây ra nếu loại bỏ tất cả các khí khác ra khỏi hỗn hợp khí.
- Áp suất riêng phần và tỷ lệ hòa trộn của khí X liên quan với nhau bởi **định luật Dalton**.

$$P_x = C_x \times P$$



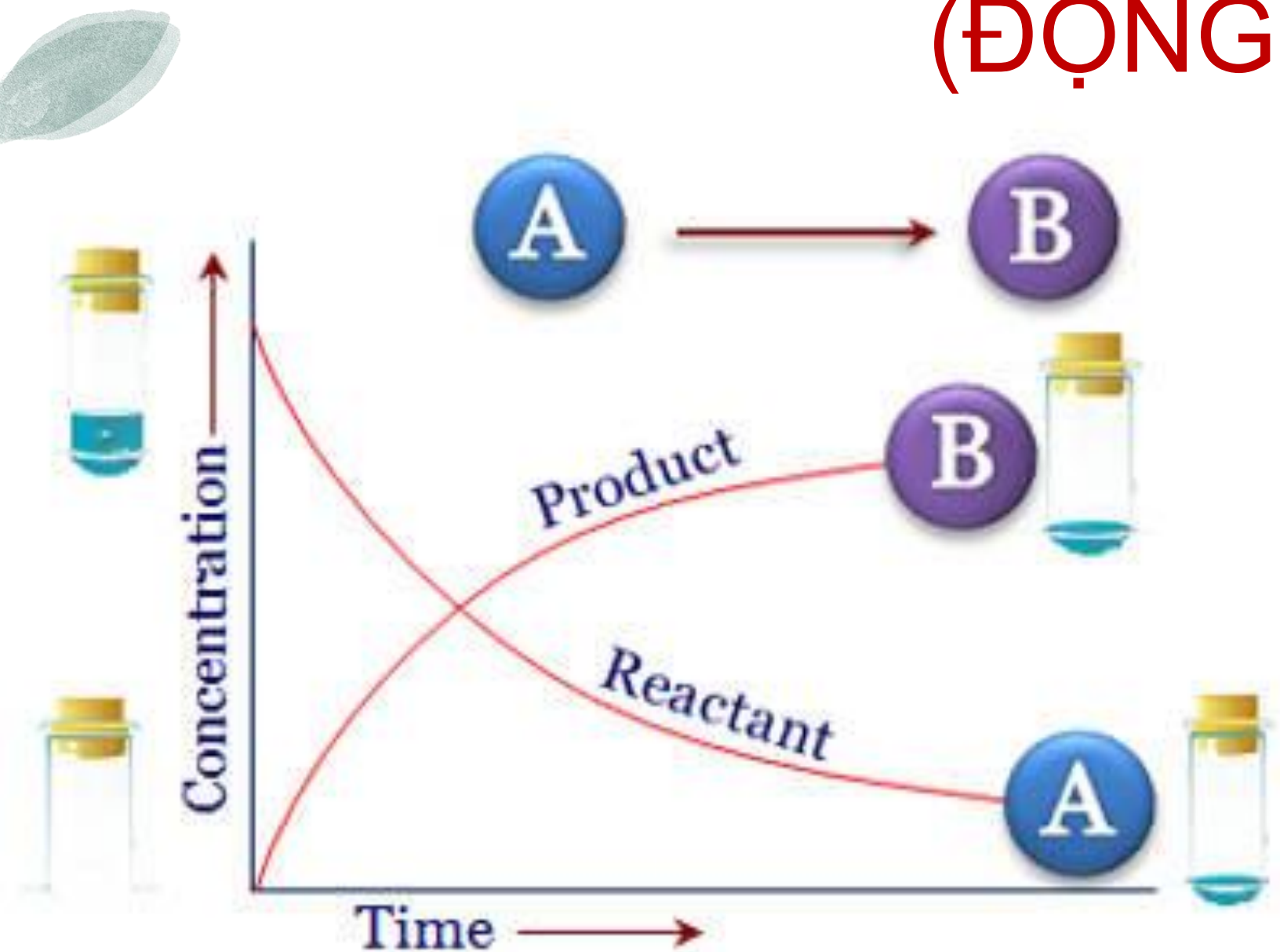
Định luật Dalton: Ở một nhiệt độ xác định, áp suất toàn phần của một hỗn hợp khí bằng tổng số áp suất riêng phần của các cấu tử của hỗn hợp. $P = \sum p_i$

Chapter 1

CHEMICAL KINETICS

(ĐỘNG HỌC HÓA HỌC)

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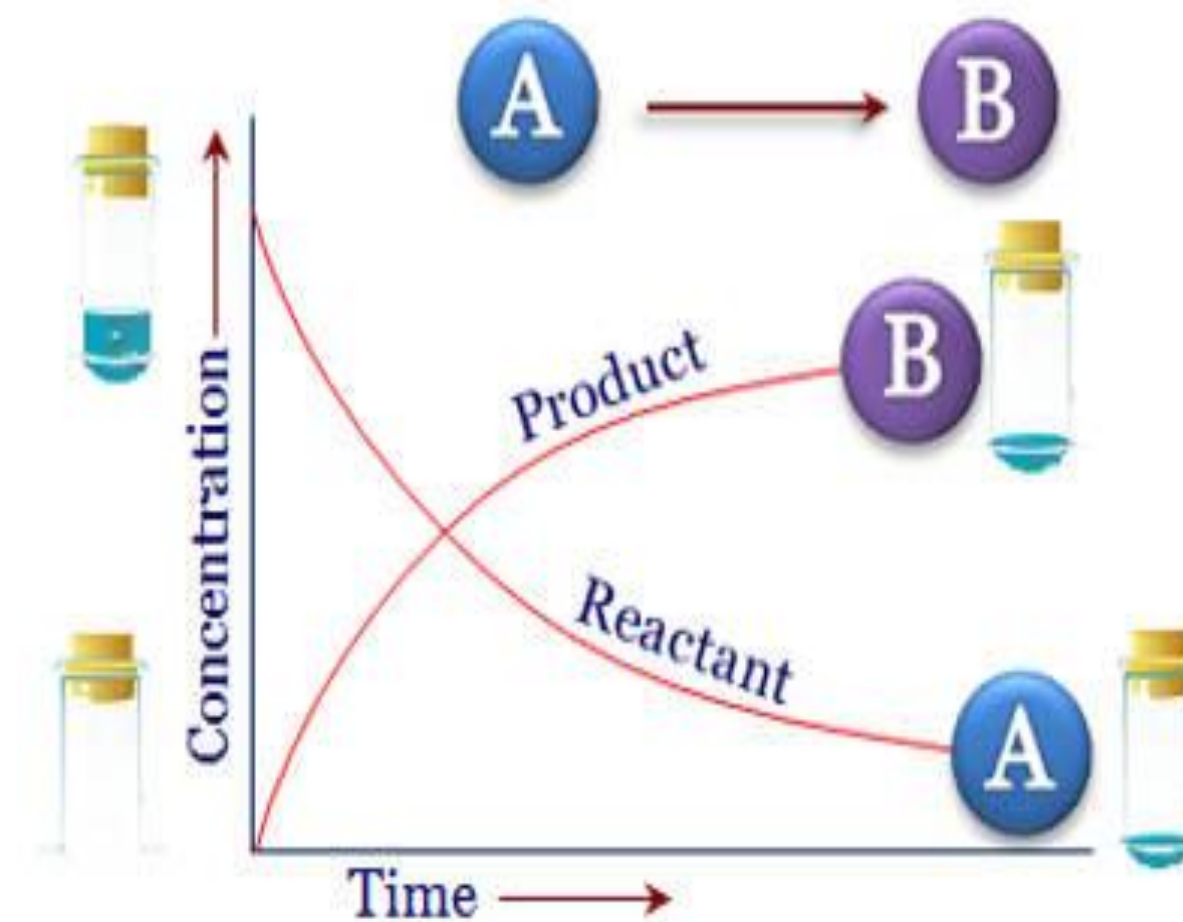


❖ Outline:

- Definition;
- Elementary and none elementary reactions;
- Rate law (example for determination of reaction rate and application of reaction rate)

❖ This chapter will answer:

- What is the chemical reaction?
- What is the meaning of reaction rate and its applications
- How to determine reaction rate?
- What are factors influencing reaction rate?



What is the chemical reaction?

❖ Chemical Kinetics:

- Deals with the **rate and mechanism** of chemical reactions; that is, **how the reaction proceeds**. (Động hóa học liên quan đến tốc độ và cơ chế của phản ứng hóa học hay nó nói lên phản ứng đó xảy ra như thế nào).

❖ Chemical reaction

A chemical reaction is, by definition, one in which **chemical substances are transformed into other substances**, which means that **chemical bonds are broken and formed** so that there are changes in **the relative positions of atoms in molecules**. (trong PUHH, các chất tham gia chuyển thành các chất khác, có nghĩa là các liên kết hóa học trong chất tham gia bị phá vỡ và hình thành các liên kết hóa học mới trong sản phẩm).



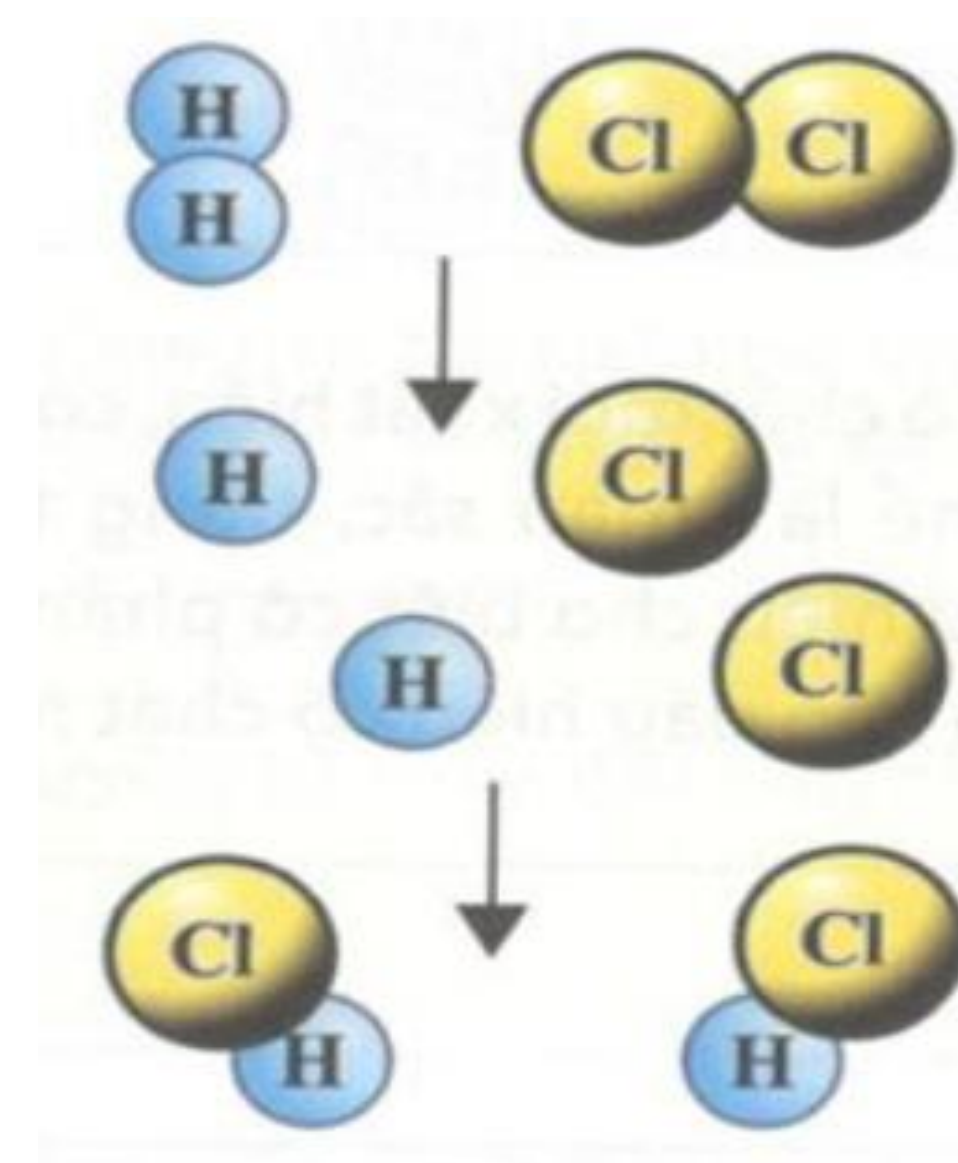
What is the chemical reaction?

❖ Chemical reaction

At the same time, there are **shifts in the arrangements of the electrons that form the chemical bonds**. (Trong thời gian xảy ra phản ứng, có sự sắp xếp lại các electron để hình thành các liên kết hóa học mới).

A description of a **reaction mechanism** must therefore deal with the **movements and speeds of atoms and electrons**. (Sự mô tả cơ chế của phản ứng phải liên quan đến sự di chuyển và tốc độ của các phân tử và electron).

The **detailed mechanism** by which a chemical process occurs is referred to as the **reaction path, or pathway**.

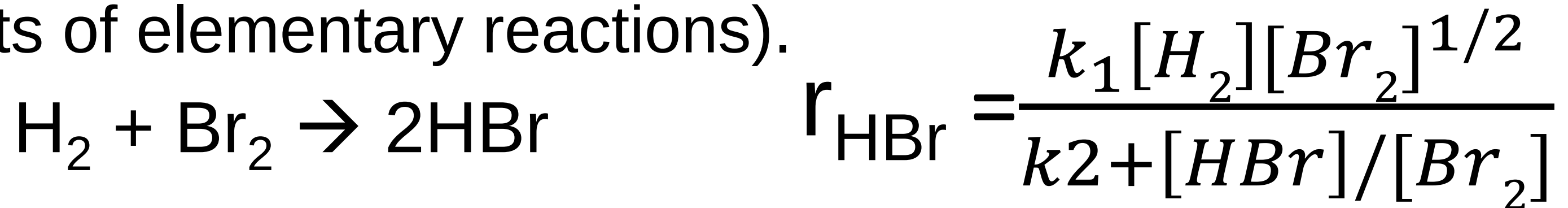


1. Elementary and nonelementary reaction

- **Elementary:** the rate equation corresponds to a stoichiometric equation (phản ứng cơ bản là phản giữa các cấu tử mà bản thân các cấu tử đó không thể chia nhỏ hơn).



- **Nonelementary:** there is no correspondence between stoichiometry and rate (includes many elementary reactions → are sets of elementary reactions).



Nếu một phản ứng phức tạp bao gồm nhiều phản ứng đơn giản (nhiều giai đoạn) xảy ra thì tập hợp các phản ứng đơn giản đó chính là cơ chế phản ứng. Và tốc độ của giai đoạn chậm nhất quyết định tốc độ phản ứng.

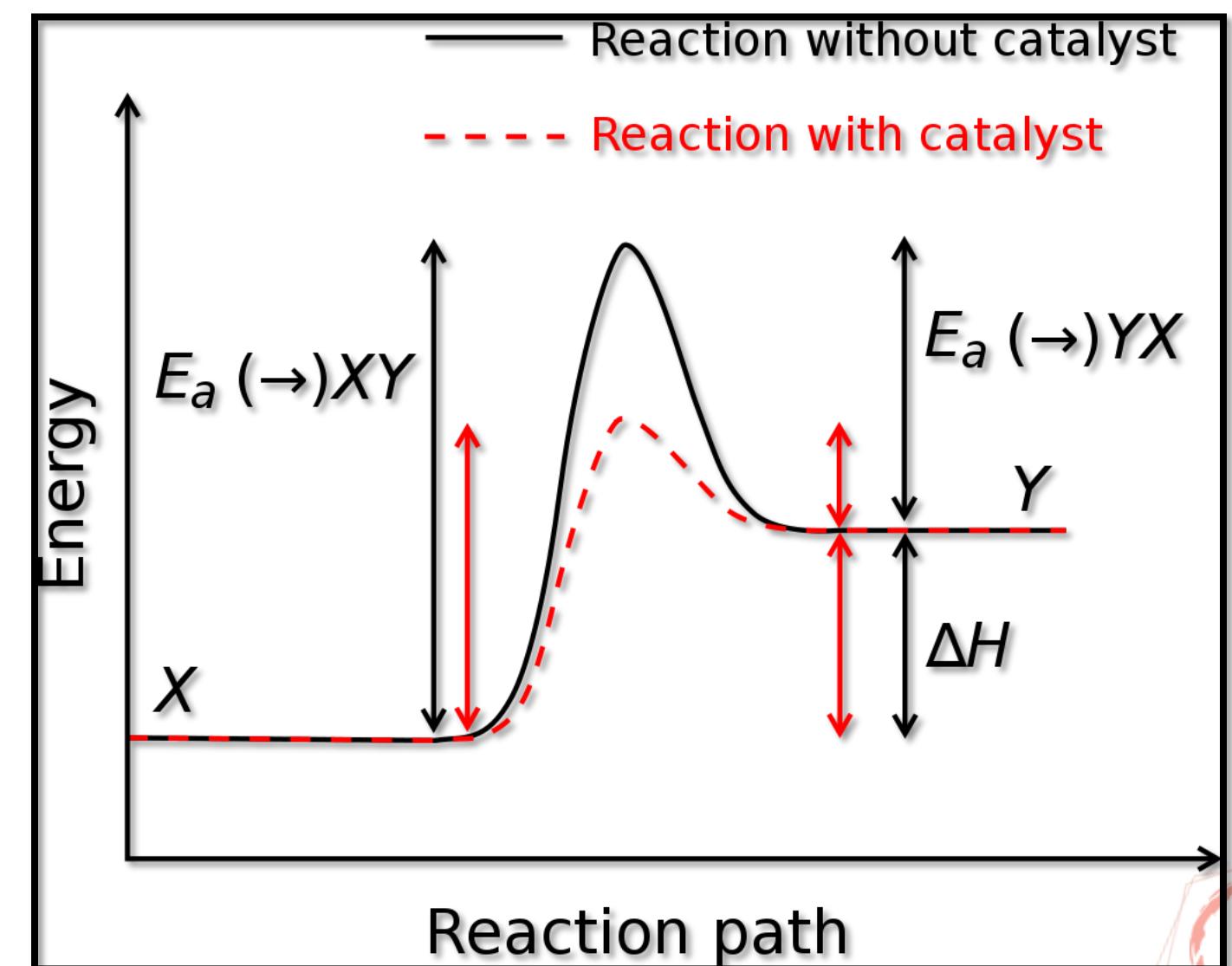
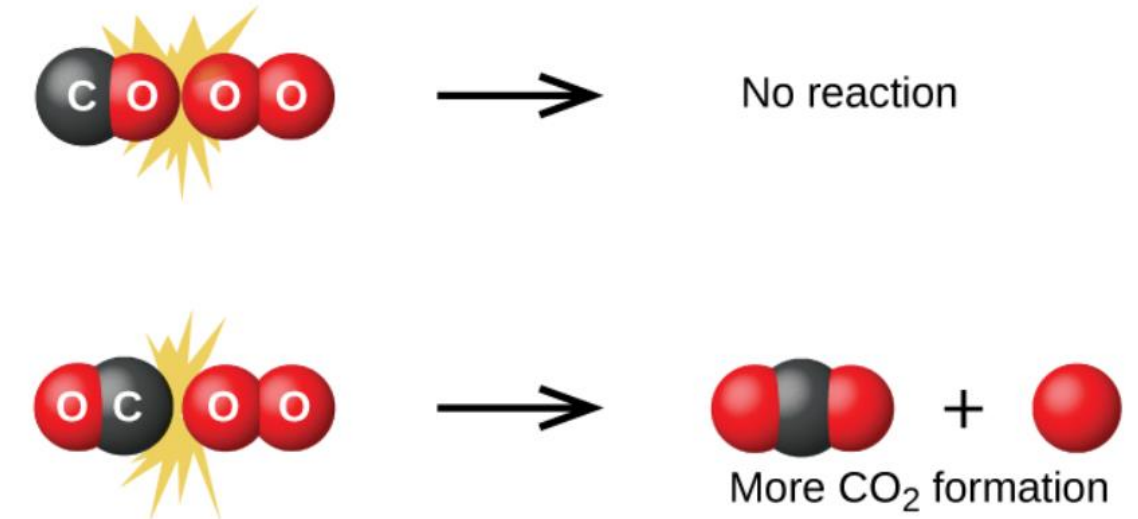
2. Reaction rate

❖ The reaction rate depends on:

- Collision frequency
- A probability or orientation factor
- Activation energy (E_a)

Một phản ứng hóa học xảy ra khi có sự tiếp xúc (**collision**) của các phân tử tham gia phản ứng ở vị trí phù hợp (**orientation**) và đủ năng lượng hoạt hóa (**activation energy**).

What is reaction rate?

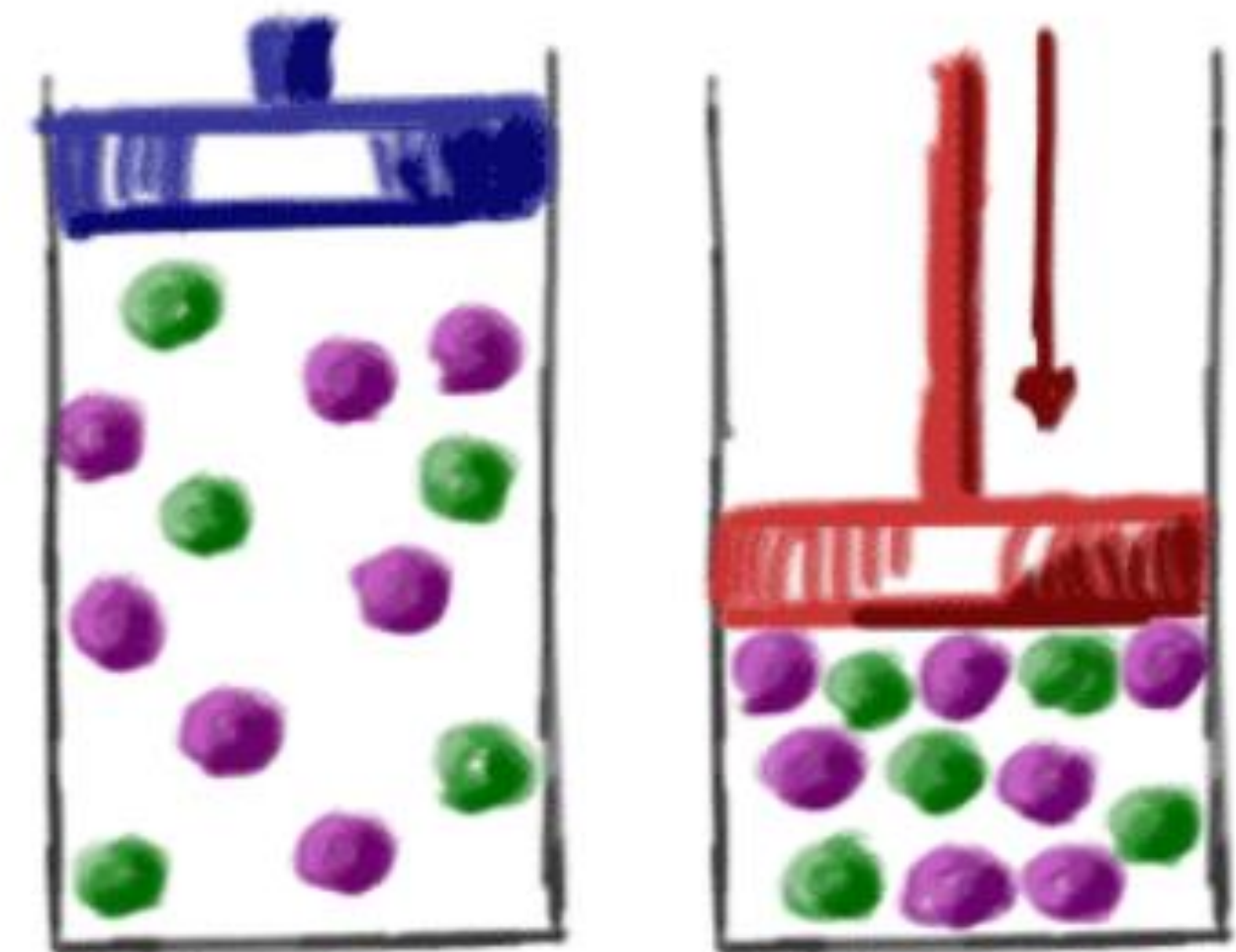
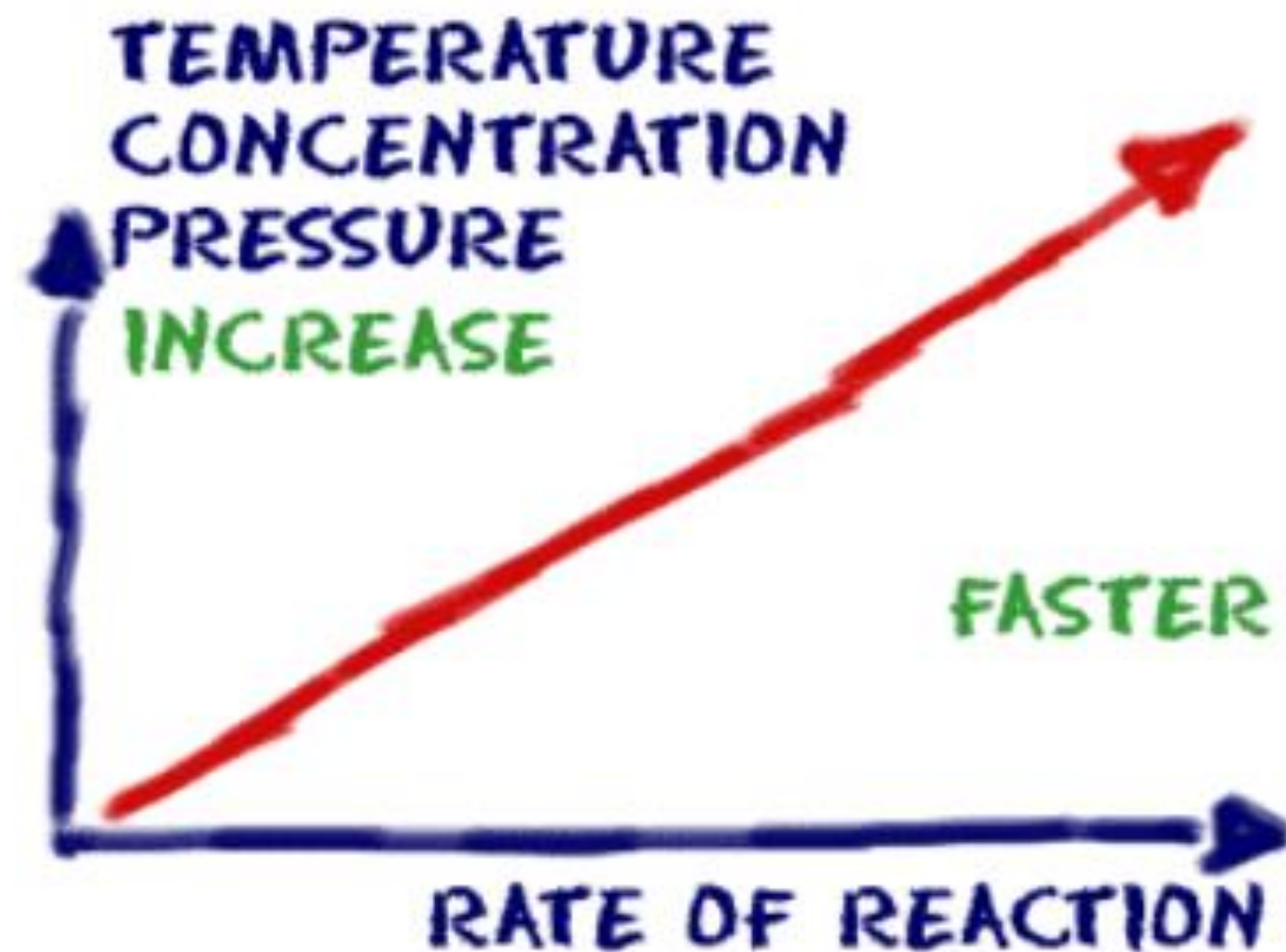


2. Reaction rate

What is reaction rate?

❖ The reaction rate increases as number of collisions between reacting species increases.

- Concentration
- Temperature

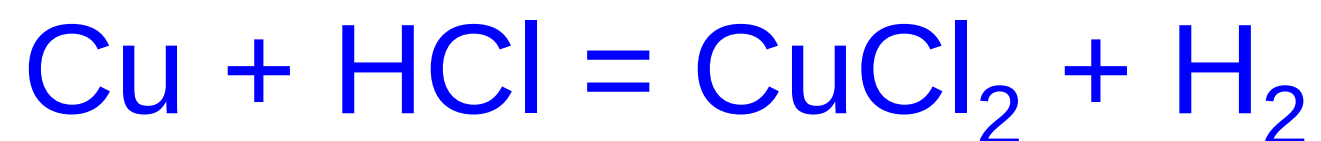


AS PRESSURE INCREASES,
THE GAS MOLECULES CAN
HAVE MORE COLLISIONS.

2. Reaction rate

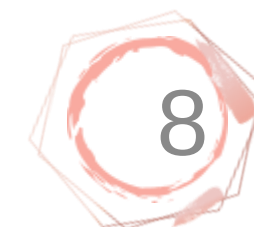
$$\text{Reaction rate} = \frac{d[A]}{dt}$$

The reaction rate is the speed at which a chemical reaction takes place, defined as proportional to the increase in the concentration of a product per unit time and to the decrease in the concentration of a reactant per unit time. (Tốc độ phản ứng là tỷ lệ của sự tăng nồng độ sản phẩm hay sự giảm nồng độ chất tham gia phản ứng trong một đơn vị thời gian, h, min, s).



t = 0s	2 M	0 M
t = 60s	0.1 M	>0 M

$$\text{Tốc độ phản ứng } r = (2-0.1)\text{M}/60\text{s} = 0.03167 \text{ mol/L.s}$$



2. Reaction rate

Rate law

For an irreversible, elementary reaction



Rate law: $\frac{d[A]}{dt} = -k[A]^a[B]^b[P][Q]^q$

Form of rate law depends on reaction mechanism

Rate relationship between each specie (vì 1 mol A phản ứng với b mol B,...):

$$\frac{d[A]}{dt} = \frac{1}{2} \frac{d[B]}{dt} = -\frac{d[P]}{dt} = -\frac{1}{q} \frac{d[Q]}{dt}$$

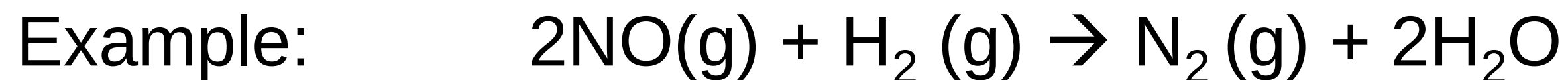
Overall order = $a+b+p+q$ (a, b, p, q là bậc phản ứng tương ứng với A, B, P, Q)

For irreversible, p, q are usually zero, then the order: $\frac{d[A]}{dt} = -k[A][B]^2$

(bậc phản ứng tương ứng với A và B lần lượt là 1 và 2, bậc phản ứng tổng quát là $1+2=3$)

2. Reaction rate

Rate law



If the rate law was determined as:

$$\text{Rate} = k[\text{NO}]^2 [\text{H}_2]$$

- NO: 2nd order
- H₂: 1st order
- Overall: 3rd order (=1+2)

The reaction order can be:

- **Zero** (when the conc dose not affect the rate)
- **Fration**, ie: 1/3
- **Negative**:

2. Reaction rate

Example

Example: Hãy xác định bậc phản ứng (overall order) của các phản ứng cho trong bảng sau:

Reaction	Rate law	Reaction order
$I_2 \rightarrow 2I$	$\text{rate} = k[I_2]$	?
$2 HI \rightarrow H_2 + I_2$	$\text{rate} = k.[HI]^2$	?
$2 NO + O_2 \rightarrow 2 NO_2$	$\text{rate} = k.[NO]^2 .[O_2]$	?
$H_2 + Cl_2 \rightarrow 2HCl$	$\text{rate} = k[H_2]^0 [Cl_2]^0$	?

2. Reaction rate

Rate law

If $\frac{d[A]}{dt} = k[A]^n$, n is the order of reaction.

k : rate constant with $(\text{time})^{-1} (\text{concentration})^{1-n}$

Đơn vị của hằng số của k luôn được viết rõ đơn vị của nồng độ và thời gian

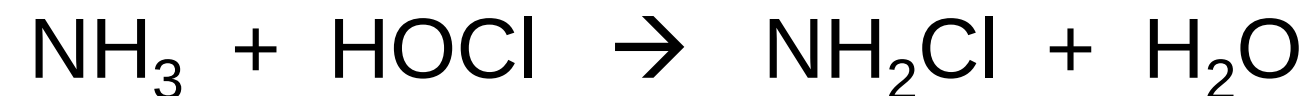
Bậc phản ứng n	Đơn vị của k (time = s, conc. = M)
1	?
2	?
3	?

Ý nghĩa và ứng dụng của hằng số k ?

2. Reaction rate

Example 2 -1

1. Amoniac – NH_3 là thành phần phổ biến của nhiều loại nước trong tự nhiên và nước thải. NH_3 sẽ phản ứng với acid HOCl trong dung dịch tạo thành NH_2Cl như phản ứng sau:



Với hằng số tốc độ $k = 5.1 \times 10^6 \text{ (L/mol.s)}$

Định luật tốc độ phản ứng: $\frac{d[\text{HOCl}]}{dt} = -k[\text{HOCl}][\text{NH}_3]$

a. Xác định bậc phản ứng tổng thể

b. Tốc độ phản ứng giảm bao nhiêu lần nếu giảm nồng độ của mỗi chất đi 50%

c. Xác định giá trị hằng số tốc độ k' mới nếu đơn vị biểu thị nồng độ là mg/L.

2. Reaction rate

[A]: nồng độ chất A tại thời điểm t

[A]₀: nồng độ chất A tại thời điểm t = 0

❖ For zero order reaction (n=0)

$$(*) \rightarrow \frac{d[A]}{dt} = -k[A]^0 = -k$$

Intergrating $\rightarrow [A] = [A]_0 - kt$

$$\text{Reaction rate } k = \frac{1}{t} ([A]_0 - [A])$$



Determine reaction rate?

❖ For 1st order reaction (n=1)

$$(*) \rightarrow \frac{d[A]}{[A]} = -kt$$

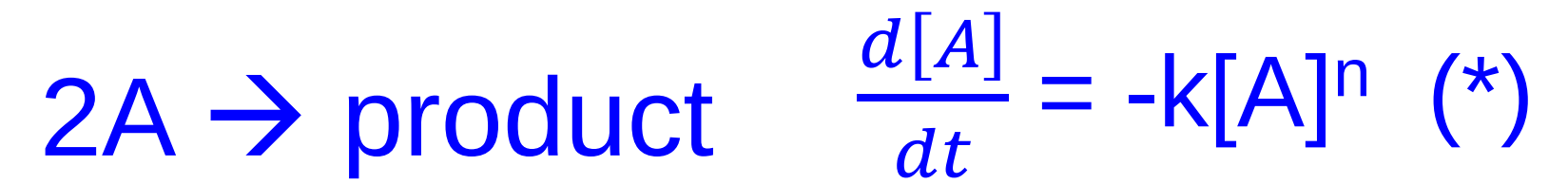
Intergrating $\rightarrow kt = \ln[A]_0 - \ln[A]$

$$\text{Reaction rate } k = \frac{1}{t} \ln \frac{[A]_0}{[A]}$$

2. Reaction rate

[A]: nồng độ chất A tại thời điểm t

[A]₀: nồng độ chất A tại thời điểm t = 0



Determine reaction rate?

❖ For 2nd order reaction (n=2)

$$(*) \rightarrow kdt = -\frac{d[A]}{[A]^2}$$

$$\text{Intergrating} \rightarrow kt = \frac{1}{[A]} - \frac{1}{[A]_0}$$

$$\text{Reaction rate } k = \frac{1}{t} \left(\frac{1}{[A]} - \frac{1}{[A]_0} \right)$$

2. Reaction rate

Half-live ($t_{1/2}$)?

$t_{1/2}$ is the time 50% of $[A]$ is consumed in the reaction.

$$[A] = \frac{1}{2} [A]_0$$

Reaction order (n)	Reaction rate (k)	Replace $[A] = \frac{1}{2} [A]_0$	$t_{1/2}$
0	$k = \frac{1}{t} ([A]_0 - [A])$	$k = \frac{1}{t_{0.5}} ([A]_0 - \frac{[A]_0}{2})$	$t_{1/2} = \frac{[A]_0}{2k}$
1	$k = \frac{1}{t} \ln \frac{[A]_0}{[A]}$	$k = \frac{1}{t_{0.5}} \ln \frac{2[A]_0}{[A]_0}$	$t_{1/2} = \frac{\ln 2}{k}$
2	$k = \frac{1}{t} \left(\frac{1}{[A]} - \frac{1}{[A]_0} \right)$	$k = \frac{1}{t_{0.5}} \left(\frac{2}{[A]_0} - \frac{1}{[A]_0} \right)$	$t_{1/2} = \frac{1}{k[A]_0}$

2. Reaction rate

Half-live ($t_{1/2}$)?

Example:

Cho phản ứng phân hủy Nito là phản ứng bậc 1, biết thời gian bán phân hủy của Nito là 4 ngày.

- Xác định hằng số tốc độ phản ứng k
- Phân tích một mẫu nước chứa Nito thu được kết quả hàm lượng Nito còn lại là 0.8%. Hỏi thời gian mẫu nước tồn tại là bao nhiêu?

2. Reaction rate

Determine reaction order and rate constant

Example: a 1st order reaction is 60% complete in half hour.
What is the rate constant?

2. Reaction rate

Determine reaction order and rate constant from experiment data

What if we have no ideal of the order and rate constant?

Example: $\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$

Time, min	0	10	20	30	40	50
$[\text{H}_2\text{O}_2]$, M	0.032	0.023	0.018	0.013	0.0099	0.0071

For 1st order reaction: $kt = \ln[A]_0 - \ln[A] \rightarrow \ln[A] = \ln[A]_0 - kt$

For 2nd order reaction: $kt = \frac{1}{[A]} - \frac{1}{[A]_0} \rightarrow 1/[A] = 1/[A]_0 + kt$

$$y = ax + b$$

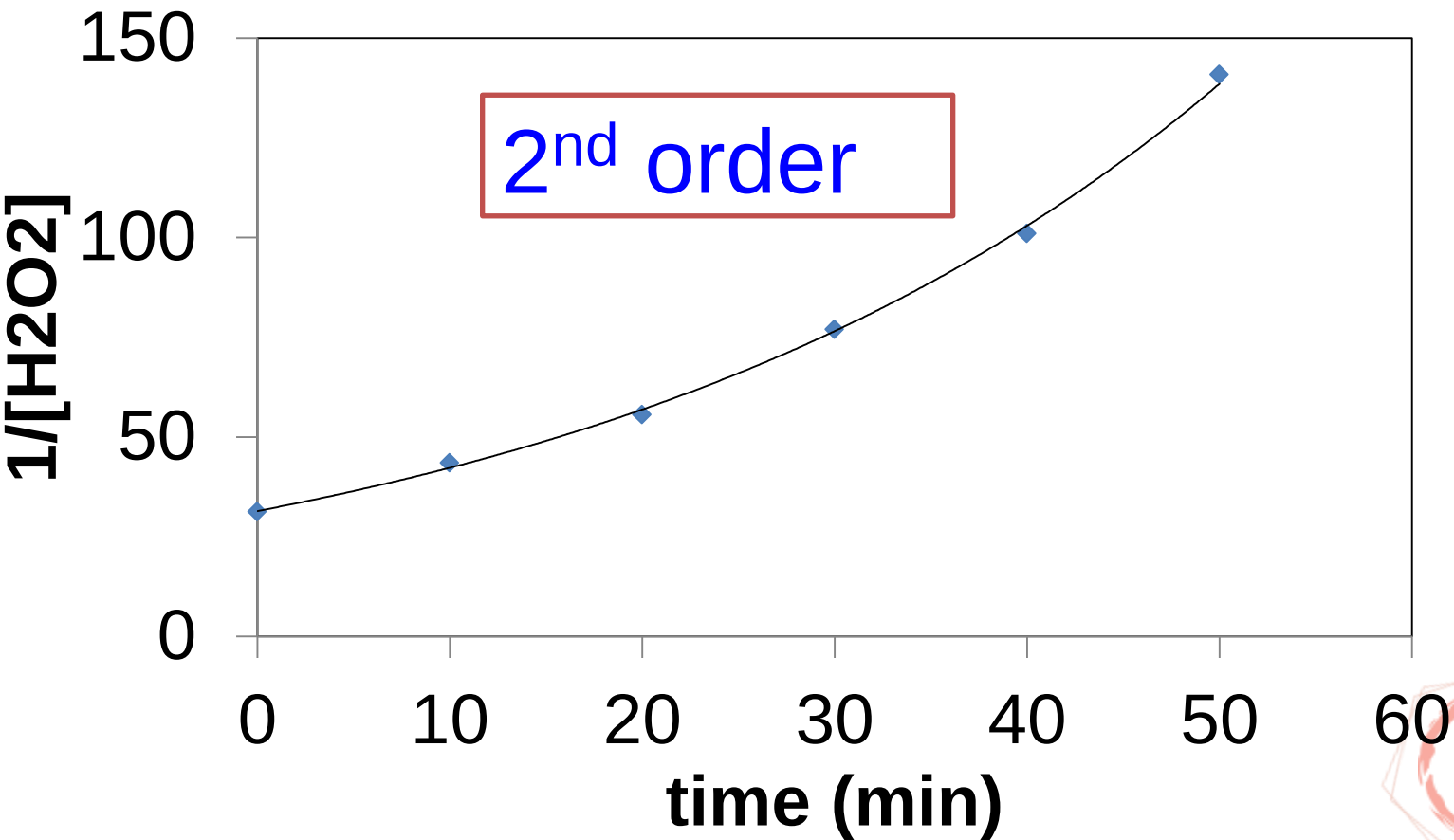
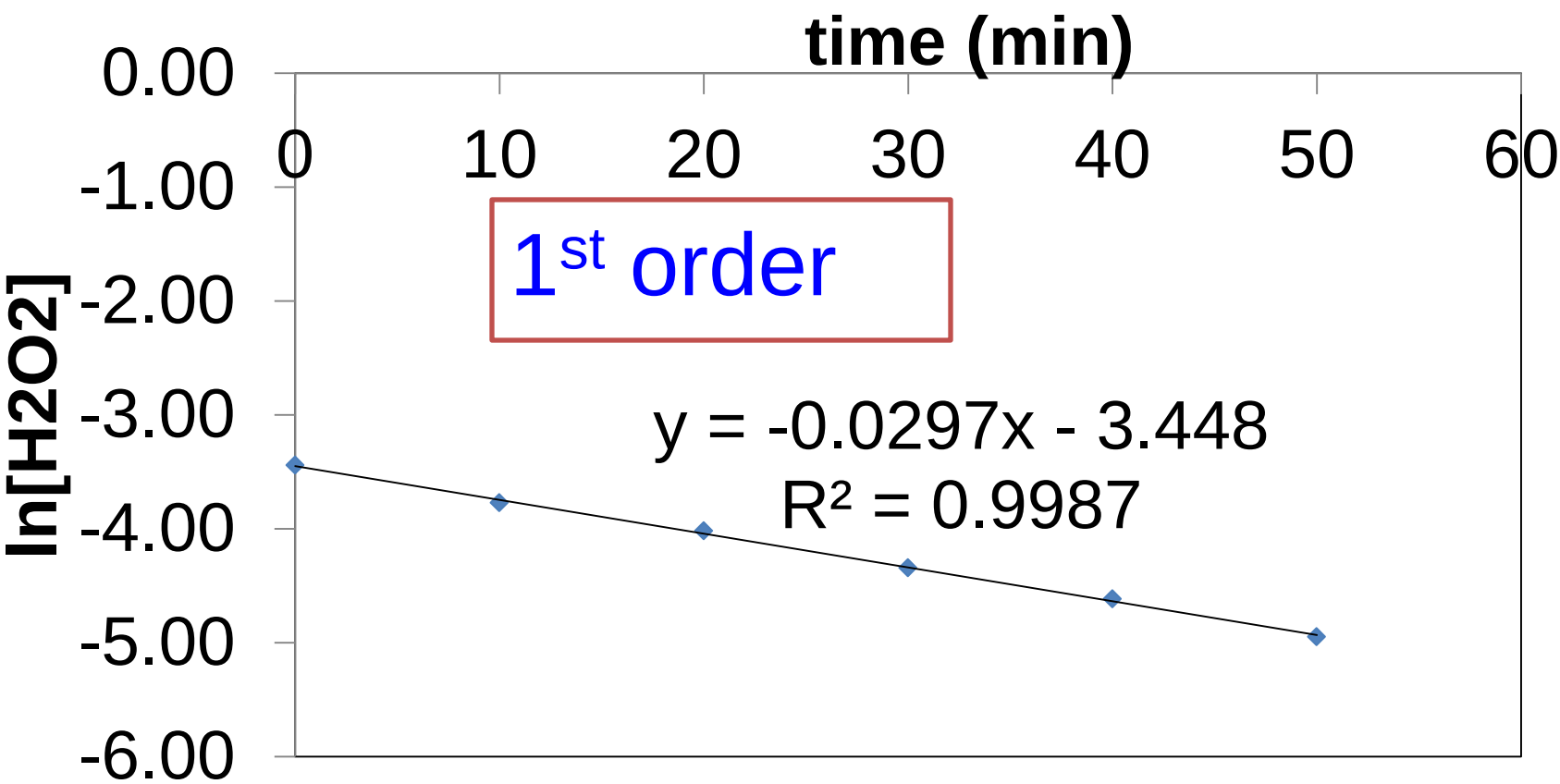
1st order: Plot $\ln[A]$ vs t ($y = \ln[A]$, $x = t$)

1st order: Plot $1/[A]$ vs t ($y = 1/[A]$, $x = t$)

Determine reaction order and rate constant from experiment data

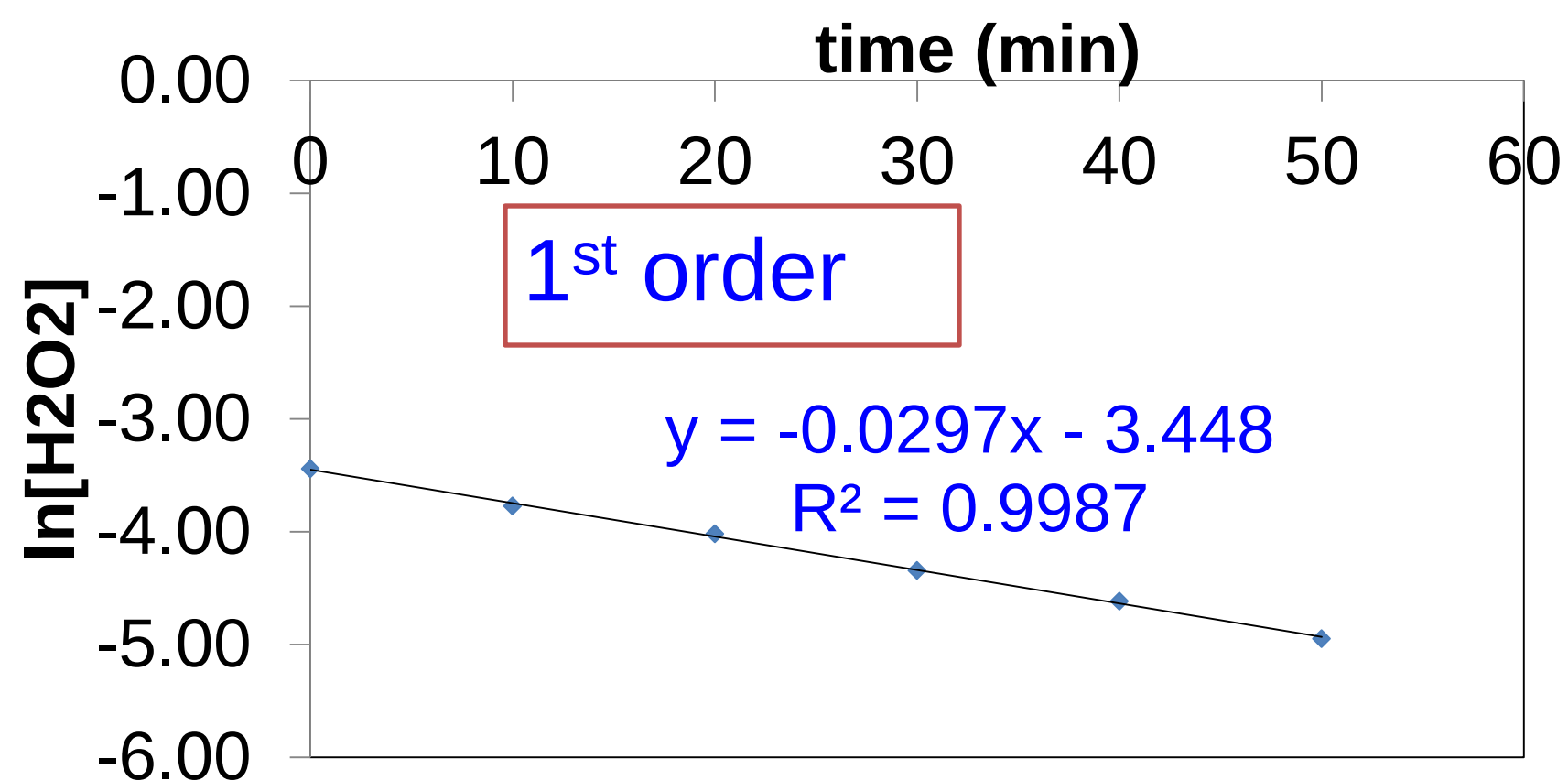
2. Reaction rate

time, min	[H ₂ O ₂], M	ln[H ₂ O ₂]	1/[H ₂ O ₂]
0	0.032	-3.44	31.25
10	0.023	-3.77	43.48
20	0.018	-4.02	55.56
30	0.013	-4.34	76.92
40	0.0099	-4.62	101.01
50	0.0071	-4.95	140.85



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Determine reaction order and rate constant from experiment data

→ This is the **1st order reaction**

→ The rate constant can be determined from the slope of the plot of ln[H₂O₂] vs t.

$$\text{Slope} = \frac{\Delta \ln[A]}{\Delta t} = -k = \frac{[-4.02 - (-3.44)]}{20 - 0} = -0.029/\text{min}$$

$$k = \frac{+0.029}{\text{min}} \times \frac{1\text{min}}{60\text{s}} = 4.83 \times 10^{-4}/\text{s}$$

3. Factors influencing on reaction

Nồng độ

- $aA + bB = cC + dD$
- $\text{Rate} = k[A]^a[B]^b \rightarrow [A], [B] \text{ tăng} \rightarrow \text{tăng tốc độ}$

Nhiệt độ

- Arrhenius' law
- $k = Ae^{-E_a/RT}$

Xúc tác

- Làm giảm giá trị $E_a \rightarrow \text{tăng } k \rightarrow \text{tăng rate}$

3. Factors influencing on reaction

Increase in temperature leads to increase in reaction rate via increase in rate constant, k

Temperature

Arrhenius' law:

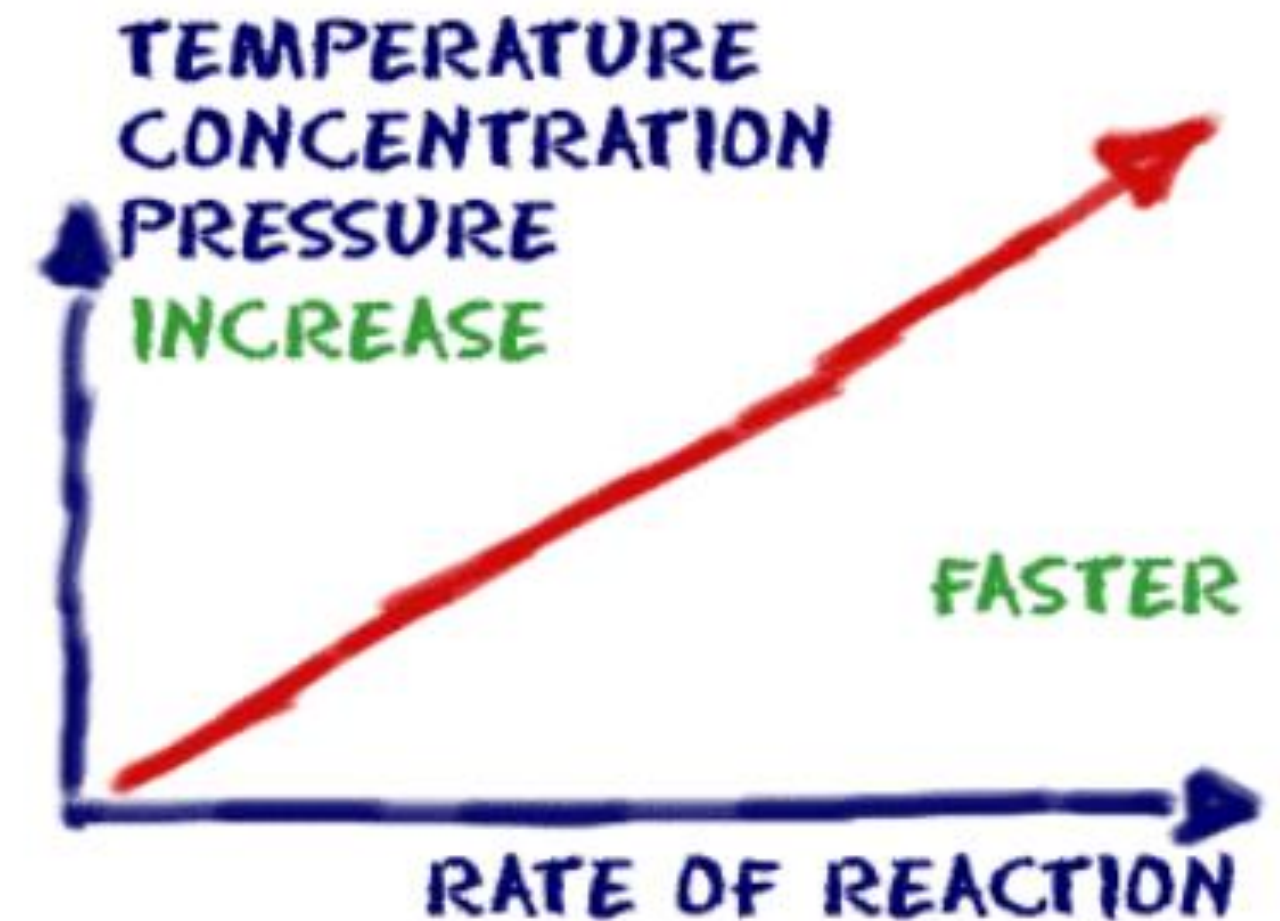
$$k = Ae^{-E_a/RT}$$

A = the pre-exponential factor, or frequency factor, and is usually treated as a constant that is independent of temperature for a particular reaction,

E_a = the activation energy and is also treated as a constant for a particular reaction

R = the ideal gas constant

T = the temperature in °K



3. Factors influencing on reaction

Arrhenius' law:

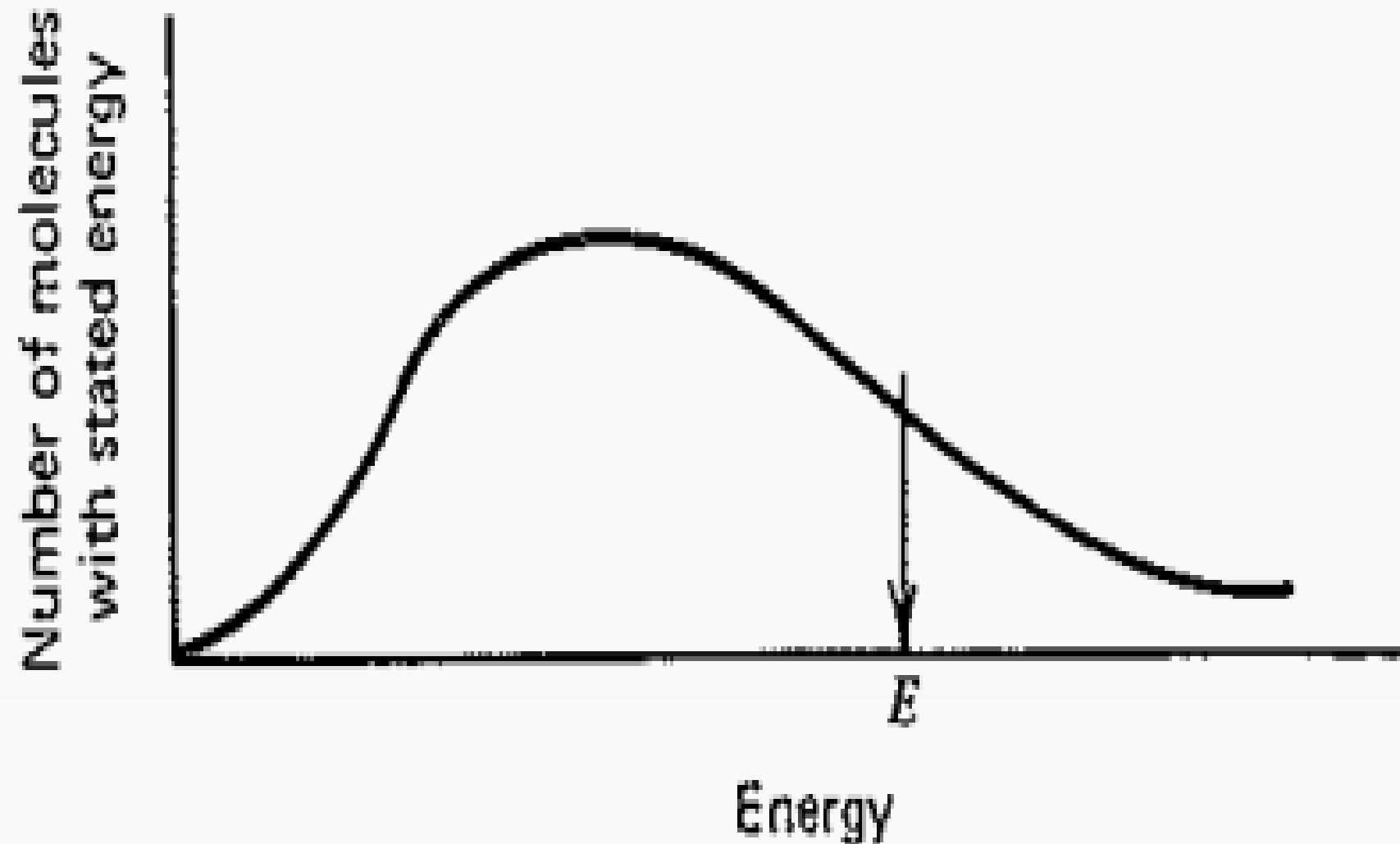
$$k = Ae^{-E_a/RT} \quad (**)$$

Temperature

$$N = N_0 D \exp\left(\frac{-E}{RT}\right)$$

$$T \uparrow \rightarrow N \uparrow$$

Temperature affects on number of molecules with energy larger than that given at a certain temperature.



3. Factors influencing on reaction

Phản ứng phân hủy chất A là $k = 5.10^{11}.e^{-2900/T}$
Năng lượng hoạt hóa của phản ứng: $E_a = ?$

Temperature

$$R=8.34 \text{ J/kmol.K}$$

$$\frac{-E_a}{R} = -29000$$

3. Factors influencing on reaction

Arrhenius' law:

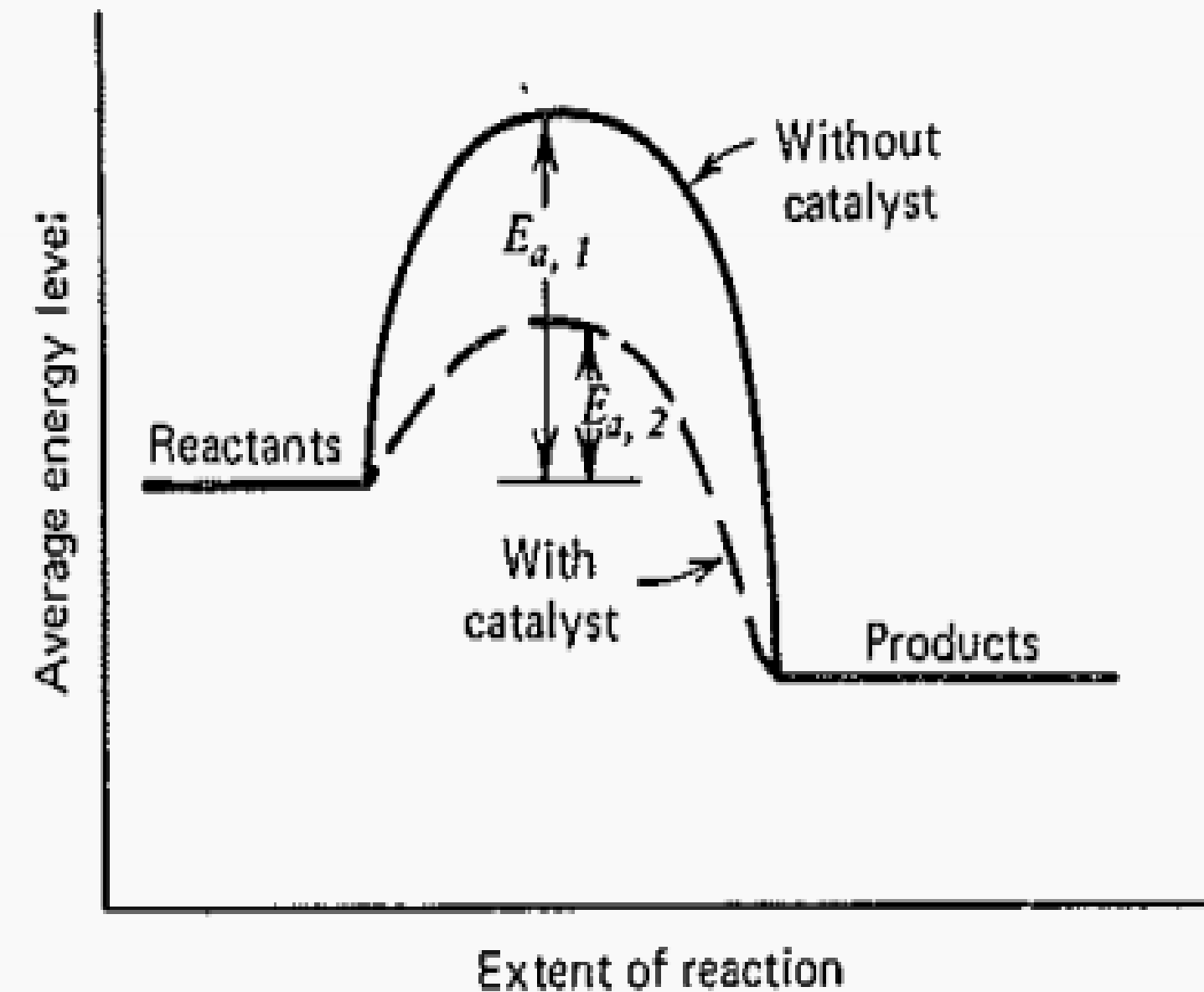
$$k = Ae^{-E_a/RT} \quad (**)$$

Giảm năng lượng hoạt hóa E_a

Catalyst

Chất xúc tác là chất:

- Làm tăng tốc độ phản ứng hoặc gây ra phản ứng về mặt nguyên tắc có thể xảy ra
- Không mất đi hay thay đổi tính chất sau phản ứng
- Có tính chất xúc tác chọn lọc
- Dùng ít hơn chất phản ứng



3. Factors influencing on reaction

Arrhenius' law:

$$k = Ae^{-E_a/RT} \quad (**)$$

Nếu tốc độ phản ứng tăng lên 2 lần (25°C) thì E_a giảm được bao nhiêu?

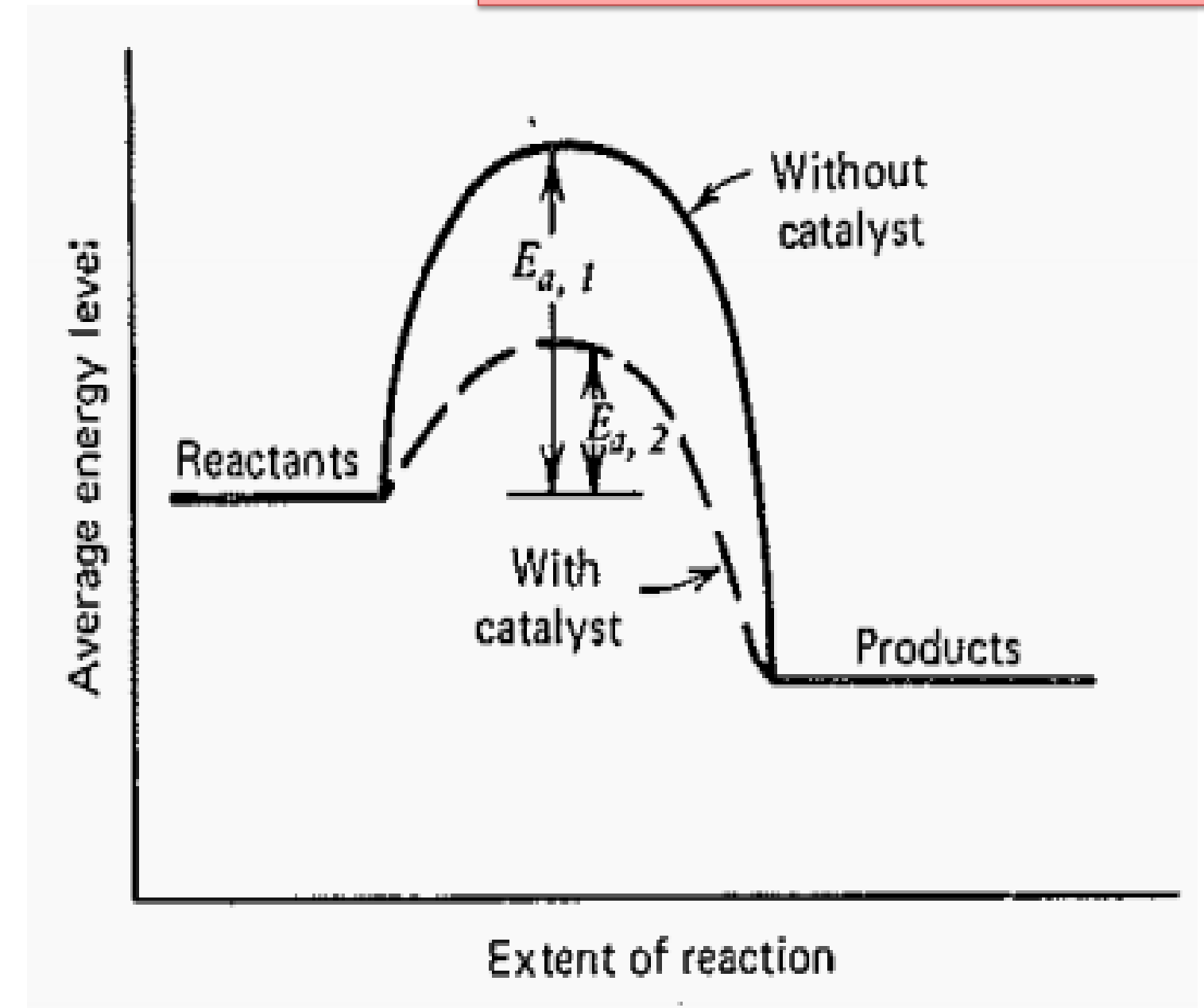
$$\ln k_1 = \ln A - \frac{E_{a,1}}{RT}$$

$$\ln k_2 = \ln (2k_1) = \ln A - \frac{E_{a,2}}{RT}$$

$$\ln k_1 - \ln 2k_1 = \frac{E_{a,2}}{RT} - \frac{E_{a,1}}{RT}$$

$$RT \ln \left(\frac{1}{2}\right) = E_{a,2} - E_{a,1}$$

Calalyst



3. Factors influencing on reaction

Calalyst

Một phản ứng có $E_a = 167.2\text{J}$ ở 500K , khi cho thêm xúc tác vào, năng lượng hoạt hóa giảm xuống còn 125.4J . Tốc độ phản ứng tăng lên bao nhiêu lần?

3. Empirical rate law

Trong thực tế, các phản ứng xảy ra phức tạp hơn nhiều so với lý thuyết, nhưng nó tuân theo các quy luật nhất định.

Xem xét việc phân hủy các chất hữu cơ trong quá trình phân tích BOD (lượng oxi cần thiết để oxi hóa các chất hữu cơ bằng VSV) tuân theo phản ứng bậc 1. $BOD_n = DO_n - DO_0$

$$\frac{dL}{dt} = -kL$$

Tích phân: $L = L_0 e^{-kt}$

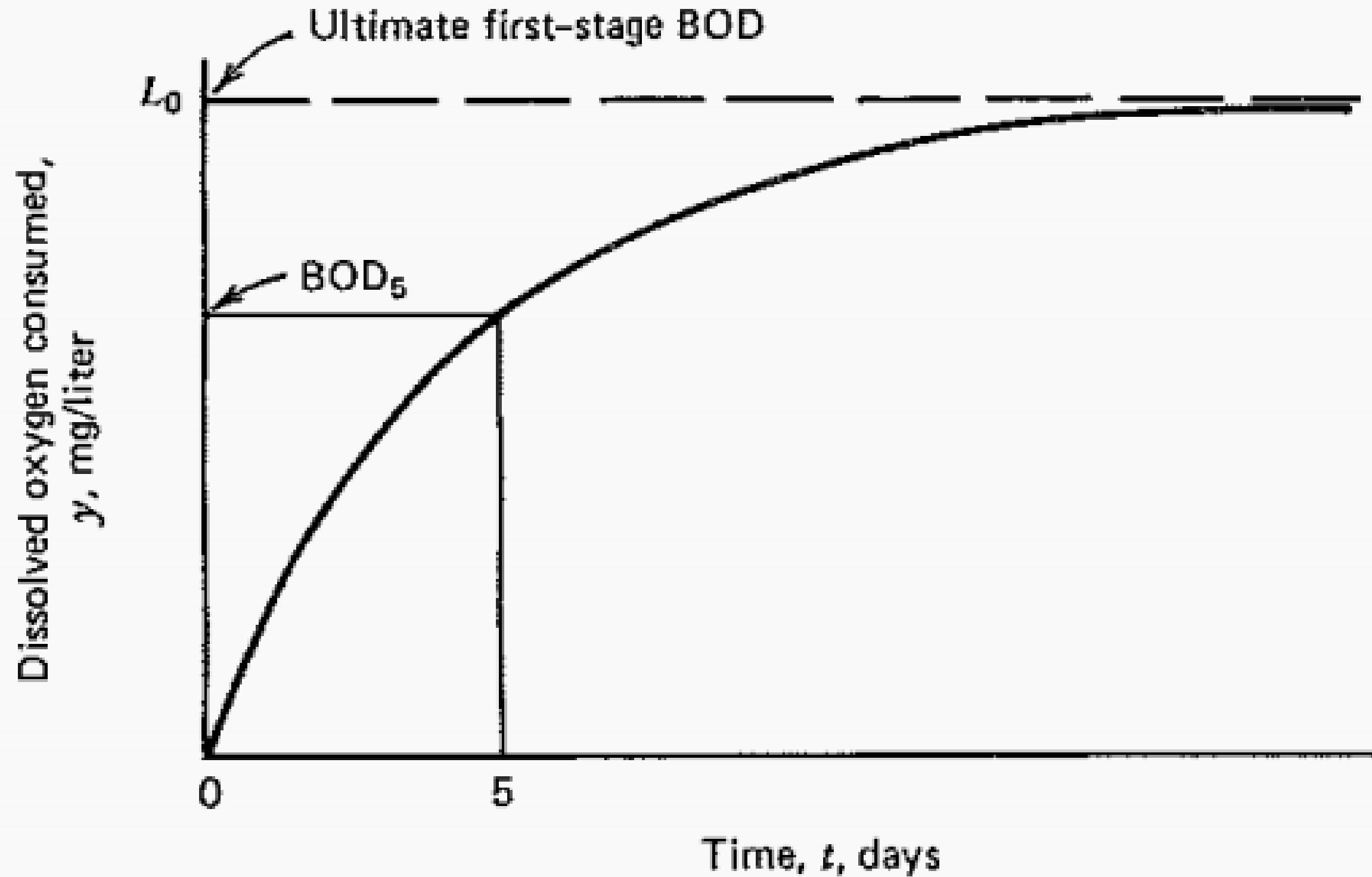
L_0 : nồng độ oxi ban đầu

$$L_0 - L = y = L_0 - L_0 e^{-kt}$$

$$y = L_0 (1 - e^{-kt})$$

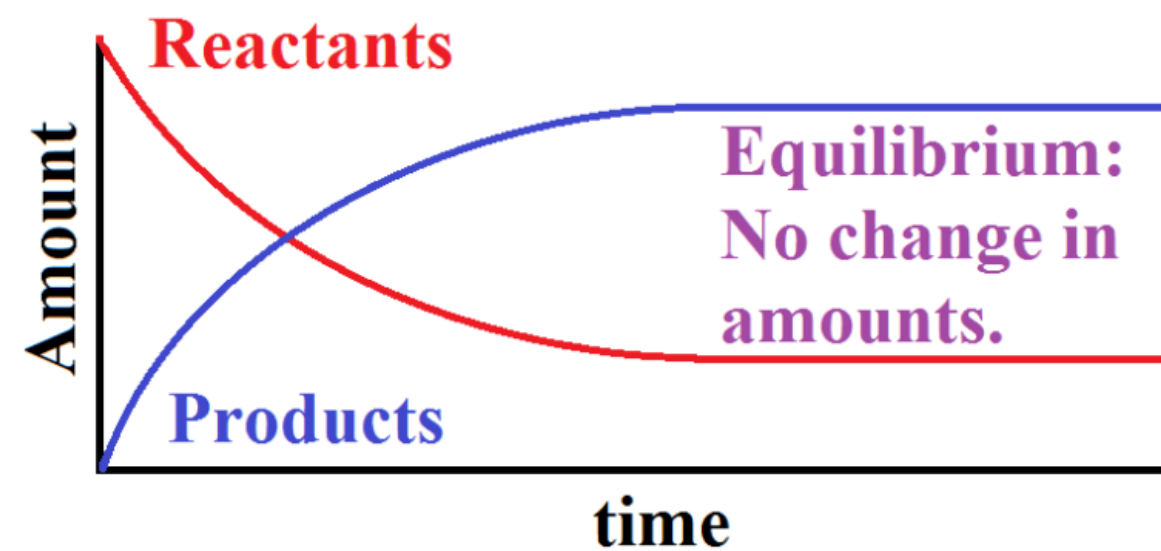
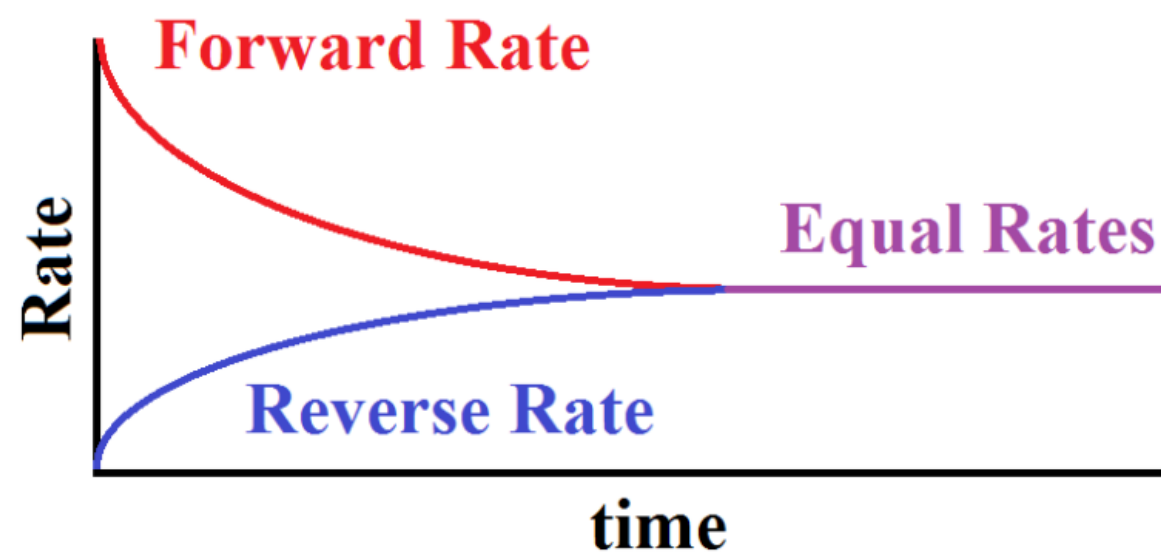
3. Empirical rate law

$$y = L_0 (1 - e^{-kt})$$



Chapter 1

CHEMICAL EQUILIBRIUM (CÂN BẰNG HÓA HỌC)



Lecturer: Nguyen Duy Dat
Email: datnd@hcmute.edu.vn

3.1 Introduction

❖ This chapter will answer:

1. Will this reaction go?



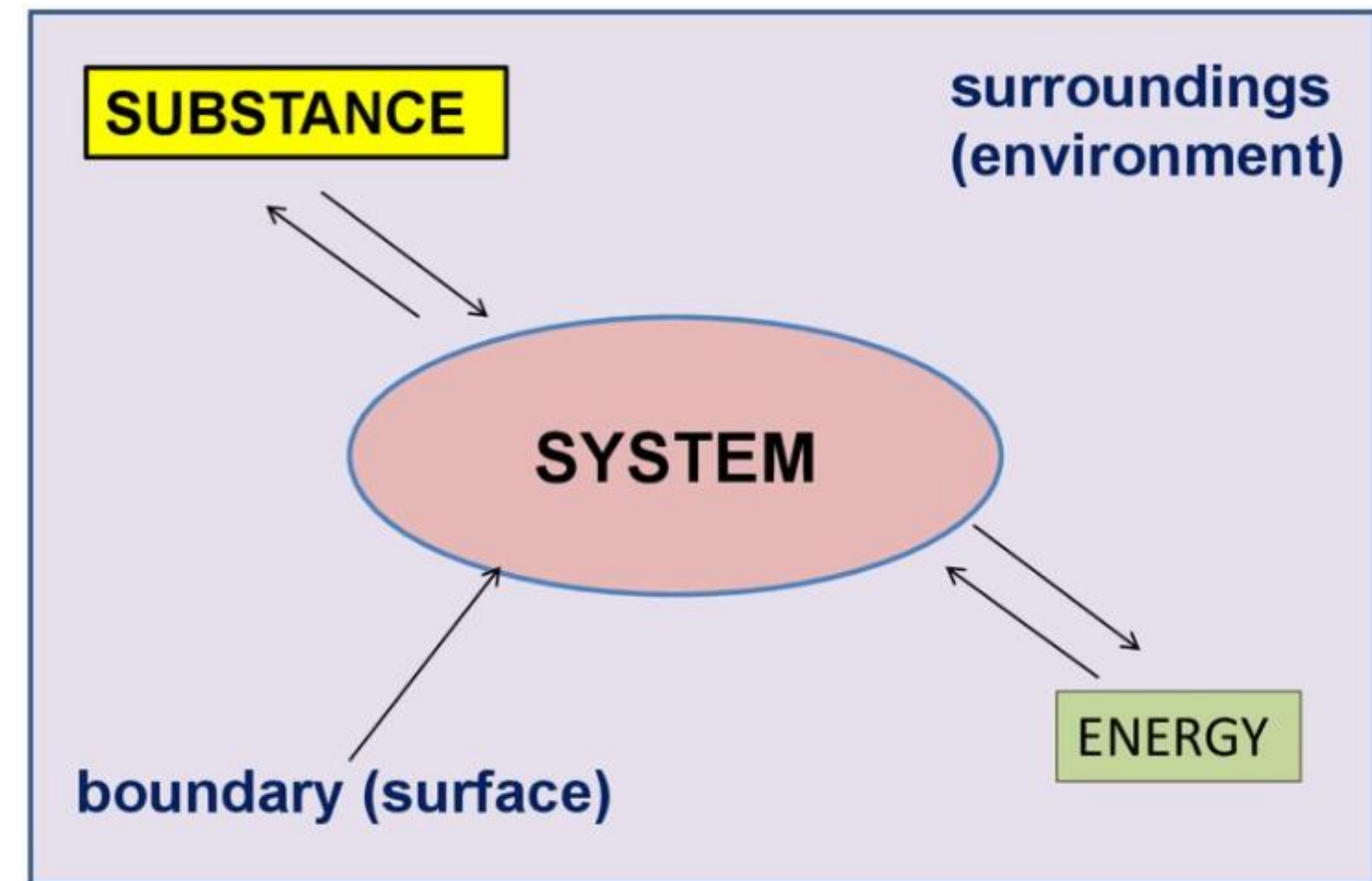
2. How far can it proceed?

→ Important questions to pose before setting the investigation of any chemical or biochemical system.

3.1 Introduction

❖ Thermodynamics (nhiệt động lực học)

- Thermodynamics will be used as techniques or a map in solving problems of chemical equilibrium.
- Chemical thermodynamics is a science of transformations of energy, amount of heat gained or released from a system, a spontaneity of a reaction, Gibbs free energy function, relationship between Gibbs Free Energy and chemical equilibrium.



3.1 Introduction

❖ Reversible and irreversible reactions

- **Reversible:** at equilibrium, the forward and backward reactions have the same rates and the concentration of each species is non-zero.



- **Irreversible:** there is no equilibrium for irreversible reaction.



3.1 Introduction

❖ Chemical speciation

- Chemical speciation comprises the **distribution of chemical compounds over the different forms** in which they are present in different phases of soil-water system.
- **Total positive charges = total negative charges**

3.1 Introduction

❖ Chemical speciation

- Compounds may be present in the form of various species, such as:

➤ in the water phase:

- Free ions: H^+ , Cd^{2+} , OH^- , PO_4^{3-} , Al^{3+}
- Complexes (ion pairs): CdCl^+ , AlOH^{2+} , H_2PO_4^- , chelates (EDTA)
- Metal-DOC complexes: Cu-DOC
(DOC = Dissolved organic carbon)

➤ in the solid phase:

- Minerals or solids: CdS(s) , $\text{CdCO}_3\text{(s)}$, $\text{AlPO}_4 \cdot 2\text{H}_2\text{O(s)}$, $\text{Fe(OH)}_3\text{(s)}$
- Organic liquids: Benzen, toluene, PAHs

➤ in the gas phase:

- $\text{NH}_3\text{(g)}$, $\text{CO}_2\text{(g)}$, $\text{H}_2\text{S(g)}$, HCN(g)

3.1 Introduction

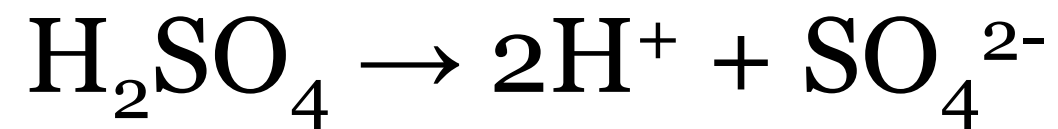
❖ Chemical speciation

❖ in a closed system: no gas, no solid



Consider
reactions/species,
charge balance?

➤ Reactions: (5)



Irreversible reaction:
axit mạnh và baso mạnh

Reversible reaction

3.1 Introduction

0.01 M H_2SO_4
 10^{-3} M Na_2CO_3

❖ Chemical speciation

➤ Species: (7) H^+ , OH^- , SO_4^{2-} , Na^+ , CO_3^{2-} , HCO_3^- , H_2CO_3^*

We have totally 7 ions, how to know concentration of 7 ions?

We need 7 equations?

(1) $\text{SO}_4^{2-} = 0.01\text{M}$

(2) $\text{Na}^+ = ?$

(3) $\text{CO}_2 \text{ total} = \text{CO}_3^{2-} + \text{HCO}_3^- + \text{H}_2\text{CO}_3^* = 10^{-3} \text{ M}$

Charge balance: $\sum \text{positive charge} = \sum \text{negative charge}$

(4) $[\text{H}^+] + [\text{Na}^+] = [\text{OH}^-] + 2[\text{SO}_4^{2-}] + [\text{HCO}_3^-] + [\text{CO}_3^{2-}]$

(5) $k_1 = \frac{[\text{H}^+][\text{HCO}_3^-]}{\text{H}_2\text{CO}_3^*}$ (6) $k_2 = \frac{[\text{H}^+][\text{CO}_3^{2-}]}{[\text{HCO}_3^-]}$ (7) $k_w = [\text{H}^+][\text{OH}^-]$

3.2 The dynamic nature of chemical equilibrium

Consider elementary, reversible reaction occurs at constant temperature:

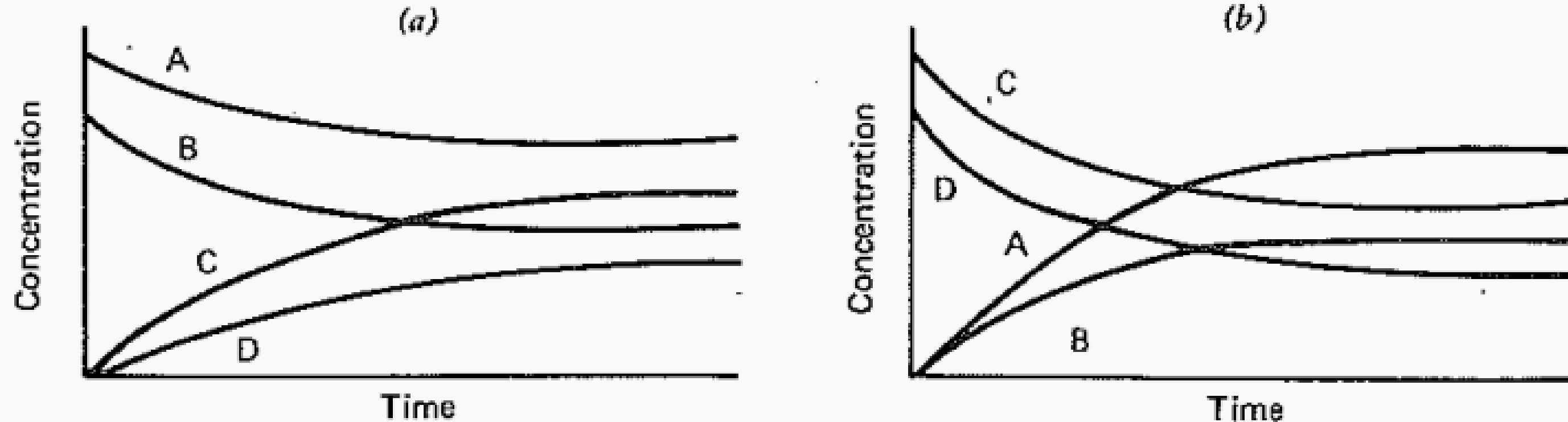
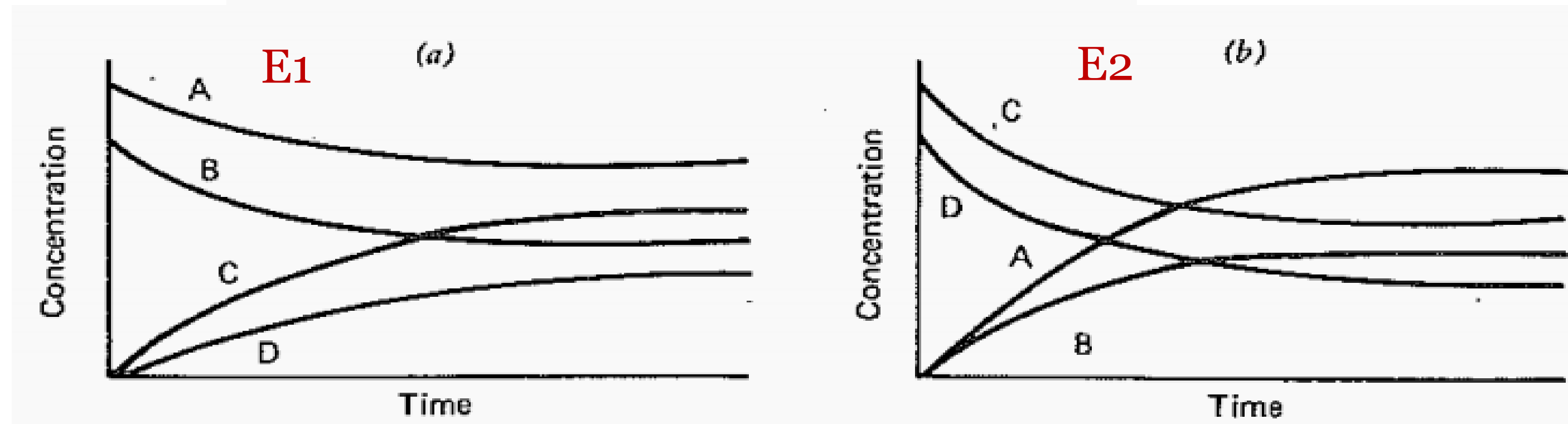


Fig. 3-1. Course of reaction between A, B, C, and D. (a) Initially only A and B are present; (b) initially only C and D are present.

3.2 The dynamic nature of chemical equilibrium



At equilibrium, concentrations of A, B, C, D = constant = nồng độ lúc cân bằng.
In E1 and E2, we will find:

$$\frac{[C]^c [D]^d}{[A]^a [B]^b} = \text{const} = K \text{ (equilibrium constant)}$$

Concentration is M, however,
other units can be used.

Equilibrium state can be approached
from both direction.

3.2 The dynamic nature of chemical equilibrium

Equilibrium condition is a dynamic process, not a static situation. The forward and reverse reactions proceed at such a constant rate of K .

$$\frac{[C]^c[D]^d}{[A]^a[B]^b} = \text{const} = K \text{ (equilibrium constant)}$$

In other words, a chemical reaction is at equilibrium if its forward rate, v_f , is equal to its reverse rate, v_r .

$$\begin{aligned} V_f &= k_1 [A]^a[B]^b \\ V_r &= k_2 [C]^c[D]^d \\ V_f &= V_r \\ \rightarrow k_1 [A]^a[B]^b &= k_2 [C]^c[D]^d \end{aligned}$$

$$\frac{[C]^c[D]^d}{[A]^a[B]^b} = \frac{k_1}{k_2} = K$$

3.3 The thermodynamic basis of chemical equilibrium

Biết được điểm cân bằng rất quan trọng, vì nó cho biết phản ứng sẽ xảy ra như thế vào khi biết nồng độ của các chất tham gia phản ứng.

Ví dụ:

- Trả lời CaCO_3 hòa tan hay kết tủa trong môi trường nước nhất định ?
- Liệu có thể dùng S^{2-} để oxi hóa NO_3^- ?

Có 2 cách để trả lời:

- Làm thí nghiệm chứng minh
- Tính toán để trả lời sử dụng các dữ liệu về cân bằng

Thermodynamic = nhiệt động lực học

3.3 The thermodynamic basis of chemical equilibrium

Biết được điểm cân bằng rất quan trọng, vì nó cho biết phản ứng sẽ xảy ra như thế vào khi biết nồng độ của các chất tham gia phản ứng.

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Có 2 cách để trả lời:

- Làm thí nghiệm chứng minh
 - Tính toán để trả lời sử dụng các dữ liệu về cân bằng
- Will this reaction go?
- Where do these reaction end up? → what species are present as equilibrium?
- Concentration of these species as equilibrium?

3.3 The thermodynamic basis of chemical equilibrium

Các phản ứng được xem xét khi đặt trong **hệ kín (close system)**. Tức là vật chất không được thêm vào hay mất đi, tuy nhiên năng lượng có thể đi qua biên của hệ. Để xét một phản ứng trong hệ kín, người ta thường xét nó trong điều kiện **nhiệt độ và áp suất không đổi**.

Biểu diễn nhiệt động lực học cho năng lượng tự do:

$$G = H - TS$$

- G: năng lượng tự do Gibbs = là năng lượng vốn có của một hệ thống, khi cần nó được dùng để thực hiện công dưới các điều kiện và áp suất nhất định. Nó là năng lượng tối đa tiềm ẩn trong hệ thống. Các chất đều có chứa năng lượng tự do. Khi xảy ra phản ứng, có sự biến đổi năng lượng tự do được ký hiệu bằng ΔG .
- H: enthalpy (Kcal) là tổng năng lượng của hợp chất.
- S: entropy (Kcal/K) = một tập hợp các biểu hiện bên trong của năng lượng, đặc trưng cho sự trật tự của hệ.
- T: nhiệt độ (K)

3.3 The thermodynamic basis of chemical equilibrium

Khi xét một phản ứng, biến thiên năng lượng tự do ΔG luôn xem xét:

$$\Delta G = \Delta H - T\Delta S$$

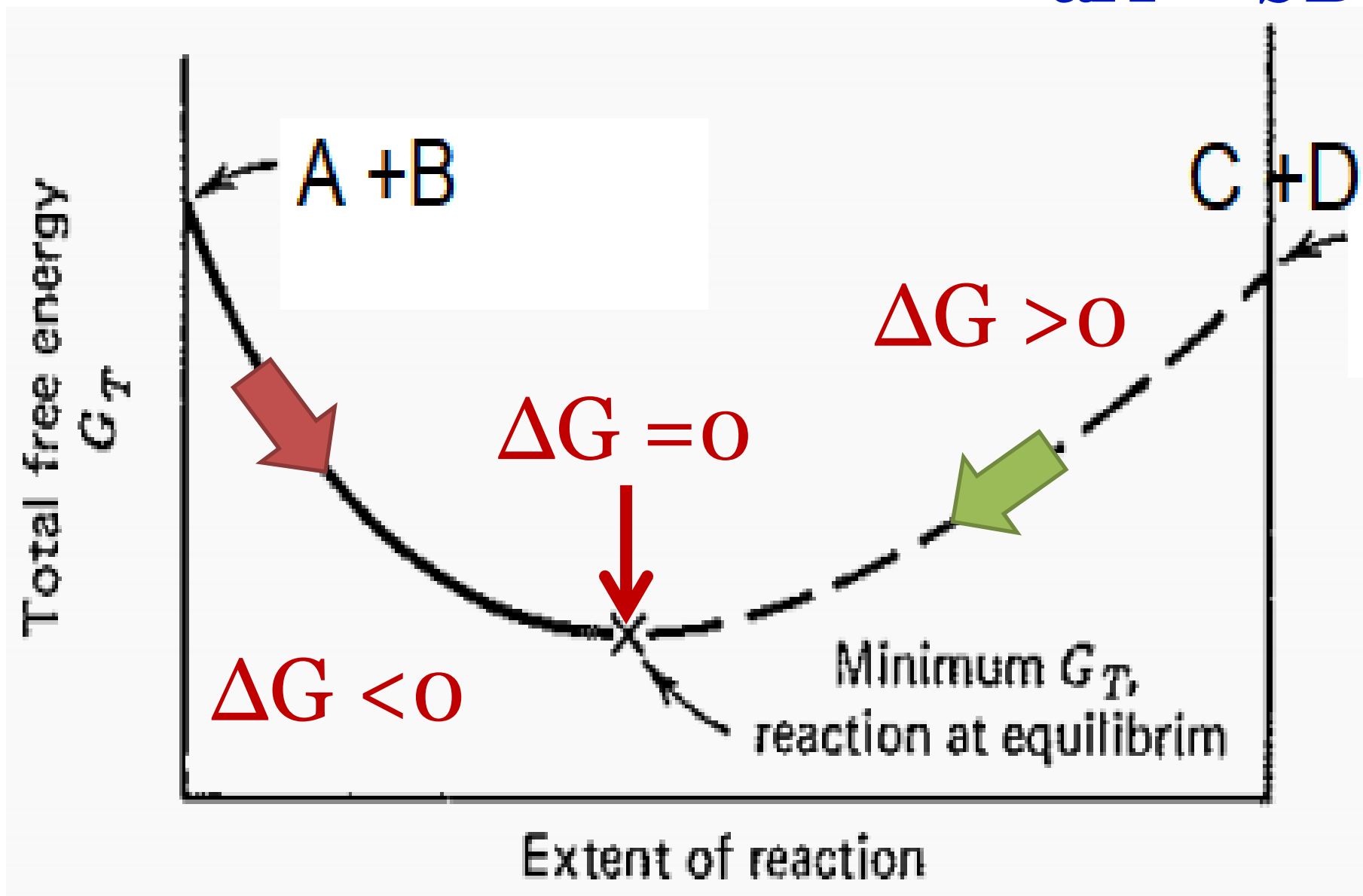
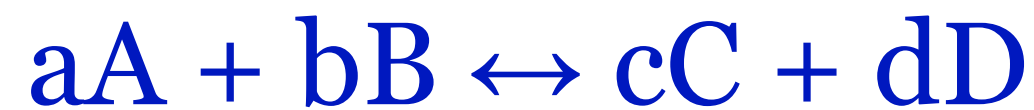
Hai định luật nhiệt động lực học:

Định luật 1: năng lượng của hệ (kín) được bảo toàn.

Định luật 2: một phản ứng tự xảy ra luôn diễn ra theo chiều giảm năng lượng tự do ($\Delta G < 0$) và tăng entropy ($\Delta S > 0$).

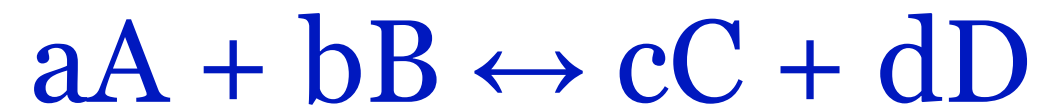
3.3 The thermodynamic basis of chemical equilibrium

Xét một phản ứng trong hệ kín, ở điều kiện nhiệt độ và áp suất không đổi, hệ đạt cân bằng khi năng lượng tự do của hệ là thấp nhất ($\Delta G = 0$).



- $G_T = n_A \bar{G}_A + n_B \bar{G}_B + n_C \bar{G}_C + n_D \bar{G}_D$
n: số mol
G: năng lượng tự do của các chất
- Phản ứng sẽ xảy ra theo chiều từ trái sang phải hay ngược lại, miễn là G_T giảm;
- Phản ứng xảy ra đến khi $G_T = G_{T \text{ min}}$, tức là đạt cân bằng ($\Delta G = 0$).

3.3 The thermodynamic basis of chemical equilibrium



Biên thiên năng lượng tự do của hệ (ΔG):

$$\Delta G = (\sum_i n_i \mu_i)_{\text{products}} - (\sum_i n_i \mu_i)_{\text{reactants}}$$

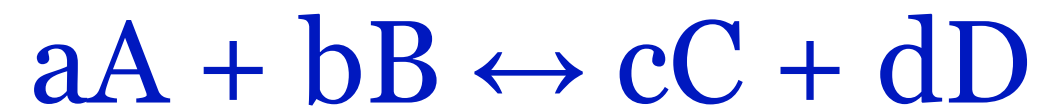
n_i : số mol (a, b, c, d)

μ_i : thế hóa của các chất (chemical potential)

- **Thế hóa** là năng lượng tự do riêng phần mol ($\frac{dG}{dn_i} = \mu_i$) ở điều kiện nhiệt độ, áp suất nhất định.

- $\Delta G < 0$: phản ứng tự phát (spontaneous rxn)
- $\Delta G = 0$: at equilibrium
- $\Delta G > 0$: phản ứng xảy ra theo chiều ngược lại.

3.3 The thermodynamic basis of chemical equilibrium



Biên thiên năng lượng tự do của hệ (ΔG): là công cụ để dự đoán phản ứng có xảy ra hay đạt cân bằng, nếu xảy ra thì xảy ra theo hướng nào.

$$\Delta G = \Delta G^\circ + RT \ln \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

ΔG° : năng lượng tự do tiêu chuẩn (standard-state free energy)/mol chất tham gia phản ứng (free energy/mol of species)

Standard state: áp suất = 1 atm, nhiệt độ: 297K

- Chất lỏng: tinh khiết, pH=7
- Chất hòa tan (trong lỏng): 1M
- Chất khí: áp suất riêng phần = 0.1MPa
- Chất rắn: tinh khiết (rắn)

Hằng số khí lý tưởng:

$$\begin{aligned} R &= 1.987 \text{ cal}/(\text{mol.K}) \\ &= 8.314 \text{ J}/(\text{mol.K}) \\ (1 \text{ cal} &= 4.187 \text{ J}) \end{aligned}$$

3.3 The thermodynamic basis of chemical equilibrium

Biến thiên năng lượng tự do tiêu chuẩn (standard): ΔG°

$$\Delta G^\circ_{\text{rxn}} = \sum \Delta \overline{G}^\circ_f \text{ products} - \sum \Delta \overline{G}^\circ_f \text{ reactants}$$

$\Delta G^\circ_{\text{rxn}}$: được tính dựa vào các $\Delta \overline{G}^\circ_f$ (năng lượng tự do riêng- energy per mol) của các chất tham gia phản ứng.

- $\Delta G^\circ < 0$: phản ứng tự phát (spontaneous rxn)
- $\Delta G^\circ > 0$: phản ứng không tự phát (non spontaneous rxn)

TABLE 3-1 Thermodynamic Constants for Species Importance in Water Chemistry.^a

Species	$\Delta\bar{H}_f^\circ$ kcal/mole	$\Delta\bar{G}_f^\circ$ kcal/mole
$\text{Ca}_{(aq)}^{2+}$	-129.77	-132.18
$\text{CaCO}_{3(s)}$, calcite	-288.45	-269.78
$\text{CaO}_{(s)}$	-151.9	-144.4
$\text{C}_{(s)}$, graphite	0	0
$\text{CO}_{2(g)}$	-94.05	-94.26
$\text{CO}_{2(aq)}$	-98.69	-92.31
$\text{CH}_{4(g)}$	-17.889	-12.140
$\text{H}_2\text{CO}_{3(aq)}^*$	-167.0	-149.00
$\text{HCO}_{3(aq)}^-$	-165.18	-140.31
$\text{CO}_{3(aq)}^{2-}$	-161.63	-126.22
CH_3COO^- , acetate	-116.84	-89.0
$\text{H}_{(aq)}^+$	0	0
$\text{H}_{2(g)}$	0	0
$\text{Fe}_{(aq)}^{2+}$	-21.0	-20.30
$\text{Fe}_{(aq)}^{3+}$	-11.4	-2.52
$\text{Fe}(\text{OH})_{3(s)}$	-197.0	-166.0
$\text{Mn}_{(aq)}^{2+}$	-53.3	-54.4
$\text{MnO}_{2(s)}$	-124.2	-111.1
$\text{Mg}_{(aq)}^{2+}$	-110.41	-108.99
$\text{Mg}(\text{OH})_{2(s)}$	-221.00	-199.27

Species	$\Delta\bar{H}_f^\circ$ kcal/mole	$\Delta\bar{G}_f^\circ$ kcal/mole
$\text{NO}_{3(aq)}^-$	-49.372	-26.43
$\text{NH}_{3(g)}$	-11.04	-3.976
$\text{NH}_{3(aq)}$	-19.32	-6.37
$\text{NH}_4(aq)^+$	-31.74	-19.00
$\text{HNO}_{3(aq)}$	-49.372	-26.41
$\text{O}_{2(aq)}$	-3.9	3.93
$\text{O}_{2(g)}$	0	0
$\text{OH}_{(aq)}^-$	-54.957	-37.595
$\text{H}_2\text{O}_{(g)}$	-57.7979	-54.6357
$\text{H}_2\text{O}_{(l)}$	-68.3174	-56.690
$\text{SO}_{4(aq)}^{2-}$	-216.90	-177.34
$\text{HS}_{(aq)}^-$	-4.22	3.01
$\text{H}_2\text{S}_{(g)}$	-4.815	-7.892
$\text{H}_2\text{S}_{(aq)}$	-9.4	-6.54

3.3 The thermodynamic basis of chemical equilibrium

Biến thiên năng lượng tự do tiêu chuẩn (standard): ΔG°

$$\Delta G^\circ_{\text{rxn}} = \sum \Delta \overline{G}^\circ_{f \text{ products}} - \sum \Delta \overline{G}^\circ_{f \text{ reactants}}$$

Vấn đề 1: dựa vào năng lượng tự do tiêu chuẩn cho phản ứng sau xét về phản ứng $\text{C (graphite)} + 3/2 \text{ O}_2 (\text{g}) + \text{Ca (s)} \rightarrow \text{CaCO}_3 (\text{s})$ (calcite)

$$\Delta \overline{G}^\circ_f = -269.78 \text{ kcal/mole}$$

- Dấu “–” thể hiện phản ứng này là phản ứng sinh năng lượng
- Để thực hiện phản ứng ngược lại, muốn chuyển (break down) 1 mol $\text{CaCO}_3 (\text{s})$ sang 1 mol C, $3/2$ mol O_2 và 1 mol Ca(s) thì cần cung cấp năng lượng + 269.78 kcal.

3.3 The thermodynamic basis of chemical equilibrium

Biến thiên năng lượng tự do tiêu chuẩn (standard): ΔG°

Vấn đề 2: xem xét phản ứng hòa tan CaCO_3 (s) (calcite) trong axit:
 $\text{CaCO}_3 \text{ (s)} + \text{H}^+ \rightarrow \text{Ca}^{2+} + \text{HCO}_3^-$

	$\Delta \bar{G}_f^\circ$
$\text{CaCO}_{3(\text{s})} \rightarrow \text{Ca} + \text{C} + \frac{3}{2}\text{O}_{2(\text{g})}$	+269.78
$\text{H}^+ \rightarrow \text{H}^+$	0
$\text{Ca} \rightarrow \text{Ca}^{2+}$	-132.18
$\text{H}^+ + \text{C} + \frac{3}{2}\text{O}_{2(\text{g})} \rightarrow \text{HCO}_3^-$	-140.31
$\text{CaCO}_{3(\text{s})} + \text{H}^+ \rightarrow \text{Ca}^{2+} + \text{HCO}_3^-;$	$\Delta G^\circ = -2.71$

- Phản ứng hòa tan CaCO_3 trong axit có $\Delta G^\circ < 0$
- Năng lượng này có thể tích bảng công thức:

$$\Delta G = \left(\sum_i n_i \mu_i \right)_{\text{products}} - \left(\sum_i n_i \mu_i \right)_{\text{reactants}}$$

$$\Delta G^\circ = \Delta \bar{G}_{f, \text{HCO}_3^-}^\circ + \Delta \bar{G}_{f, \text{Ca}^{2+}}^\circ - \Delta \bar{G}_{f, \text{H}^+}^\circ - \Delta \bar{G}_{f, \text{CaCO}_3}^\circ$$

3.3 The thermodynamic basis of chemical equilibrium

Điều kiện phản ứng không là điều kiện chuẩn (standard state)

$$\bar{G}_i = \Delta \bar{G}_{i,1}^\circ + RT \ln \{i\}$$

$\bar{G}_i$: năng lượng tự do/mol chất i

$\{i\}$: nồng độ hoạt độ của chất i (activity)

- Trong một dung dịch, $\{i\} = \gamma_i [i]$ với γ_i là hệ số hoạt độ. Với dung dịch loãng $\gamma_i \rightarrow 1$ và $\{i\} \rightarrow [i]$
- Cho chất rắn tinh khiết, $\{i\} = 1$.
- Cho chất khí, $\{i\} = \gamma_i P_i$ (P_i = áp suất riêng phần, P tổng giảm $\gamma_i \rightarrow 1$, nếu phản ứng xảy ra ở trong khí quyển, $\{i\} = P_i$)
- Cho dung dịch hỗn hợp, $\{i\} = X_i$ (nồng độ phần mol)

3.3 The thermodynamic basis of chemical equilibrium

Điều kiện phản ứng không là điều kiện chuẩn (standard state)

$$\Delta G = \Delta \bar{G}_{f,\text{HCO}_3^-}^\circ + RT \ln \{ \text{HCO}_3^- \} + \Delta \bar{G}_{f,\text{Ca}^{2+}}^\circ + RT \ln \{ \text{Ca}^{2+} \} \\ - \Delta \bar{G}_{f,\text{H}^+}^\circ - RT \ln \{ \text{H}^+ \} - \Delta \bar{G}_{f,\text{CaCO}_3(\text{s})}^\circ - RT \ln \{ \text{CaCO}_3(\text{s}) \}$$

Collecting terms, we obtain

$$\Delta G = \Delta \bar{G}_{f,\text{HCO}_3^-}^\circ + \Delta \bar{G}_{f,\text{Ca}^{2+}}^\circ - \Delta \bar{G}_{f,\text{H}^+}^\circ - \Delta \bar{G}_{f,\text{CaCO}_3(\text{s})}^\circ \\ + RT \ln \{ \text{HCO}_3^- \} + RT \ln \{ \text{Ca}^{2+} \} - RT \ln \{ \text{H}^+ \} - RT \ln \{ \text{CaCO}_3(\text{s}) \}$$

Since, from Eq. 3-13,

$$\Delta G^\circ = \Delta \bar{G}_{f,\text{HCO}_3^-}^\circ + \Delta \bar{G}_{f,\text{Ca}^{2+}}^\circ - \Delta \bar{G}_{f,\text{H}^+}^\circ - \Delta \bar{G}_{f,\text{CaCO}_3(\text{s})}^\circ$$

we can state

$$\Delta G = \Delta G^\circ + RT \ln \frac{\{ \text{HCO}_3^- \} \{ \text{Ca}^{2+} \}}{\{ \text{H}^+ \} \{ \text{CaCO}_3(\text{s}) \}} \quad (3-16)$$

3.3 The thermodynamic basis of chemical equilibrium

Trong dung dịch loãng, với độ hoạt động chất rắn tinh khiết là 1, và giả sử hệ số hoạt độ các chất là 1:

$$\Delta G = \Delta G^\circ + RT \ln \frac{[\text{HCO}_3^-][\text{Ca}^{2+}]}{[\text{H}^+]}$$

$$Q = \frac{\{\text{HCO}_3^-\} \{\text{Ca}^{2+}\}}{\{\text{H}^+\} \{\text{CaCO}_{3(s)}\}} = \frac{[\text{HCO}_3^-][\text{Ca}^{2+}]}{[\text{H}^+]}$$

Khi phản ứng chưa đặt cân bằng: $Q \neq K$ ($Q = K$ @ equilibrium)

3.3 The thermodynamic basis of chemical equilibrium

Khái quát:



$$Q = \frac{\{C\}^c \{D\}^d}{\{A\}^a \{B\}^b}$$

$$Q = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

$$\Delta G = \Delta G^\circ + RT \ln Q$$

Khi phản ứng đạt cân bằng: $\Delta G = 0$

$$0 = \Delta G^\circ + RT \ln K$$

$$K = \exp(-\Delta G^\circ/RT)$$

$$\Delta G^\circ = -RT \ln K$$

$$\Delta G = -RT \ln K + RT \ln Q = RT \ln \frac{Q}{K}$$

→ tỷ số Q/K sẽ quyết định phản ứng có xảy ra hay không.

3.3 The thermodynamic basis of chemical equilibrium

1. If $Q/K > 1$, then ΔG is positive and the reaction is impossible as written.
2. If $Q/K = 1$, then $\Delta G = 0$ and the system is at equilibrium.
3. If $Q/K < 1$, then ΔG is negative and the reaction is spontaneous as written.

3.3 The thermodynamic basis of chemical equilibrium



Example 3.1:

1. Xác định hằng số cân bằng của phản ứng phân ly của nước thành H^+ và OH^- ở 25°C
2. Khi $[\text{H}^+] = 10^{-6}\text{M}$ và $[\text{OH}^-] = 5 \times 10^{-8}\text{M}$, phản ứng trên sẽ xảy ra theo chiều nào?

3.3 The thermodynamic basis of chemical equilibrium

Example 3.2:

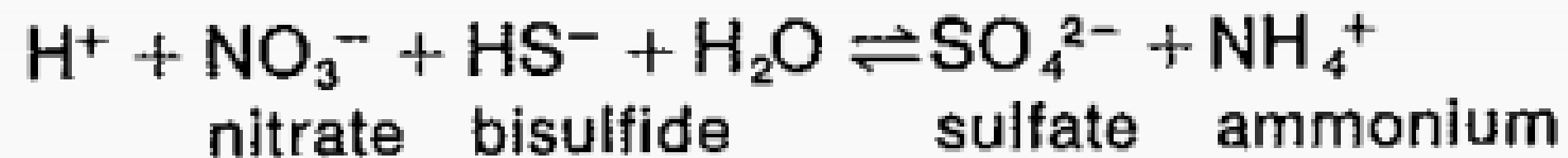
Tìm hằng số cân bằng phản ứng Fe^{2+} bị oxi hóa bởi O_2 trong không khí ở 25°C . Biết O_2 hòa tan bão hòa trong nước. Giả sử hệ số hoạt độ các ion là 1.



3.3 The thermodynamic basis of chemical equilibrium

Example 3.3:

Is it possible at 25°C to oxidize sulfide in natural waters with nitrate? Typical concentrations that exist are $10^{-4} M$ of reacting species and pH 8 ($[H^+] = 10^{-8} M$). Assume that ionic strength effects are negligible (these effects will be discussed later in this chapter). The reaction is



3.3 The thermodynamic basis of chemical equilibrium

Note: ΔG° có tính cộng gộp. Ví dụ trong phản ứng sau, sản phẩm của phản ứng này lại tham gia phản ứng khác. Khi đó, ΔG° của phản ứng tổng có thể được tính bằng tổng ΔG° của 2 phản ứng.

	ΔG° (kcal)
$\text{CaCO}_{3(s)} \rightleftharpoons \text{Ca}^{2+} + \text{CO}_3^{2-}$	+11.38
$\text{CO}_3^{2-} + \text{H}^+ \rightleftharpoons \text{HCO}_3^-$	-14.09
	-2.71 kcal

Nếu ΔG của phản ứng tổng < 0 , thì phản ứng đó vẫn xảy ra, và không cần phản ứng thành phần có $\Delta G < 0$.

3.4 Enthalpy and the temperature dependence of K

Sự biến thiên enthalpy (ΔH) của phản ứng chính là lượng nhiệt mà sinh ra hay hấp thu vào của phản ứng.

$$\Delta H^\circ = \Delta G^\circ + T \Delta S^\circ$$

$$\Delta H^\circ = \left(\sum_i \nu_i \bar{H}_i^\circ \right)_{\text{products}} - \left(\sum_i \nu_i \bar{H}_i^\circ \right)_{\text{reactants}}$$

ν_i = the stoichiometric coefficient

$\bar{H}_i^\circ$ = the enthalpy of species i in kcal/mole at standard conditions of 25°C and 1 atm pressure

For the general reaction, $aA + bB \rightleftharpoons cC + dD$, ΔH° is equal to the amount of heat taken up or released when a moles of A and b moles of B , each in their standard states, are completely converted to c moles of C and d moles of D , each in their standard states.

3.4 Enthalpy and the temperature dependence of K

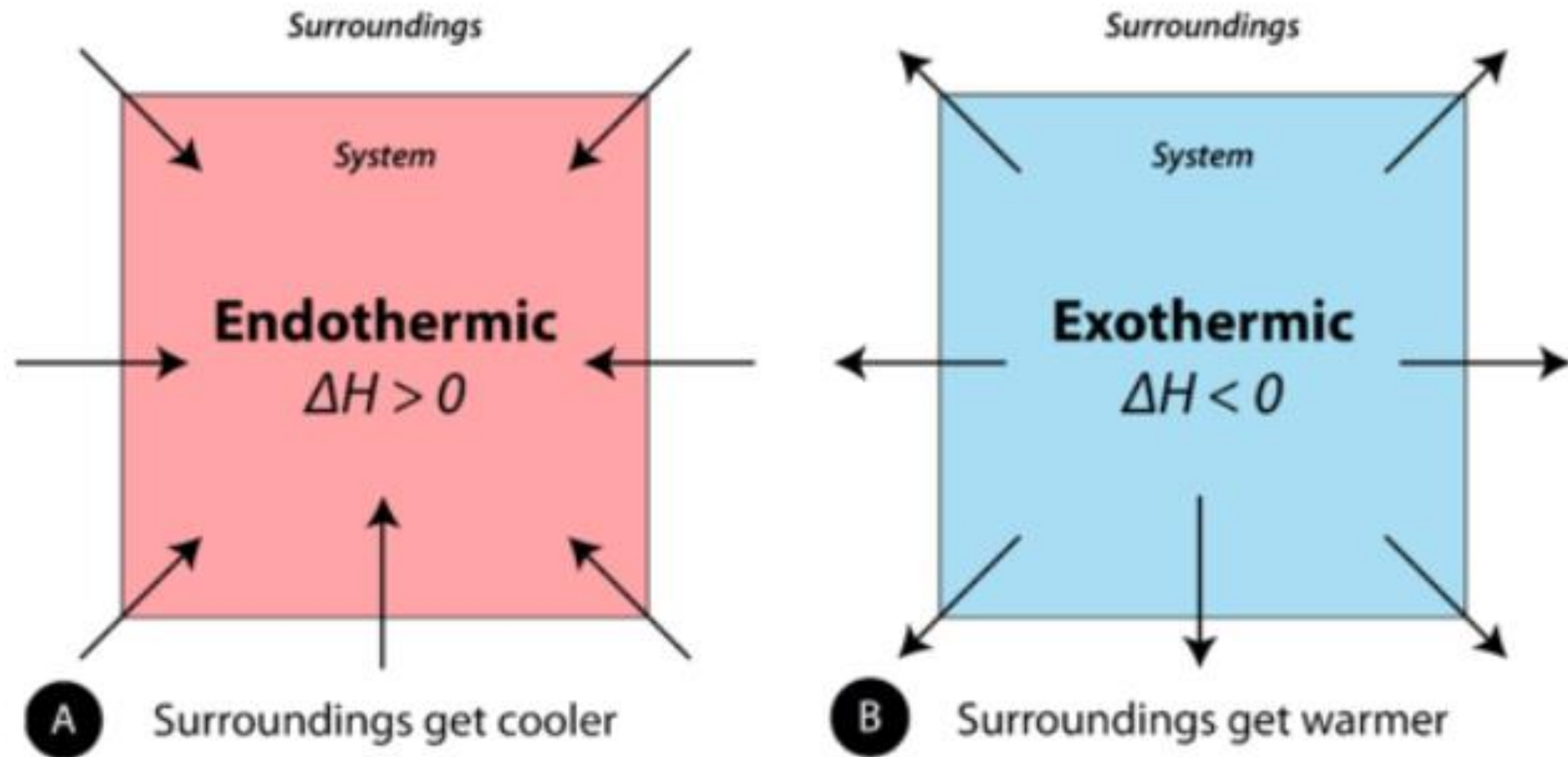


Figure 8.7.1: (A) Endothermic reaction. (B) Exothermic reaction.

3.4 Enthalpy and the temperature dependence of K

$$\Delta G^\circ_1 = -RT_1 \ln K_1 = \Delta H^\circ - T_1 \Delta S^\circ$$

$$\Delta G^\circ_2 = -RT_2 \ln K_2 = \Delta H^\circ - T_2 \Delta S^\circ$$

Giả sử ΔH° và ΔS° không phụ thuộc nhiệt độ.

$$\ln K_1 = -\frac{\Delta H^\circ}{RT_1} + \frac{\Delta S^\circ}{R} \quad (1)$$

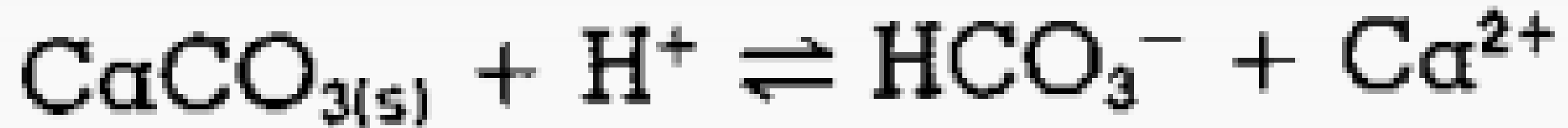
$$\ln K_2 = -\frac{\Delta H^\circ}{RT_2} + \frac{\Delta S^\circ}{R} \quad (2)$$

$$(1) - (2) = \ln K_2/K_1 = \frac{\Delta H^\circ}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

3.4 Enthalpy and the temperature dependence of K

Example 3.4:

A municipal water supply enters a residence at 15°C and is heated to 60°C in the home water heater. If the water is just saturated with respect to $\text{CaCO}_{3(s)}$ at 25°C, what will be the condition of the water (i.e., oversaturated or undersaturated) with respect to $\text{CaCO}_{3(s)}$ (1) as it enters the residence and (2) as it leaves the water heater?



Phân tích đề: ở 25 °C, nước bão hòa $\text{CaCO}_3 \rightarrow$ phản ứng trên đạt cân bằng tại 25 °C.

$\rightarrow$ có thể tính được K ở điều kiện 25 °C ($\Delta G^\circ = -2.71$ kcal).

So sánh K ở 25 °C với K ở 2 điều kiện nhiệt độ khác nhau

$\rightarrow$ chiều phản ứng (chưa bão hòa hay quá bão hòa).

3.4 Enthalpy and the temperature dependence of K

Example 3.5:

Find the standard enthalpy of formation, ΔH° for $\text{H}_2\text{SO}_{4(l)}$ given the following reactions:

	ΔH° (kcal)
1. $\text{S}_{(s)} + \text{O}_{2(g)} \rightleftharpoons \text{SO}_{2(g)}$	-70.96
2. $\text{SO}_{2(g)} + \frac{1}{2}\text{O}_2 \rightleftharpoons \text{SO}_{3(g)}$	-23.5
3. $\text{SO}_{3(g)} + \text{H}_2\text{O}_{(l)} \rightleftharpoons \text{H}_2\text{SO}_{4(l)}$	-31.14
4. $\text{H}_{2(g)} + \frac{1}{2}\text{O}_{2(g)} \rightleftharpoons \text{H}_2\text{O}_{(l)}$	-68.32

Solution

Adding reactions (1) through (4) and the ΔH° values, we obtain



where $\Delta H^\circ = -70.96 - 23.5 - 31.4 - 68.32 = -193.92$ kcal. Applying Eq. 3-25, we obtain

$$\Delta \bar{H}_f^\circ \text{ for } \text{H}_2\text{SO}_{4(l)} = -193.92 \text{ kcal}$$

3.5 Ví dụ làm thêm

Ví dụ 3.5.1:

At 3000K, the equilibrium partial pressures of CO_2 , CO , and O_2 are 0.6, 0.4, and 0.2 atm, respectively.

For the reaction: $2\text{CO}_2 \leftrightarrow 2\text{CO} + \text{O}_2$. Calculate (a) ΔG° (standard state: 1 atm) and (b) equilibrium constant K .

3.5 Ví dụ làm thêm

Ví dụ 3.5.2:

Ammonia, NH_3 , is a base and will readily accept a proton in accordance with the following reaction



- (a) Calculate the equilibrium constant, K , for this reaction at 25°C .
- (b) If, at some time, $\text{pH} = 9.0$, $[\text{NH}_3] = 10^{-5} \text{ M}$, and $[\text{NH}_4^+] = 10^{-6} \text{ M}$, is the reaction at equilibrium? If not, in which direction is the reaction going?

Chapter 4

AXID-BASE CHEMISTRY (HÓA HỌC AXIT-BASO)

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4.1 Introduction

❖ Acid-base chemistry

- Essential for understanding the carbonate system that has an important influence on the pH of natural water and may govern the solubility of certain metal ions.
- Many precipitation-dissolution, oxidation-reduction, and complexation reactions involve acid-base reactions.
- The concentrations of metal ions in water are governed to a great extent by acid-base phenomena.
- In water treatment, dosages of chemicals required to reduce hardness (Ca, Mg) are partially governed by the acid-base properties of treated solution.

4.1 Introduction

❖ This chapter will

- Consider in detail acid-base reactions using relevant examples from water treatment processes and natural water chemistry to illustrate the various principles.
- Given techniques for calculating the composition of solution of acids and bases.
- This chapter follows and deals with precipitation, coordination chemistry, and oxidation-reduction chemistry, which will use many of principles developed in this study.

4.2 Definition of terms

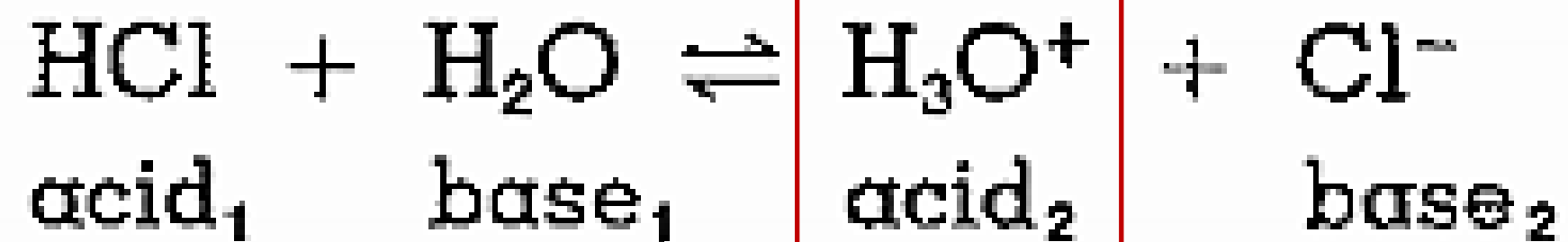
Conjugate acid-base pair

- Acid: a substance can donate H^+ (proton) $\rightarrow$ HA or HB
- Base: a substance can accept proton $\rightarrow$ A^- or B^-



- Water can be an acid or an base

Base:



Acid:



$$K = \frac{\{\text{H}_3\text{O}^+\}\{\text{OH}^-\}}{\{\text{H}_2\text{O}\}^2}$$

or when $\{\text{H}_2\text{O}\} = 1$ and neglecting ionic strength effects,

$$K = [\text{H}_3\text{O}^+][\text{OH}^-] = K_w$$

$$\text{pH} + \text{pOH} = \text{p}K_w = 14$$

$$\begin{aligned} \text{pH} &= -\log \{\text{H}^+\} \\ \text{pOH} &= -\log \{\text{OH}^-\} \\ \text{p}K_w &= -\log K_w \end{aligned}$$

4.2 Definition of terms

Conjugate acid-base pair



$$K = \frac{\{\text{H}_3\text{O}^+\} \{\text{OH}^-\}}{\{\text{H}_2\text{O}\}^2}$$

or when $\{\text{H}_2\text{O}\} = 1$ and neglecting ionic strength effects,

$$K = [\text{H}_3\text{O}^+][\text{OH}^-] = K_w$$

$$\text{pH} + \text{pOH} = \text{p}K_w = 14$$

$$\begin{aligned}\text{pH} &= -\log \{\text{H}^+\} \\ \text{pOH} &= -\log \{\text{OH}^-\} \\ \text{p}K_w &= -\log K_w\end{aligned}$$

4.2 Definition of terms

Conjugate acid-base pair

$$\begin{aligned}\ln K_{w,50^\circ} &= \ln K_{w,25^\circ} + \frac{\Delta H^\circ}{R} \left(\frac{1}{298} - \frac{1}{323} \right) \\ &= \ln (1.0 \times 10^{-14}) + \frac{13.36}{1.987 \times 10^{-3}} \left(\frac{1}{298} - \frac{1}{323} \right)\end{aligned}$$

$$K_{w,50^\circ} = 5.73 \times 10^{-14}$$

TABLE 4-2 Ion Product of Water

°C	K_w	pK_w	pH of a "neutral" solution ($\{H^+\} = \{OH^-\}$)
0	0.12×10^{-14}	14.93	7.47
15	0.45×10^{-14}	14.35	7.18
20	0.68×10^{-14}	14.17	7.08
25	1.01×10^{-14}	14.00	7.00
30	1.47×10^{-14}	13.83	6.92
40	2.95×10^{-14}	13.53	6.76

Source. H. S. Harned and B. B. Owen, *The Physical Chemistry of Electrolyte Solutions*, 3rd ed., Reinhold, New York, 1958.

4.2 Definition of terms

Strong and weak acid



$$K_a = \frac{\{\text{H}^+\}\{\text{A}^-\}}{\{\text{HA}\}}$$

Assume: $\gamma=1$

$$K_a = \frac{[\text{H}^+][\text{A}^+]}{[\text{HA}]}$$

Acidity constant, K_a : the equilibrium constant for an acid association reaction

$$\text{p}K_a = -\log K_a$$

$$K_a = 10^{-4.7}$$
$$\text{p}K_a = 4.7$$

4.2 Definition of terms

Strong and weak acid



If: $[\text{H}^+] = K_a \rightarrow [\text{A}^-] = [\text{HA}]$

$$[\text{H}^+] \gg K_a \rightarrow [\text{HA}] \gg [\text{A}^-]$$

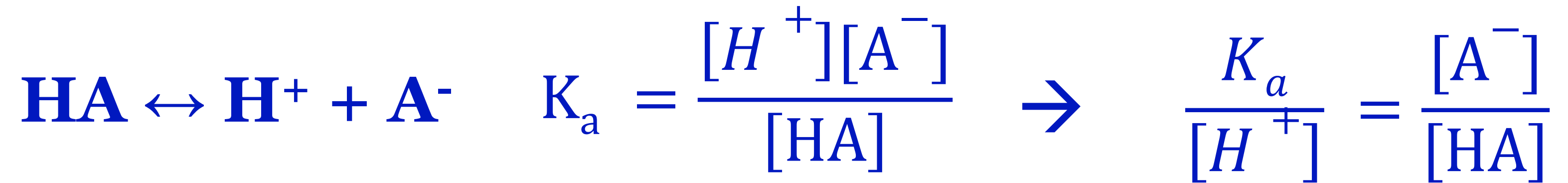
$$[\text{H}^+] \ll K_a \rightarrow [\text{HA}] \ll [\text{A}^-]$$

$$K_a \gg 1 \rightarrow \text{p}K_a < 0$$

$K_{a1} > K_{a2}$: acid 1 is stronger acid than acid 2 ($\text{p}K_{a1} < \text{p}K_{a2}$)

4.2 Definition of terms

Strong and weak acid



In dilute solution: $1 \leq \text{pH} \leq 14$

So if (1) $\text{pH} > \text{pK}_a \rightarrow [\text{H}^+] < K_a$

(2) $[\text{A}^-] \gg [\text{HA}]$: HA is strong acid

4.2 Definition of terms

Strong and weak acid

TABLE 4-1 Acidity and Basicity Constants for Substances in Aqueous Solution at 25°C

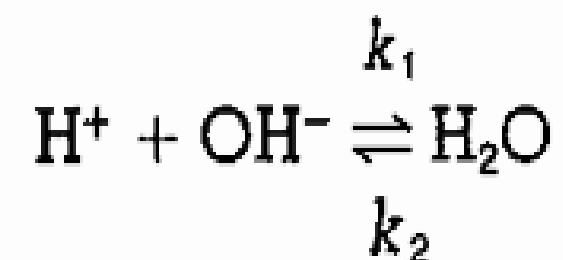
Acid		$-\log K_a$ = $\text{p}K_a$	Conjugate Base		$-\log K_b$ = $\text{p}K_b$
HClO ₄	Perchloric acid	-7	ClO ₄ ⁻	Perchlorate ion	21
HCl	Hydrochloric acid	~-3	Cl ⁻	Chloride ion	17
H ₂ SO ₄	Sulfuric acid	~-3	HSO ₄ ⁻	Bisulfate ion	17
HNO ₃	Nitric acid	-0	NO ₃ ⁻	Nitrate ion	14
H ₃ O ⁺	Hydronium ion	0	H ₂ O	Water	14
HIO ₃	Iodic acid	0.8	IO ₃ ⁻	Iodate ion	13.2
HSO ₄ ⁻	Bisulfate ion	2	SO ₄ ²⁻	Sulfate ion	12
H ₃ PO ₄	Phosphoric acid	2.1	H ₂ PO ₄ ⁻	Dihydrogen phosphate ion	11.9
Fe(H ₂ O) ₆ ³⁺	Ferric ion	2.2	Fe(H ₂ O) ₅ OH ²⁺	Hydroxo iron(III) complex	11.8
HF	Hydrofluoric acid	3.2	F ⁻	Fluoride ion	10.8
HNO ₂	Nitrous acid	4.5	NO ₂ ⁻	Nitrite ion	9.5
CH ₃ COOH	Acetic acid	4.7	CH ₃ COO ⁻	Acetate ion	9.3
Al(H ₂ O) ₆ ³⁺	Aluminum ion	4.9	Al(H ₂ O) ₅ OH ²⁺	Hydroxo aluminum(III) complex	9.1

4.2 Definition of terms

Strong and weak acid

H_2CO_3^*	Carbon dioxide and carbonic acid	6.3	HCO_3^-	Bicarbonate ion	7.7
H_2S	Hydrogen sulfide	7.1	HS^-	Bisulfide ion	6.9
H_2PO_4^-	Dihydrogen phosphate	7.2	HPO_4^{2-}	Monohydrogen phosphate ion	6.8
HOCl	Hypochlorous acid	7.5	OCl^-	Hypochlorite ion	6.4
HCN	Hydrocyanic acid	9.3	CN^-	Cyanide ion	4.7
H_3BO_3	Boric acid	9.3	B(OH)_4^-	Borate ion	4.7
NH_4^+	Ammonium ion	9.3	NH_3	Ammonia	4.7
H_4SiO_4	Orthosilicic acid	9.5	H_3SiO_4^-	Trihydrogen silicate ion	4.5
$\text{C}_6\text{H}_5\text{OH}$	Phenol	9.9	$\text{C}_6\text{H}_5\text{O}^-$	Phenolate ion	4.1
HCO_3^-	Bicarbonate ion	10.3	CO_3^{2-}	Carbonate ion	3.7
HPO_4^{2-}	Monohydrogen phosphate	12.3	PO_4^{3-}	Phosphate ion	1.7
H_3SiO_4^-	Trihydrogen silicate	12.6	$\text{H}_2\text{SiO}_4^{2-}$	Dihydrogen silicate ion	1.4
HS^-	Bisulfide ion	14	S^{2-}	Sulfide ion	0
H_2O	Water	14	OH^-	Hydroxide ion	0
NH_3	Ammonia	~23	NH_2^-	Amide ion	-9
OH^-	Hydroxide ion	~24	O^{2-}	Oxide ion	-10

4.3 Rate of reaction



$$\text{Forward rate} = k_1 [\text{H}^+][\text{OH}^-]$$

$$\text{Reverse rate} = k_2 [\text{H}_2\text{O}]$$

with $k_1 = 1.4 \times 10^{11}$ liter/mole, sec. and $k_2 = 2.5 \times 10^{-5}$ /sec. at 25°C .

At equilibrium, forward rate = reverse rate; therefore,

$$[\text{H}^+][\text{OH}^-] k_1 = k_2 [\text{H}_2\text{O}]$$

or

$$\frac{k_2 [\text{H}_2\text{O}]}{k_1} = [\text{H}^+][\text{OH}^-] \text{ (mole}^2\text{/liter}^2\text{)}$$

Substituting for k_1 and k_2 and using 55.5 moles/liter as the concentration of water,³ we obtain the concentration product of water,

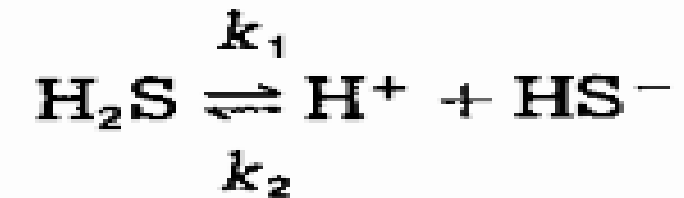
$$\frac{(2.5 \times 10^{-5}/\text{sec})(55.5 \text{ mole/liter})}{1.4 \times 10^{+11} \text{ liter/mole, sec.}} = 0.99 \times 10^{-14} \cong 10^{-14}$$

$$1\text{L H}_2\text{O} = 1000\text{g} \rightarrow 55.5 \text{ moles/ } 1\text{L H}_2\text{O}$$

4.3 Rate of reaction

Example 4-1

The dissociation of H_2S is given by



where $k_1 = 4.3 \times 10^3 \text{ sec}^{-1}$.⁴ Assuming that the reaction is irreversible, how long will it take for 50 percent of the H_2S to dissociate?

❖ Solution

$$\frac{d[\text{H}_2\text{S}]}{dt} = -k_1[\text{H}_2\text{S}] \quad \longrightarrow \quad \ln \frac{[\text{H}_2\text{S}]_t}{[\text{H}_2\text{S}]_0} = -k_1 t$$

where

$[\text{H}_2\text{S}]_0$ = concentration at $t = 0$, and
 $[\text{H}_2\text{S}]_t$ = concentration at $t = t$.

$$\frac{\ln \frac{1}{2}}{-4.3 \times 10^3} = t = 1.6 \times 10^{-4} \text{ sec}$$

H_2S khi tan vào nước sẽ tan rất nhanh trong nước

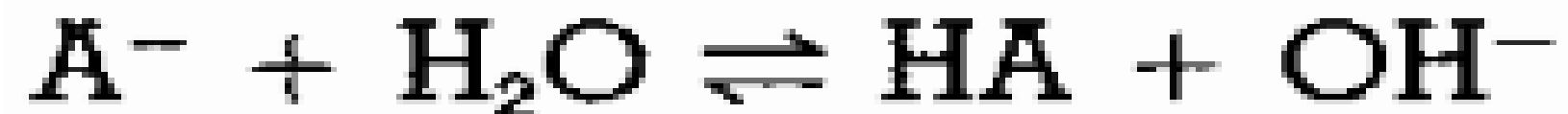
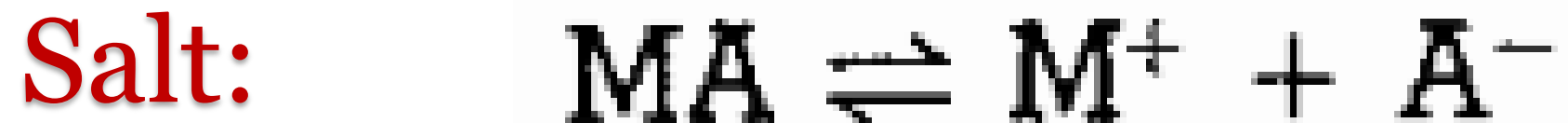
4.4 Equilibrium calculations

4.4.1 Mass balances



Assume that the system is homogeneous (đồng nhất) and closed (no species containing A can enter or leave the system, no precipitation/dissolution).

A mass balance on all species containing A: $C_{r,A} = [\text{HA}] + [\text{A}^-]$



$$C = C_{T,M} = [\text{M}^+] + [\text{MA}]$$

MA dissociates completely: $C_{T,M} = [\text{M}^+]$ and $C_{T,A} = [\text{HA}] + [\text{A}^-]$

When MA and HA both added to a solution:

$$C_{T,A} = [\text{HA}] + [\text{A}^-] = \text{sum of moles HA and MA}$$

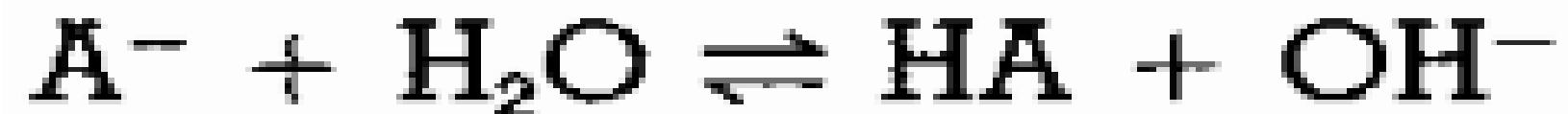
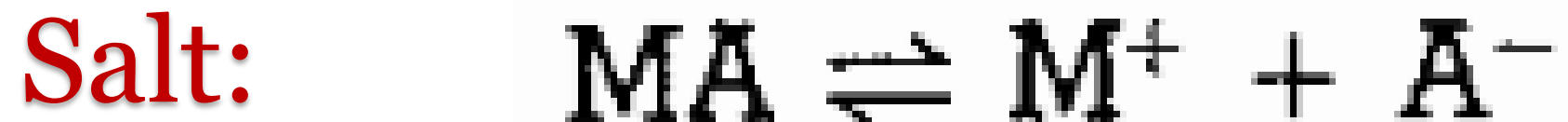
4.4 Equilibrium calculations

4.4.1 Mass balances



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When MA and HA both added to a solution:

$$C_{T,A} = [\text{HA}] + [\text{A}^-] = \text{sum of moles HA and MA}$$

4.4 Equilibrium calculations

4.4.2 Equilibrium relationships

Example: HA added to pure water, in aqueous solution, neglecting ionic strength, we obtain for the dissociation of water:

$$K_w = [H^+] + [OH^-] = 10^{-14} \text{ at } 25^\circ\text{C}$$



The dissociation of HA is described by:

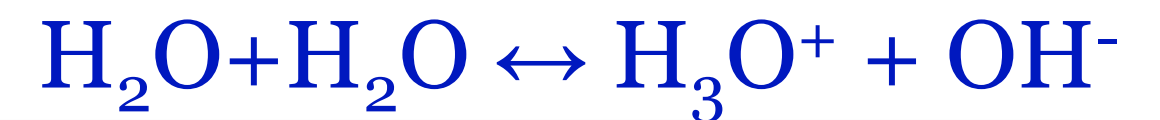
$$K_a = \frac{[H^+][A^-]}{[HA]}$$

4.4 Equilibrium calculations

Định luật bảo toàn proton

4.4.3 The proton condition

- Nhằm xem xét những cân bằng phản ứng có liên quan đến H^+ hoặc OH^-
- Người ta thiết lập cân bằng proton dựa vào việc thiết lập 1 mức tham chiếu (reference) là mức **không proton** (zero level) hay còn gọi là **PRL** (proton reference level). **Những gốc chứa proton nhiều hơn PRL phải bằng nhóm có ít proton hơn PRL.**



Example 4-2



Determine the proton condition when HA is added to water.

Solution

The species present are H_3O^+ , H_2O , OH^- , HA , and A^- . All species are involved in reactions with H^+ or OH^- .

PRL = HA , H_2O ,

Species with protons $>$ PRL = H_3O^+ ,

Species with protons $<$ PRL = OH^- , A^- .

Proton Condition

$$[H_3O^+] = [OH^-] + [A^-]$$

Note that HA and H_2O , the species at the PRL do not appear in the proton condition.

4.4 Equilibrium calculations

4.4.3 The proton condition

Example 4-3

Determine the proton condition when the salt MA is added to water.

Solution

The species present are H_3O^+ , H_2O , OH^- , M^+ , A^- , and HA . The species that involve reactions with H^+ or OH^- are H_3O^+ , H_2O , OH^- , A^- , and HA .

$\text{PRL} = \text{H}_2\text{O}$ and MA (or, equivalently, A^- , since M^+ completely dissociates from A^- and does not affect the proton condition).

Species with protons $> \text{PRL} = \text{HA}$, H_3O^+ ,

Species with protons $< \text{PRL} = \text{OH}^-$.

Proton Condition

$$[\text{HA}] + [\text{H}_3\text{O}^+] = [\text{OH}^-]$$

Again the species at the PRL, A^- , and H_2O , do not appear in the proton condition.

4.4 Equilibrium calculations

4.4.3 The proton condition

Summary:

the total of proton excess w.r.t reference species
= the total of proton deficit w.r.t reference species

Ex1: There are NaCl, NaOH in water

If refer water: $\cancel{[\text{Na}^+]} + [\text{H}^+] = \cancel{[\text{Cl}^-]} + [\text{OH}^-]$

Ex2: There are NaCl, NaHCO₃ in water

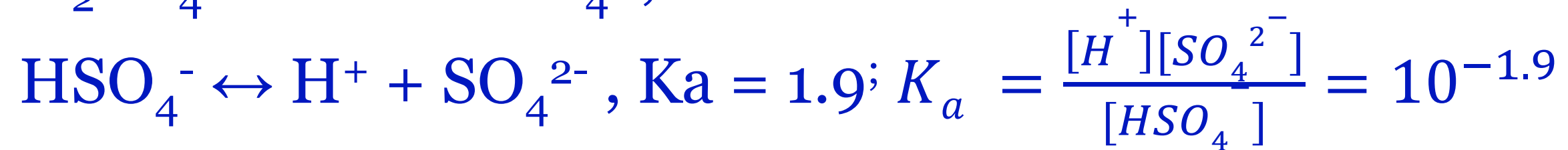
If refer HCO₃⁻: $\cancel{[\text{Na}^+]} + [\text{H}^+] + [\text{H}_2\text{CO}_3] = \cancel{[\text{Cl}^-]} + [\text{OH}^-] + [\text{CO}_3^{2-}]$

(w.r.t =with respect to)

4.4 Equilibrium calculations

4.4.4 Example

A rain water sample is collected and the following measurements are made: pH = 4.2, $(\text{SO}_4)_T = 5.0 \times 10^{-5} \text{ M}$. Speciation of sulfate?



$$\begin{aligned} C_{\text{SO}_4, T} &= [\text{SO}_4^{2-}] + [\text{HSO}_4^-] \\ &= [\text{SO}_4^{2-}] + \frac{[\text{H}^+][\text{SO}_4^{2-}]}{[K_a]} = 5.0 \times 10^{-5} \text{ M} \\ &= [\text{SO}_4^{2-}] \left(\frac{10^{-4.2}}{10^{-1.9}} + 1 \right) = 5.0 \times 10^{-5} \text{ M} \end{aligned}$$

$$\begin{aligned} \rightarrow [\text{SO}_4^{2-}] &\approx ? \\ [\text{HSO}_4^-] &= ? \end{aligned}$$

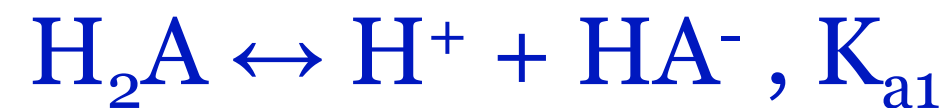
4.5 pC –pH Diagram

4.5.1 Basic information

Monoprotic acid/base



Diprotic acid



Define: $\text{pC} \equiv -\log\{i\} \approx -\log[i]$

How dose the speciation of acid/base pair HA & A- change with pH?

- Neglect activity correction (γ)
- Assume that we can add strong acid or base to get whatever pH we want

4.5 pC –pH Diagram

Essential Guide

4.5.2 Plot of pC-pH diagram

❖ Summary

1. How to plot pC-pH diagram
 - a. Plot C_T (constant across the whole figure)
 - b. Plot pH and pOH as function of pH
 - c. Plot each species by
 - i. Rearrang equilibrium as a function of C_T , K_a , and H^+ .
 - ii. Discuss significance of H^+ - species concentration at different zones in the figure.
 - d. Correct pHA and pA⁻ at the intersection

4.5 pC –pH Diagram

Importance

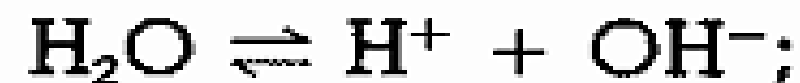
4.5.2 Plot of pC-pH diagram

Problem: HCN 10^{-3} M solution in close system, $K_a = 10^{-9.3}$

Equilibria



$$K_a = \frac{[\text{H}^+][\text{CN}^-]}{[\text{HCN}]} = 10^{-9.3} \quad (\text{Table 4-1})$$



$$K_w = [\text{H}^+][\text{OH}^-] = 1 \times 10^{-14}$$

Mass Balance

$$C_{\text{T,CN}} = [\text{HCN}] + [\text{CN}^-] = 10^{-3}$$

Proton Condition (Charge Balance)

$$[\text{H}^+] = [\text{CN}^-] + [\text{OH}^-]$$

4.5 pC –pH Diagram

$$\text{HCN } 10^{-3} \text{ M}, K_a = 10^{-9.3}$$

4.5.2 Plot of pC-pH diagram

- Biểu thức đầu tiên được dùng để vẽ là cân bằng vật chất:

$$-\log C_{T,\text{CN}} = pC_{T,\text{CN}} = 3$$

- Vẽ đường số (1), là đường song song với trục hoành ở vị trí $pC = 3$.
- Biểu thức thứ 2 được vẽ là biểu diễn K_w :

$$\log K_w = \log [H^+] + \log [OH^-]$$

$$pOH = 14 - pH$$

hệ số góc: $\frac{dpOH}{dpH} = -1$

Khi $pH = 0$, $pOH = 14$, vẽ đường (2) với hệ số góc = 1

4.5 pC –pH Diagram

HCN 10^{-3} M, $K_a = 10^{-9.3}$

4.5.2 Plot of pC-pH diagram

- Ta có:

$$\begin{aligned} -\log [H^+] &= \text{pH} \\ \frac{d(-\log [H^+])}{d\text{pH}} &= 1 \end{aligned}$$

- Khi $\text{pH}=0$, $-\log[H^+]=0$. Ta vẽ được đường số (3).
- Có thể vẽ đường số 2, và 3 trước đường số 1.
- Bước kế tiếp là vẽ các đường cho $[\text{CN}^-]$ và $[\text{HCN}]$. Để vẽ các đường này trên biểu đồ pC-pH diagram. Đầu tiên ta phải biểu diễn nồng độ những chất này theo pH và các hằng số.

Đường số 1

Đường số 3

Đường số 2

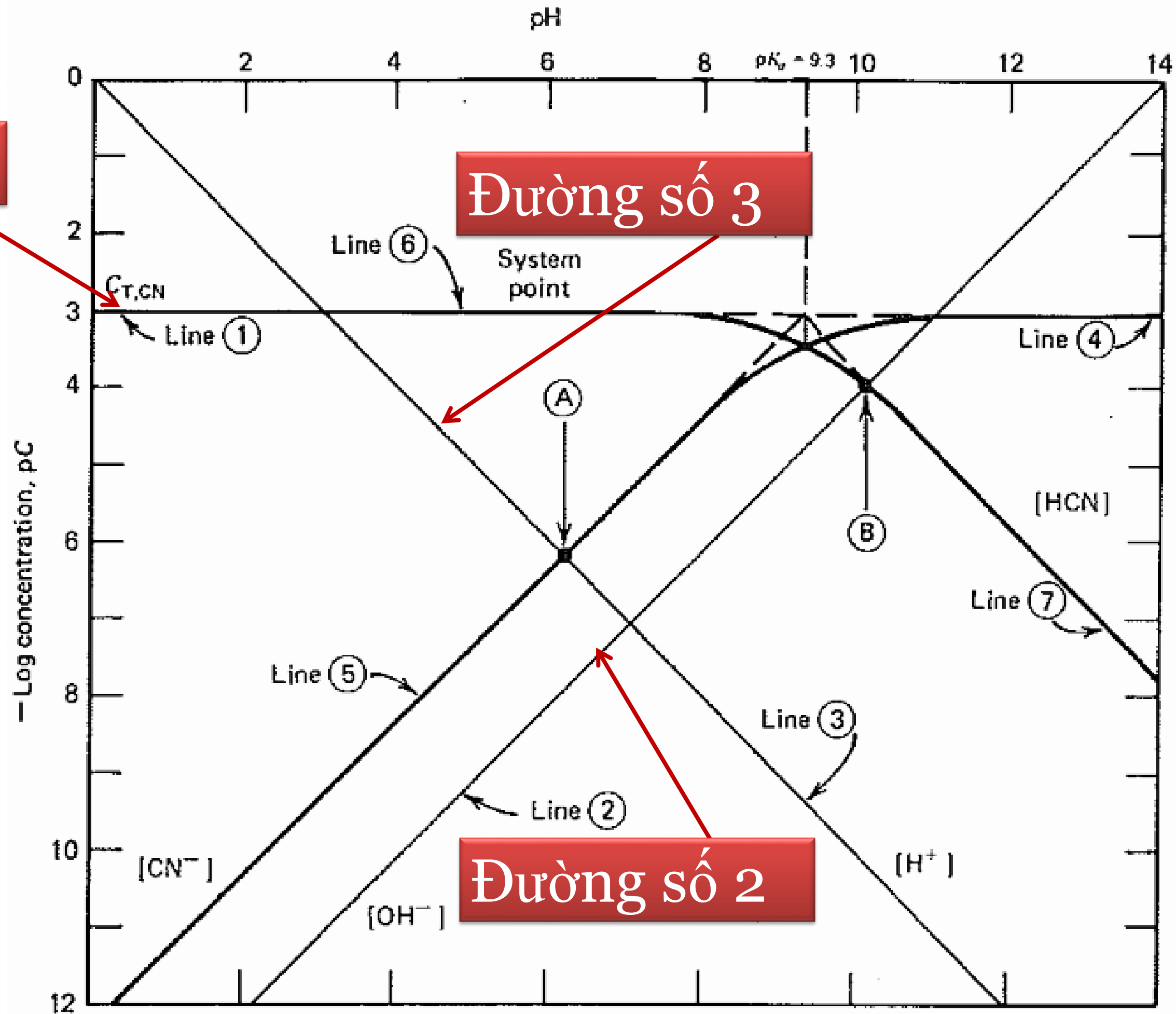


Fig. 4-1. The pC-pH diagram for cyanide-hydrocyanic acid; $C_{T,CN} = 10^{-3} M$; $pK_a = 9.3$ at $25^\circ C$.

4.5 pC –pH Diagram

$$\text{HCN } 10^{-3} \text{ M}, K_a = 10^{-9.3}$$

4.5.2 Plot of pC-pH diagram

$$K_a = \frac{[\text{H}^+][\text{CN}^-]}{[\text{HCN}]} \Rightarrow K_a = \frac{[\text{H}^+][\text{CN}^-]}{(C_{\text{T,CN}} - [\text{CN}^-])}$$

$$\Rightarrow [\text{CN}^-] = \frac{K_a C_{\text{T,CN}}}{([\text{H}^+] + K_a)} \quad (*)$$

- Vùng của biểu đồ mà $K_a \gg [\text{H}^+]$, hoặc $\text{pH} > \text{p}K_a$. Bởi vì $\text{p}K_a = 9.3$, đây là vùng có $\text{pH} > 9.3$. Nên (*) có thể trở thành $[\text{CN}^-] C_{\text{T,CN}} = 10^{-3}$
- Do đó, trong vùng $\text{pH} > 9.3$, ta vẽ đường số (4).

4.5 pC –pH Diagram

$$\text{HCN } 10^{-3} \text{ M}, K_a = 10^{-9.3}$$

4.5.2 Plot of pC-pH diagram

- Trong vùng $[\text{H}^+] \gg K_a$, hoặc $\text{p}K_a > \text{pH}$. (*) trở thành

$$\frac{K_a C_{\text{T,CN}}}{[\text{H}^+]} = [\text{CN}^-] \Rightarrow \log C_{\text{T,CN}} + \log K_a - \log [\text{H}^+] = + \log [\text{CN}^-]$$

$$\text{p}[\text{CN}^-] = \text{p}C_{\text{T,CN}} + \text{p}K_a - \text{pH}$$

Hệ số góc: $\frac{d(\text{p}[\text{CN}^-])}{d\text{pH}} = -1$

Khi $\text{pH} = \text{p}K_a = 9.3$: $\text{p}[\text{CN}^-] = \text{p}C_{\text{T,CN}} = 3$

- Vẽ đường số (5) trong vùng $\text{pH} < 9.3$

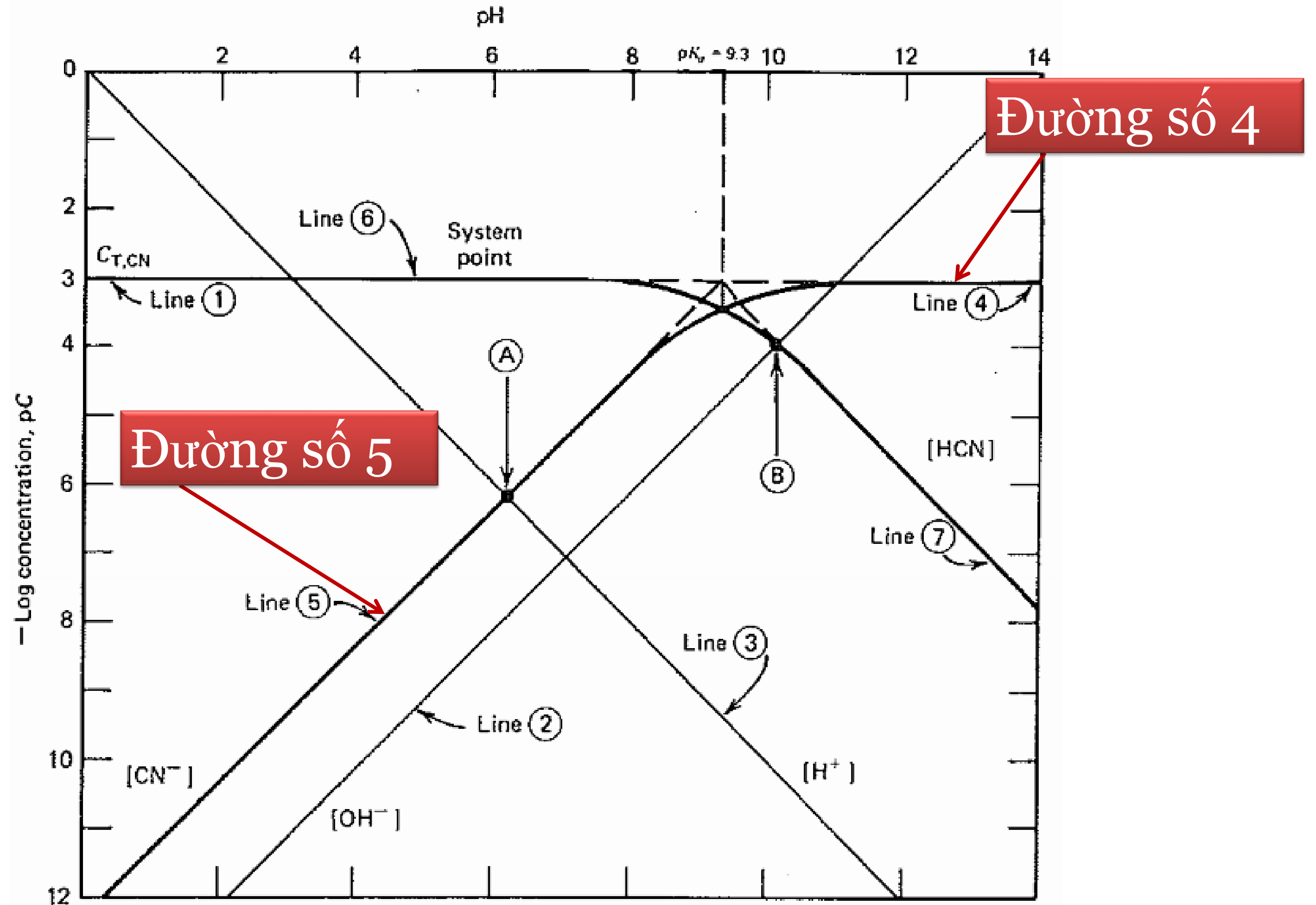


Fig. 4-1. The pC-pH diagram for cyanide-hydrocyanic acid; $C_{T,CN} = 10^{-3} M$; $pK_a = 9.3$ at $25^\circ C$.

4.5 pC –pH Diagram

$$\text{HCN } 10^{-3} \text{ M}, K_a = 10^{-9.3}$$

4.5.2 Plot of pC-pH diagram

- Vì $[\text{CN}^-]$ không nằm trên đường số 4 và 5 ở gần vị trí $\text{pH} = \text{pK}_a = 9.3$. Nên ở gần vị trí này, đường số 4 và 5 nên vẽ bằng nét đứt (dashed line). Đường vẽ cho $[\text{CN}^-]$ sẽ được hoàn thiện sau khi vẽ đường $[\text{HCN}]$

$$K_a = \frac{[\text{H}^+][\text{CN}^-]}{[\text{HCN}]} \quad \Rightarrow \quad K_a = \frac{[\text{H}^+](C_{\text{T,CN}} - [\text{HCN}])}{[\text{HCN}]}$$

$$\Rightarrow [\text{HCN}] = \frac{C_{\text{T,CN}}[\text{H}^+]}{[\text{H}^+] + K_a} \quad (**)$$

- Vùng của biểu đồ mà $[\text{H}^+] \gg K_a$, hoặc $\text{pK}_a > \text{pH}$, (**) trở thành:

$$[\text{HCN}] = C_{\text{T,CN}} \text{ hay } \text{p}[\text{HCN}] = \text{p}C_{\text{T,CN}} = 3$$

- Vẽ đường số (6) tại $\text{pC} = 5$ cho HCN trong vùng $\text{pH} < 9.3$

4.5 pC –pH Diagram

$$\text{HCN } 10^{-3} \text{ M}, K_a = 10^{-9.3}$$

4.5.2 Plot of pC-pH diagram

$$[\text{HCN}] = \frac{C_{\text{T,CN}}[\text{H}^+]}{[\text{H}^+] + K_a} \quad (**)$$

- Vùng của biểu đồ mà $[\text{H}^+] \ll K_a$, hoặc $\text{pH} > \text{p}K_a$, (**) trở thành:

$$[\text{HCN}] = \frac{C_{\text{T,CN}}[\text{H}^+]}{K_a} \quad \Rightarrow \quad \text{p}[\text{HCN}] = \text{pH} + \text{p}C_{\text{T,CN}} - \text{p}K_a$$

Hệ số góc: $\frac{d(\text{p}[\text{HCN}])}{d\text{pH}} = +1$

Khi $\text{pH} = \text{p}K_a = 9.3$, $\text{p}[\text{HCN}] = C_{\text{T,CN}} = 3$

- Vẽ đường số (7) trong vùng $\text{pH} > 9.3$, có hệ số góc là 1 và đi qua điểm có $\text{pH} = \text{p}K_a = 9.3$

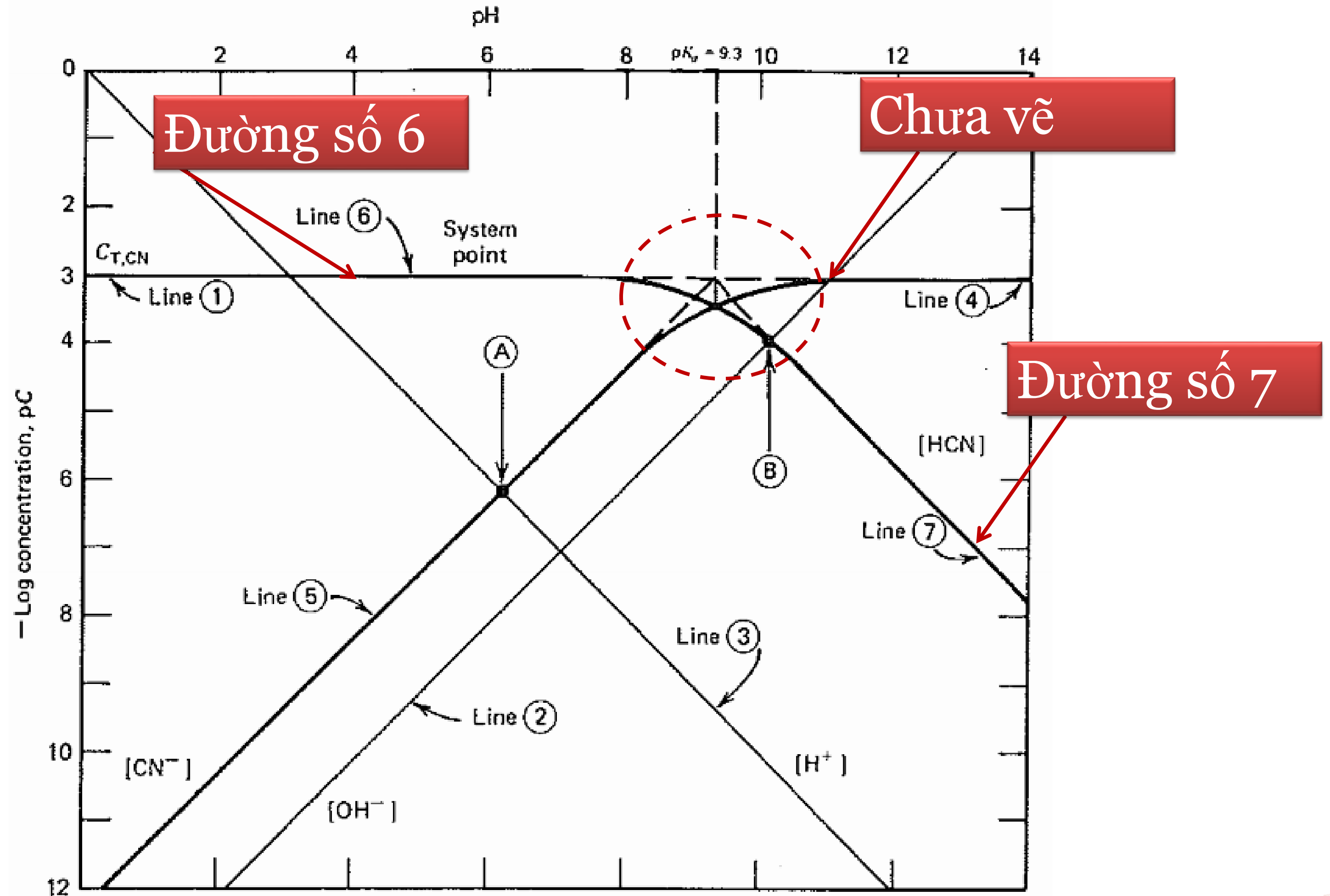


Fig. 4-1. The pC-pH diagram for cyanide-hydrocyanic acid; $C_{T,CN} = 10^{-3} M$; $\text{p}K_a = 9.3$ at 25°C .

4.5 pC –pH Diagram

$$\text{HCN } 10^{-3} \text{ M}, K_a = 10^{-9.3}$$

4.5.2 Plot of pC-pH diagram

- Khi $\text{pH} = \text{pK}_a$ hay $[\text{H}^+] = K_a$, ta có: $K_a = \frac{[\text{H}^+][\text{CN}^-]}{[\text{HCN}]} \Rightarrow [\text{CN}^-] = \frac{C_{\text{T,CN}}}{2}$
- $\Rightarrow \text{p}[\text{CN}^-] = \text{p}C_{\text{T,CN}} + 0.3$
- Khi $\text{pH} = \text{pK}_a$, ta cũng có được: $\text{p}[\text{HCN}] = \text{p}C_{\text{T,CN}} + 0.3$
- Dựa vào $\text{pC} = \text{p}C_{\text{T,CN}} + 3$, ta xác định được 1 điểm uốn của cả 2 đường $[\text{CN}^-]$ và $[\text{HCN}]$ khi $\text{pK}_a = 9.3$, $\text{pC} = 3 + 0.3 = 3.3$.
- Tuy nhiên, cần xác định điểm bắt đầu uốn trên 2 phần đường thẳng của 2 đường $[\text{CN}^-]$ và $[\text{HCN}]$.

4.5 pC –pH Diagram

$$\text{HCN } 10^{-3} \text{ M}, K_a = 10^{-9.3}$$

4.5.2 Plot of pC-pH diagram

- Tuy nhiên, cần xác định điểm bắt đầu uốn trên 2 phần đường thẳng của 2 đường $[\text{CN}^-]$ và $[\text{HCN}]$.
- Để cho tăng độ chính xác, người ta thường xác định điểm bắt đầu uốn ở vị trí $\text{pH} = \text{pK}_a - 1$

$$K_a = \frac{[\text{H}^+][\text{CN}^-]}{[\text{HCN}]} \quad C_{\text{T,CN}} = [\text{HCN}] + [\text{CN}^-] = 10^{-3}$$

$$\text{p}[\text{HCN}] = \text{p}C_{\text{T,CN}} + 0.04$$

- và $\text{pH} = \text{pK}_a + 1$.

$$\text{p}[\text{CN}^-] = \text{p}C_{\text{T,CN}} + 0.04$$

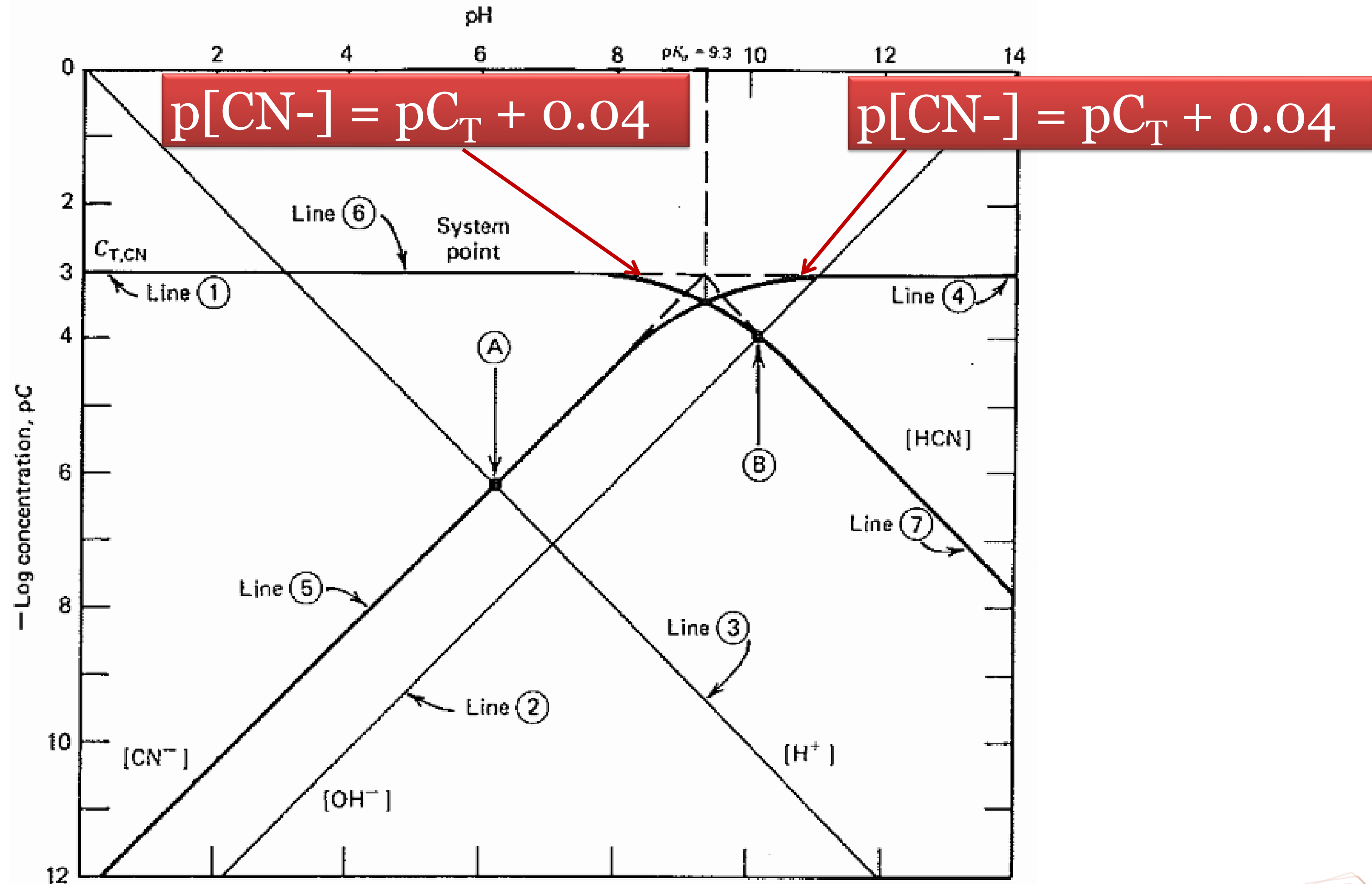


Fig. 4-1. The pC-pH diagram for cyanide-hydrocyanic acid; $C_{T,\text{CN}} = 10^{-3} \text{ M}$; $\text{p}K_a = 9.3$ at 25°C .

4.5 pC –pH Diagram

Essential Guide

4.5.3 Plot of pC-pH diagram

❖ How to use pC-pH diagram

- Simplify charge balance equation by assuming negligible ions.
- The **distribution of each species will be obtained.**

For further reference on how to develop diprotic acid:

https://www.youtube.com/watch?v=Ty3cR4_996U

4.5 pC –pH Diagram

4.5.3 Plot of pC-pH diagram

- Ví dụ áp dụng pC-pH diagram

What is the pH, pOH, $[\text{CN}^-]$, and $[\text{HCN}]$ in a 10^{-3} M HCN solution at 25°C which is closed to the atmosphere so that no HCN can volatilize? Neglect ionic strength effects.

Solution

The proton condition for HCN in H_2O is

$$[\text{H}^+] = [\text{CN}^-] + [\text{OH}^-]$$

We must use the same technique of approximation that was employed in the computation procedure to find the point in the diagram where the proton condition is satisfied. We are not, however, flying blind in this instance because we have the diagram to guide us in our selection of assumptions. Examining point A in Fig. 4-1, we can see that $[\text{CN}^-] \gg [\text{OH}^-]$ and $[\text{H}^+] = [\text{CN}^-]$. This point satisfies our simplified proton condition, which when neglecting $[\text{OH}^-]$ becomes $[\text{H}^+] = [\text{CN}^-]$. Point A occurs at $\text{pH} = 6.2$ and we can read off all the other species concentrations at pH 6.2: $[\text{HCN}] = 10^{-3}$, $[\text{H}^+] = 10^{-6.2}$, $[\text{CN}^-] = 10^{-6.2}$, and $[\text{OH}^-] = 10^{-7.8}$. Our assumption that $[\text{CN}^-] \gg [\text{OH}^-]$ is justified, since $10^{-6.2} \gg 10^{-7.8}$.

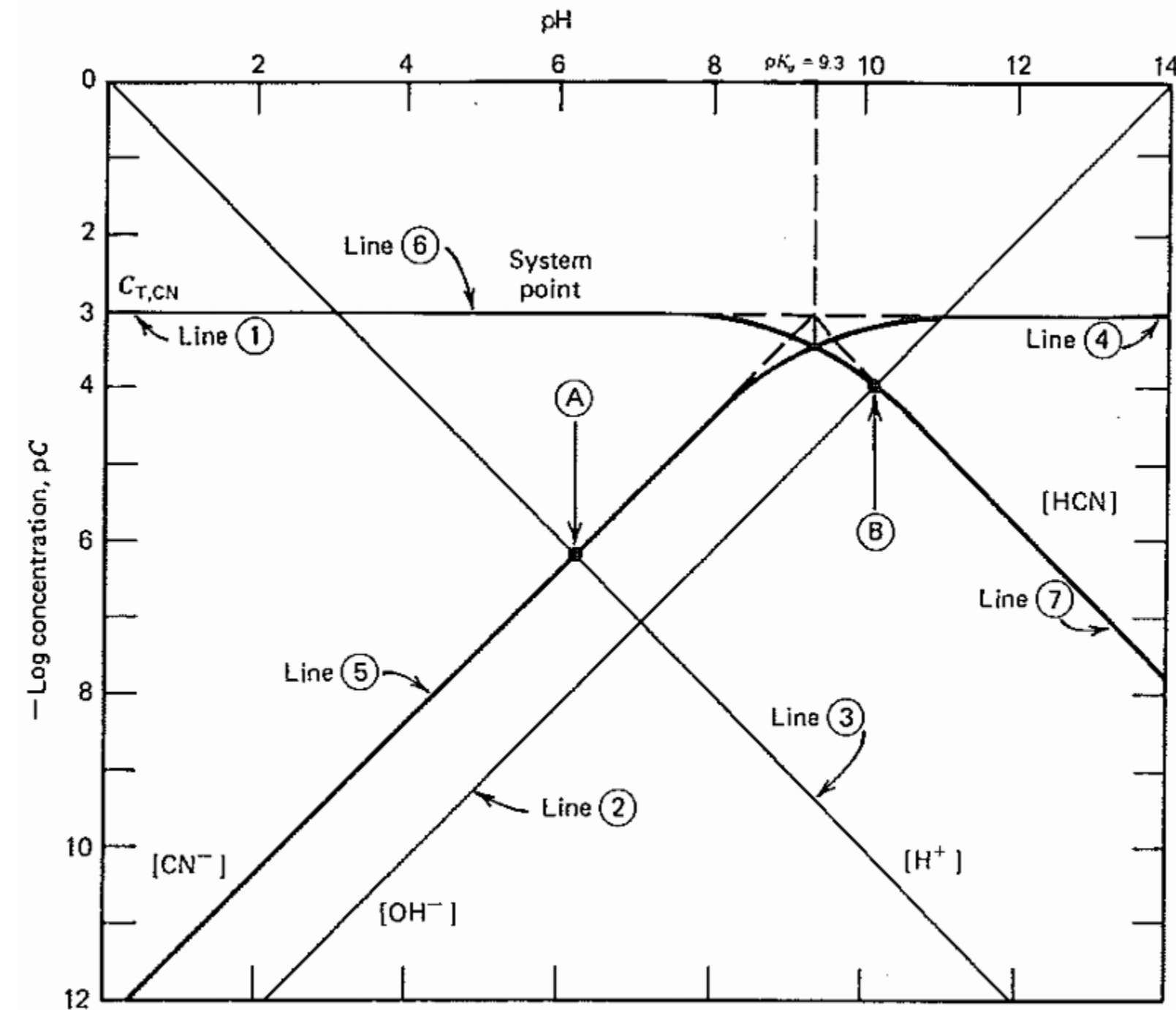


Fig. 4-1. The pC-pH diagram for cyanide-hydrocyanic acid; $C_{\text{T,CN}} = 10^{-3} \text{ M}$; $\text{pK}_a = 9.3$ at 25°C .

4.5 pC –pH Diagram

4.5.3 Plot of pC-pH diagram

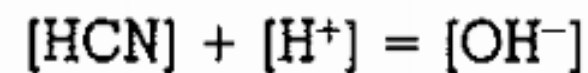
- Ví dụ áp dụng pC-pH diagram

Example 4-17

Find the pH, pOH, [HCN], [CN⁻], and [Na⁺] in a 10⁻³ M NaCN solution. The temperature is 25°C; neglect ionic strength effects.

Solution

The proton condition for NaCN in H₂O is



Examining Fig. 4-1 at point B, we see that [HCN] >> [H⁺] and the approximate proton condition [HCN] = [OH⁻], is satisfied. This point occurs at pH = 10.1. At this pH, [H⁺] = 10^{-10.1} M, [OH⁻] = 10^{-3.9} M, [HCN] = 10^{-3.9} M, [CN⁻] = 10^{-3.05} M. From a mass balance on sodium, [Na⁺] = C_{T,Na} = 10⁻³ M.

Checking our assumption in the proton condition, we find that

$$10^{-10.1} + 10^{-3.9} = 10^{-3.9}$$

The assumption is acceptable, since the error is a small fraction of 1 percent.

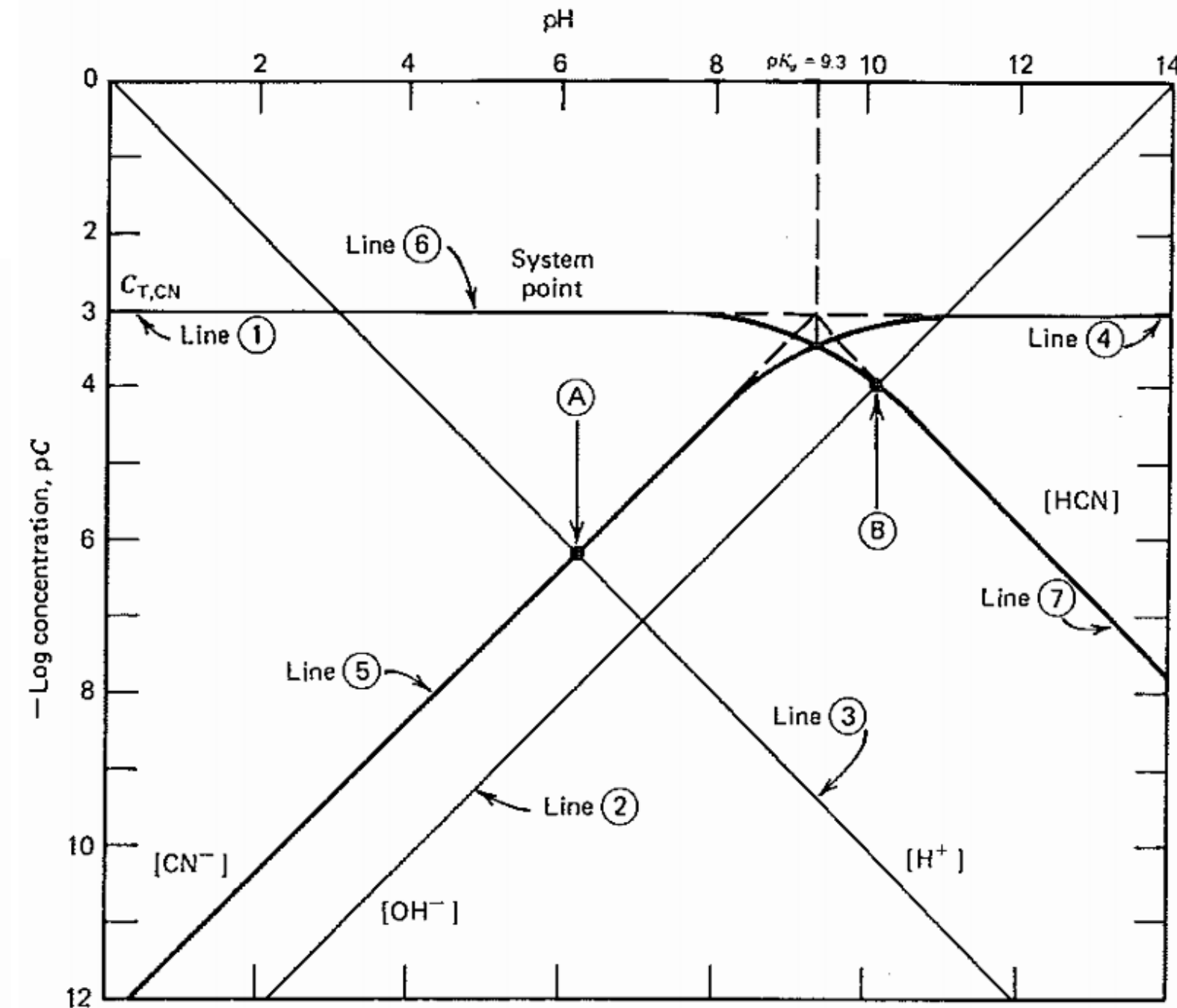


Fig. 4-1. The pC-pH diagram for cyanide-hydrocyanic acid; C_{T,CN} = 10⁻³ M; pK_a = 9.3 at 25°C.

4.5 pC –pH Diagram

4.5.3 Plot of pC-pH diagram

- Ví dụ áp dụng pC-pH diagram

Example 4-18

Find the pH, [HOCl], and [OCl⁻] in a solution to which 10⁻⁴ mole of sodium hypochlorite, NaOCl, and 5 × 10⁻⁵ mole of hypochlorous acid, HOCl, pK_a = 7.5, have been added per liter of solution. The temperature is 25°C; neglect ionic strength effects.

Solution

1. Construct a pC-pH diagram for HOCl following the procedure described previously. In the diagram, Fig. 4-2,

$$C_{T,OCl} = [HOCl] + [OCl^-] = 5 \times 10^{-5} + 10^{-4} = 1.5 \times 10^{-4}$$
$$pC_{T,OCl} = 3.8$$

2. Because we added both an acid and a conjugate base we must account for the cation (Na⁺) accompanying the conjugate base in our diagram.
A mass balance on Na⁺ yields

$$C_{T,Na} = [Na^+] = 10^{-4} M$$
$$pC_{T,Na} = 4$$

We can draw the [Na⁺] line in the diagram as a horizontal line at pC_T = 4.

To solve for the species present, we must examine the diagram for the point at which electroneutrality is met. For situations where an acid and its conjugate base are added, the charge balance is preferred over the proton condition for finding the point of neutrality because it is easier to write.

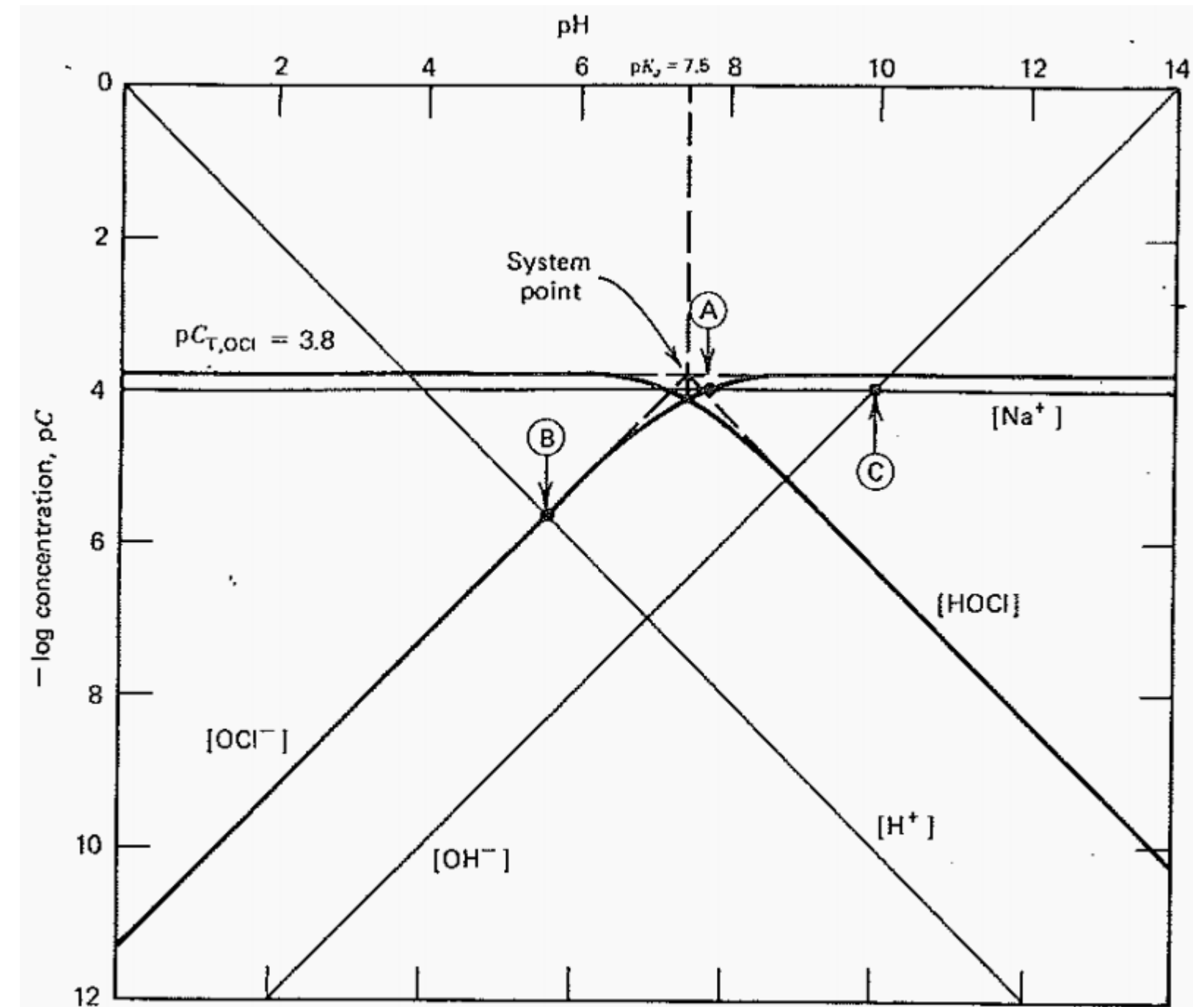


Fig. 4-2. The pC-pH diagram for hypochlorite-hypochlorous acid: $C_{T,OCl} = 1.5 \times 10^{-4} M$, 25°C.

4.5 pC –pH Diagram

4.5.3 Plot of pC-pH diagram

- Ví dụ áp dụng pC-pH diagram

The charge balance for our mixture is

$$[\text{Na}^+] + [\text{H}^+] = [\text{OH}^-] + [\text{OCl}^-]$$

Examining Fig. 4-2, we can see that at point A, $[\text{Na}^+] \gg [\text{H}^+]$ and $[\text{OCl}^-] \gg [\text{OH}^-]$ so that the charge balance simplifies to $[\text{Na}^+] = [\text{OCl}^-]$, a condition that is met at pH 7.7. At this pH, $[\text{Na}^+] = 10^{-4} \text{ M}$, $[\text{OCl}^-] = 10^{-4} \text{ M}$, $[\text{H}^+] = 10^{-7.7} = 1.99 \times 10^{-8} \text{ M}$, $[\text{OH}^-] = 10^{-6.3} = 5.0 \times 10^{-7} \text{ M}$, and $[\text{HOCl}] = 10^{-4.2} = 6.3 \times 10^{-5} \text{ M}$.

Checking the charge balance, we obtain,

$$1 \times 10^{-4} + 1.99 \times 10^{-8} = 1 \times 10^{-4} + 5.0 \times 10^{-7}$$

$$\text{error} = \frac{(1 \times 10^{-4} + 5.0 \times 10^{-7}) - (1 \times 10^{-4} + 1.99 \times 10^{-8})}{1 \times 10^{-4} + 1.99 \times 10^{-8}} \times 100$$

$$= 0.5 \text{ percent}$$

so that the assumptions are acceptable.

We can prove that these are the only viable set of assumptions to make in the charge balance by using the pC-pH diagram to check the validity of other possible assumptions. For example, if we assume $[\text{H}^+] \gg [\text{Na}^+]$ and $[\text{OCl}^-] \gg [\text{OH}^-]$, then the charge balance becomes $[\text{H}^+] = [\text{OCl}^-]$, a condition that is met at point B in Fig. 4-2. However, at this pH of 5.8, $[\text{H}^+] = 10^{-5.8}$ and $[\text{Na}^+] = 10^{-4}$ so that $[\text{Na}^+] \gg [\text{H}^+]$ and our assumption is not valid. If we make the assumptions that $[\text{Na}^+] \gg [\text{H}^+]$ and $[\text{OH}^-] \gg [\text{OCl}^-]$, then the charge balance reduces to $[\text{Na}^+] = [\text{OH}^-]$, a condition that is satisfied at point C in Fig. 4-2. At this pH of 10, $[\text{OH}^-] = 10^{-4}$ and $[\text{OCl}^-] = 10^{-3.8}$ so that the assumption of $[\text{OH}^-] \gg [\text{OCl}^-]$ is not acceptable. We have now made all possible combinations of assumptions in the charge balance and could by this process of elimination eventually find the correct set of assumptions.

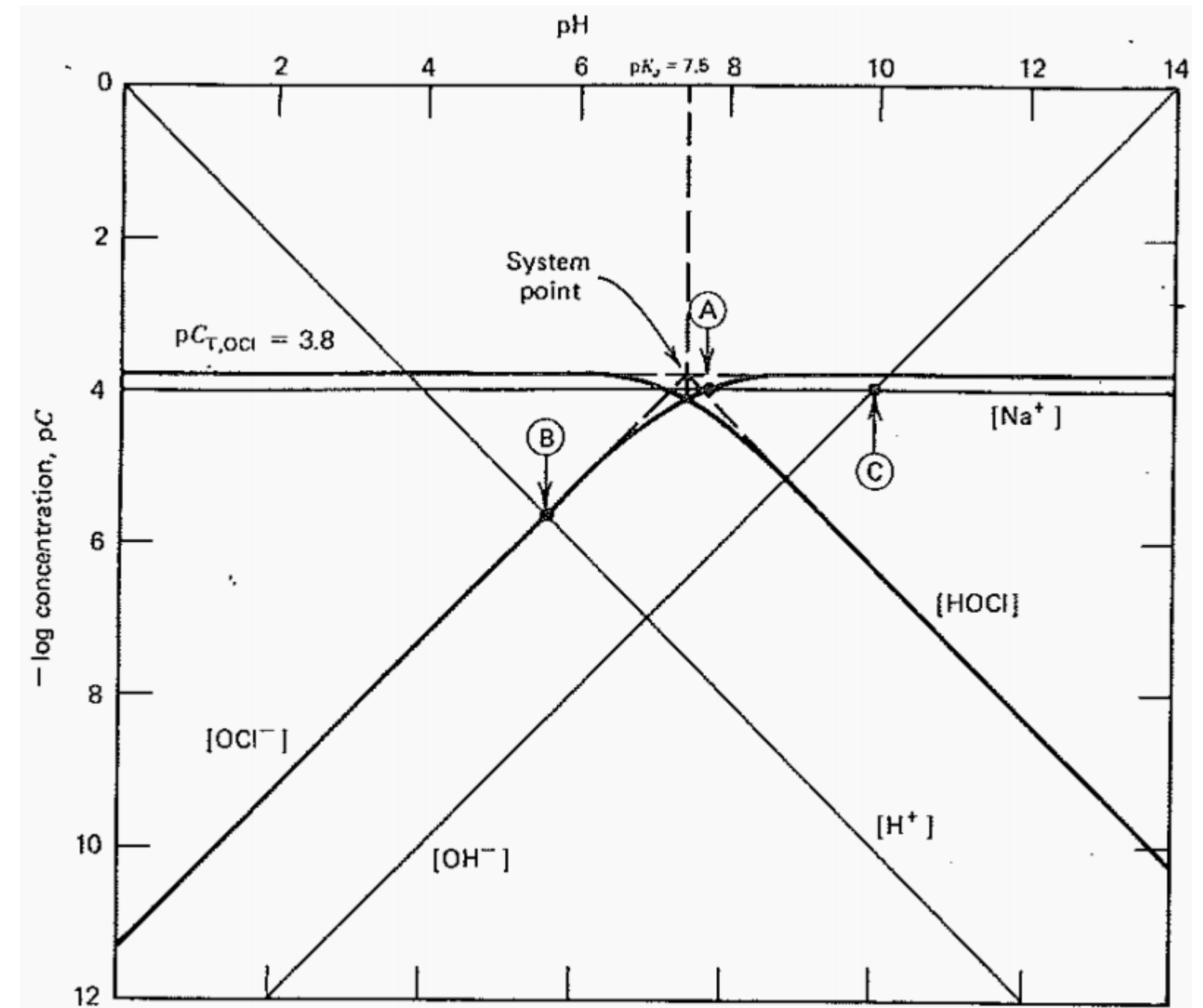


Fig. 4-2. The pC-pH diagram for hypochlorite-hypochlorous acid: $C_{\text{T,OCI}} = 1.5 \times 10^{-4} \text{ M}$, 25°C .

4.5 pC –pH Diagram

4.5.3 Plot of pC-pH diagram

- Ví dụ áp dụng pC-pH diagram

Hệ $\text{NH}_3(\text{aq}), \text{NH}_4^+$ vs pH

Cho $\text{NH}_4\text{OH}, \text{NH}_4\text{Cl}, \dots$ vào nước

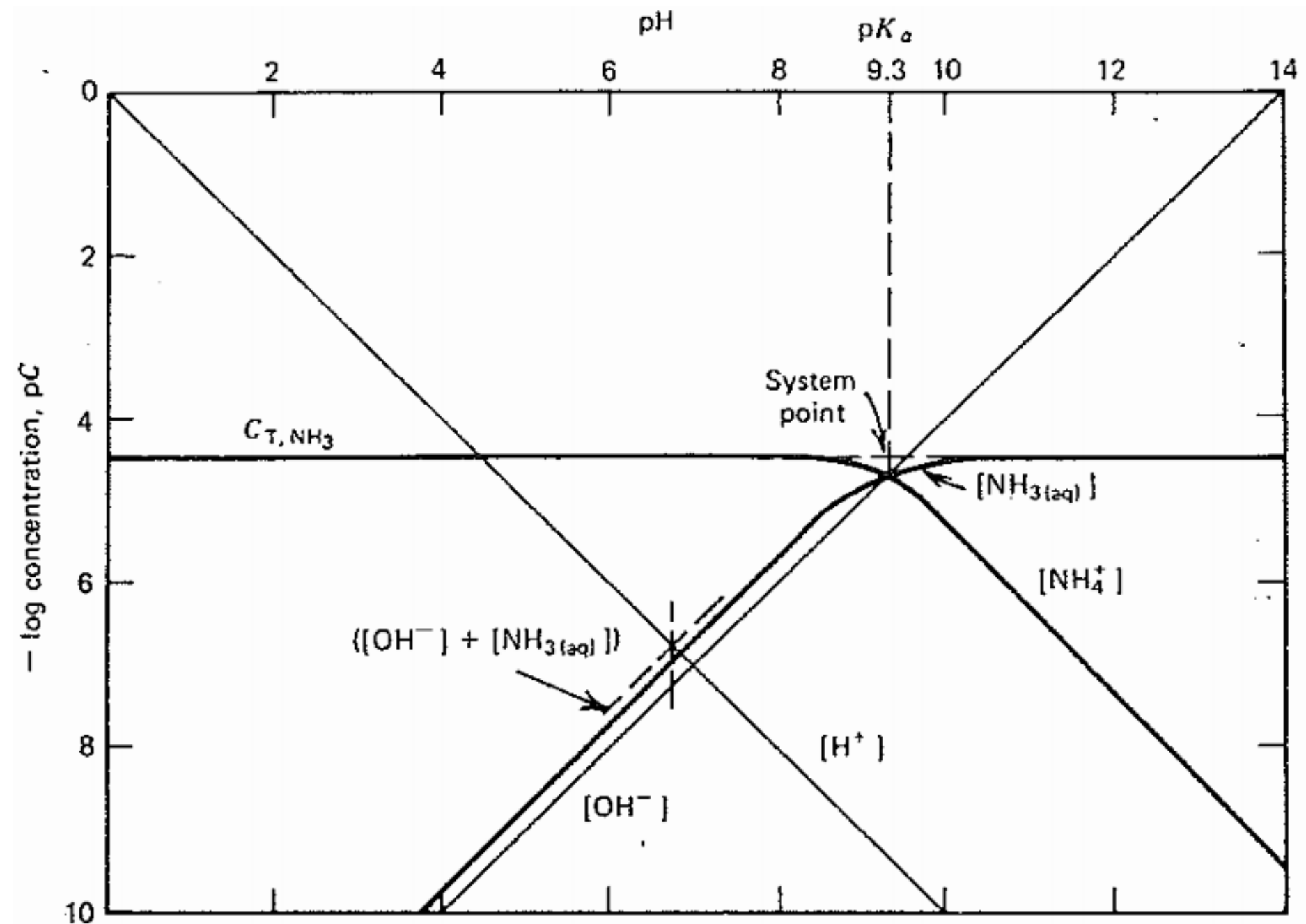
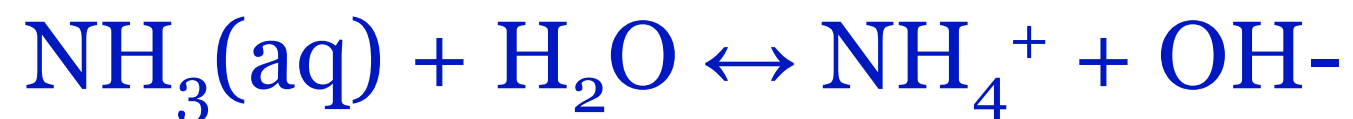


Fig. 4-3. The pC-pH diagram for a $10^{-4.5} \text{ M}$ NH_4Cl solution at 25°C .

4.5 pC –pH Diagram

4.5.3 Plot of pC-pH diagram

- Ví dụ áp dụng pC-pH diagram

Hệ HOCl, OCl⁻ vs pH

Cho Cl₂, NaOCl, Ca(OCl)₂ vào nước:

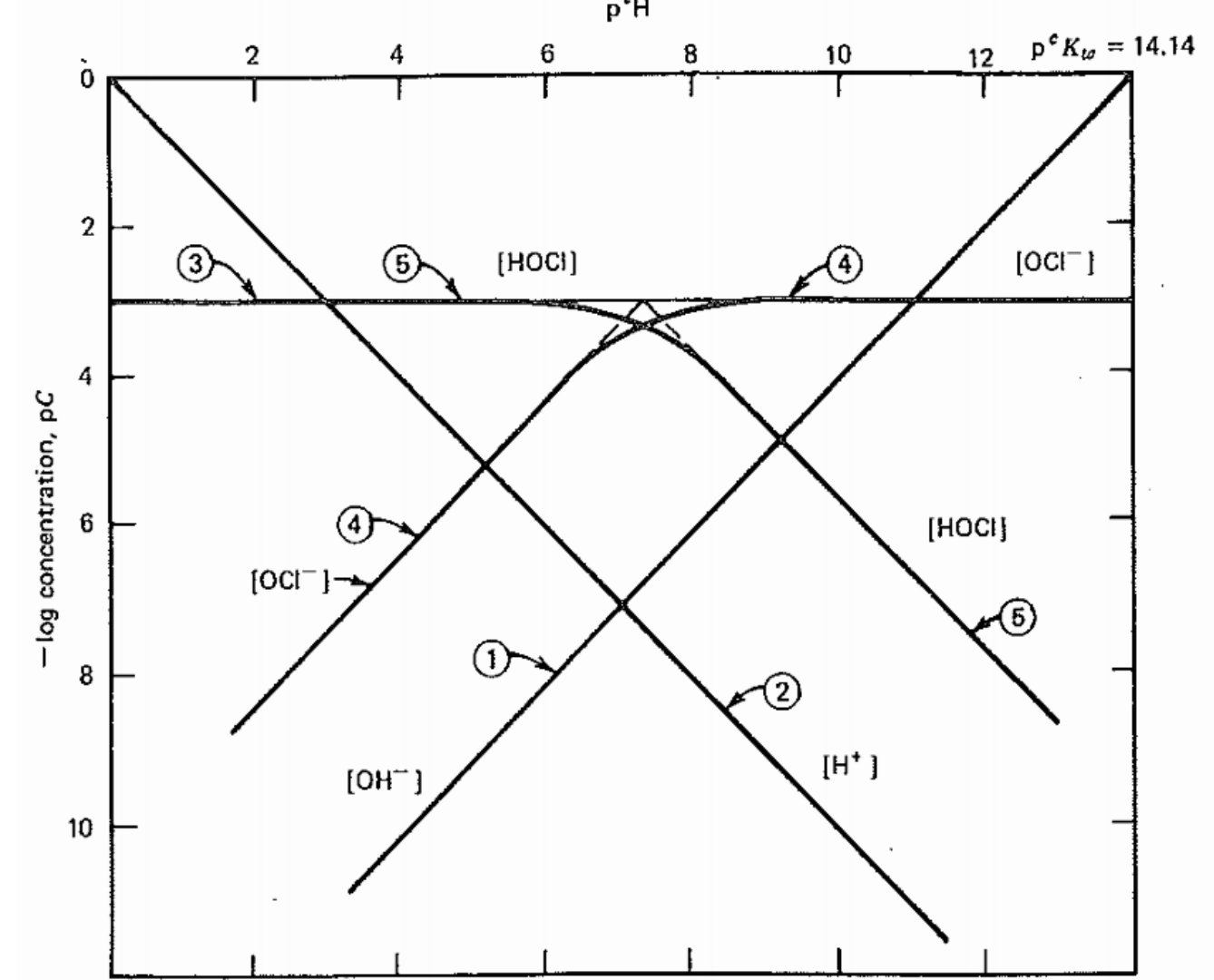
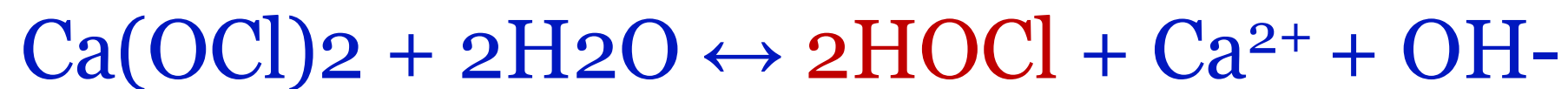
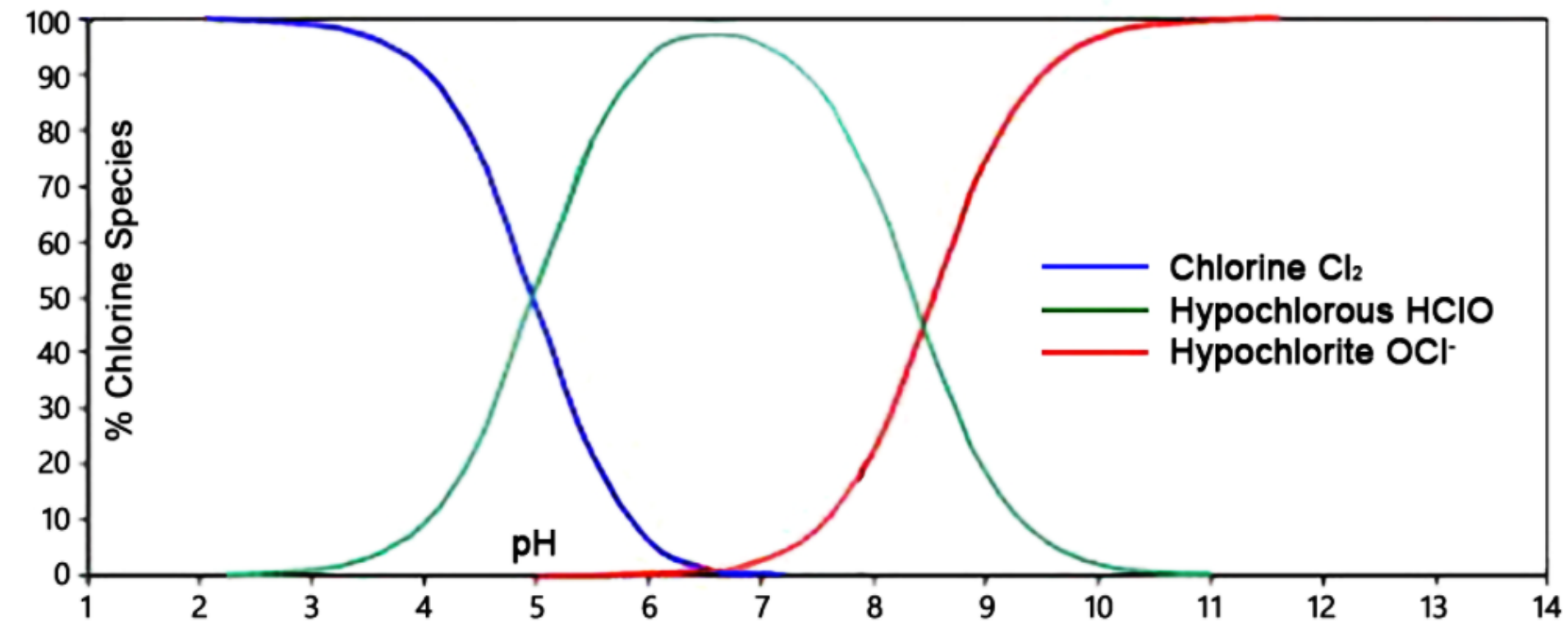


Fig. 4-4. The pC-pH diagram for HOCl—OCl⁻ at 15°C, $\mu = 0.1$.



4.6 Carbonate system

- Water is seldom free of CO₂.
- Acidity and alkalinity are largely based on carbonate species.
- Precipitation of CaCO₃ is the main process in water softening.
- The presence of CO₂ in water would affect the solution pH, precipitation of metal ions, and ozonation processes.

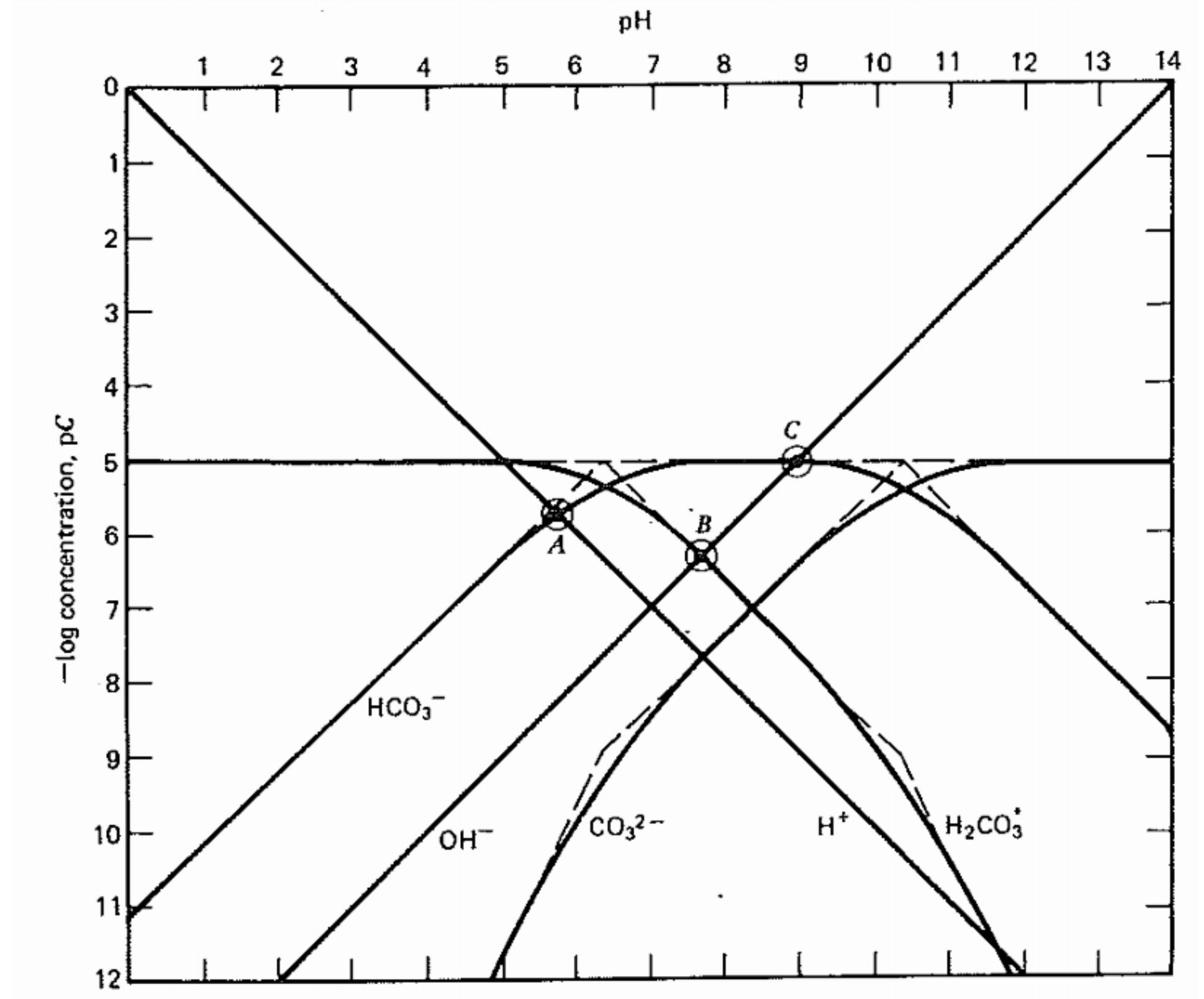
Normal atmosphere, $P_{\text{CO}_2} = 10^{-3.5}$ atm



4.6 Carbonate system

Closed system at 25 °C.

$$C_T = 10^{-5} = \text{constant}$$



4.6 Carbonate system

Open system at 25 °C.

$$[\text{H}_2\text{CO}_3^*] = P \cdot K_H = 10^{-3.5} \cdot 10^{-1.5} \\ = 10^{-5} = \text{constant}$$

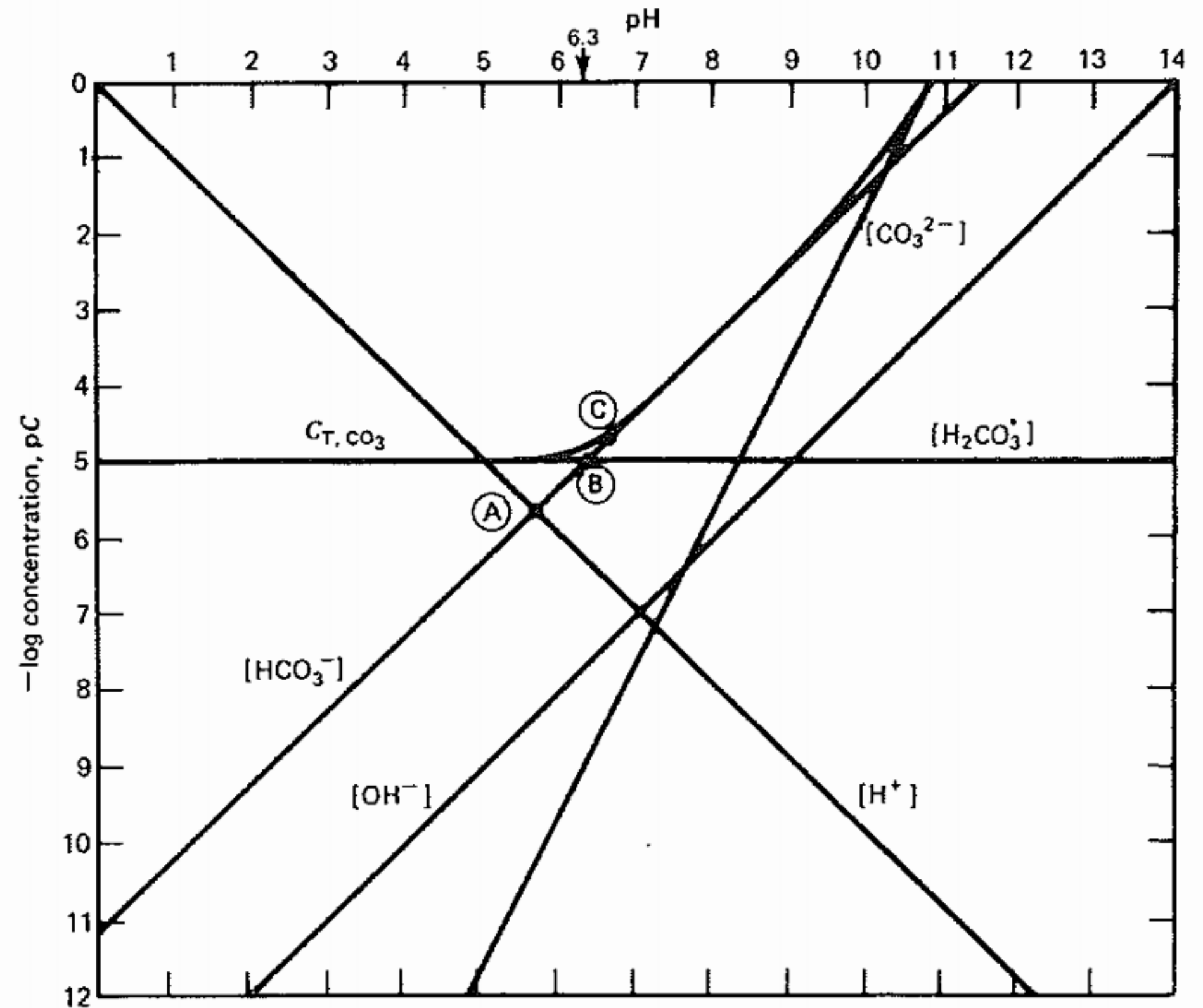


Fig. 4-16. Total concentration of carbonates, C_{T,CO_3} , versus pH for an open system at 25°C.



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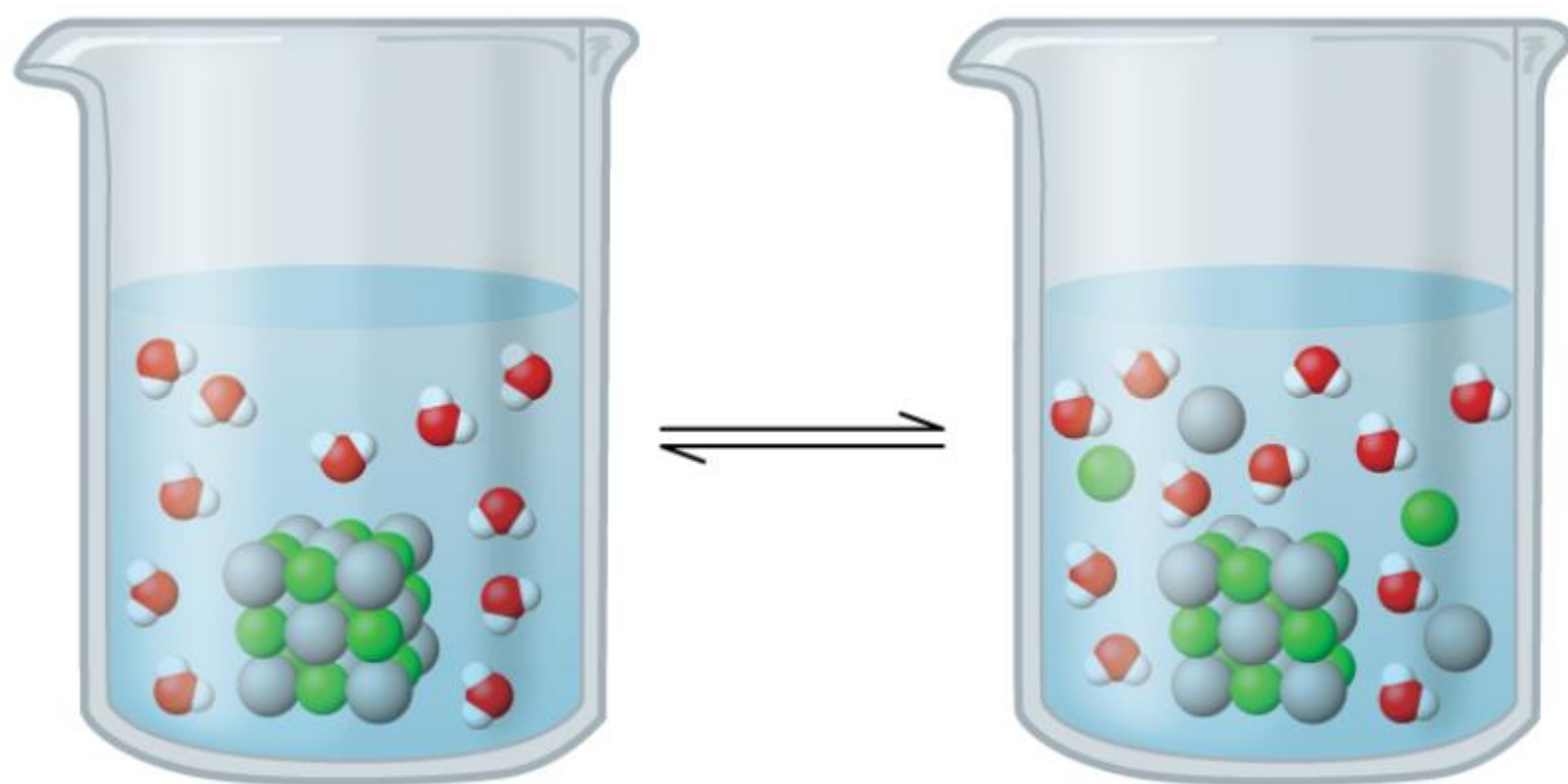
Faculty of Chemical and Food Technology



Chapter 5

PRECIPITATION AND DISSOLUTION

(KẾT TỦA VÀ HÒA TAN)



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4.1 Introduction

❖ Precipitation and dissolution

- Two phenomenon are extremely important in both natural waters and water treatment processes.
- Dissolution of minerals is a prime factor in determining the chemical composition can be altered by precipitation of minerals and the subsequent sedimentation of these solids from supersaturated solutions.
- Water and wastewater treatment processes such as lime-soda softening, iron removal, coagulation with hydrolyzing metal salts, and phosphate precipitation are based on precipitation phenomena.

4.1 Introduction

❖ Precipitation and dissolution

- Based on knowledge of equilibrium relationship, concentrations of species at equilibrium are calculated and thus the amount of precipitate of material that dissolves can be calculated.
- This chapter will deal with the kinetics and equilibrium calculations of heterogeneous systems.
- Graphical and computational methods for solving equilibrium problems will be presented.
- The precipitation of calcium carbonate will be discussed in some detail.

4.2 Precipitation

❖ Precipitation and dissolution kinetics

Precipitation can be observed to occurred in three steps:

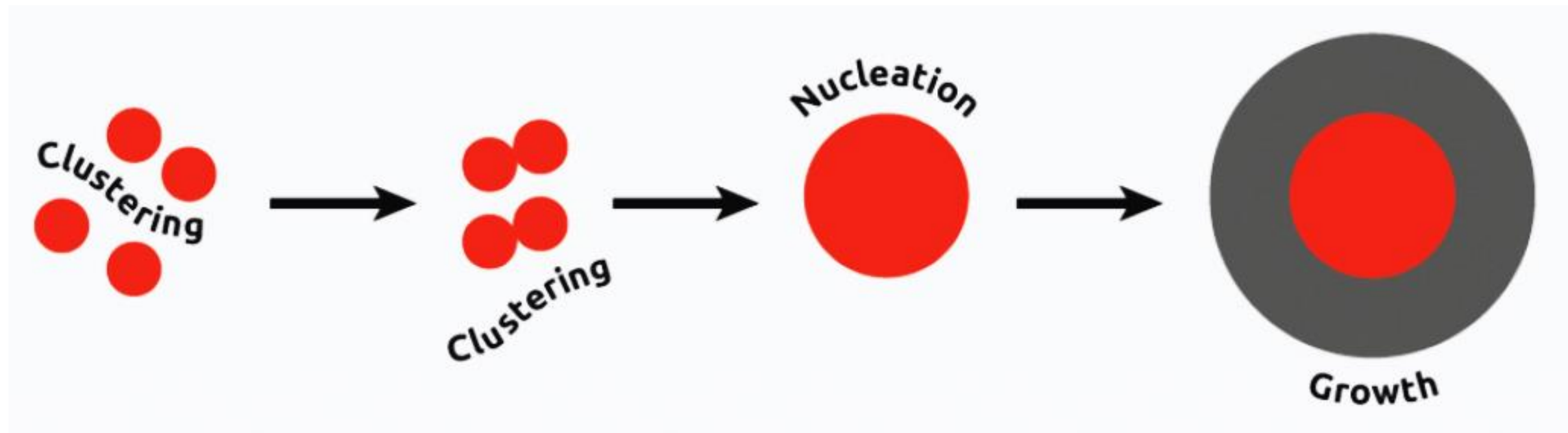
1. Nucleation
2. Crystal growth
3. Agglomeration (kết tụ) and ripening (chín) of the solids

When the degree of supersaturation is increased or when fine particles of a substance are mixed with this solution, precipitation occurs.

4.2 Precipitation

4.2.1. Nucleation

- A nucleus (hạt nhân) is a fine particle on which spontaneous formation or precipitation of a solid phase can take place.
- Nuclei can be formed from clusters of a few molecules or ion pairs of component ions of precipitate.



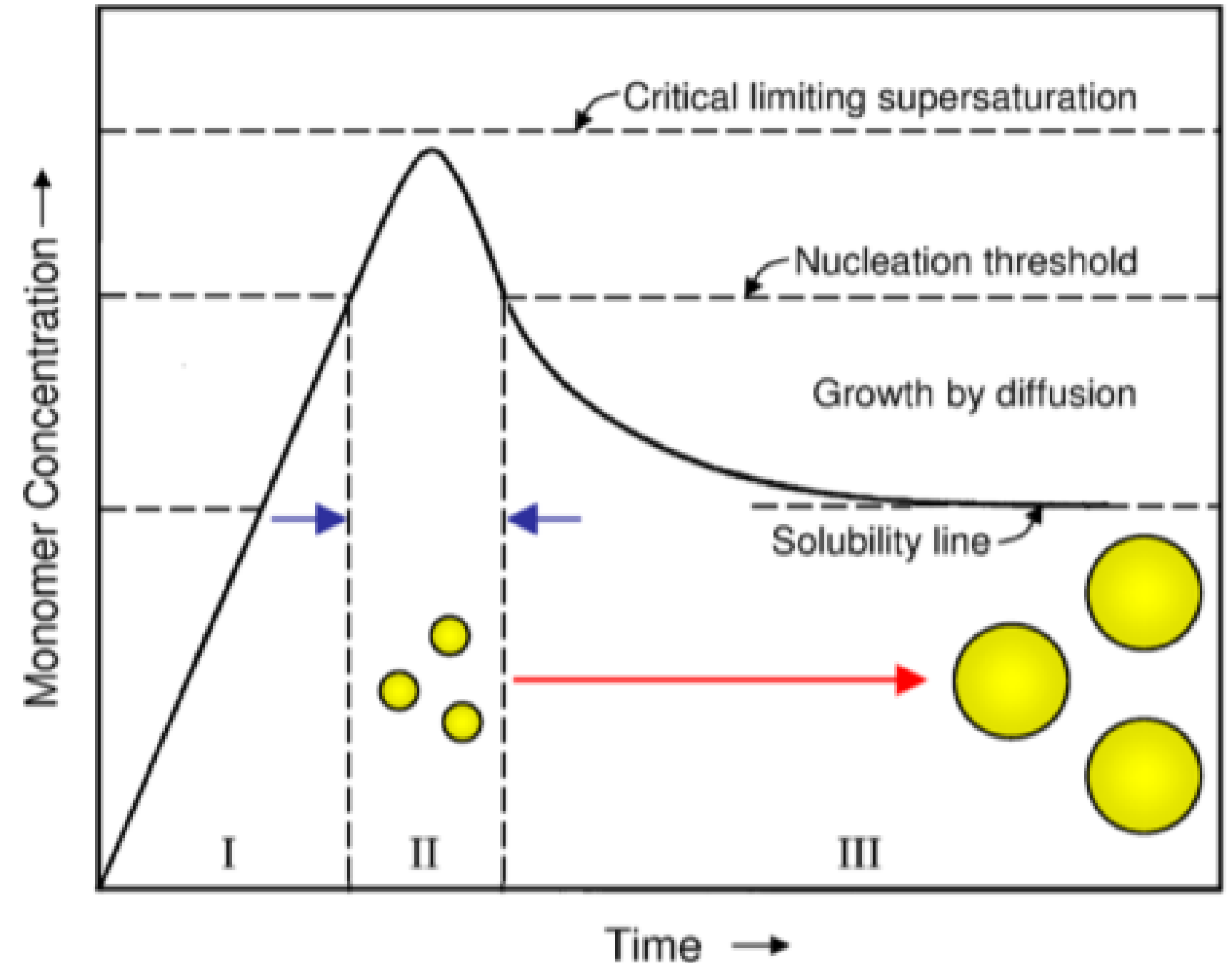
<https://www.syrris.com/>

- Nuclei = precipitate: homogeneous nucleation
- Nuclei = foreign particle: heterogeneous nucleation.

4.2 Precipitation

4.2.1. Nucleation

- The monomer concentration steadily rises in time zone I, and passes the threshold concentration for nucleation in time-zone II.
- Nucleation and subsequent growth decreases the monomer concentration below the nucleation threshold, such that nucleation ceases and growth upon the existing nuclei occurs in time zone III.



4.2 Precipitation

4.2.2. Crystal growth

Crystal forms by deposition of the precipitate constituent ions onto nuclei.

Since water and wastewater treatment processes involving precipitation often do not reach equilibrium, the rate of crystal growth is of critical importance. Crystal growth rate can be expressed as:

$$\frac{dC}{dt} = -kS(C - C^*)^n$$

C^* = saturation concentration, mole/liter

C = actual concentration of limiting ion, mole/liter

k = rate constant, liter ^{n} time⁻¹ mg⁻¹, mole^(1- n)

S = surface area available for precipitation, mg/liter of
a given particle size

n = constant

For example, Nancollas and Reddy⁴ found that the precipitation of calcium carbonate from a particular solution corresponded to the rate law:

$$\frac{d[Ca^{2+}]}{dt} = kS ([Ca^{2+}] - [Ca^{2+}]^*)^2 \quad (6-2)$$

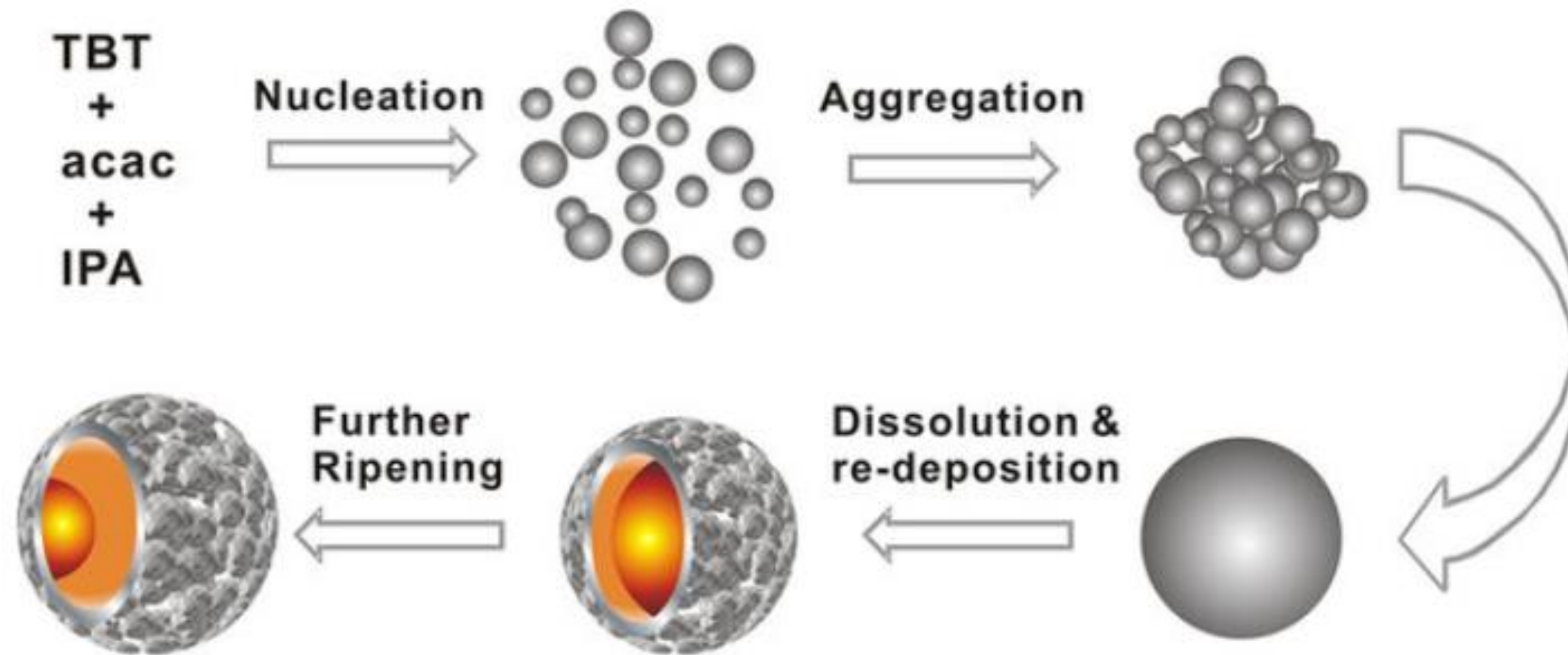
where $k = 5.76 \text{ liter}^2 \text{ mg}^{-1} \text{ mole}^{-1} \text{ min}^{-1}$. Of course, a different rate equation is expected for solutions with other properties.

When the diffusion rate of ions to the surface of the crystal controls the crystal growth rate, the exponent n has a value of unity; when other processes such as the reaction rate at the crystal surface are rate limiting, the exponent may have a value other than unity. The value of k depends on solution conditions and the nature of the solid being precipitated.

4.2 Precipitation

4.2.3. Agglomeration and ripening of the solids

Initial solid (may be not stable) → stable phase → change of crystal structure (aging) → ripening (crystal size increases)



4.2 Precipitation

4.2.4. Dissolution

According to Walton,⁵ dissolution is nearly always controlled by the rate of diffusion of the species away from the solids. Accordingly, the rate law,

$$\frac{dC}{dt} = kS (C^* - C) \quad (6-3)$$

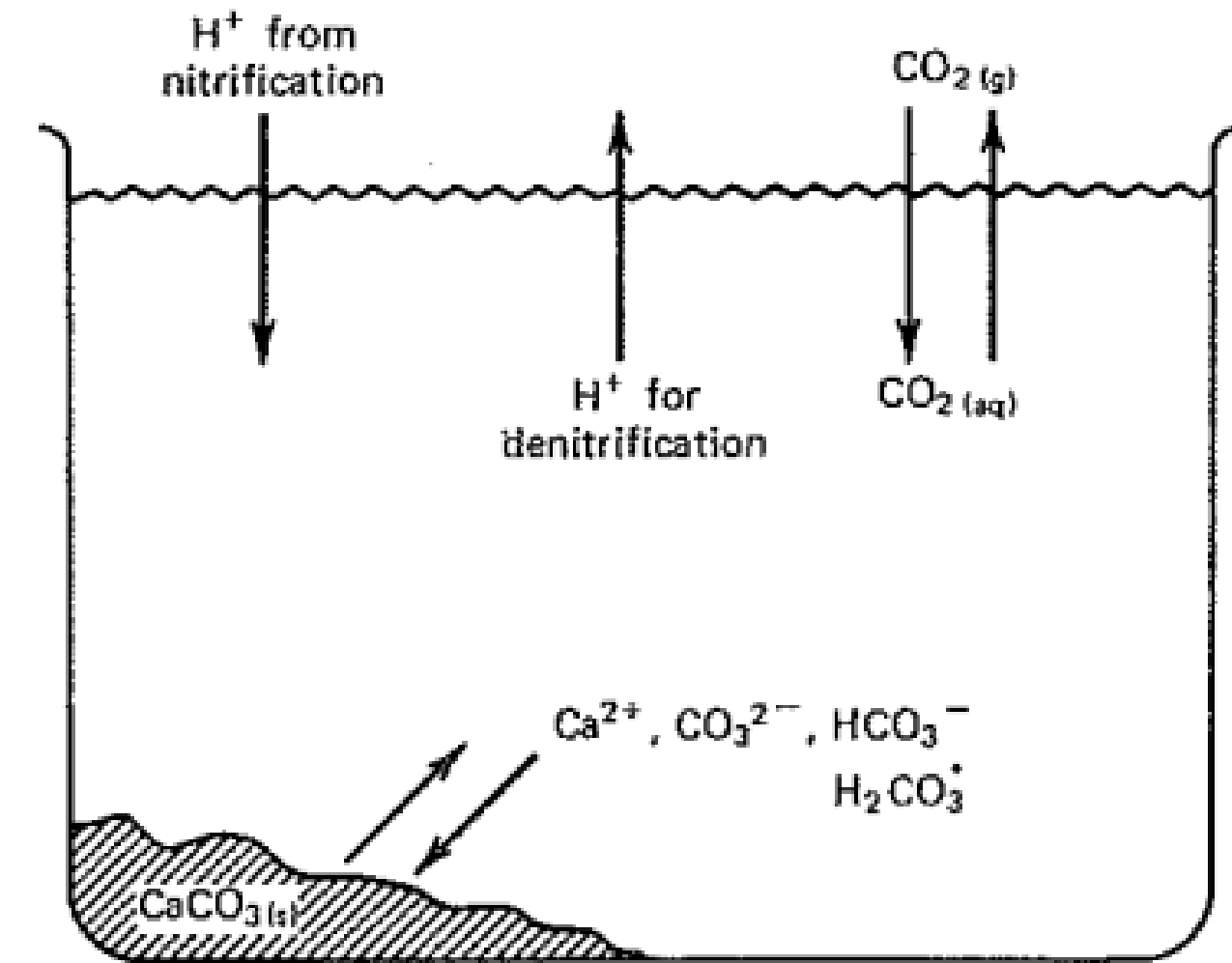
is followed. On occasion the rate of dissolution of the substance may be affected by the formation of surface complexes.

4.3 Equilibrium calculations

Heterogeneous calculations only apply when there is an excess of the solid of interest present. If solid is absent, the equilibrium calculations are not valid for predicting solution composition.

4.3.1. The solubility product

Solubility product is the name given to the equilibrium constant that describes the reaction by which a precipitate dissolves in pure water to form its constituent ions,



4.3 Equilibrium calculations

4.3.1. The solubility product

Generally stated, the equilibrium constant, K_{so} , is

$$K_{so} = \frac{\{A^{y+}\}^z \{B^{z-}\}^y}{\{A_z B_{y(s)}\}}$$

The activity of the solid phase can be taken as unity (see Section 3-3). The concentration product, $^cK_{so}$, has the same form as the equilibrium constant except that the concentrations of the species in solution are used instead of their activities. Accordingly,

$$^cK_{so} = [A^{y+}]^z [B^{z-}]^y = \frac{K_{so}}{(\gamma_{A^{y+}})^z (\gamma_{B^{z-}})^y} \quad (6-5)$$

4.3 Equilibrium calculations

4.3.1. The solubility product

TABLE 6-1 Solubility Product Constants at 25°C^a

Solid	pK _{so}	Solid	pK _{so}
Fe(OH) ₃ (amorph)	38	BaSO ₄	10
FePO ₄	17.9	Cu(OH) ₂	19.3
Fe ₃ (PO ₄) ₂	33	PbCl ₂	4.8
Fe(OH) ₂	14.5	Pb(OH) ₂	14.3
FeS	17.3	PbSO ₄	7.8
Fe ₂ S ₃	88	PbS	27.0
Al(OH) ₃ (amorph)	33	MgNH ₄ PO ₄	12.6
AlPO ₄	21.0	MgCO ₃	5.0
CaCO ₃ (calcite)	8.34	Mg(OH) ₂	10.7
CaCO ₃ (aragonite)	8.22	Mn(OH) ₂	12.8
CaMg(CO ₃) ₂ (dolomite)	16.7	AgCl	10.0
CaF ₂	10.3	Ag ₂ CrO ₄	11.6
Ca(OH) ₂	5.3	Ag ₂ SO ₄	4.8
Ca ₃ (PO ₄) ₂	26.0	Zn(OH) ₂	17.2
CaSO ₄	4.59	ZnS	21.5
SiO ₂ (amorph)	2.7		

^a Equilibrium constants for the reaction $A_zB_{y(s)} \rightleftharpoons zA^{r+} + yB^{x-}$.

4.3 Equilibrium calculations

4.3.1. The solubility product

Example 6-1

Calculate the solubility in mg/liter of CaF_2 in pure water at 25°C , neglecting ionic strength effects.



Solution

$$(1) \quad K_{so} = 5 \times 10^{-11} = {}^cK_{so} = [\text{Ca}^{2+}][\text{F}^-]^2$$

Each mole of CaF_2 that dissolves in pure water produces 1 mole of Ca^{2+} and 2 moles of F^- . Therefore, if we set S = solubility in moles/liter, we can write

$$(2) \quad [\text{Ca}^{2+}] = S$$

and

$$(3) \quad [\text{F}^-] = 2S$$

From (1), (2), and (3)

$$S(2S)^2 = 4S^3 = 5 \times 10^{-11}$$

Therefore, $S = 2.32 \times 10^{-4}$ moles/liter.

The molecular weight of CaF_2 is 78; therefore,

$$S = 78 \text{ g/mole} \times 2.32 \times 10^{-4} \text{ moles/liter} \times 10^3 \text{ mg/g} = 18.1 \text{ mg/liter.}$$

4.3 Equilibrium calculations

4.3.2. Effect of temperature on solubility

Temperature affects both the equilibrium position of the precipitation reaction and the reaction rate. In general, solubility increases with increasing temperature with a few notable exceptions such as CaCO_3 , $\text{Ca}_3(\text{PO}_4)_2$, CaSO_4 , and FePO_4 , which are of importance in water chemistry. The effect of temperature on equilibrium is presented in Section 3-4 and the following example illustrates the effect on a typical precipitation reaction.

4.3 Equilibrium calculations

4.3.2. Effect of temperature on solubility

Example 6-2

Find K_{so} for FePO_4 at 50°C given the following information at 25°C for the reaction,



$$\Delta H^\circ = -18.7 \text{ kcal}$$

$$\Delta G^\circ = 24.4 \text{ kcal}$$

Solution

Knowing $\Delta G^\circ = -RT \ln K$, we find

$$K_{so,25^\circ\text{C}} = \exp\left[\frac{-\Delta G^\circ}{RT}\right] = 1.19 \times 10^{-18}$$

We can then use the relationship derived by Van't Hoff from Chapter 3,

$$\ln \frac{K_1}{K_2} = \frac{\Delta H^\circ}{R} \left[\frac{1}{T_2} - \frac{1}{T_1} \right]$$

Setting

$$K_1 = K_{so,50^\circ\text{C}} \quad \text{and} \quad K_2 = K_{so,25^\circ\text{C}}$$

Then

$$T_1 = 323^\circ\text{K} \quad \text{and} \quad T_2 = 298^\circ\text{K}$$

$$\ln \frac{K_{so,50^\circ\text{C}}}{K_{so,25^\circ\text{C}}} = \left[\frac{-18.7}{1.987 \times 10^{-3}} \left[\frac{1}{298} - \frac{1}{323} \right] \right] = -2.444$$

$$K_{so,50^\circ\text{C}} = 1.03 \times 10^{-19}$$

Thus the $\text{FePO}_{4(s)}$ solubility decreases with increasing temperature.

4.3 Equilibrium calculations

4.3.3. The common ion effect

- When a solution contains an ion that is the same as one of the ions which result from the dissolution of the solid, the solubility of the solid will be less than that when the solid dissolves in pure water. The modification of solubility is called the *common ion effect*. When calculating the solubility of the solid in such a solution, we need to remember that the solubility is based on ions or species that come only from the dissolution of the solid.

4.3 Equilibrium calculations

4.3.3. The common ion effect

Example 6-3

The solubility of stannous fluoride, $\text{SnF}_{2(s)}$, in water at 20°C is $0.012 \text{ g}/100 \text{ ml}$. What is the solubility of $\text{SnF}_{2(s)}$ in a 0.08 M NaF solution neglecting ionic strength effects?

Solution

Let us first calculate the concentration product for the reaction,



$$K_{so} = {}^cK_{so} = [\text{Sn}^{2+}][\text{F}^-]^2$$

The molecular weight of SnF_2 is 156.6

$$\begin{aligned} S &= 0.012 \frac{\text{g}}{100 \text{ ml}} \times \frac{1000 \text{ ml}}{1 \text{ liter}} \times \frac{1}{156.6} \frac{\text{mole}}{\text{g}} \\ &= 7.7 \times 10^{-4} \text{ moles/liter} \end{aligned}$$

Therefore,

$$\begin{aligned} {}^cK_{so} &= (7.7 \times 10^{-4})(7.7 \times 10^{-4} \times 2)^2 \\ &= 1.8 \times 10^{-9} \end{aligned}$$

Now let S' = moles/liter of SnF_2 , which will dissolve. If 8×10^{-2} mole of F^- /liter is originally present,

$$\begin{aligned} [\text{F}^-] &= (8 \times 10^{-2} + 2S') \\ [\text{Sn}^{2+}] &= S' \end{aligned}$$

and

$${}^cK_{so} = (8 \times 10^{-2} + 2S')^2(S') = 1.8 \times 10^{-9}$$

For this example let us assume that $2S'$ is negligible compared to 8×10^{-2} . Then

$${}^cK_{so} = [S'] [0.08]^2 = 1.8 \times 10^{-9}$$

$$S' = \frac{1.8 \times 10^{-9}}{(0.08)^2} = 2.8 \times 10^{-7} \text{ moles/liter}$$

Checking our assumption that $8 \times 10^{-2} > 2S'$, we see that $2S' = 5.6 \times 10^{-7} \ll 8 \times 10^{-2}$. The assumption is valid.

4.3 Equilibrium calculations

4.3.3. The common ion effect

Example 6-4

What is the solubility of SnF_2 in a 10^{-4} mole/liter NaF solution? Neglect ionic strength effects.

Solution

From Example 6-3,

$$K_{so} = 3.5 \times 10^{-9}$$

Let $[\text{Sn}^{2+}] = S'$. Then

$$[\text{F}^-] = 10^{-4} + 2S'$$

and

$$K_{so} = (S')(10^{-4} + 2S')^2$$

Assuming that $2S' \ll 10^{-4}$, we obtain

$$S' = \frac{3.5 \times 10^{-9}}{(10^{-4})^2} = 0.35$$

Checking the assumption that $2S' \ll 10^{-4}$, we find $2S' = 7 \times 10^{-4}$, which is not $\ll 10^{-4}$. The assumption is not valid. We will therefore proceed as we did in the solution of acid-base equilibrium problems and make the opposite assumption, that is, assume that $2S' \gg 10^{-4}$. Then

$$S' = 9.6 \times 10^{-4}$$

Checking the assumption, we find $2S' = 19.2 \times 10^{-4} \gg 10^{-4}$ so that the assumption is acceptable and the solubility is 9.6×10^{-4} mole/liter.

4.3 Equilibrium calculations

4.3.4. Conditional solubility product

The conditional solubility product has the form

$$P_s = (C_{T,M})(C_{T,A})$$

where

P_s = conditional solubility product

$C_{T,M}$ = total concentration of the metal ion M in all of its complex forms and

$C_{T,A}$ = total concentration of the anion A in all of its forms

The value of P_s may change as a function of solution properties such as pH and concentration of complex-forming species because the fractions of $C_{T,M}$ and $C_{T,A}$ present as the free cation and anion may vary as a function of these properties.

The conditional solubility product and K_{so} , the solubility product, are related. For any set of conditions, if α_M = fraction of $C_{T,M}$ present as the free metal ion, $[M^+] = \alpha_M (C_{T,M})$, and if α_A = fraction of $C_{T,A}$ present as free anion, $[A^-] = \alpha_A (C_{T,A})$. Then

$$K_{so} = [M^+][A^-]$$

$$K_{so} = \alpha_M \alpha_A (C_{T,M})(C_{T,A})$$

$$K_{so} = \alpha_M \alpha_A P_s$$

4.3 Equilibrium calculations

4.3.4. Conditional solubility product

Example 6-5

Will $\text{CaCO}_{3(s)}$ precipitate at pH 9.1 from a solution containing $C_{T,\text{Ca}} = 10^{-2} \text{ M}$, $C_{T,\text{CO}_3} = 10^{-3} \text{ M}$? The ionic strength is 10^{-2} and $P_s = 10^{-6.7}$ at pH 9.1.

Solution

$$(C_{T,\text{Ca}})(C_{T,\text{CO}_3}) = 10^{-2} \times 10^{-3} = 10^{-5}$$

Since $P_s = 10^{-6.7} < 10^{-5}$, the solution is supersaturated and $\text{CaCO}_{3(s)}$ will precipitate.

4.3 Equilibrium calculations



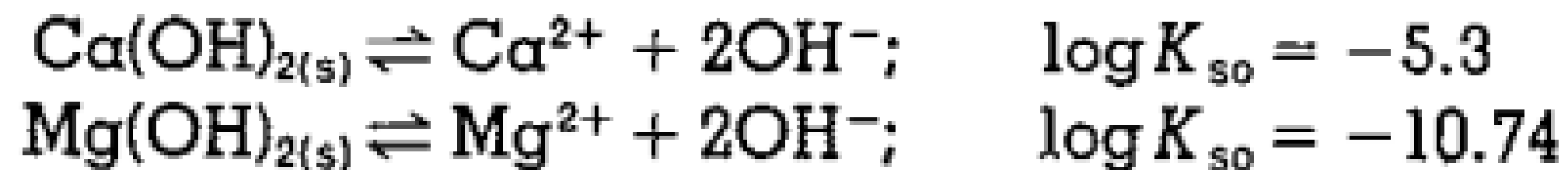
Độ tan của 1 chất có liên quan đến pH hay không?
Nếu có thì liên quan như thế nào?



4.3 Equilibrium calculations

4.3.5. Log concentration diagrams

The presentation of precipitation equilibria on log concentration diagrams greatly aids the solution of problems and the understanding of these equilibria. For example, consider $\text{Ca(OH)}_{2(s)}$ and sparingly soluble $\text{Mg(OH)}_{2(s)}$, both of which are important in water treatment; $\text{Ca(OH)}_{2(s)}$ or hydrated (slaked) lime is commonly used to bring about the precipitation of Ca^{2+} as $\text{CaCO}_{3(s)}$ and Mg^{2+} as $\text{Mg(OH)}_{2(s)}$. The equilibrium relationships at 25°C are



Sparingly: ít tan
Slaked lime: vôi sống

4.3 Equilibrium calculations

4.3.5. Log concentration diagrams

Assuming that the activity effects are negligible, that is, activity coefficients are unity:⁷

$$[\text{Ca}^{2+}][\text{OH}^-]^2 = 10^{-5.3}$$

or

$$\log [\text{Ca}^{2+}] + 2 \log [\text{OH}^-] = -5.3$$

or

$$-\log [\text{Ca}^{2+}] + 2\text{pOH} = 5.3 \quad (6-7)$$

Similarly, for $\text{Mg}(\text{OH})_{2(s)}$,

$$-\log [\text{Mg}^{2+}] + 2\text{pOH} = 10.74 \quad (6-8)$$

Equations 6-7 and 6-8 can be plotted as a log concentration diagram (Fig. 6-3). This diagram can be used handily to solve a variety of problems.

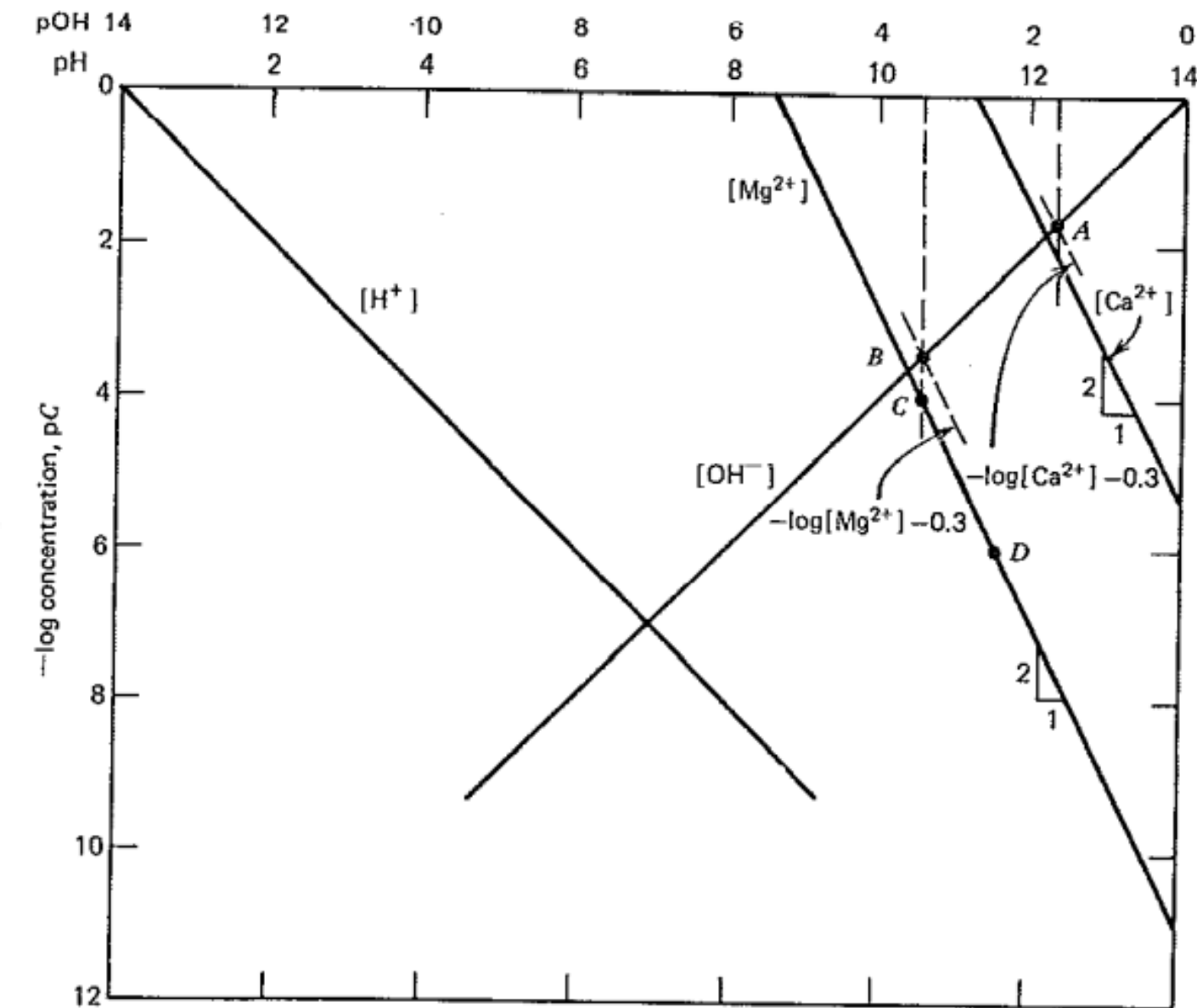


Fig. 6-3. The pC-pH diagram for Ca^{2+} in equilibrium with $\text{Ca}(\text{OH})_{2(s)}$ and Mg^{2+} in equilibrium with $\text{Mg}(\text{OH})_{2(s)}$ at 25°C.

4.3 Equilibrium calculations

4.3.5. Log concentration diagrams

Example 6-6

Find the solubility of (1) $\text{Ca}(\text{OH})_{2(s)}$ and (2) $\text{Mg}(\text{OH})_{2(s)}$ in distilled water.

Solution

(1) $\text{Ca}(\text{OH})_{2(s)}$

In addition to the equilibrium relationship for $\text{Ca}(\text{OH})_{2(s)}$, Eq. 6-7, and K_w for water, which are presented in Fig. 6-3, the charge balance equation must be satisfied, that is,

$$2[\text{Ca}^{2+}] + [\text{H}^+] = [\text{OH}^-]$$

Assuming that $[\text{H}^+] \ll 2[\text{Ca}^{2+}]$,

$$2[\text{Ca}^{2+}] = [\text{OH}^-]$$

$$0.3 + \log [\text{Ca}^{2+}] = \log [\text{OH}^-]$$

or

$$\text{pOH} = -\log [\text{Ca}^{2+}] - 0.3$$

Plotting the line $(-\log [\text{Ca}^{2+}] - 0.3)$, we find this equation is satisfied at point A in Fig. 6-3 ($\text{pOH} = 1.7$). The assumption that $[\text{H}^+] \ll 2[\text{Ca}^{2+}]$ is valid at this point.

When $\text{pOH} = 1.7$, from Fig. 6-3 we find $-\log [\text{Ca}^{2+}] = 2$. Since $S = [\text{Ca}^{2+}]$,

$$S = 10^{-2} \text{ M}$$

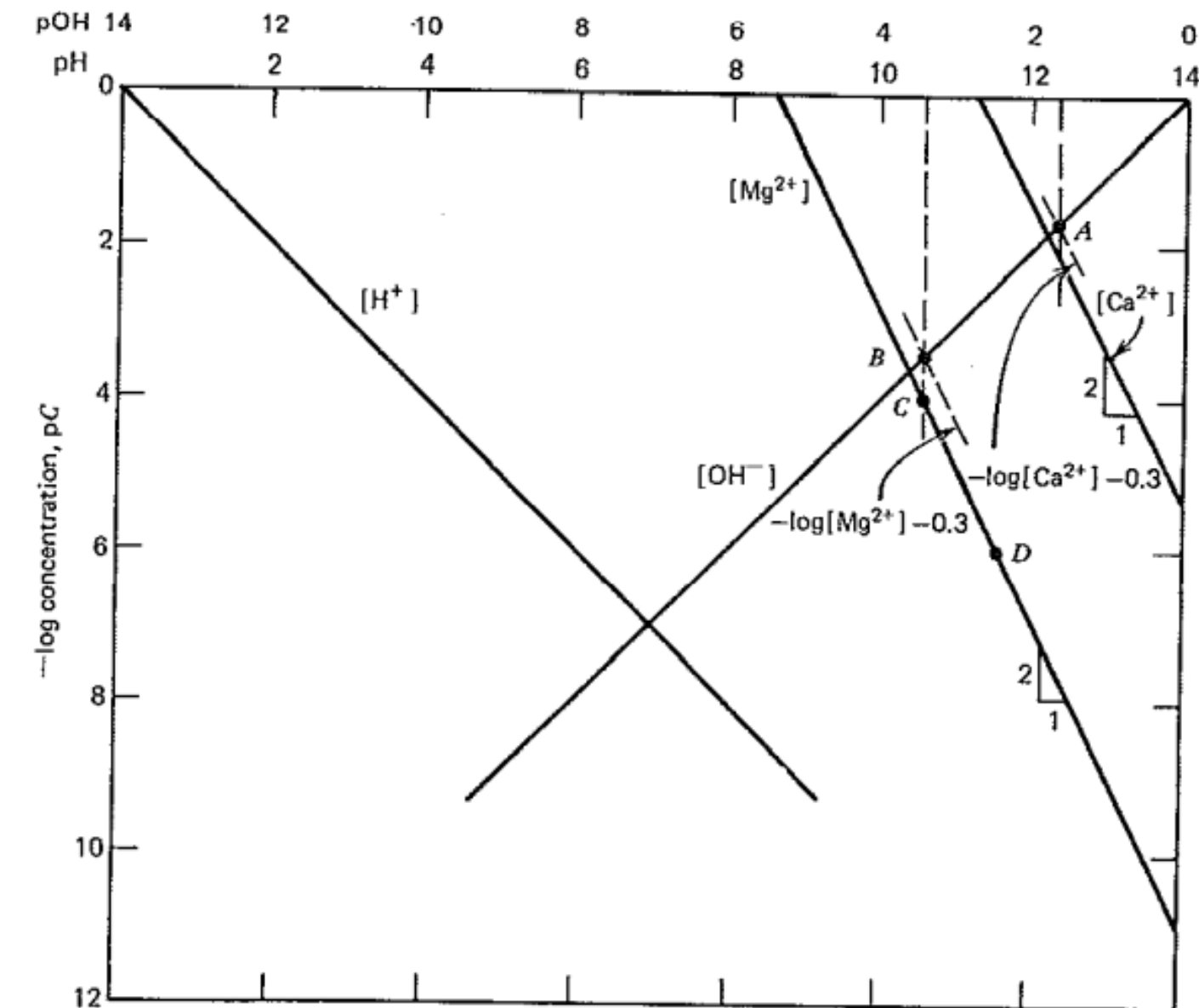


Fig. 6-3. The pC-pH diagram for Ca^{2+} in equilibrium with $\text{Ca}(\text{OH})_{2(s)}$ and Mg^{2+} in equilibrium with $\text{Mg}(\text{OH})_{2(s)}$ at 25°C .

4.3 Equilibrium calculations

4.3.5. Log concentration diagrams

Example 6-6

Find the solubility of (1) $\text{Ca}(\text{OH})_{2(s)}$ and (2) $\text{Mg}(\text{OH})_{2(s)}$ in distilled water.

(2) $\text{Mg}(\text{OH})_{2(s)}$

Similarly, the charge balance for dissolution of $\text{Mg}(\text{OH})_{2(s)}$ in distilled water is

$$2[\text{Mg}^{2+}] + [\text{H}^+] = [\text{OH}^-]$$

which is satisfied at point B in Fig. 6-3. At this point, $\text{pOH} = 3.5$. When $\text{pOH} = 3.5$, $-\log [\text{Mg}^{2+}] = 3.8$. Thus

$$S = [\text{Mg}^{2+}] = 10^{-3.8}$$

or

$$S = 1.6 \times 10^{-4} \text{ M}$$

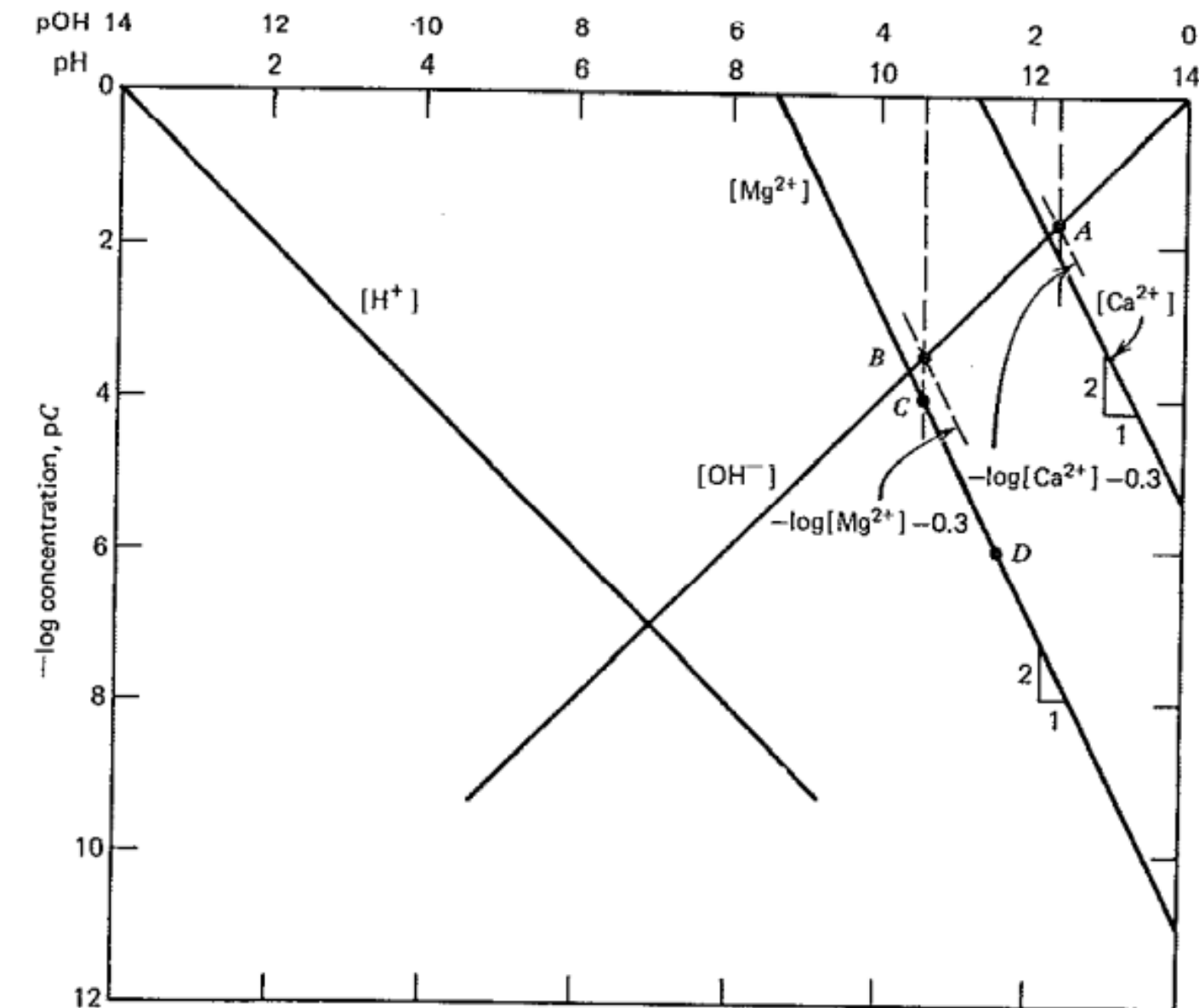


Fig. 6-3. The pC-pH diagram for Ca^{2+} in equilibrium with $\text{Ca}(\text{OH})_{2(s)}$ and Mg^{2+} in equilibrium with $\text{Mg}(\text{OH})_{2(s)}$ at 25°C.

4.3 Equilibrium calculations

4.3.5. Log concentration diagrams

Example 6-7

Sodium hydroxide is added to a groundwater containing $10^{-4} \text{ M Mg}^{2+}$ and $10^{-3} \text{ M Ca}^{2+}$. At what pH will $\text{Mg}(\text{OH})_{2(s)}$ begin to precipitate? Will $\text{Ca}(\text{OH})_{2(s)}$ precipitate at this pH? At what pH will $[\text{Mg}^{2+}]$ be reduced to 10^{-6} M ? The temperature is 25°C .

Solution

From Fig. 6-3, we find that the $10^{-4} \text{ M Mg}^{2+}$ solution is in equilibrium with $\text{Mg}(\text{OH})_{2(s)}$ only at pH 10.65 (point C). Thus $\text{Mg}(\text{OH})_{2(s)}$ will precipitate when the pH exceeds 10.65. At this pH, the solution is very much undersaturated with $\text{Ca}(\text{OH})_{2(s)}$. (From the diagram we see that more than 1 M Ca^{2+} is required for a saturated solution.)

Again from the diagram in Fig. 6-3, when $[\text{Mg}^{2+}] = 10^{-6} \text{ M}$ and the solution is in equilibrium with $\text{Mg}(\text{OH})_{2(s)}$, the pH is 11.65 (point D).

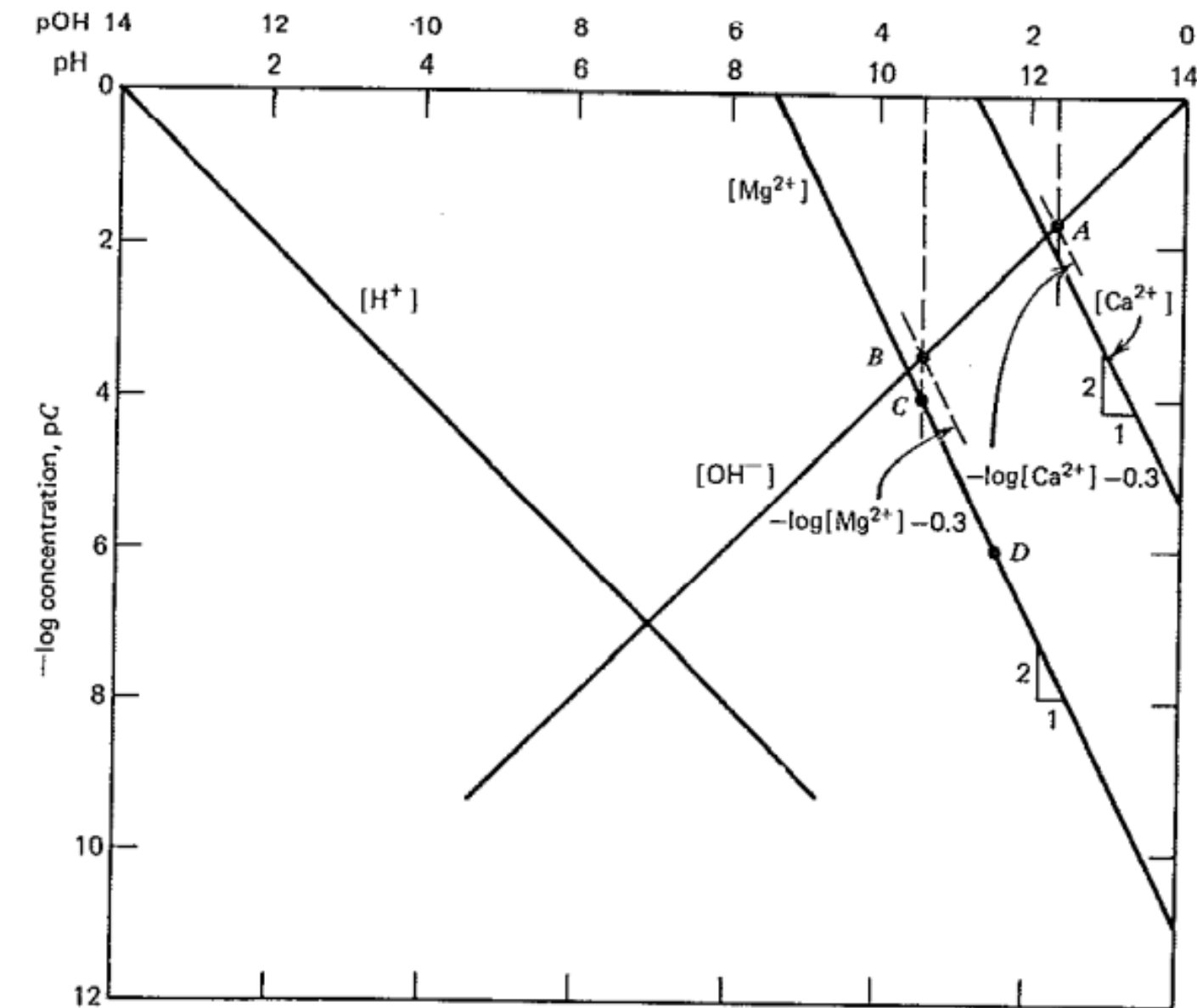
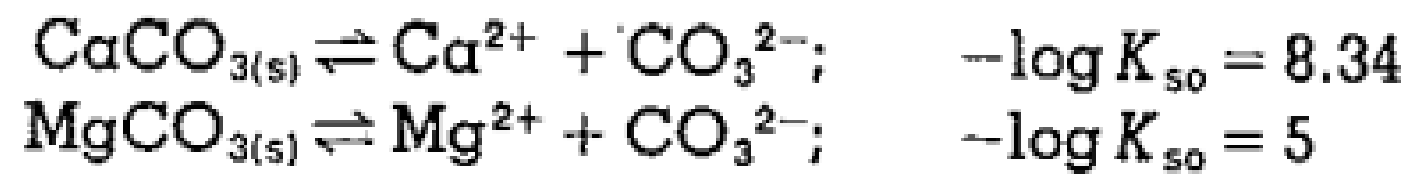


Fig. 6-3. The pC-pH diagram for Ca^{2+} in equilibrium with $\text{Ca}(\text{OH})_{2(s)}$ and Mg^{2+} in equilibrium with $\text{Mg}(\text{OH})_{2(s)}$ at 25°C .

4.3 Equilibrium calculations

4.3.5. Log concentration diagrams

Master variables other than pH and pOH can be used in log concentration diagrams. Diagrams using the negative logarithm of concentrations of other ions are often useful in the solution of solubility problems. As an example consider the carbonate salts of Ca^{2+} and Mg^{2+} :



These equilibria can be represented in linear form as follows:

$$-\log [\text{Ca}^{2+}] - \log [\text{CO}_3^{2-}] = 8.34$$

or

$$-\log [\text{Ca}^{2+}] + \text{pCO}_3 = 8.34 \quad (6-9)$$

and

$$-\log [\text{Mg}^{2+}] + \text{pCO}_3 = 5 \quad (6-10)$$

where $\text{pCO}_3 = -\log [\text{CO}_3^{2-}]$. Since the carbonate species is common to each of the precipitates, it is useful as a master variable. Equations 6-9 and 6-10 are plotted in a log concentration diagram (Fig. 6-4), which can be used to solve problems such as those in Examples 6-8 and 6-9.

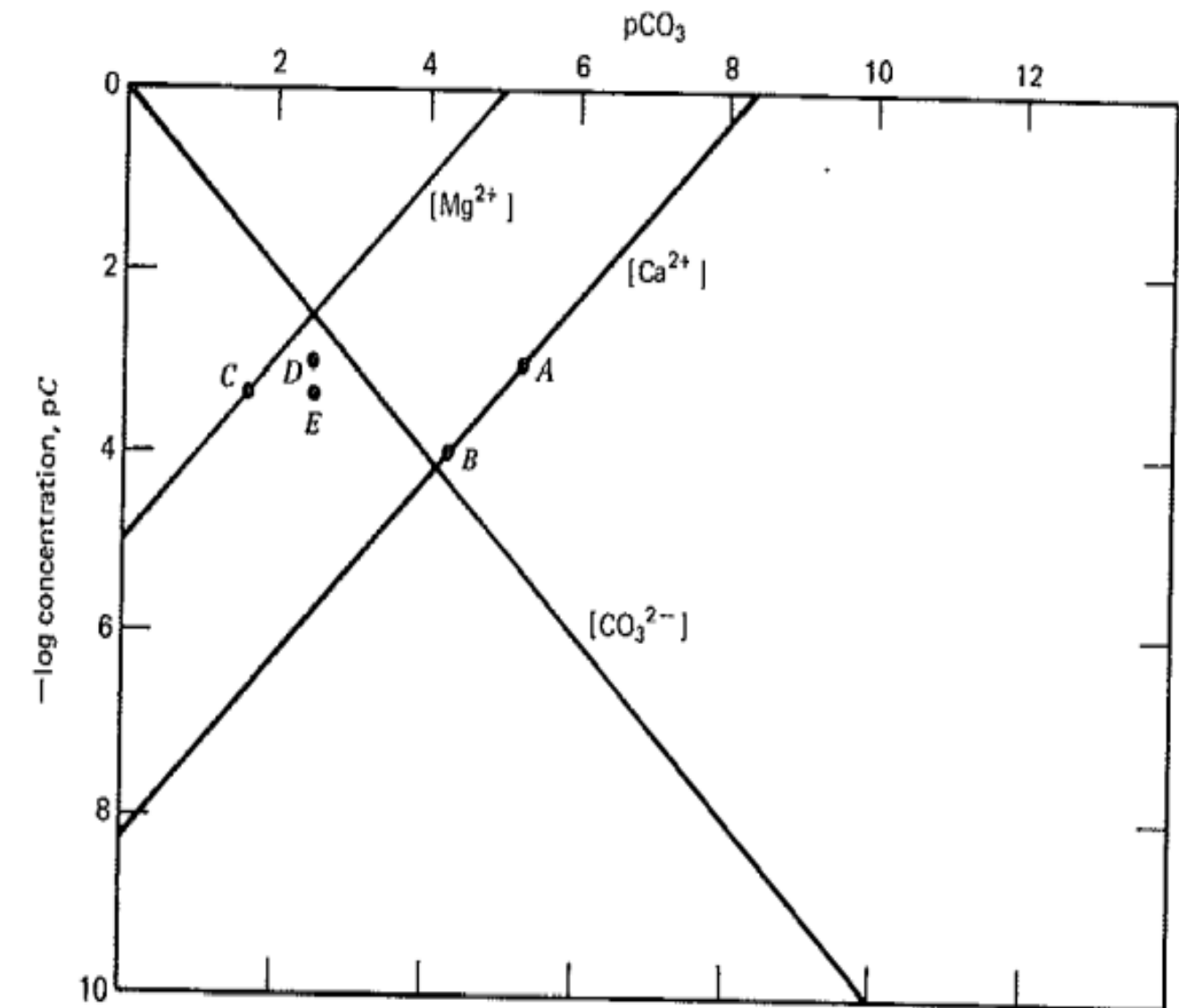


Fig. 6-4. The pC-pH diagram for Mg^{2+} in equilibrium with $\text{MgCO}_{3(s)}$ and Ca^{2+} in equilibrium with $\text{CaCO}_{3(s)}$.

4.3 Equilibrium calculations

4.3.5. Log concentration diagrams

Example 6-8

A surface water contains $[\text{Ca}^{2+}] = 10^{-3} \text{ M}$ and $[\text{Mg}^{2+}] = 5 \times 10^{-4} \text{ M}$. At what concentration of CO_3^{2-} will (1) $[\text{Ca}^{2+}]$, and (2) $[\text{Mg}^{2+}]$ be reduced by one order of magnitude at equilibrium?

Solution

1. 10^{-3} M Ca^{2+} is in equilibrium with $\text{CaCO}_{3(s)}$ when $\text{pCO}_3 = 5.3$ (point A, Fig. 6-4). To reduce the Ca^{2+} concentration to 10^{-4} , pCO_3 must be reduced to 4.3, that is, the CO_3^{2-} concentration must be increased to $10^{-4.3} \text{ M}$ at equilibrium (point B, Fig. 6-4).
2. Similarly for Mg^{2+} , if $[\text{Mg}^{2+}] = 5 \times 10^{-4}$ at equilibrium, $\text{pCO}_3 = 1.7$ ($[\text{CO}_3^{2-}] = 10^{-1.7} \text{ M}$, (point C, Fig. 6-4). When $\text{pCO}_3 = 0.7$, $[\text{Mg}^{2+}] = 5 \times 10^{-5}$.

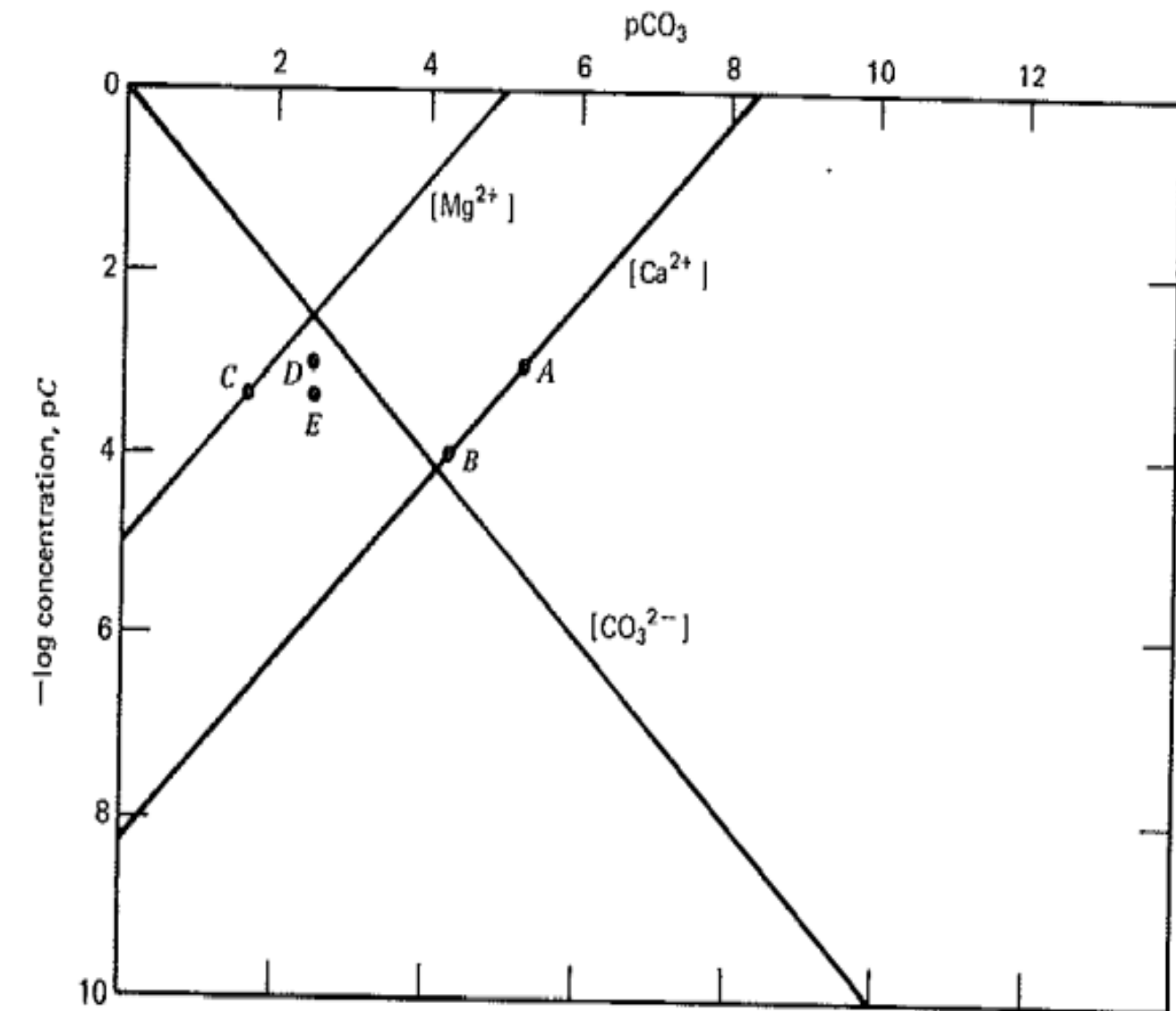


Fig. 6-4. The pC-pH diagram for Mg^{2+} in equilibrium with $\text{MgCO}_{3(s)}$ and Ca^{2+} in equilibrium with $\text{CaCO}_{3(s)}$.

4.3 Equilibrium calculations

4.3.5. Log concentration diagrams

Example 6-9

The pH of the initial solution in Example 6-8 is 8.3 and the total alkalinity is 3×10^{-3} eq/liter. All of the alkalinity is converted to CO_3^{2-} by adding OH^- . After this conversion, is there sufficient CO_3^{2-} present to begin to precipitate the Ca^{2+} and the Mg^{2+} ?

Solution

After conversion of 3×10^{-3} eq/liter of alkalinity (all present as HCO_3^-) to CO_3^{2-} , $\text{pCO}_3 = -\log(3 \times 10^{-3}) = 2.5$. From Fig. 6-4, when $[\text{Ca}^{2+}] = 10^{-3}$ and $\text{pCO}_3 = 2.5$, point D, the solution is supersaturated with CaCO_3 and it will precipitate. When $[\text{Mg}^{2+}] = 5 \times 10^{-4}$ ($-\log[\text{Mg}^{2+}] = 3.3$) and $\text{pCO}_3 = 2.5$, point E, the solution is undersaturated with MgCO_3 and thus it will not precipitate.

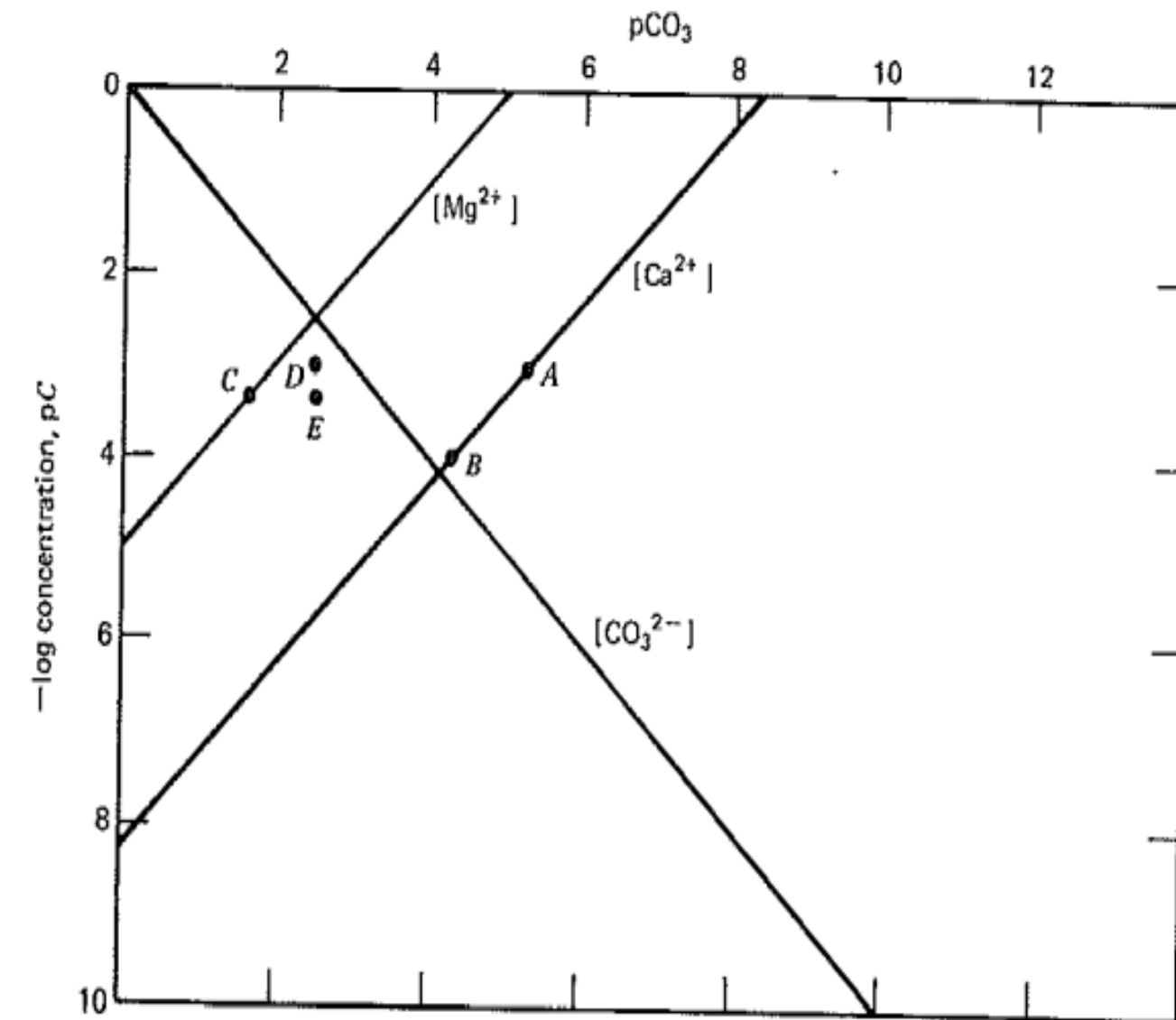


Fig. 6-4. The pC-pH diagram for Mg^{2+} in equilibrium with $\text{MgCO}_{3(s)}$ and Ca^{2+} in equilibrium with $\text{CaCO}_{3(s)}$.

4.4 Solubility of salts of weak acids and bases

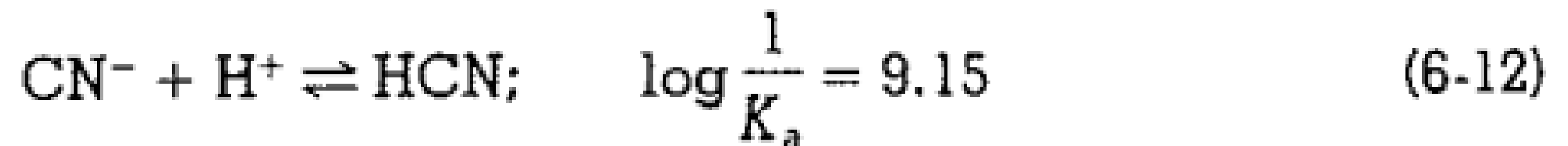
Precipitation and dissolution phenomena become more complicated if the cation of the salt is a weak acid, such as NH_4^+ , or if the anion is a weak base, such as CN^- or CO_3^{2-} . $\text{AgCN}_{(s)}$, for example, dissolves as follows:



and the CN^- ion reacts with water



or alternatively,



Example 6-10 illustrates the effect of protonation of CN^- on the solubility of AgCN .

Further reading

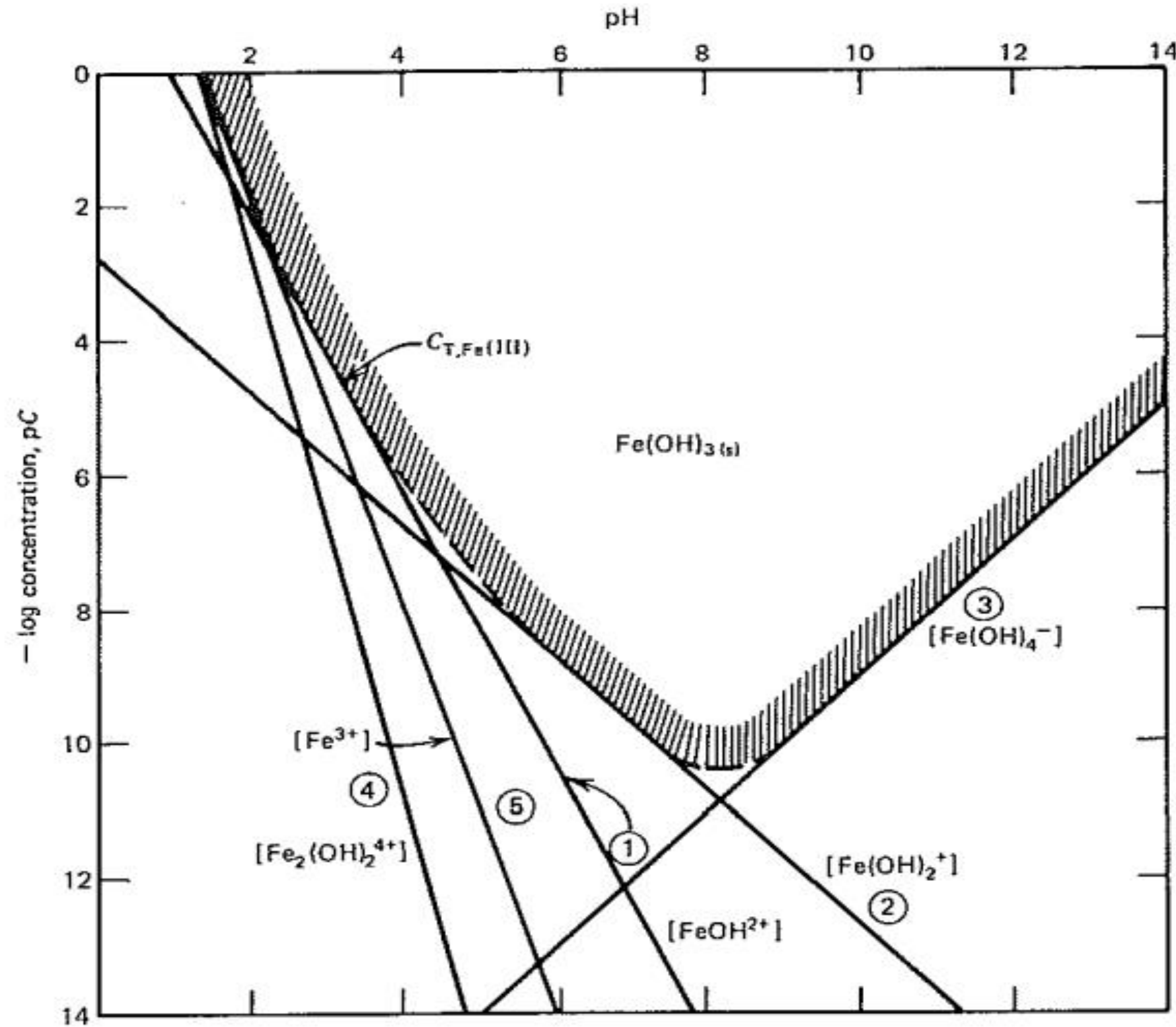


Fig. 6-7. Equilibrium concentrations of hydroxo iron(III) complexes in a solution in contact with freshly precipitated $\text{Fe}(\text{OH})_3(\text{s})$ at 25°C .

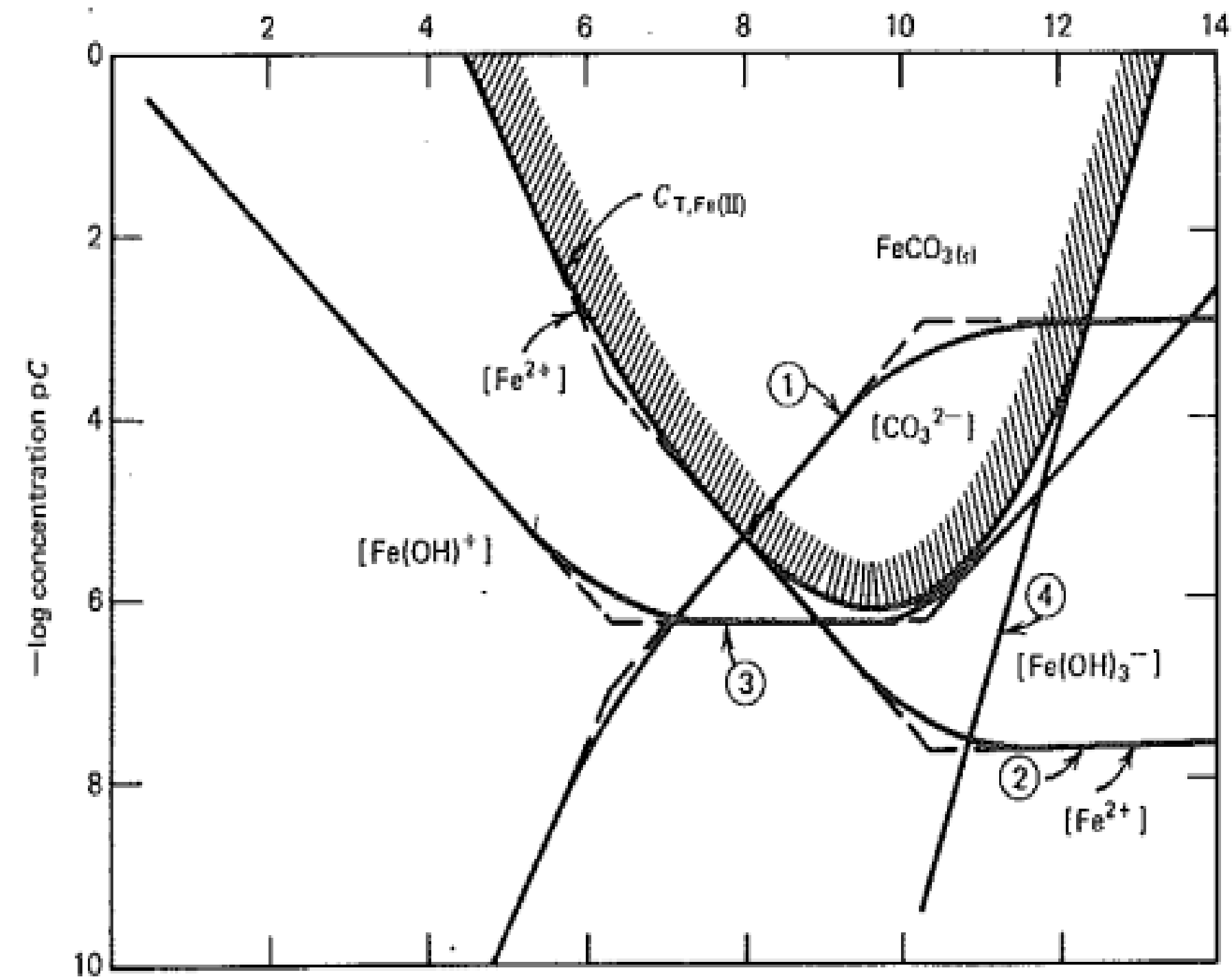


Fig. 6-10. The pC-pH diagram for soluble Fe(II) in equilibrium with $\text{FeCO}_3(\text{s})$. $C_{\text{T,CO}_3} = 10^{-3} \text{ M}$, $T = 25^\circ\text{C}$.

Further reading

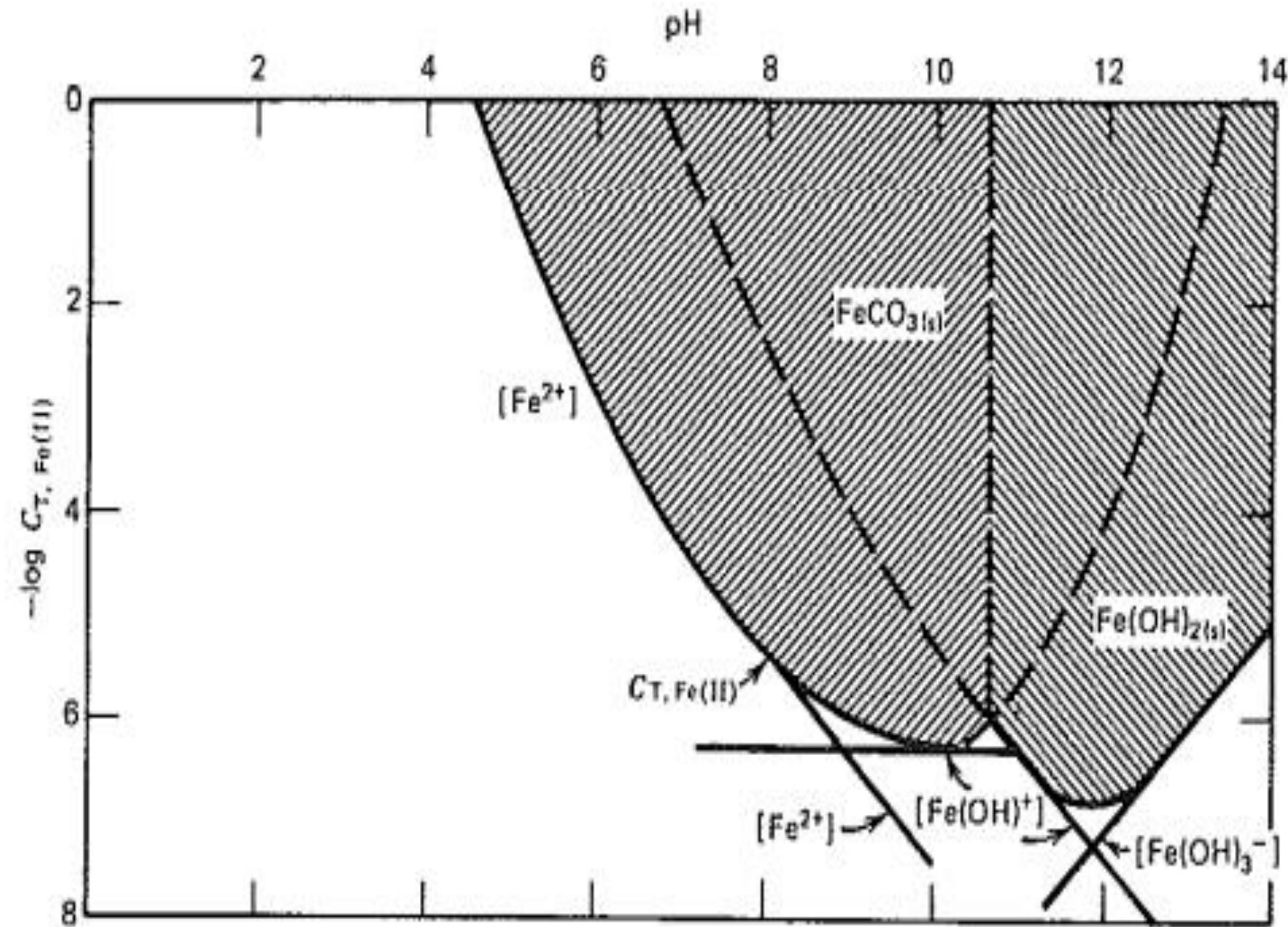


Fig. 6-11. The pC-pH diagram for soluble Fe(II) in equilibrium with $FeCO_{3(s)}$ or $Fe(OH)_{2(s)}$.

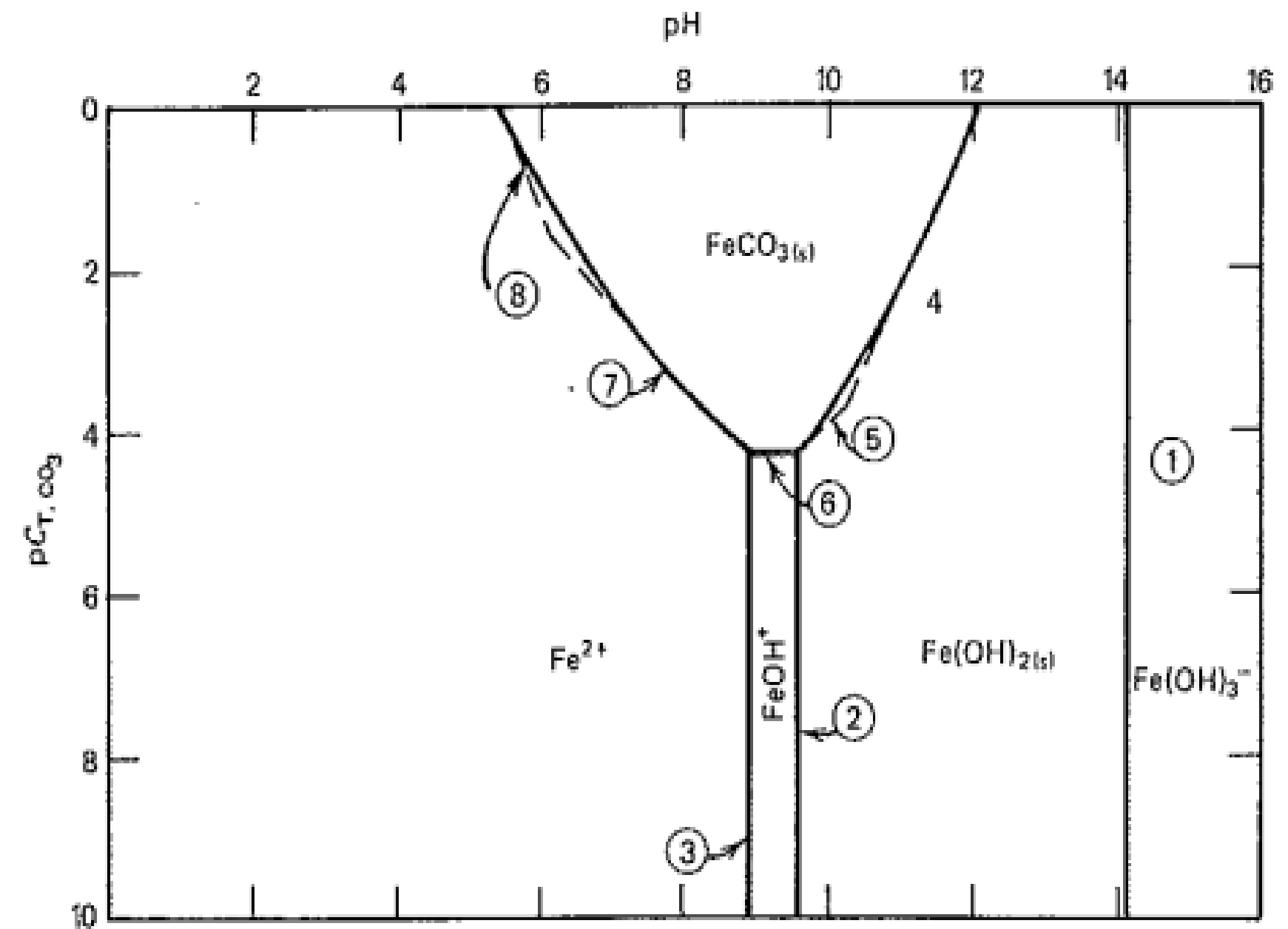


Fig. 6-12. Predominance area diagram for $FeCO_{3(s)}/Fe(OH)_{2(s)}/H_2O$ system at $25^\circ C$, $C_{T, Fe(II)} = 10^{-5} M$.

Further reading

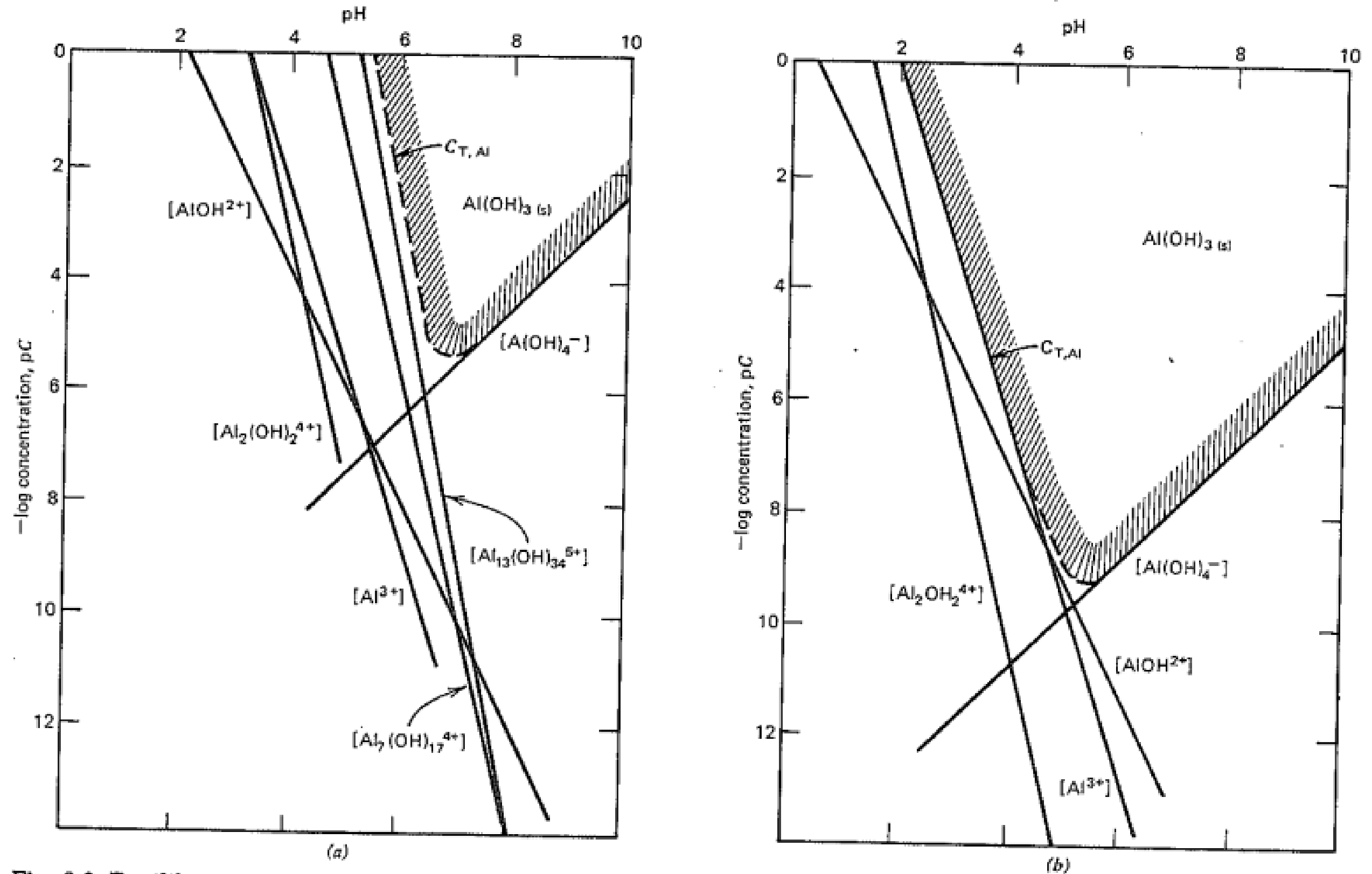


Fig. 6-8. Equilibrium concentrations of hydroxo aluminum(III) complexes in a solution in contact with $\text{Al}(\text{OH})_3(\text{s})$ at 25°C. (a) Freshly precipitated $\text{Al}(\text{OH})_3(\text{s})$; (b) aged $\text{Al}(\text{OH})_3(\text{s})$, gibbsite ($\text{Al}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$).

Further reading

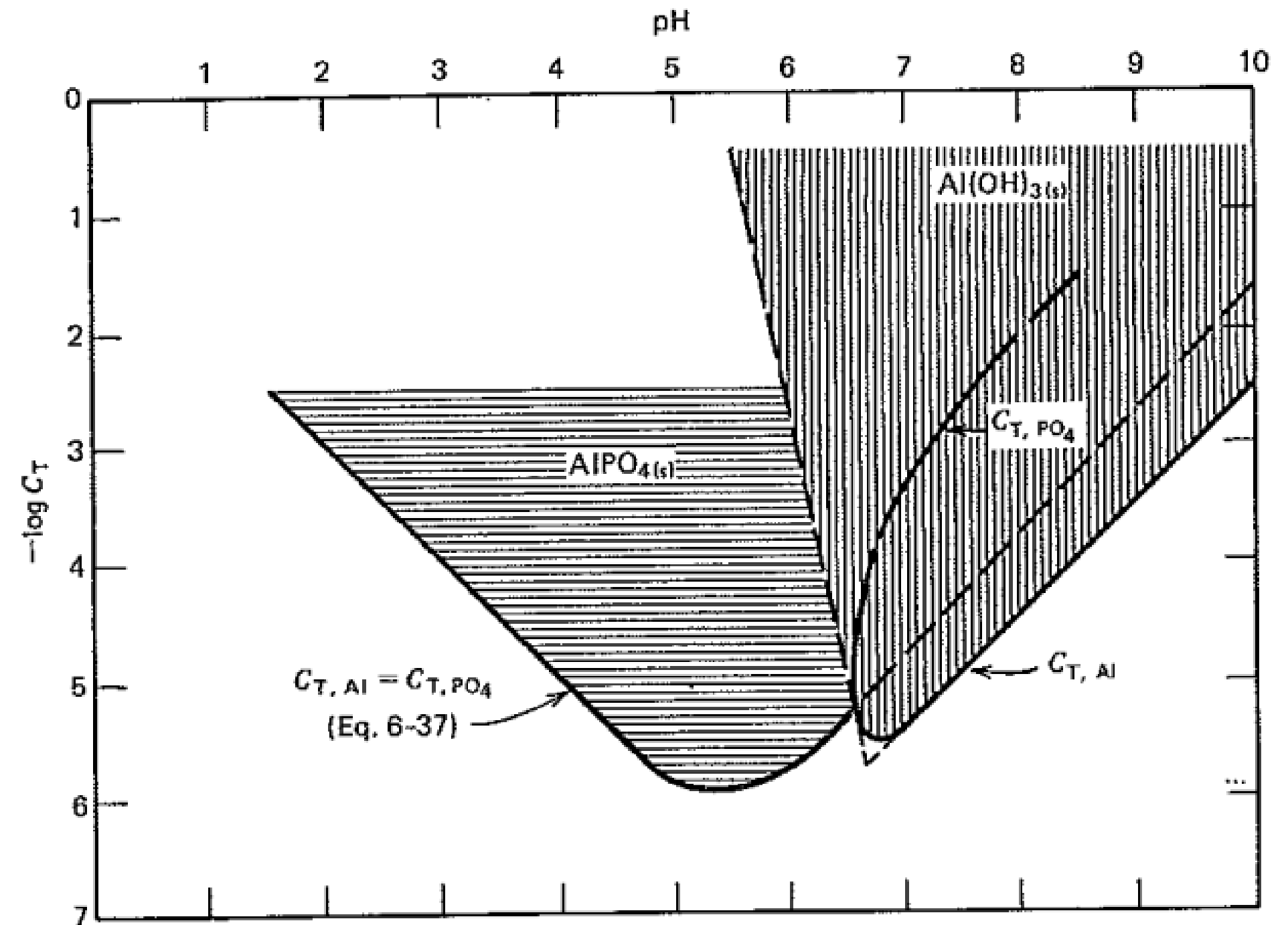
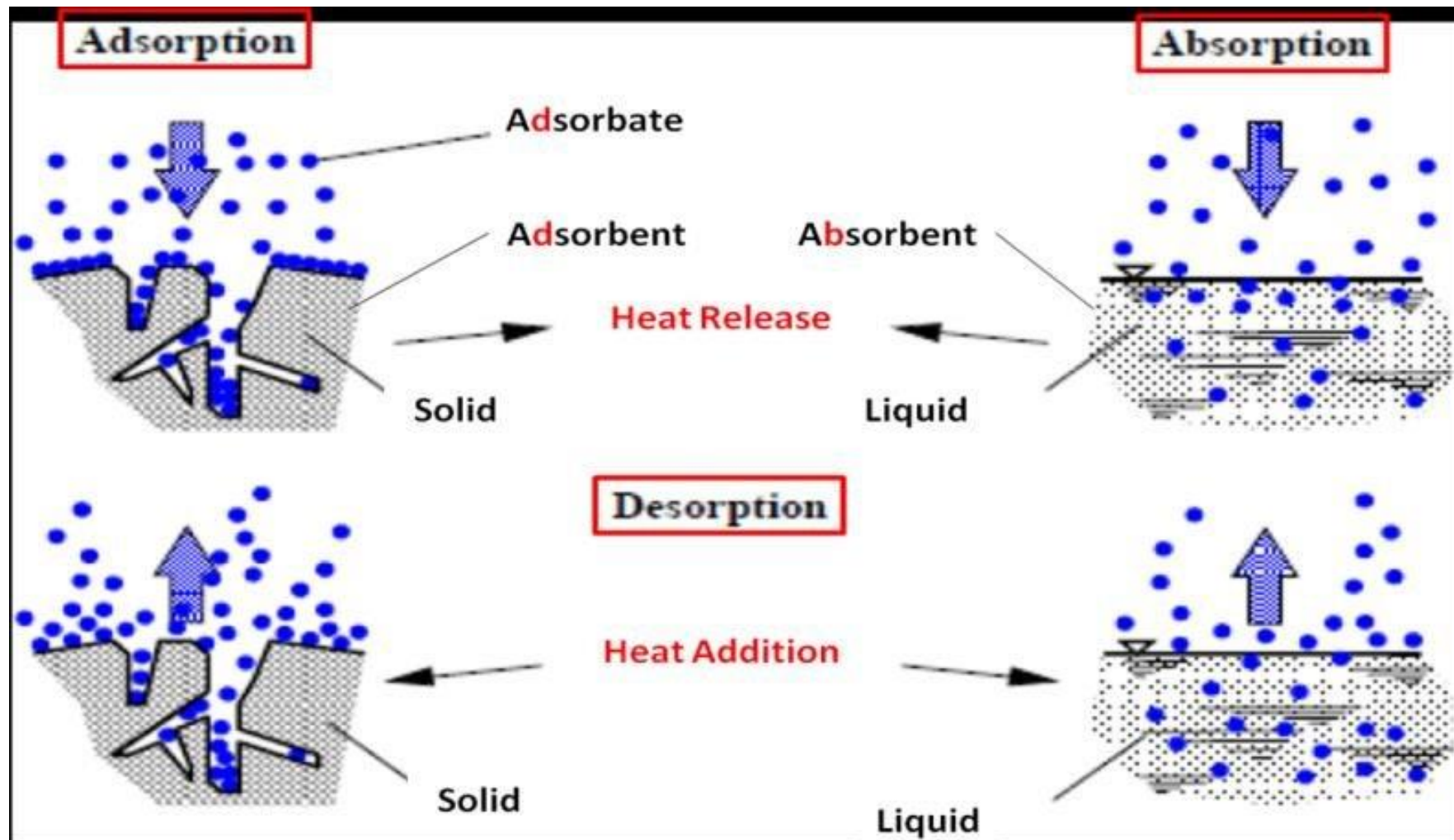


Fig. 6-13. Concentration of phosphate and aluminum that can exist at various pH values when $\text{AlPO}_{4(s)}$ is precipitated from or dissolved in pure water. Concentrations of phosphate and aluminum are controlled by $\text{AlPO}_{4(s)}$ solubility below pH 5.4. Above pH 5.4 $C_{T, \text{Al}}$ is controlled by $\text{Al(OH)}_{3(s)}$ solubility, and thus C_{T, PO_4} concentrations in this range are greater than $C_{T, \text{Al}}$.

Sự kết tủa và hòa tan của các chất phụ thuộc vào những yếu tố nào?

Chapter 6

ABSORPTION AND ADSORPTION (HẤP THU VÀ HẤP PHỤ)



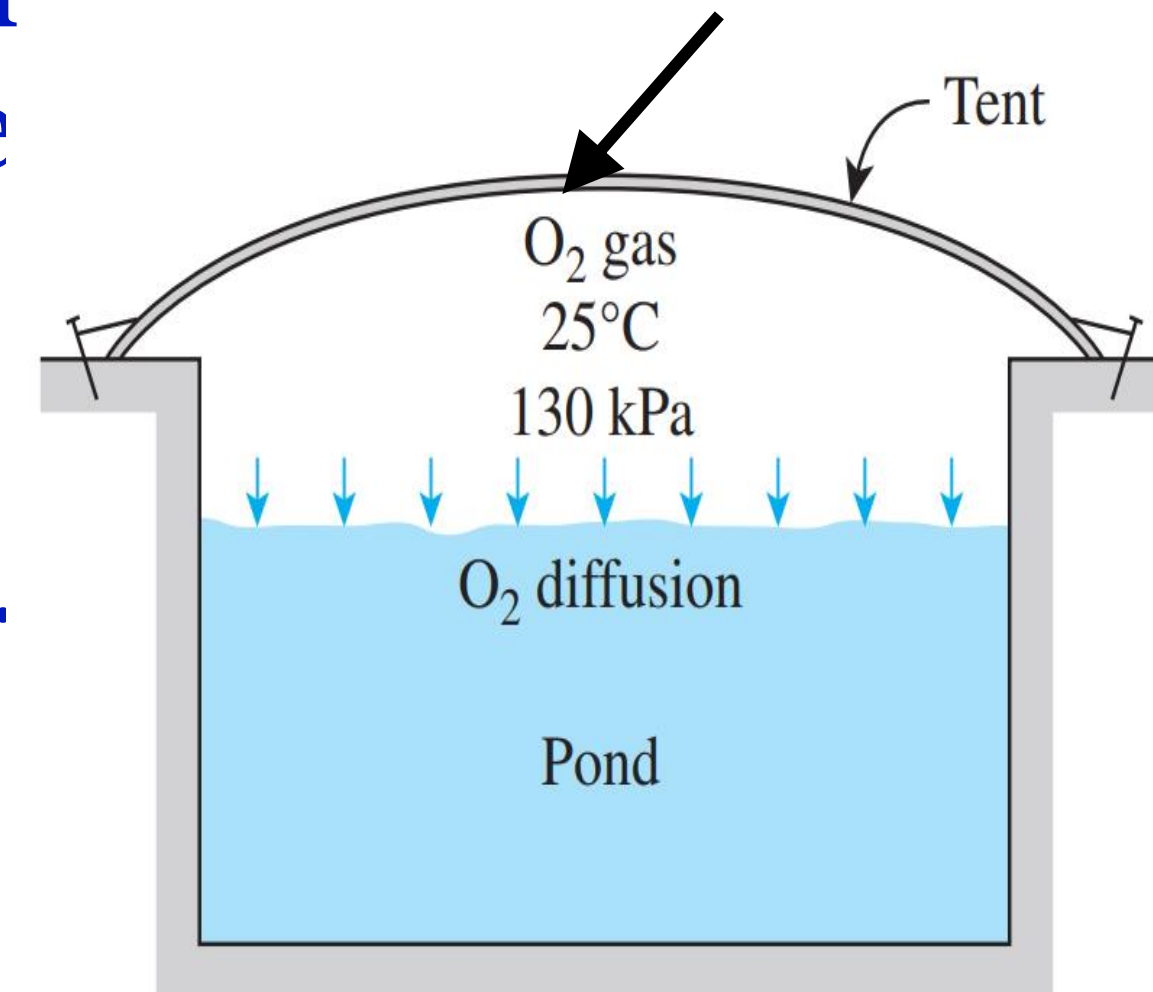
Lecturer: Nguyen Duy Dat
Email: datnd@hcmute.edu.vn

4.1 Definitions

Sorption = Absorption + Adsorption

❖ Absorption

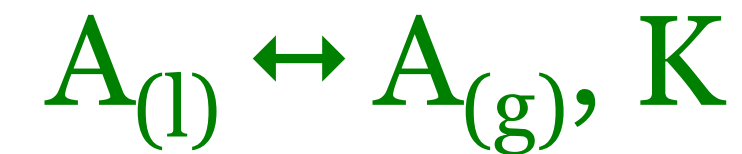
- In chemistry, absorption is the process in which atoms, molecules or ions enter some bulk phase (liquid or solid material) from gas phase.
- Absorption can be chemical or physical process.
 - **Chemical absorption** involves reactions between the absorbing component and the absorbed substance: $\text{CO}_2, \text{SO}_2 \rightarrow \text{Ca}(\text{OH})_2$
 - **Physical process**: A non-reactive process (dissolution of a gas into water): $\text{CO}_2, \text{SO}_2 \rightarrow \text{H}_2\text{O}$



4.1 Definitions

Sorption = Absorption + Adsorption

❖ Absorption



- Physical absorption of gas into water is governed by **Henry's Law**.

Henry's Law:

In physical chemistry, Henry's law is a gas law that states that the amount of dissolved gas in a liquid is proportional to its partial pressure above the liquid.

$$P_i = C_i \times K_H$$
$$\text{or } K_H = \frac{P_i}{C_i}$$

C_i : concentration of a substance in water (M)
 P_i : partial pressure of a substance in gas phase (mol/mol) or atm, mmHg, ...
 K_H : Henry constant (depends on units of C and P)

Henry constant depends on the nature of gas, temperature and pressure.

4.1 Definitions

Sorption = Absorption + Adsorption

❖ Absorption

$$P_i = C_i \times K_H, \text{ or } K_H = \frac{P_i}{C_i}$$

Henry's law constants (gases in water at 298.15 K)

Gas	$\left(\frac{\text{L} \cdot \text{atm}}{\text{mol}}\right)$	$\left(\frac{\text{mol}}{\text{L} \cdot \text{atm}}\right)$	(atm)	(dimensionless)
O ₂	770	1.3×10^{-3}	4.3×10^4	3.2×10^{-2}
H ₂	1300	7.8×10^{-4}	7.1×10^4	1.9×10^{-2}
CO ₂	29	3.4×10^{-2}	1.6×10^3	8.3×10^{-1}
N ₂	1600	6.1×10^{-4}	9.1×10^4	1.5×10^{-2}
He	2700	3.7×10^{-4}	1.5×10^5	9.1×10^{-3}
Ne	2200	4.5×10^{-4}	1.2×10^5	1.1×10^{-2}
Ar	710	1.4×10^{-3}	4.0×10^4	3.4×10^{-2}
CO	1100	9.5×10^{-4}	5.8×10^4	2.3×10^{-2}

When $K_H \uparrow$, solubility
of gas into water $\downarrow$

4.1 Definitions

Sorption = Absorption + Adsorption

❖ Absorption

$$P_i = C_i \times K_H, \text{ or } K_H = \frac{P_i}{C_i}$$

T °C	Gases									
	CO ₂	CO	C ₂ H ₆	C ₂ H ₄	He	H ₂	H ₂ S	CH ₄	N ₂	O ₂
0	0.0728	3.52	1.26	0.552	12.9	5.79	0.0268	2.24	5.29	2.55
10	0.104	4.42	1.89	0.768	12.6	6.36	0.0367	2.97	6.68	3.27
20	0.142	5.36	2.63	1.02	12.5	6.83	0.0483	3.76	8.04	4.01
30	0.186	6.20	3.42	1.27	12.4	7.29	0.0609	4.49	9.24	4.75
40	0.233	6.96	4.23	-	12.1	7.51	0.0745	5.20	10.4	5.35

When Temp ↑, solubility of gas into water ↓

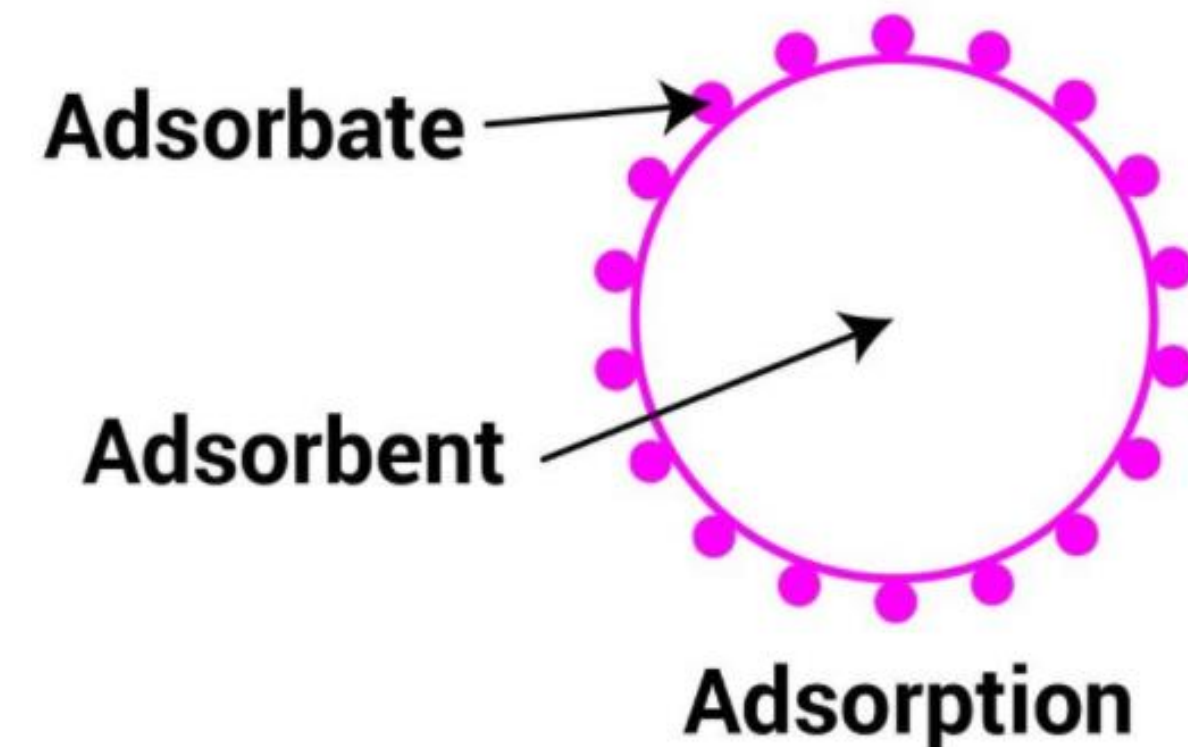
4.1 Definitions

$$\text{Sorption} = \text{Absorption} + \text{Adsorption}$$

❖ Adsorption

Adsorption is the process in which matter is extracted from one phase (gas, liquid) and concentrated at the surface of a solid phase.

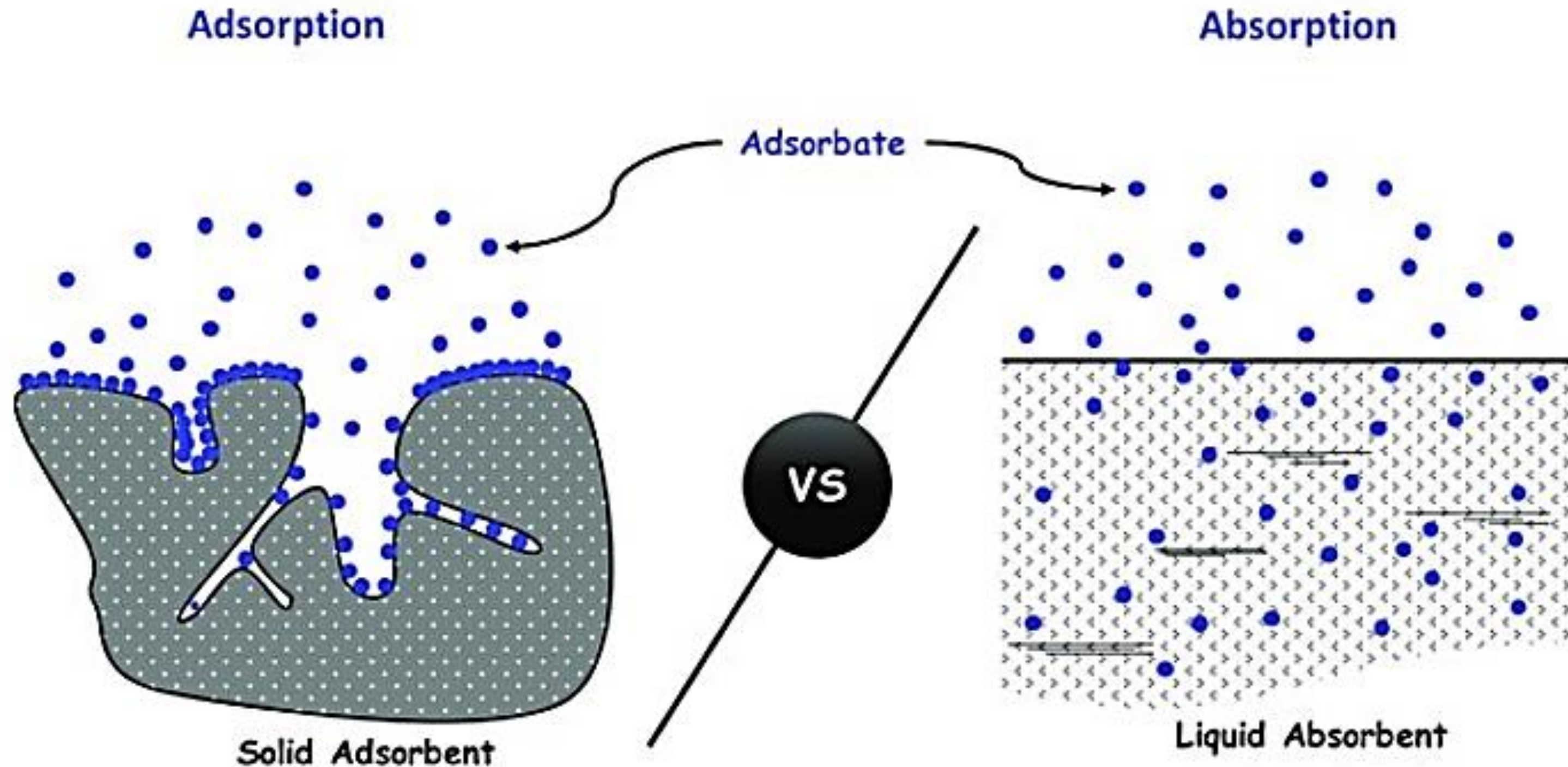
- Adsorbent (chất hấp phụ): solid
- Adsorbate (chất bị hấp phụ): gas or solutes



4.1 Definitions

Sorption = Absorption + Adsorption

❖ Adsorption vs absorption



4.2 Classification

Based on the nature of force existing between adsorbate molecule and adsorbent

Physical adsorption:

If the forces of attraction existing between adsorbate and adsorbent are Vander Waal's forces, the adsorption is called physical adsorption. This type of adsorption is also known as **physisorption or Vander Waal's adsorption**. It can be easily reversed by heating or decreasing the pressure.

Chemical adsorption:

If the forces of attraction existing between adsorbate particles and adsorbent are almost of the same strength as chemical bonds, the adsorption is called chemical adsorption. This type of adsorption is also called as **chemisorption**. This type of adsorption cannot be easily reversed.

4.2 Classification

Characteristics of physical adsorption:

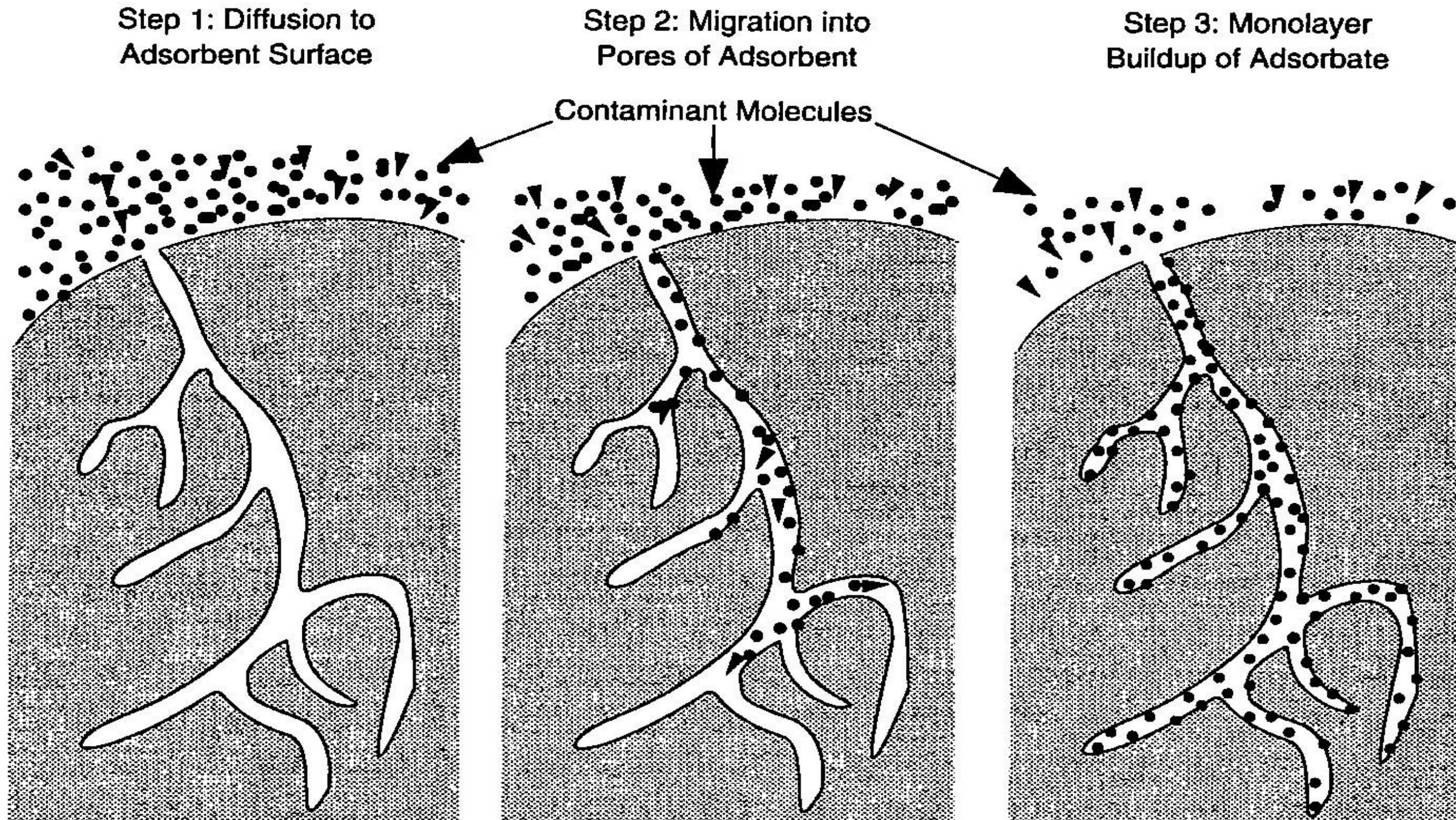
- ✓ The gas molecules adhere to the surface of the solid adsorbent as a result of **intermolecular attractive forces (van der Waals forces)** between them
- ✓ The process is **exothermic (tỏa nhiệt)**: the heat liberated is in the order of the the enthalpy of condensation of vapor (**2-20 kJ/gmole**)
- ✓ The process is **reversible** (recovery of adsorbent material or adsorbed gas is possible) by **increasing the temperature or lowering the adsorbate concentration**.
- ✓ Physical adsorption usually directly **proportional** to the amount of **solid surface area**
- ✓ Adsorbate can be adsorbed on a **monolayer** or a **number of layers**

4.2 Classification

Characteristics of chemical adsorption:

- ✓ Results from a **chemical interaction** between the adsorbate and adsorbent.
Therefore formed bond is much **stronger than that for physical adsorption**
- ✓ Heat liberated during chemisorption is in the range of **20-400 kJ/g mole**
- ✓ It is frequently **irreversible**. On desorption the chemical nature of the **original adsorbate** will have **undergone a change**.
- ✓ **Only a monomolecular layer** of adsorbate appears on the adsorbing medium

4.2 Adsorption process



4.2 Adsorption process



Dose diffusion of adsorbate affect adsorption?

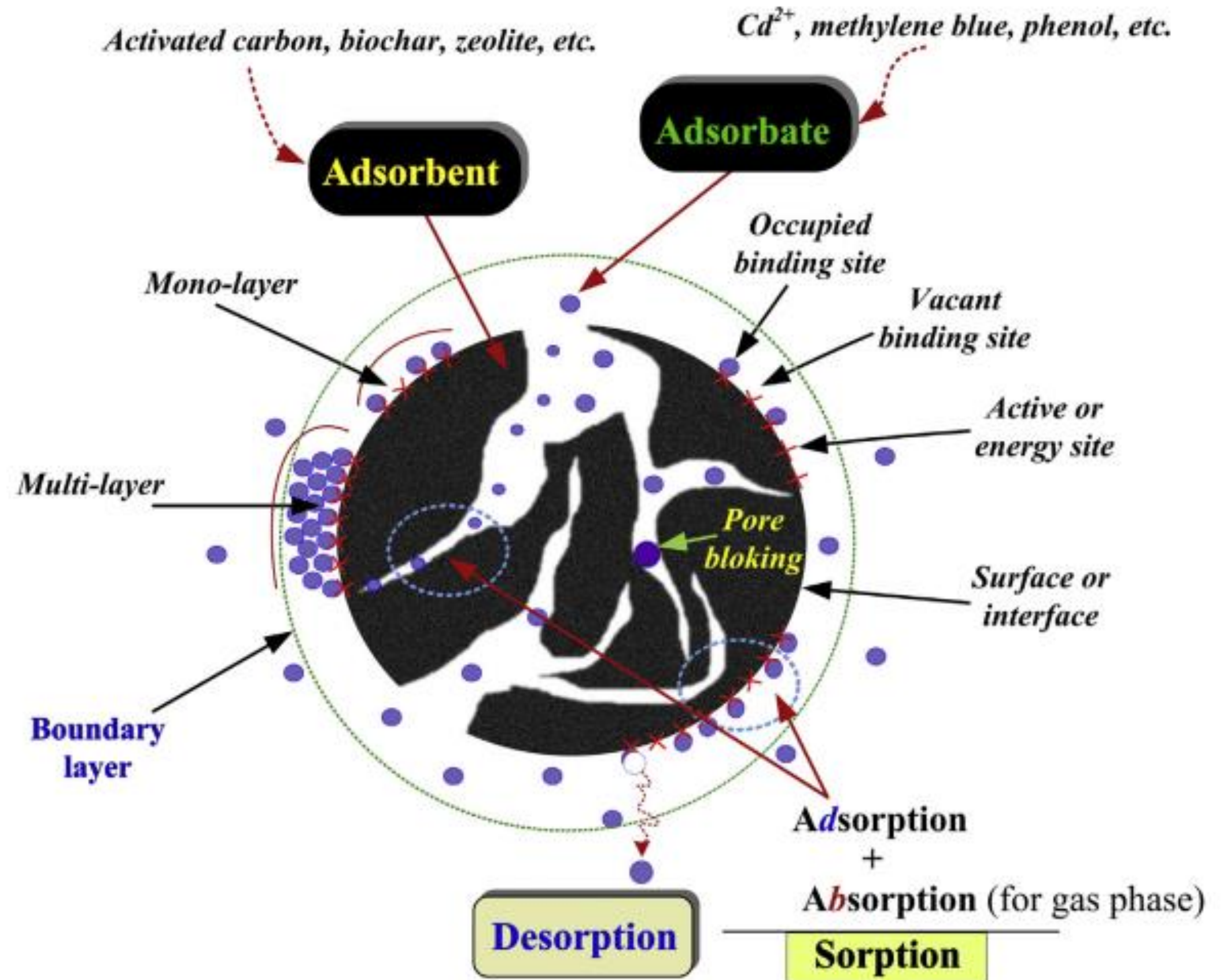
Classification of pores according to the IUPAC definition.

Pore class	Range of pore radius
Macropores	$> 25 \text{ nm}$
Mesopores	$1 \text{ nm} \dots 25 \text{ nm}$
Micropores	$< 1 \text{ nm}$



4.3 Basic terms

- Mono-layer
- Multi-layer
- Boundary layer
- Binding site/active site
(occupied/vacant)
- Surface or interface
- Desorption

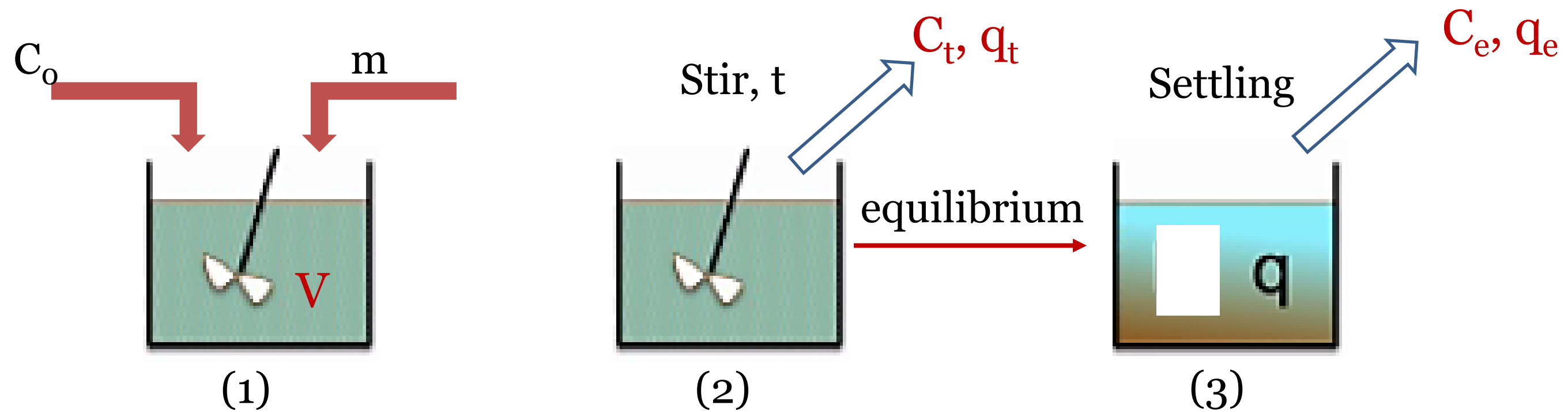


4.3 Basic terms

- Adsorption capacity is the amount of adsorbate taken up by the adsorbent per unit mass (or volume) of the adsorbent (mg/g, mol/g).
- Initial conc.: C_0 or C_{in}
- Equilibrium conc.: C_e
- Conc. at a certain time: C_t
- Mass of adsorbent: m
- Volume of adsorbed solution: V

4.3 Basic terms

- Batch adsorption (hấp phụ theo mẻ):

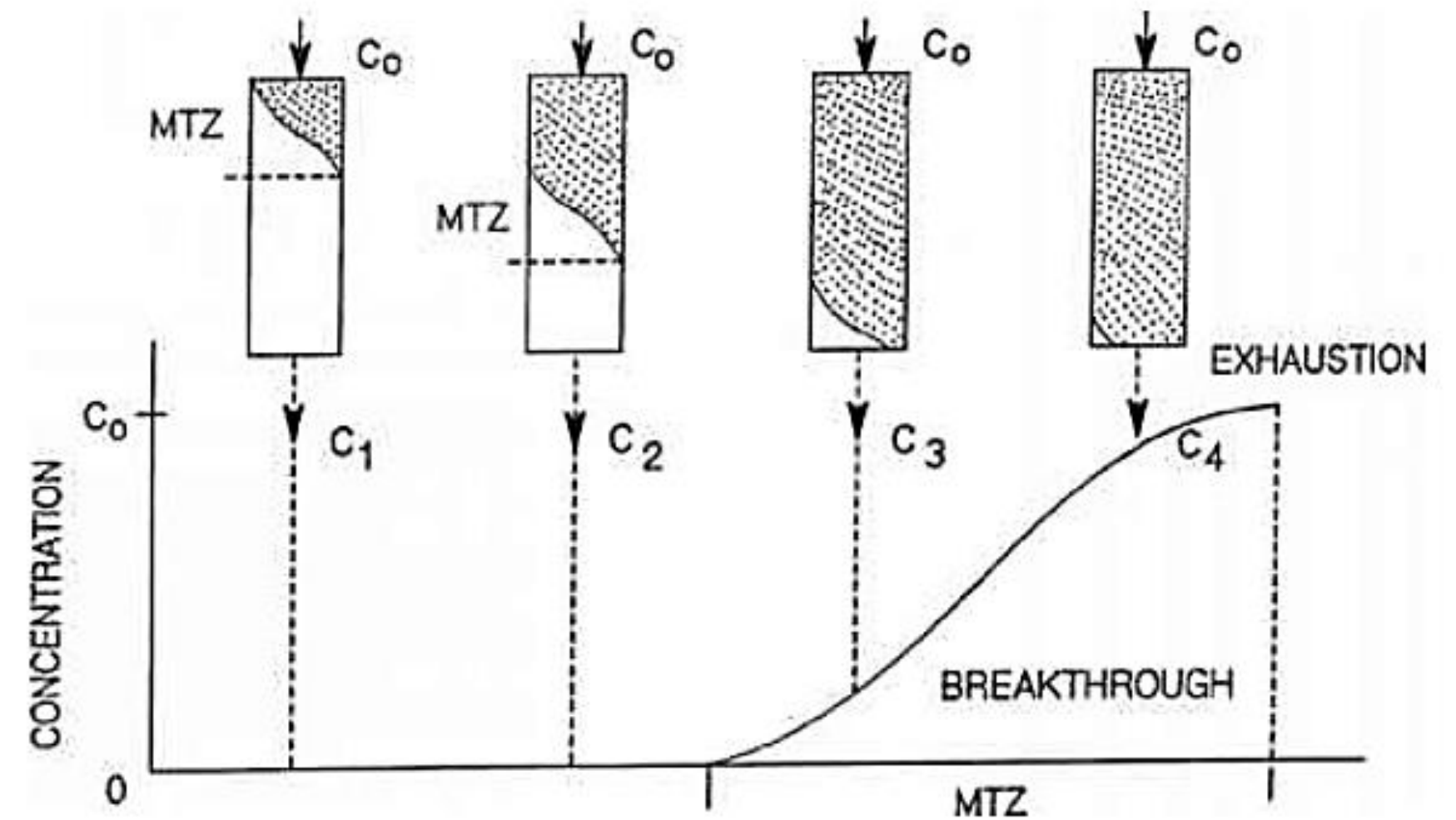
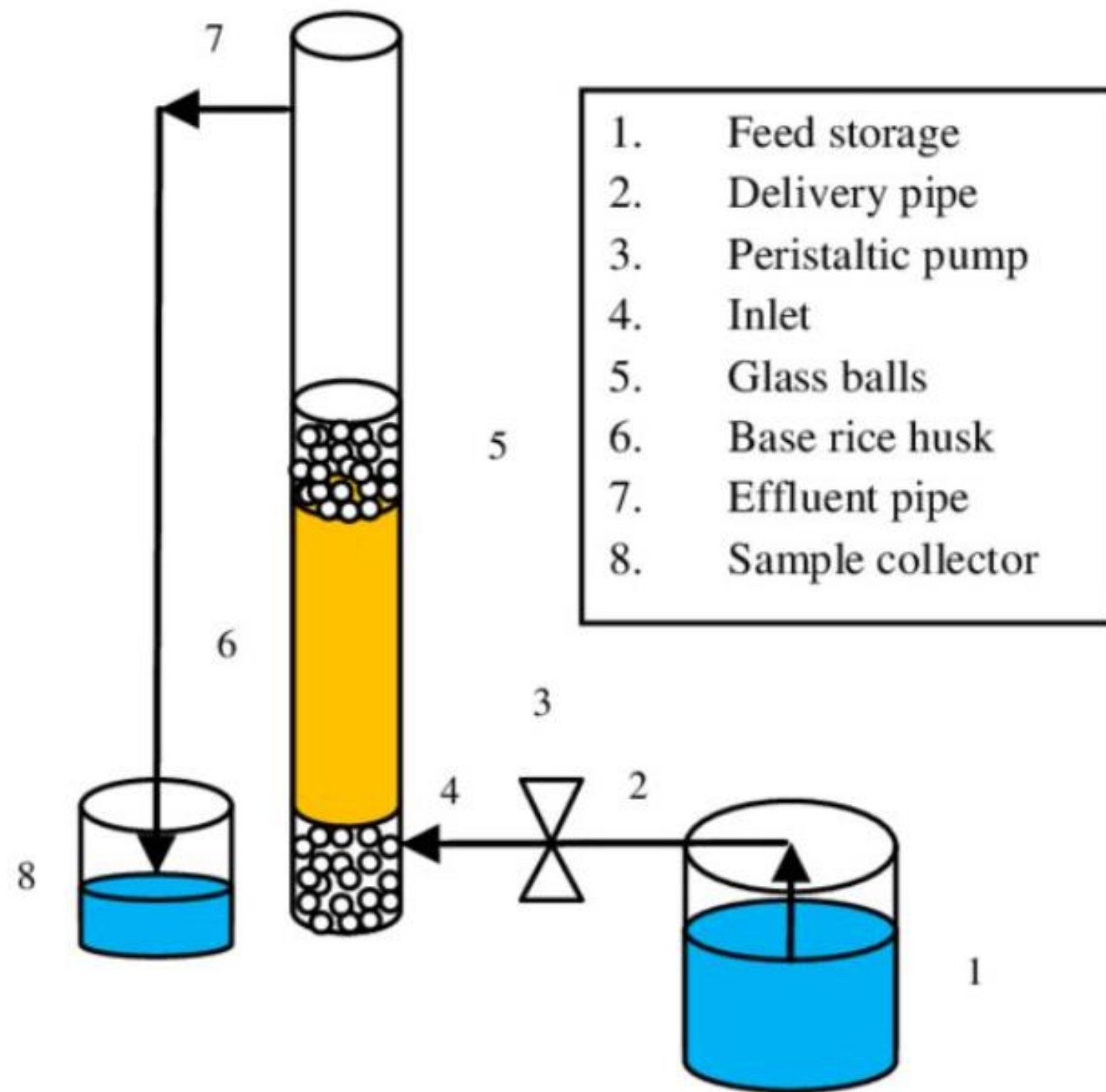


$$\text{Adsorption capacity, } q_t = \frac{(C_0 - C_t)}{m} \times V$$

$$\text{Adsorption efficiency: } E = \frac{C_0 - C_t}{C_0} \times 100\%$$

4.3 Basic terms

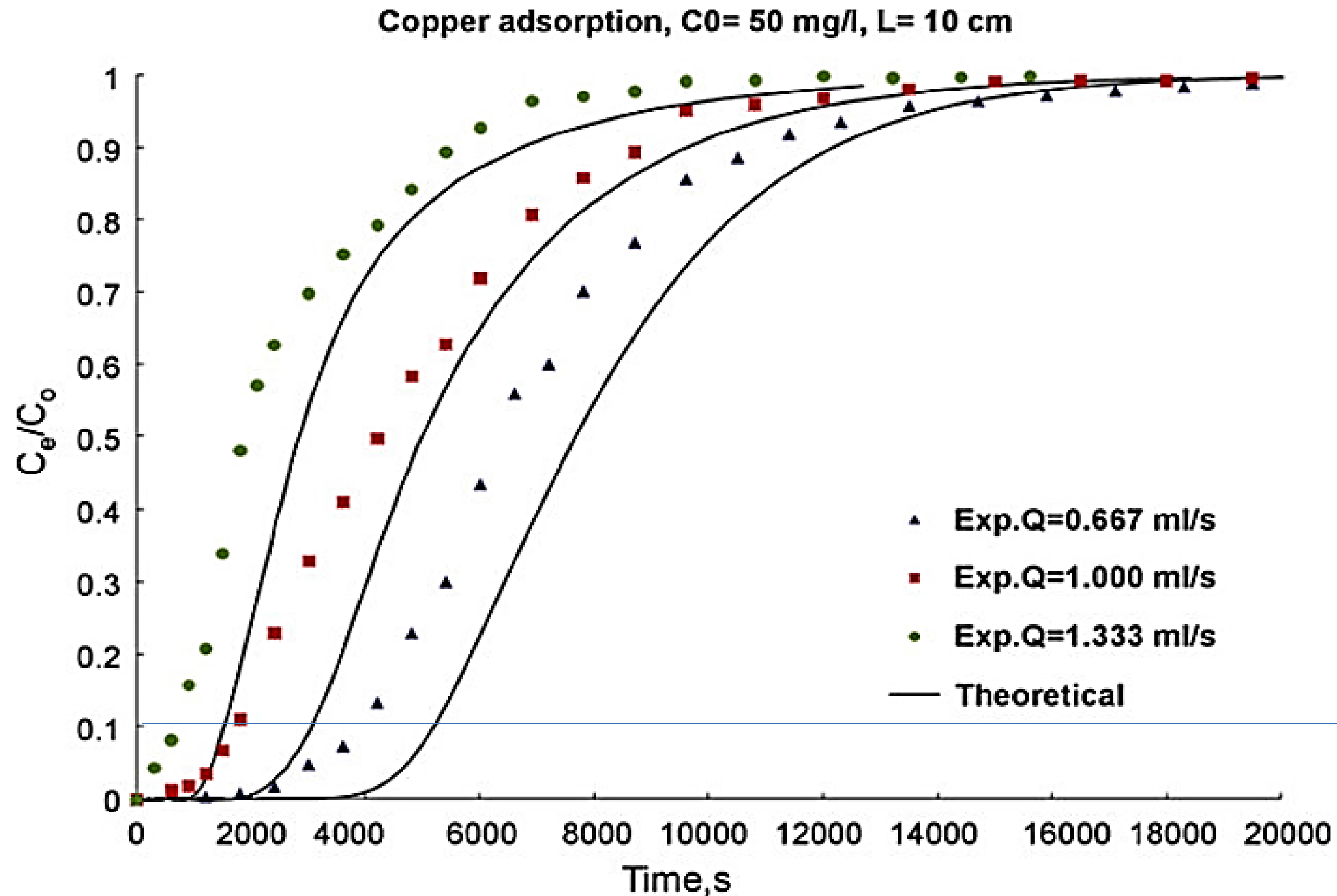
- Fixed-bed column adsorption (hấp phụ cột):



MTZ: mass transfer zone

4.3 Basic terms

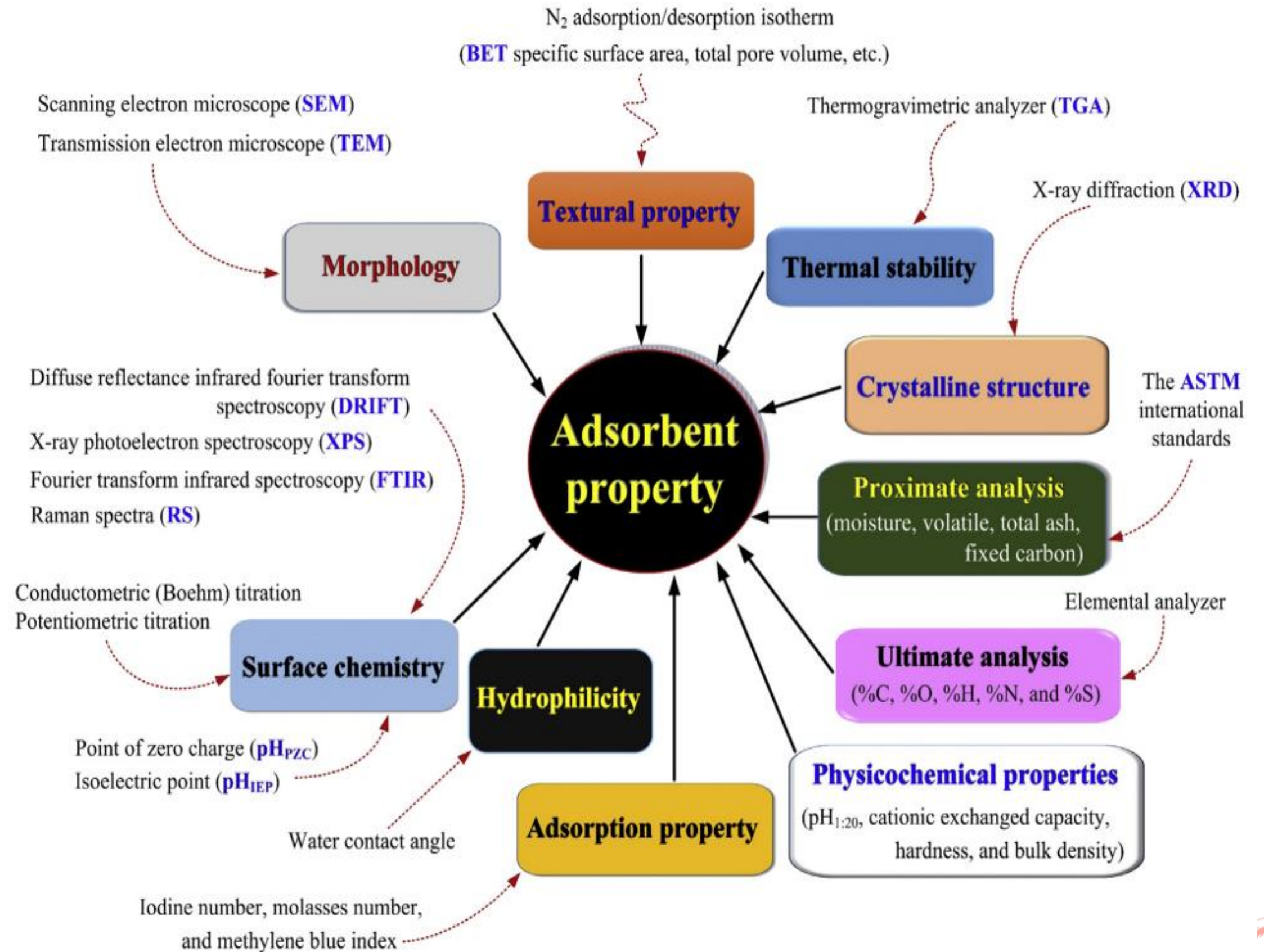
- Column adsorption (hấp phụ cột):



4.4 Factors influencing adsorption

Properties of adsorbent:

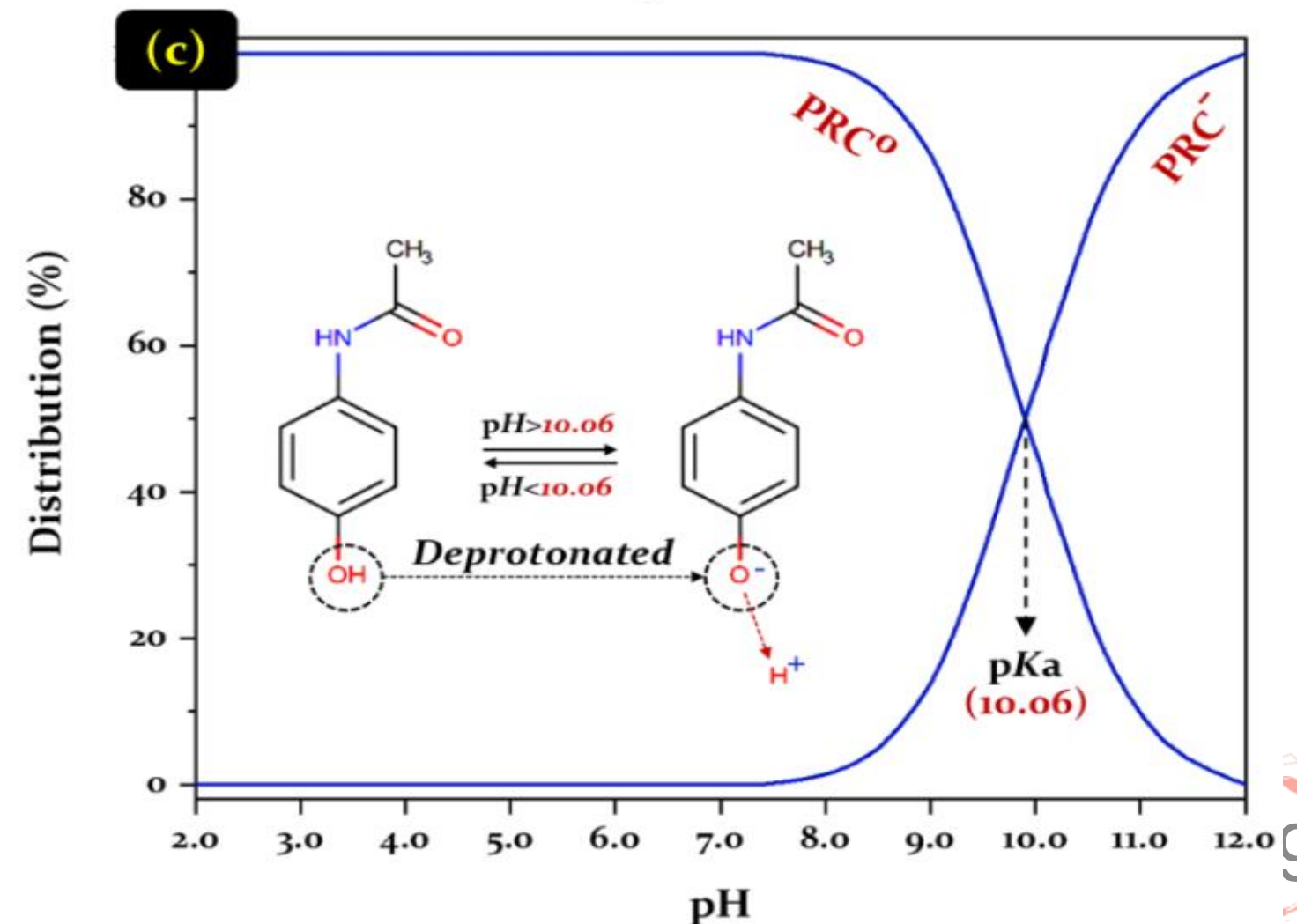
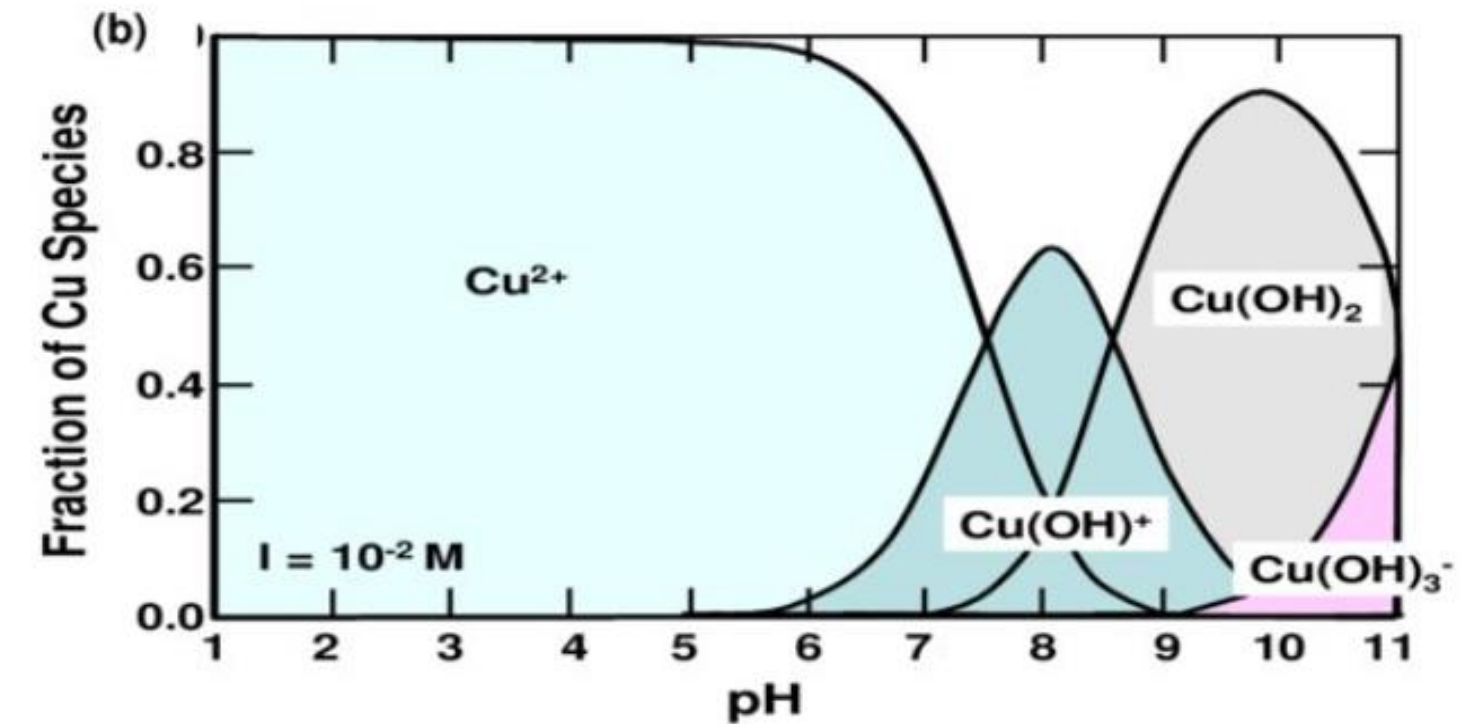
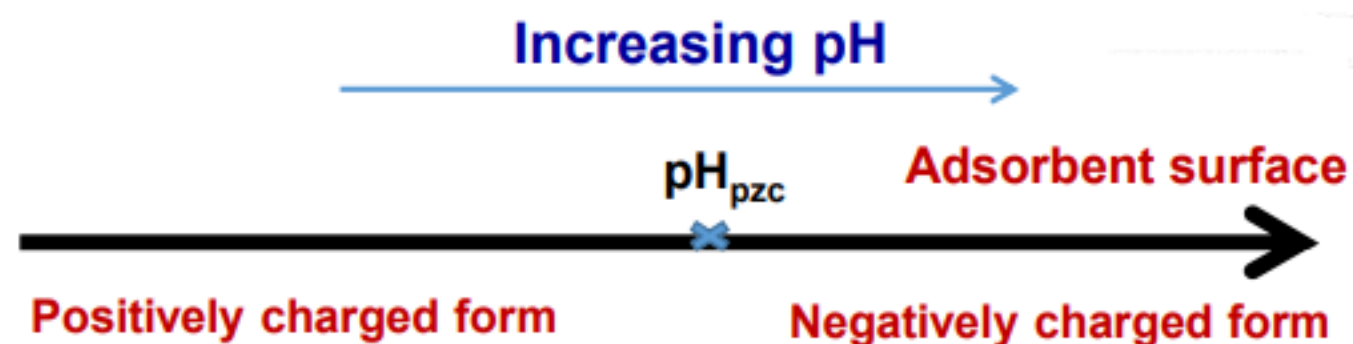
Surface area (S_{BET})
Micro volume
Functional group
Bulk density
....



4.4 Factors influencing adsorption

Environmental factors:

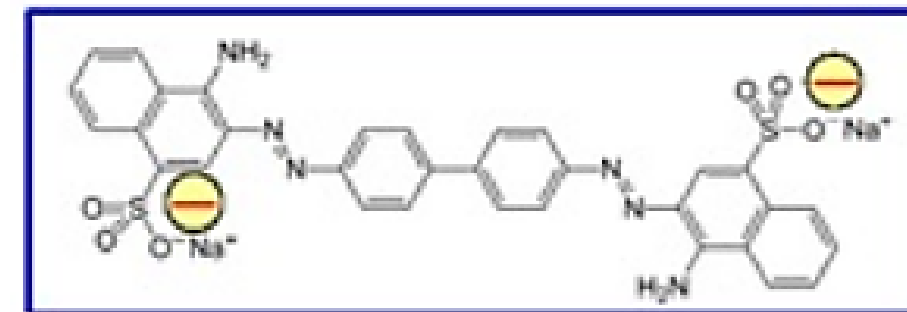
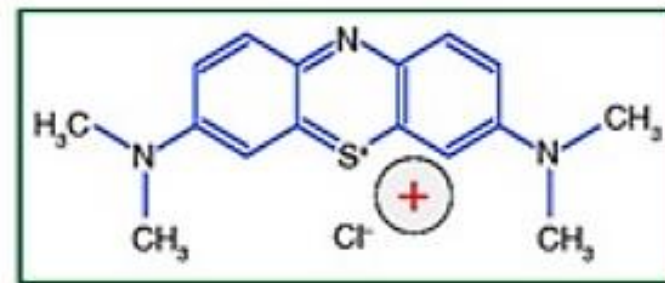
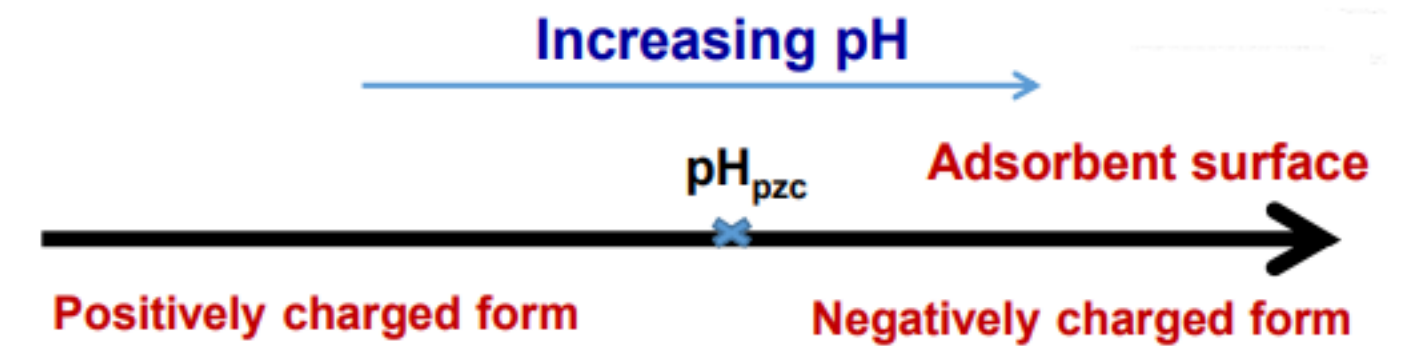
- ❖ Solution pH and pH_{PZC} of adsorbent
- ✓ Solution pH affects the ionization of species in water
- ✓ pH and pH_{PZC} of adsorbent effect the charge of adsorbent surface
- ✓ Electrostatic force occurs if adsorbent and adsorbate are oppositely charged.



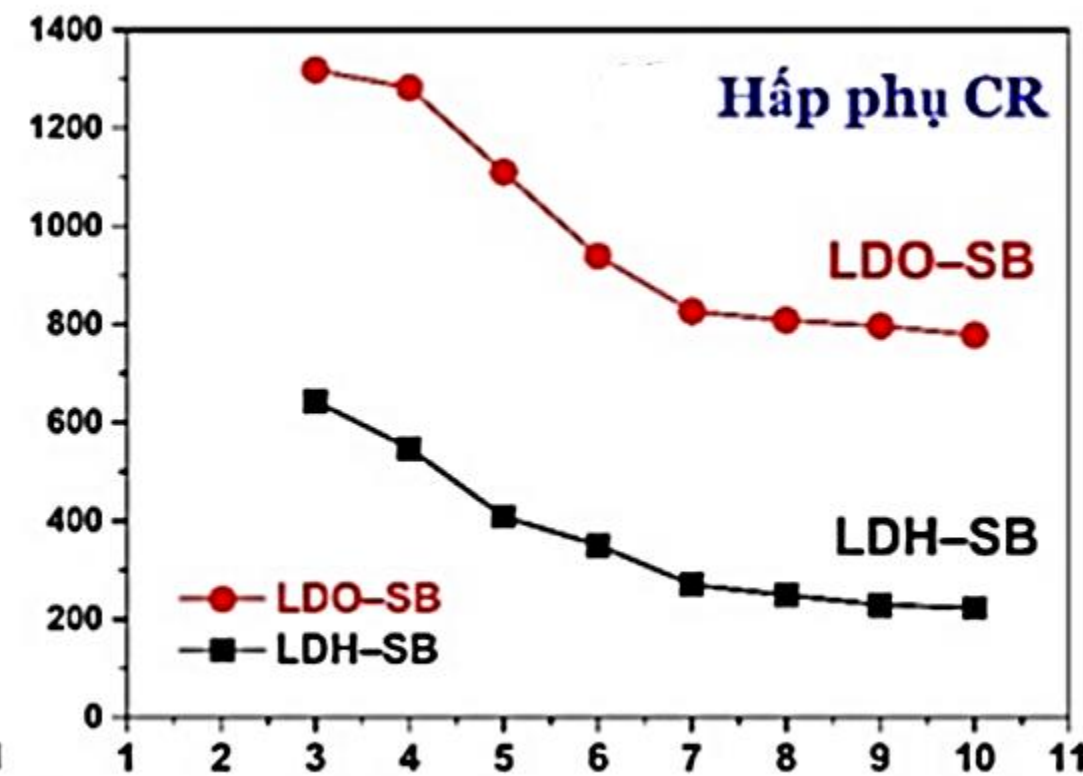
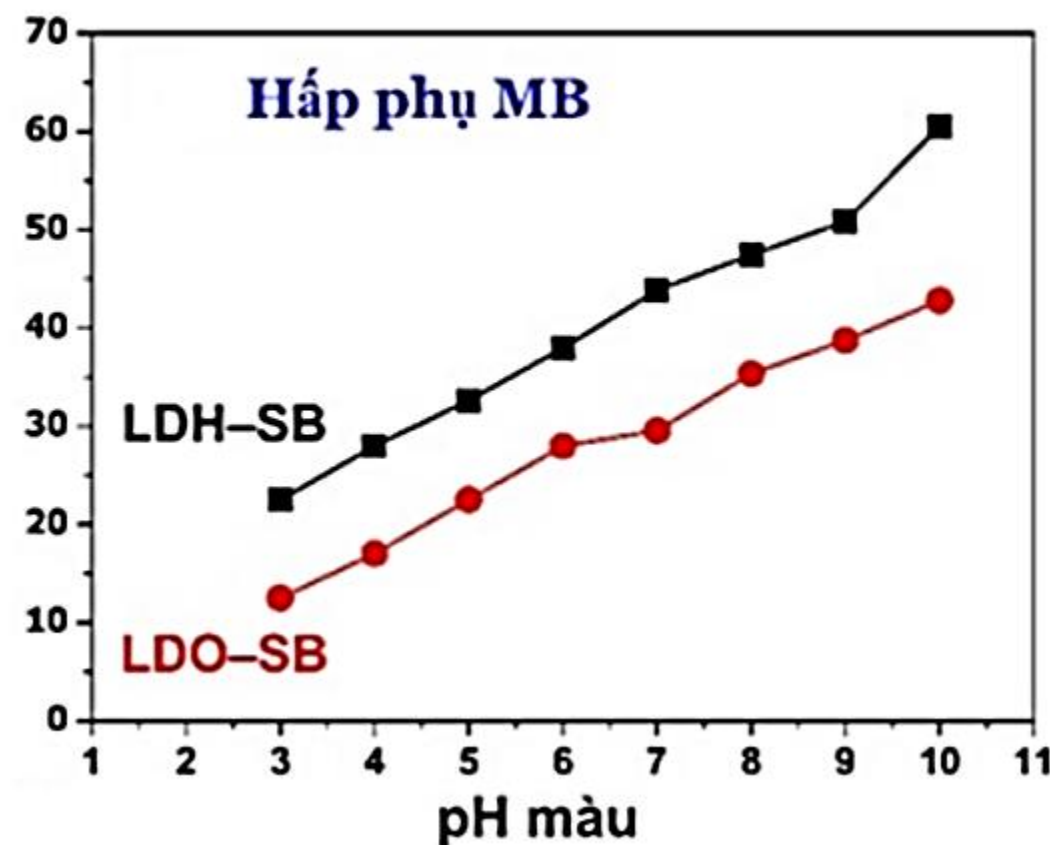
4.4 Factors influencing adsorption

Environmental factors:

❖ Solution pH and pH_{PZC} of adsorbent



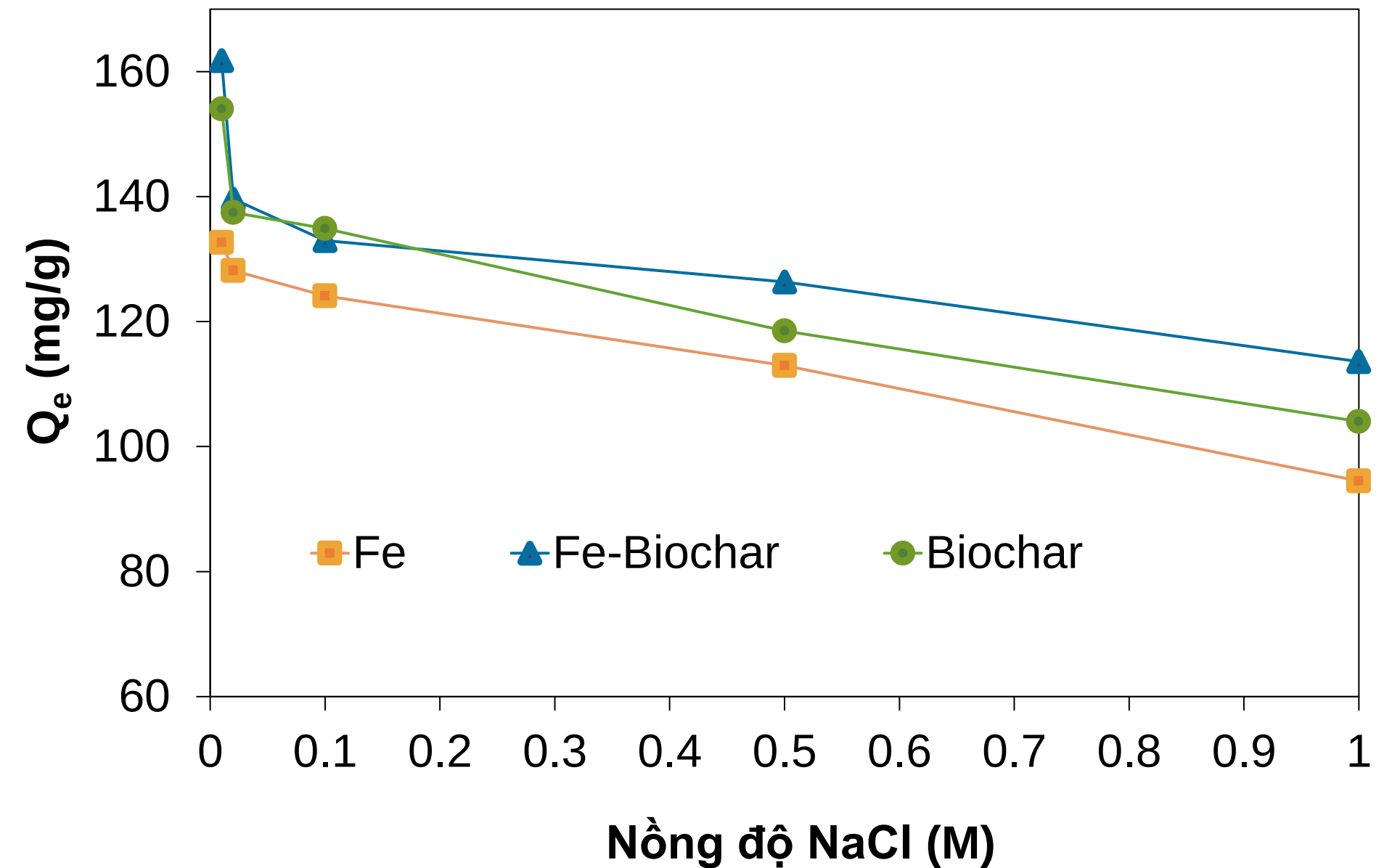
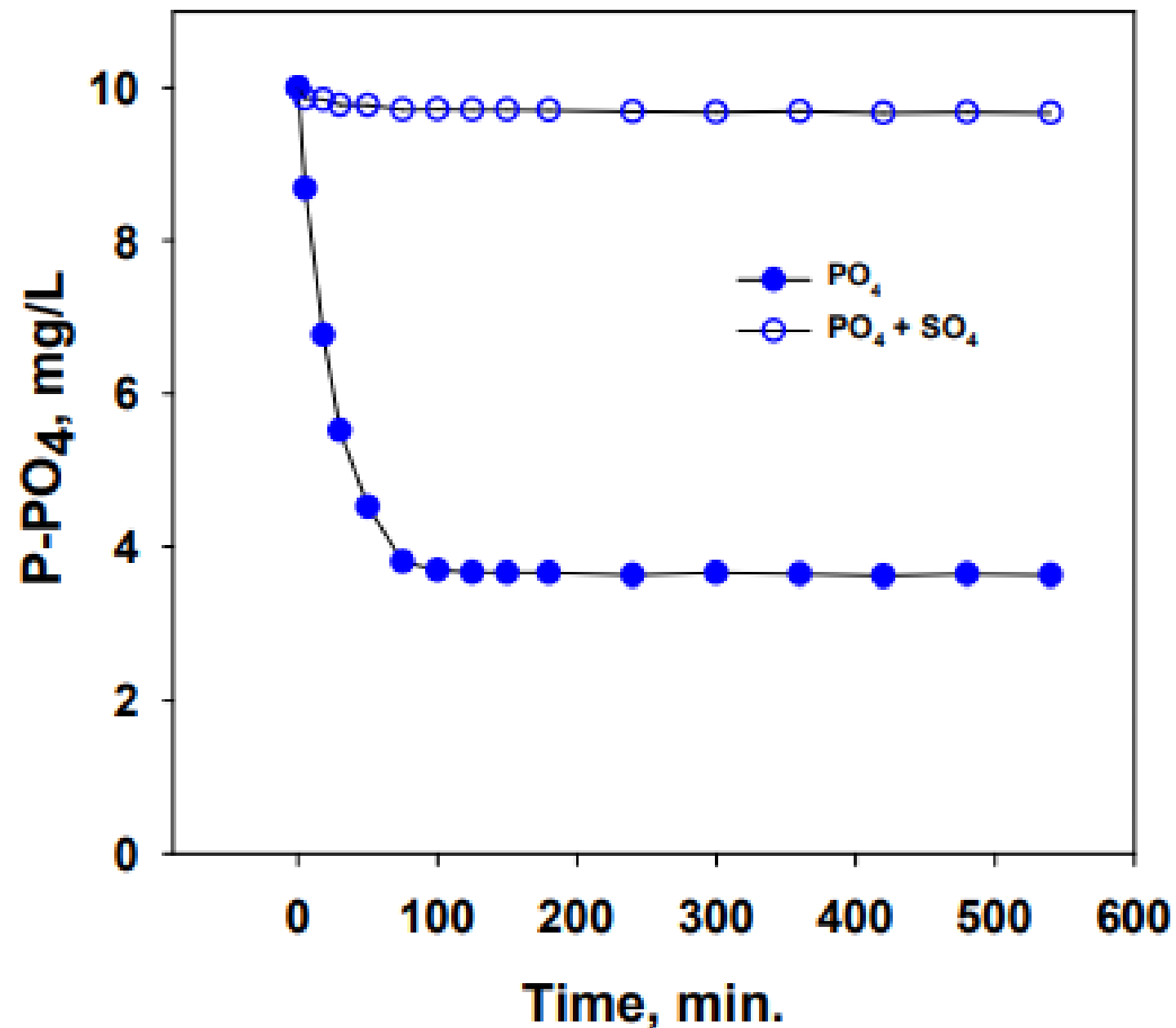
Dung lượng hấp phụ (q_e ; mg/g)



4.4 Factors influencing adsorption

Environmental factors:

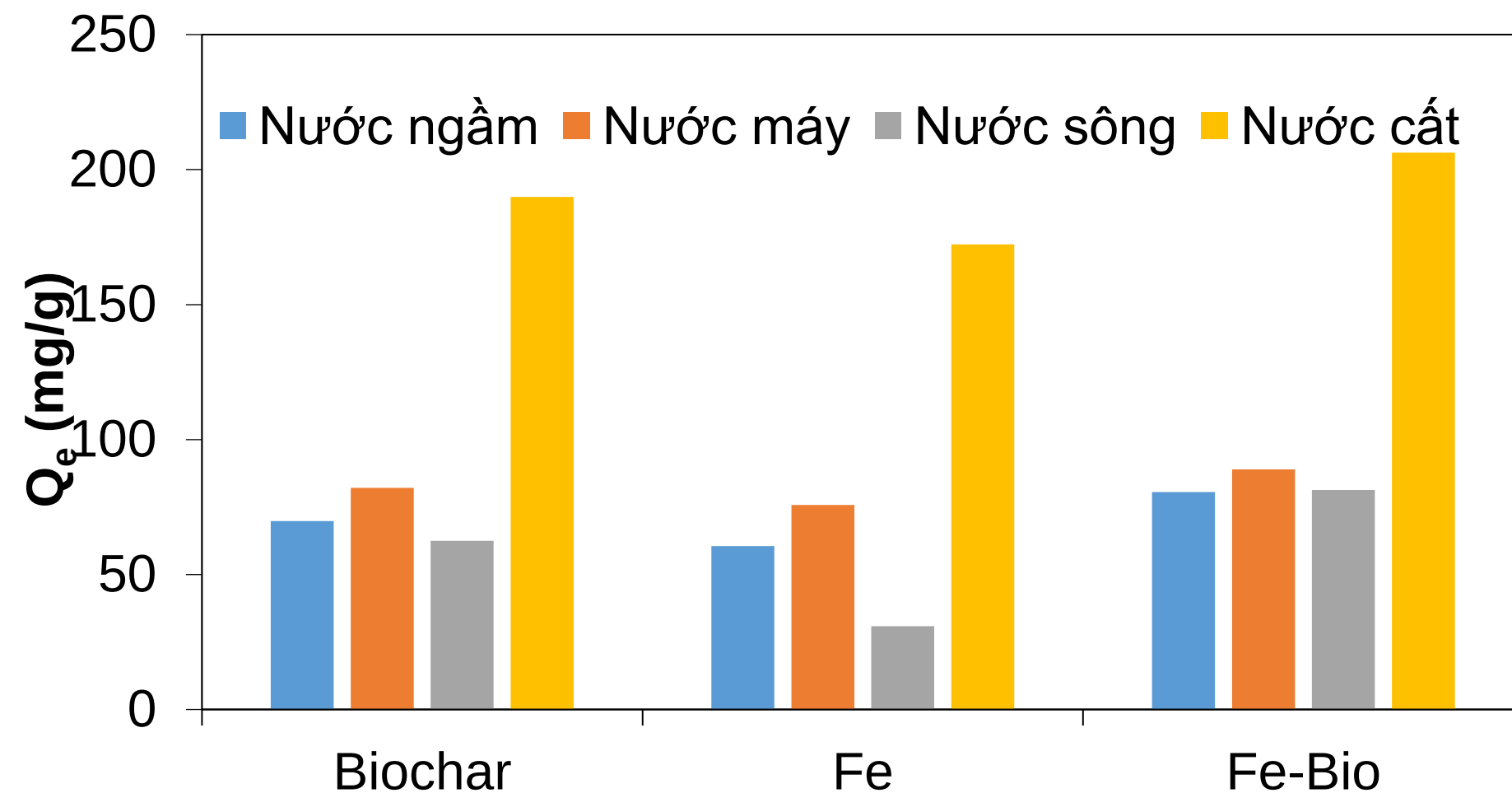
❖ Component of solution



4.4 Factors influencing adsorption

Environmental factors:

❖ Component of solution

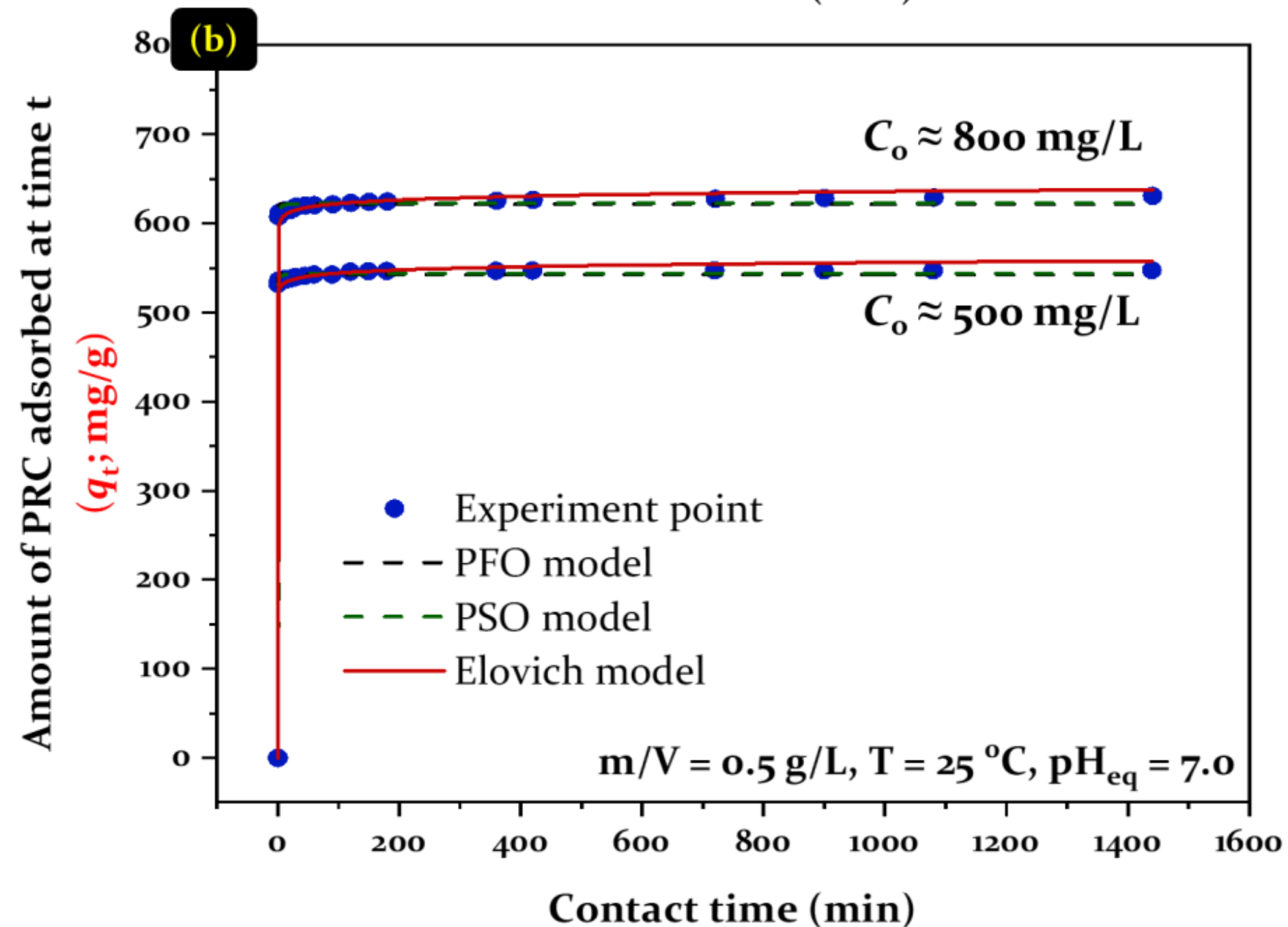


Chỉ tiêu	Nước cất	Nước máy	Nước ngầm	Nước sông
Độ dẫn điện ($\mu\text{S/cm}$)	0,42	2,65	11,63	17,96

4.4 Factors influencing adsorption

Environmental factors:

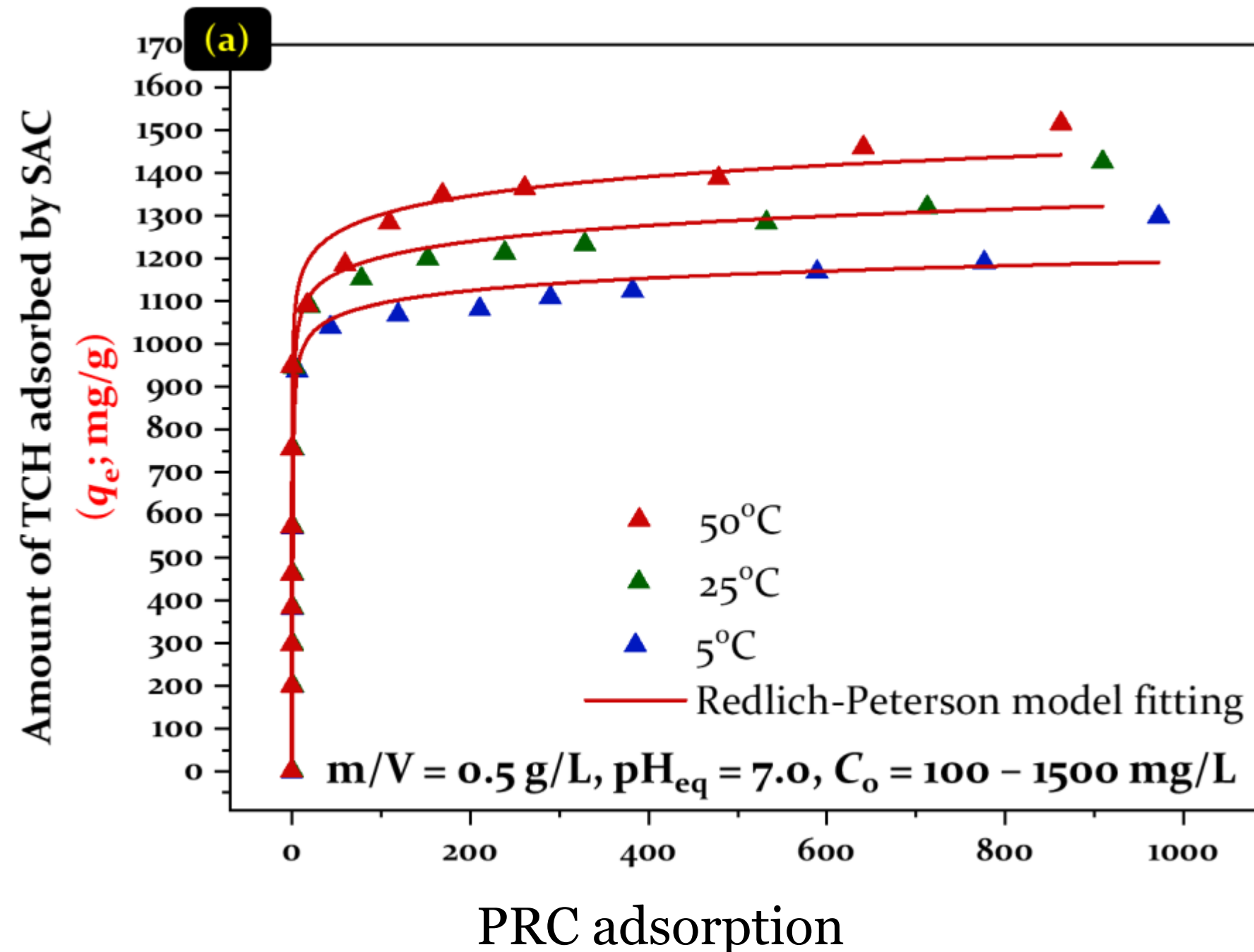
❖ Concentration of adsorbate



4.4 Factors influencing adsorption

Environmental factors:

❖ Temperature



4.5 Adsorbents

❖ Major types of adsorbents:

activated alumina



silica gel



activated carbon (AC)



❖ Other types of adsorbents: molecular sieve carbon, molecular sieve zeolites, polymeric adsorbents, modified adsorbents

4.5 Adsorbents

❖ Applications:

Adsorbent	Applications
Silica Gel	Drying of gases, refrigerants, organic solvents, transformer oils Desiccant in packings and double glazing Dew point control of natural gas
Activated Alumina	Drying of gases, organic solvents, transformer oils Removal of HCl from hydrogen Removal of fluorine in alkylation process
Carbons	Nitrogen from air Hydrogen from syngas Ethene from methane and hydrogen Vinyl chloride monomer (VCM) from air Removal of odours from gases Recovery of solvent vapours Removal of SOX and NOX Purification of helium Clean-up of nuclear off-gases Water purification
Clay	Treatment of edible oils Removal of organic pigments Refining of mineral oils Removal of polychlorinated biphenyls (PCBs)

Adsorbent	Applications
Zeolites	Oxygen from air Drying of gasses Removing water from azeotropes Sweetening sour gases and liquids Purification of hydrogen Separation of ammonia and hydrogen Recovery of carbon dioxide Separation of oxygen and argon Removal of acetylene, propane and butane from air Separation of xylenes and ethyl benzene Separation of normal from branched paraffins Separation of olefins and aromatics from paraffins Recovery of carbon monoxide from methane and hydrogen Drying of refrigerants and organic liquids Pollution control, including removal of Hg, NO _x and SO _x Recovery of fructose from corn syrup
Polymers & Resins	Water purification Recovery and purification of steroids, amino acids Separation of fatty acids from water and toluene Separation of aromatics from aliphatics Recovery of proteins and enzymes Removal of colours from syrups Removal of organics from Hydrogen peroxide

4.5 Adsorbents

❖ Criteria of adsorbents: two points of view

1. A good adsorbent material should possessed the following characteristics:

- 1) Ease and low-cost production.
- 2) Sufficient high surface area and pore volumes.
- 3) Good stability under harsh environment of moisture, thermal and chemical attack.
- 4) High adsorption capacity.
- 5) Ease of regeneration and reused.
- 6) Good Repeatability and reproducibility.

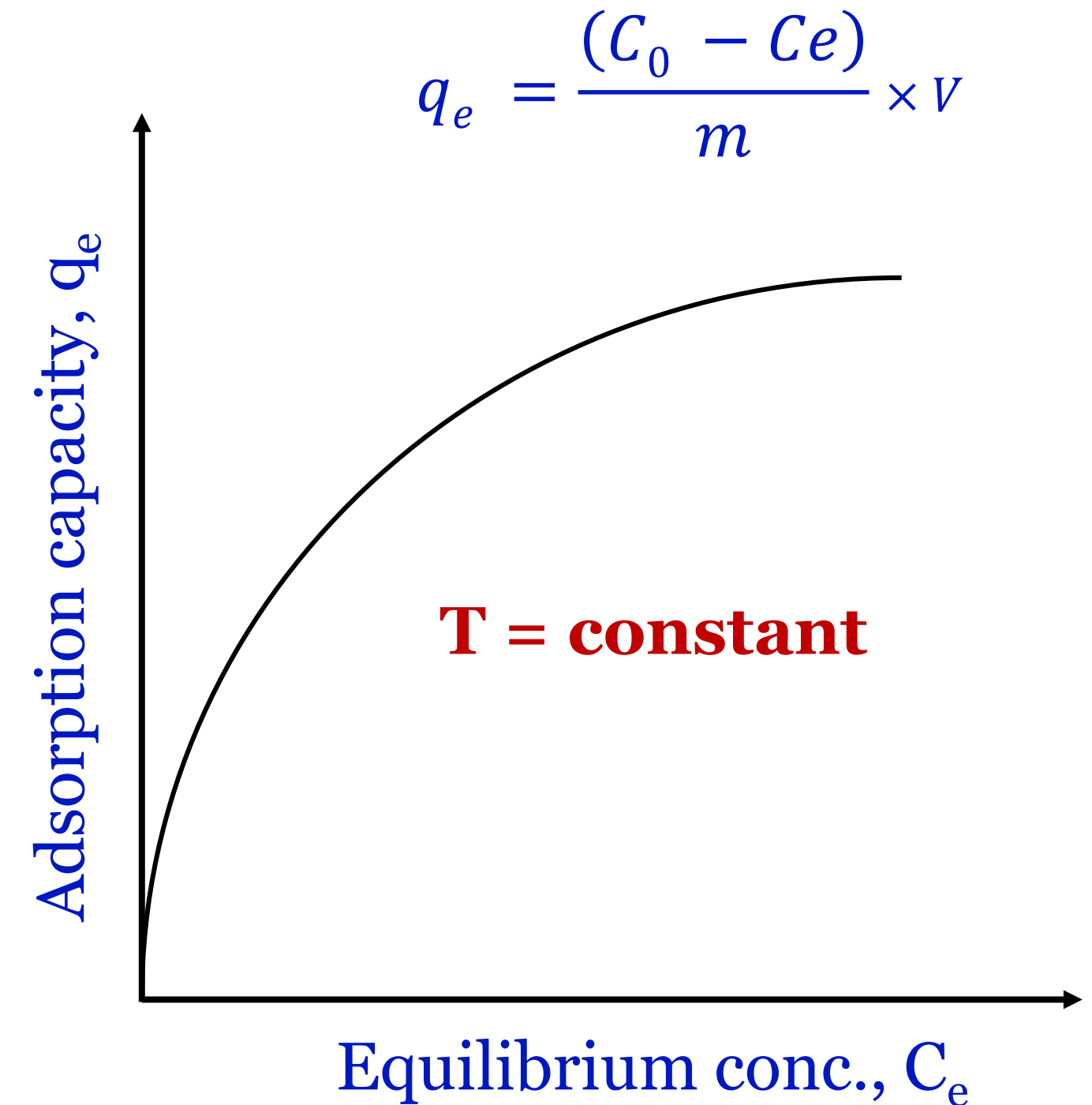
2. A good sorbent should have:

- 1) High breakthrough capacity (working capacity).
- 2) High adsorption and desorption rates (fast kinetics).
- 3) High selectivity toward the targeted species.
- 4) Can be regenerated easily.
- 5) Low cost.
- 6) Good stability.

4.6 Adsorption isotherm (đẳng nhiệt hấp phụ)

- For the adsorption process, an analysis of the isotherm data is important because the isotherms describe equilibrium relationships between adsorbent and adsorbate, usually the ratio between the quantity adsorbed and that remaining in the solution at a fixed temperature at equilibrium.

- An adsorption isotherm is a graph that represents the variation in the amount of adsorbate (x) adsorbed on the surface of the adsorbent with the change in pressure (air)/concentration at a constant temperature.



4.6 Adsorption isotherm

- Popular isotherm models
 - Langmuir isotherm
 - Freundlich isotherm
 - Redlich–Peterson (R–P) isotherm

4.6 Adsorption isotherm

4.6.1 Langmuir isotherm model

- Developed in 1918 by Irving Langmuir
- Applies to gases adsorbed on solid surfaces
- A semi-empirical isotherm with a kinetic basis and statistical thermodynamics
- The most common isotherm equation to use due to its simplicity and its ability to fit a variety of adsorption data
- Based on **four assumptions**:

- (1) monolayer adsorption;
- (2) distribution of adsorption sites is homogeneous;
- (3) adsorption energy is constant;
- (4) interaction between adsorbate molecules is negligible

→ Called as chemical adsorption model

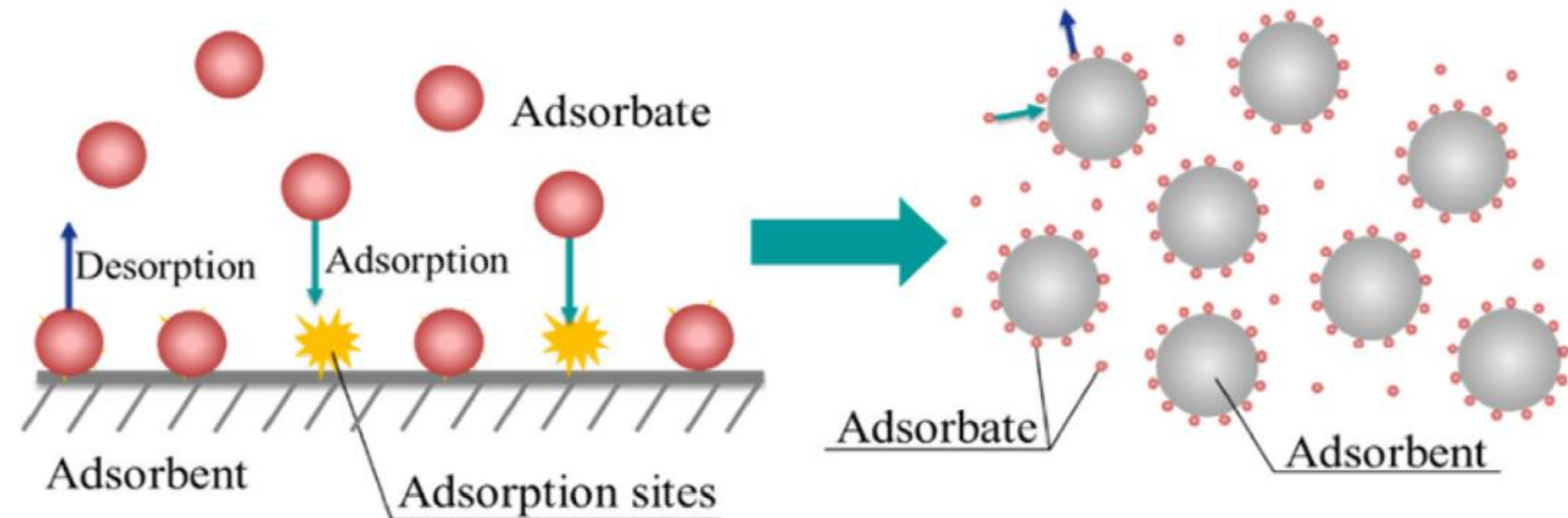


Fig. 4. The adsorption mechanisms revealed by the Langmuir isotherm model.

4.6 Adsorption isotherm

4.6.1 Langmuir isotherm model

$$q_e = \frac{q_m K_L C_e}{1 + K_L C_e} \quad (1)$$

where K_L (L mg^{-1}) is the ratio of the adsorption rate and desorption rate, q_m (mg g^{-1}) is the **maximum adsorption capacity** estimated by the Langmuir model.

K_L : hằng số (cân bằng) hấp phụ Langmuir (L/mg)

q_e : dung lượng hấp phụ cân bằng (mg/g)

$q_{\max}$: dung lượng hấp thụ tối đa của chất hấp phụ (mg/g)

C_e : nồng độ dung dịch sau hấp phụ (mg/L)

4.6 Adsorption isotherm

4.6.1 Langmuir isotherm model

$$q_e = q_{max} \frac{K_L \cdot C_e}{1 + K_L \cdot C_e} \quad (1)$$

Phương trình (1) có thể viết lại dưới dạng:

$$q_e = q_{max} \frac{C_e}{1/K_L + C_e} \quad (2)$$

Chuyển (2) về dạng tuyến tính (linear Langmuir models):

$$\frac{C_e}{q_e} = \frac{C_e}{q_m} + \frac{1}{K_L q_m} \quad (3)$$

$$y \text{ (Ce/qe)} = ax \text{ (Ce)} + b$$

4.6 Adsorption isotherm

4.6.2 Freundlich isotherm model

- ❑ The most widely used isotherm in adsorption is used to represent nonlinear adsorption phenomenon.
- ❑ In many published papers, the Freundlich isotherm was applied to represent the **multi-layer adsorption** on **heterogeneous surfaces**.
- ❑ The linear and nonlinear forms of the Freundlich model is given by the following equations:

$$q_e = K_F C_e^{1/n} \quad (4)$$

$$\log q_e = \log K_F + \frac{1}{n} \log C_e \quad (5)$$

Where K_F ($L^{1/n} mg^{1-1/n} g^{-1}$) and n , the Freundlich model will reduce to the linear model when $n=1$. are constants

4.6 Adsorption isotherm

4.6.3 Redlich–Peterson (R–P) isotherm model

The R-P model is an empirical hybrid model of the Langmuir and Freundlich models, which has been frequently applied in the homogeneous or heterogeneous adsorption processes.

$$q_e = \frac{K_{RP} C_e}{1 + a_{RP} C_e^g} \quad (6)$$

where K_{RP} (Lg^{-1}) and a_{RP} ($L^g mg^{-g}$) are constants, g is the exponent ($0 < g \leq 1$).

- when $g = 1$, the R-P model reduces to the Langmuir model (Eq. (1))
- when $g = 0$ or C_e approaches to 0, it will reduce to the linear model
- if C_e approaches to infinite (vô cực), $q_e \approx (K_{RP}/a_{RP}) C_e^{(1-g)}$, which reduces to the Freundlich model (Eq. (4)).

Linear form:

$$\ln \left(K_R \frac{C_e}{q_e} - 1 \right) = b_R \ln C_e + \ln a_R$$

Table 1: Adsorption isotherm models and their linear forms

Isotherm	Non-Linear Form	Linear Form	Plot	Constants obtained from plot
Henry		$q_e = K_{HE} C_e$	q_e vs C_e	K_{HE}
Langmuir	$q_e = q_m K_L \frac{C_e}{1 + K_L C_e}$	$\frac{C_e}{q_e} = \frac{1}{q_m K_L} + \frac{C_e}{q_m}$	$\frac{C_e}{q_e}$ vs C_e	q_m and K_L
Freundlich	$q_e = K_F C_e^{\frac{1}{n}}$	$\log q_e = \log K_F + \frac{1}{n} \log C_e$	$\log q_e$ vs $\log C_e$	K_F and n
Dubinin-Radushkevich	$q_e = q_s \exp(-K_{DR} \varepsilon^2)$	$\ln q_e = \ln q_s - K_{DR} \varepsilon^2$ $\varepsilon = RT \ln(1 + \frac{1}{C_e})$	$\ln (q_e)$ vs ε^2	K_{DR} and q_s
Tempkin	$q_e = \frac{RT}{b} \ln A C_e$	$q_e = \frac{RT}{b} \ln A + \frac{RT}{b} \ln C_e$ or $q_e = B \ln A + B \ln C_e$ Where $B = \frac{RT}{b}$	q_e vs $\ln(C_e)$	A and b
Harkin-Jura		$\frac{1}{q_e^2} = \frac{B}{A} - \left(\frac{1}{A}\right) \log C_e$	$\frac{1}{q_e^2}$ vs $\log C_e$	A and B
Elovich		$\ln \left(\frac{q_e}{C_e}\right) = \ln K_e q_m - \frac{q_e}{q_m}$	$\ln \left(\frac{q_e}{C_e}\right)$ vs q_e	K_e and q_m
Redlich-Peterson	$q_e = \frac{A C_e}{1 + B C_e^\beta}$	$\ln \left(\frac{C_e}{q_e}\right) = \beta \ln C_e - \ln A$	$\ln \left(\frac{C_e}{q_e}\right)$ vs $\ln C_e$	A and β
Jossens	$C_e = \frac{q_e}{H} \exp(F q_e^p)$	$\ln \left(\frac{C_e}{q_e}\right) = -\ln(H) + F q_e^p$	$\ln \left(\frac{C_e}{q_e}\right)$ vs q_e	H and F

4.6 Adsorption isotherm

□ Application of adsorption isotherm

- Langmuir isotherm $\rightarrow q_{\max}$ $\rightarrow$ compare adsorption ability of different adsorbents
- Suggest adsorption mechanism based on physical meaning or assumption of the models
- K_L will be used for other calculation
- **Input data** for designing adsorption processes

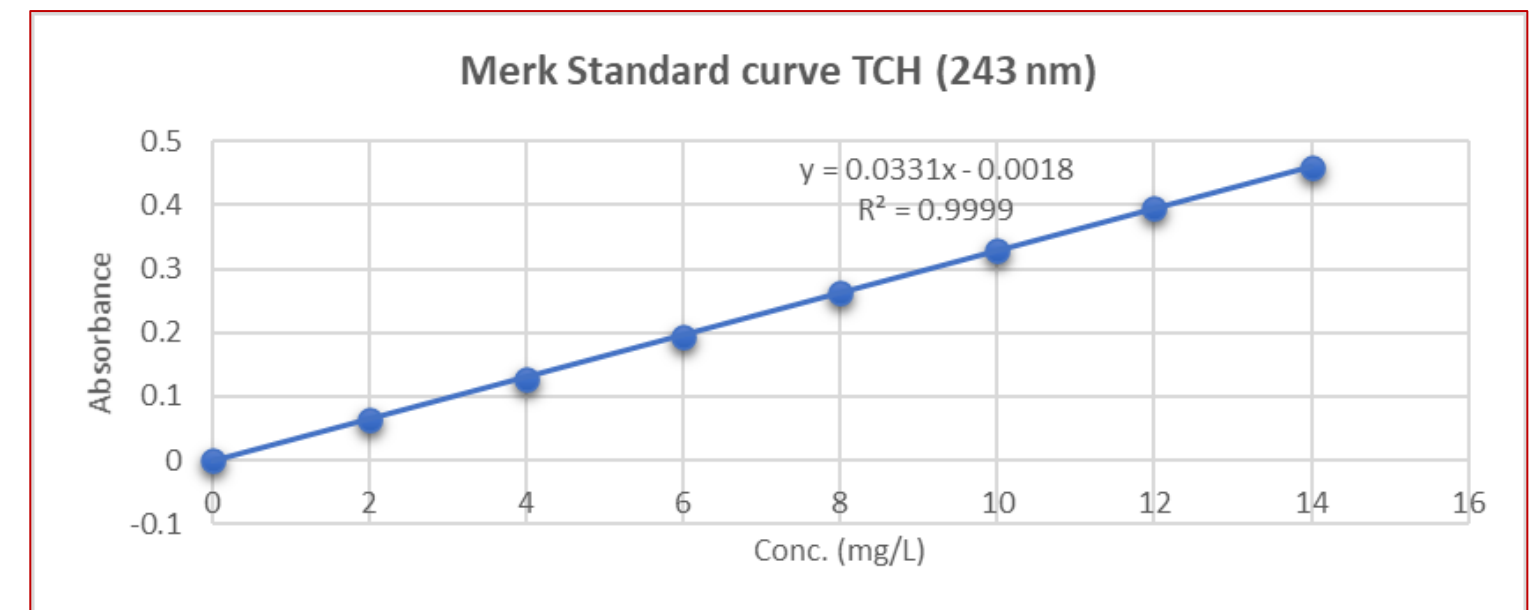
4.7 Examples

□ Example 1

Một SV làm thí nghiệm hấp phụ theo mẻ, dùng than hoạt tính từ bã đậu hấp phụ kháng sinh Tetracycline (TCH) trong nước ở 25 °C cho kết quả sau, hãy

1. Tính dung lượng hấp phụ của các mẫu đã cho.
2. Đăng nhiệt hấp phụ tuân theo mô hình nào?

C mong muốn	Ao	Pha lỏng	m	V	T (K)	A	Pha lỏng
0	0				278K	0.0000	
100	0.0644	50	0.02	0.04		0.0028	1
150	0.0964	50	0.02	0.04		0.0045	1
200	0.1248	50	0.02	0.04		0.0061	1
250	0.1511	50	0.02	0.04		0.0101	1
300	0.1876	50	0.02	0.04		0.0141	1
400	0.2482	50	0.02	0.04		0.0233	1
500	0.3123	50	0.02	0.04		0.1839	1
600	0.1844	100	0.02	0.04		0.0265	50
700	0.2143	100	0.02	0.04		0.0768	50
800	0.2468	100	0.02	0.04		0.1375	50
900	0.2775	100	0.02	0.04		0.1900	50
1000	0.3106	100	0.02	0.04		0.2510	50
1200	0.1923	200	0.02	0.04		0.1931	100
1400	0.2252	200	0.02	0.04		0.2552	100
1600	0.2665	200	0.02	0.04		0.3363	100



Mong muốn: C0: 0 → 1600 mg/L

Đo lại A0 → **C0 thật** = C0 X Hso PL

m = 0.02 (g)

V = 0.04 L

Ce: từ A → Ce = Ce x HSPL

→ $Q_e = (C_0 - C_e)V/m$

4.7 Examples

□ Example 2

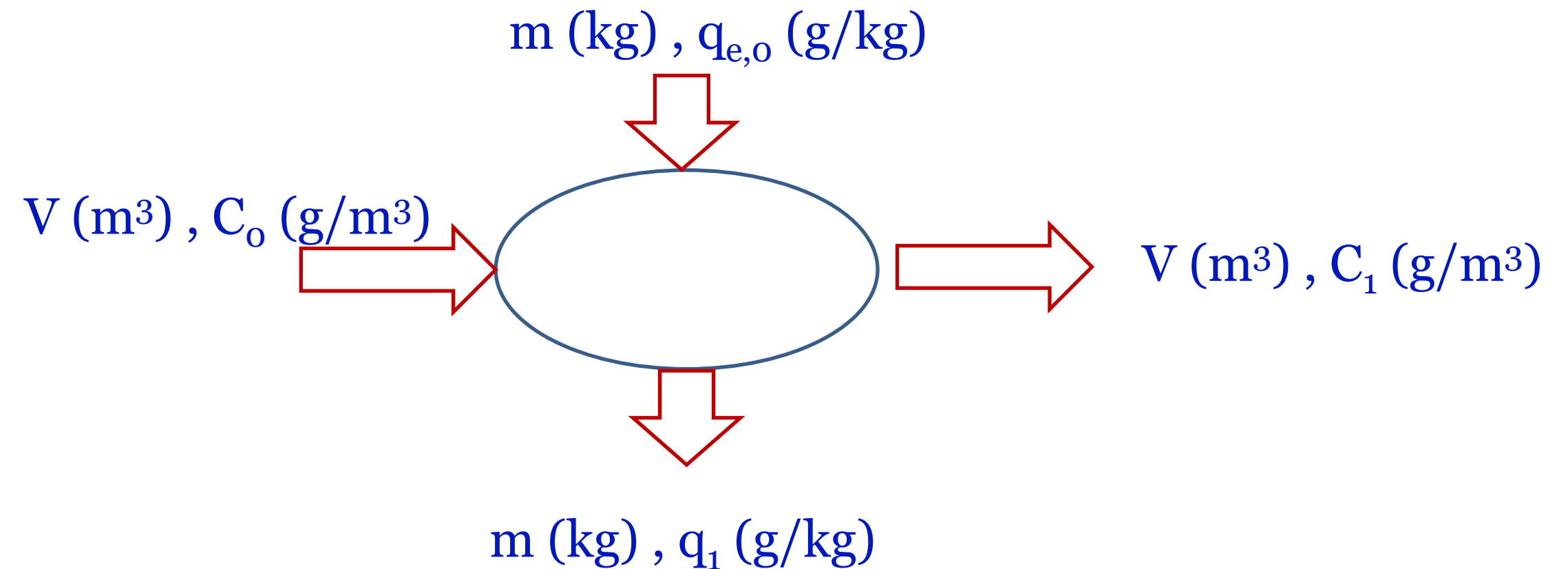
Nghiên cứu đẳng nhiệt hấp phụ ion Cd^{2+} có nồng độ ban đầu C_o (g/m^3) và nồng độ còn lại trong dung dịch lúc cân bằng là C_e (g/m^3) trong nước thải xi mạ bằng chitosan ở 30°C cho biết quá trình hấp phụ tuân theo mô hình hấp phụ đẳng nhiệt Langmuir với các tham số trong mô hình Langmuir tìm được là $K_L = 0,65$ và $q_{\max} = 56 \text{ mg/g}$. Kết quả đạt được được áp dụng cho thiết kế bể hấp phụ gián đoạn cho xử lý nước thải của nhà máy xi mạ.

- Vẽ sơ đồ bể hấp phụ một giai đoạn. Đưa ra công thức biểu diễn mối liên hệ giữa thể tích nước thải, V (m^3) và khối lượng chất hấp phụ sử dụng, m (kg) với các tham số C_o , C_e và các tham số của mô hình Langmuir.
- Tính khối lượng chất hấp phụ sử dụng (kg) để hấp phụ 90% Cd^{2+} trong 100 m^3 nước thải có nồng độ Cd^{2+} ban đầu, $C_o = 10 \text{ g}/\text{m}^3$.

4.7 Examples

□ Examples 2

a)



$$V(m^3)(C_0 - C_1)\left(\frac{g}{m^3}\right) = m(kg)(q_1 - q_{e,o})\left(\frac{g}{kg}\right)$$

$$q_{e,o} = 0 \text{ (chất hấp phụ sạch): } V(C_0 - C_1) = m(q_1)$$

Khi đạt cân bằng: $C_1 \rightarrow C_e$ và $q_1 \rightarrow q_e$

PTCBVC trong hệ hấp phụ gián đoạn một giai đoạn trở thành:

$$V(C_0 - C_e) = m(q_e)$$

$$\rightarrow \frac{m}{V} = \frac{C_0 - C_e}{q_e}$$

4.7 Examples

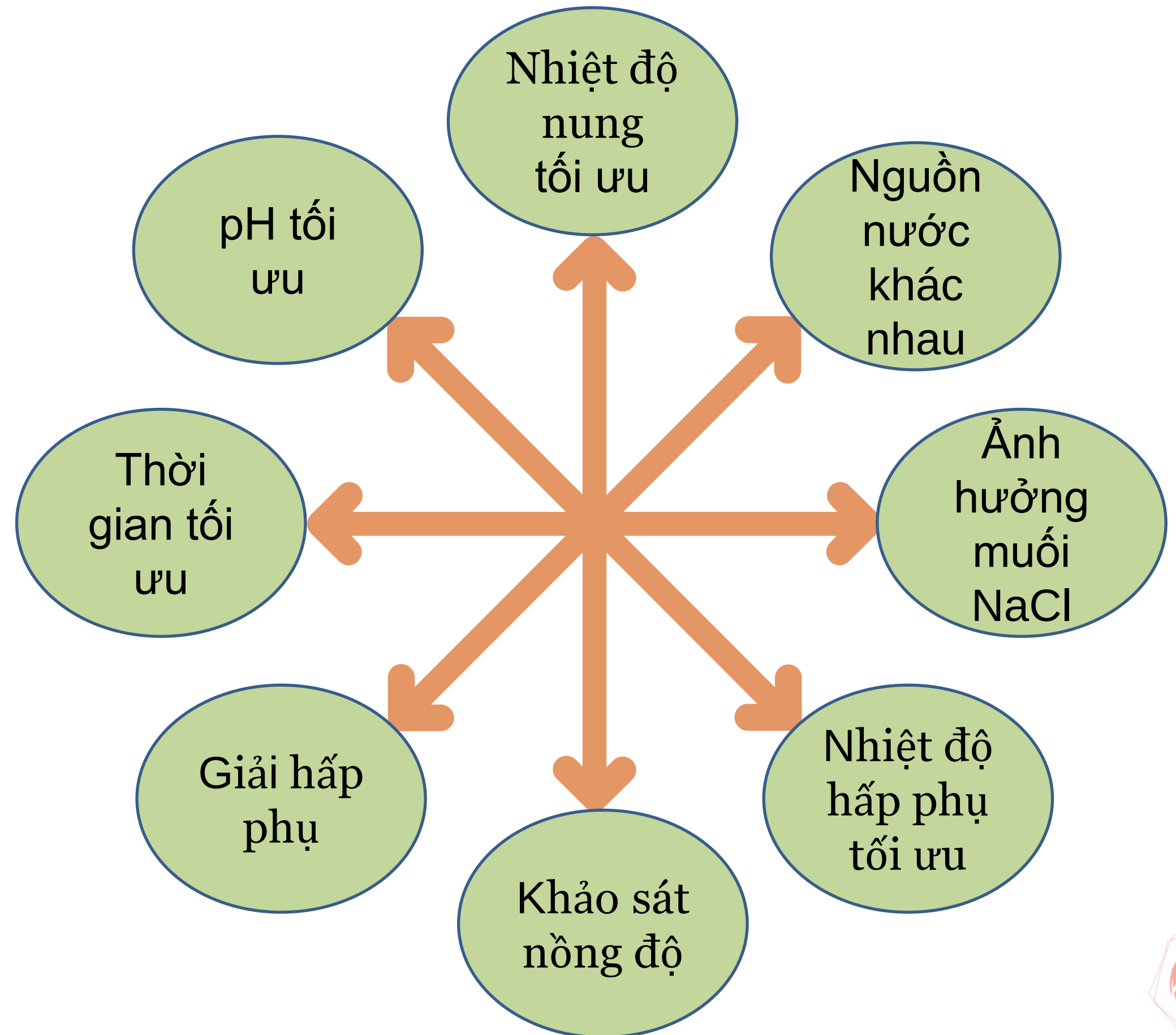
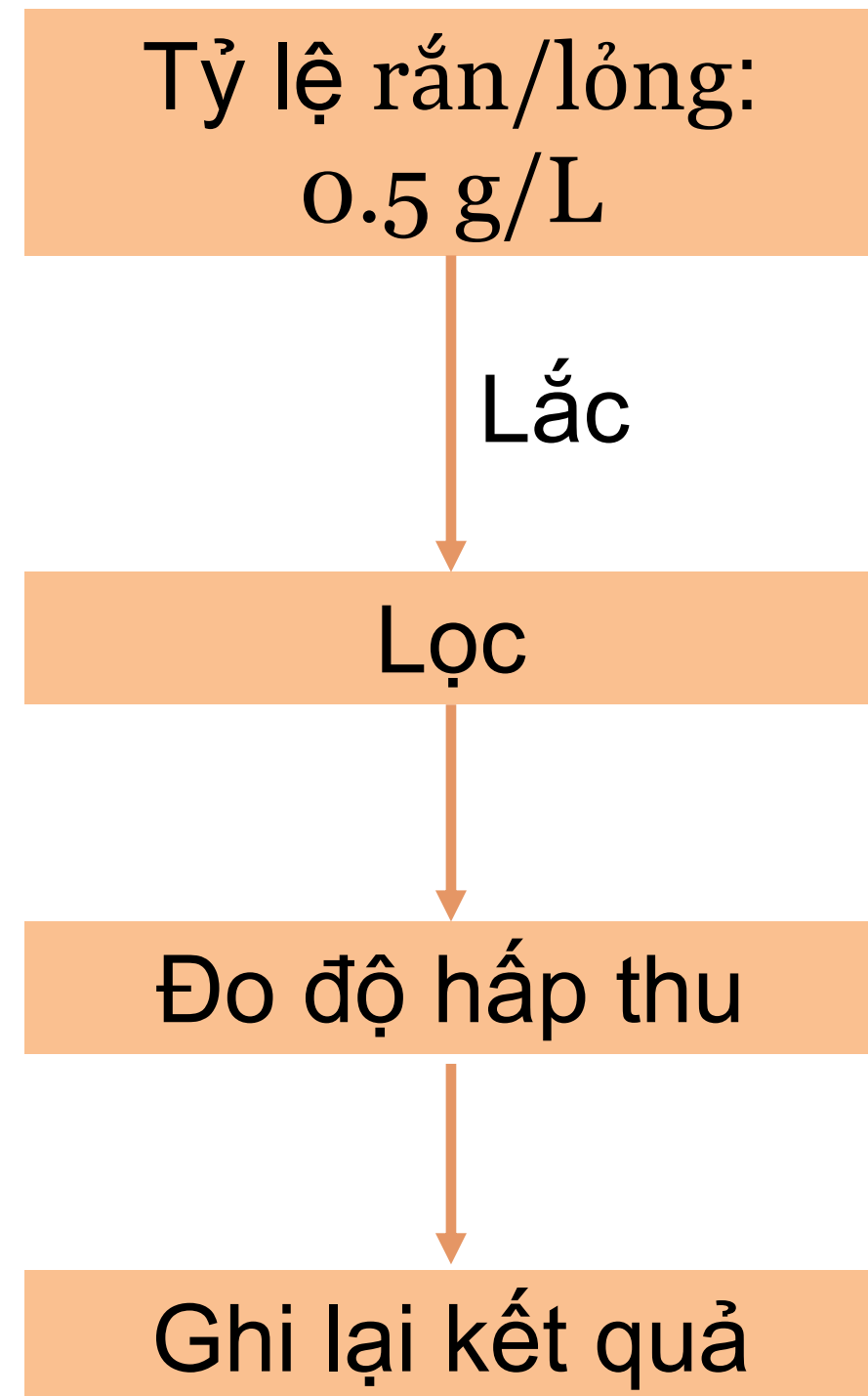
□ Examples 2

b) $C_e = 10(1-0,9) = 1,0 \text{ g/m}^3$; Vì quá trình HP tuân theo đẳng nhiệt Langmuir

$$q_e = \frac{K_L Q_{max} C_e}{1 + K_L C_e}$$
$$q_e = \frac{0,65 \times 56 \times 1,0}{1 + 0,65 \times 1,0} = 22,06 \text{ g/m}^3$$

$$\frac{m}{V} = \frac{(C_0 - C_e)}{q_e}$$
$$\frac{m}{100} = \frac{10 - 1}{22,06} \rightarrow m = 40,8 \text{ kg}$$

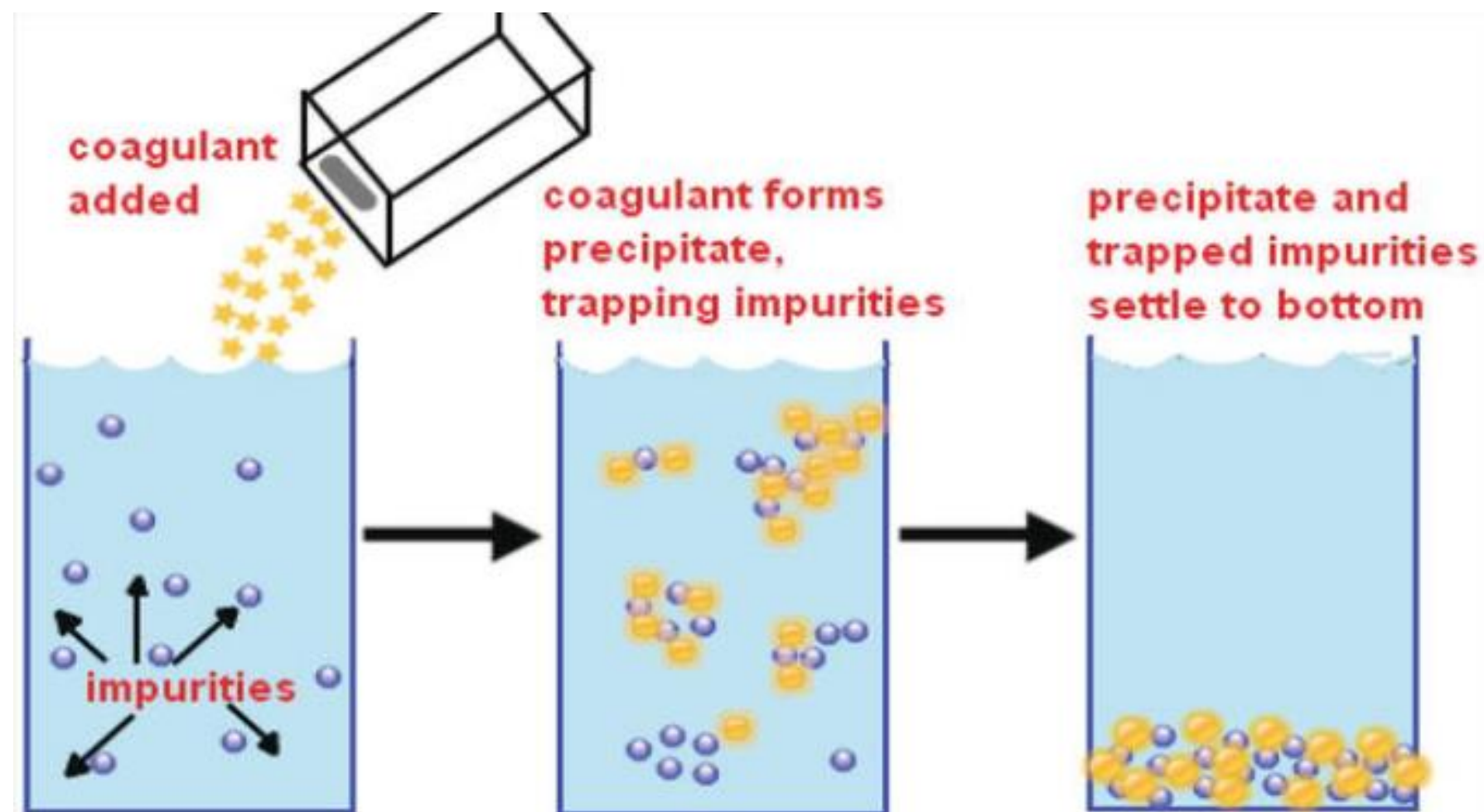
4.8 Common experimental design for bath adsorption



Chapter 7

COAGULATION AND FLOCCULATION

(KEO TỤ VÀ TẠO BÔNG)

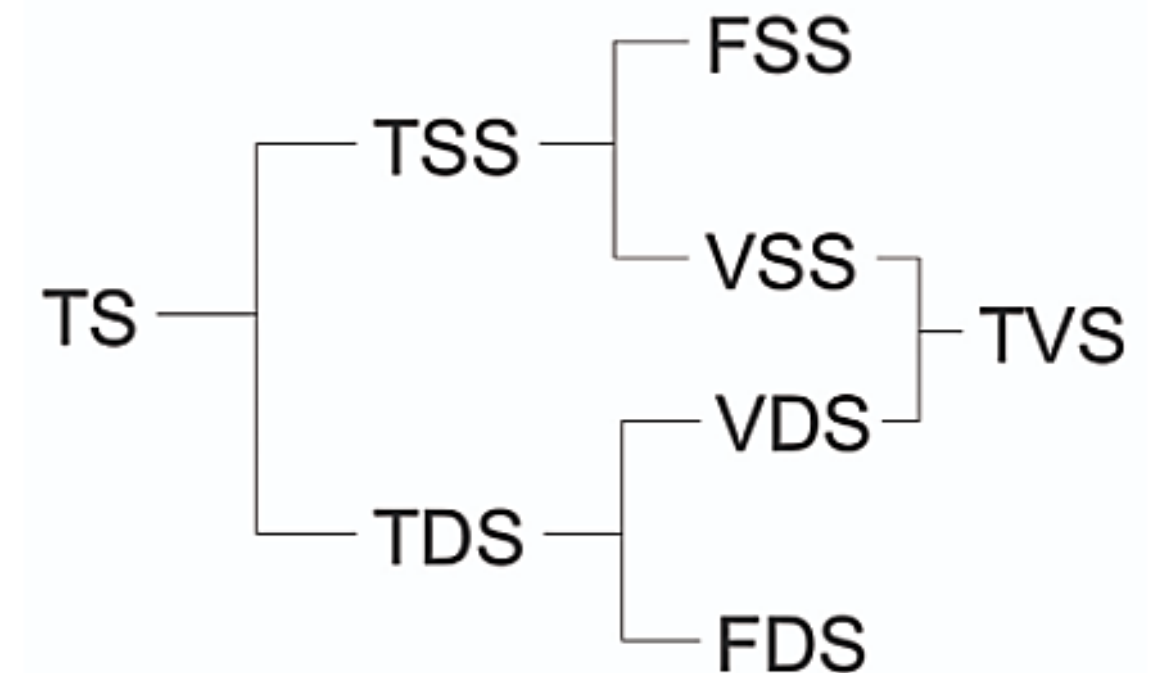


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7.1 Introduction

The need of solid removal from water

- ❑ Aesthetics (tính thẩm mỹ) and health
- ❑ Colloids – impact **color** and **turbidity** to water – aesthetical acceptability
- ❑ Microbes are colloids



QCVN 02:2009/BYT

TT	Tên chỉ tiêu	Đơn vị tính	Giới hạn tối đa cho phép		Phương pháp thử	Mức độ giám sát
			I	II		
1	Màu sắc ^(*)	TCU	15	15	TCVN 6185 - 1996 (ISO 7887 - 1985) hoặc SMEWW 2120	A
2	Mùi vị ^(*)	-	Không có mùi vị lạ	Không có mùi vị lạ	Cảm quan, hoặc SMEWW 2150 B và 2160 B	A
3	Độ đục ^(*)	NTU	5	5	TCVN 6184 - 1996 (ISO 7027 - 1990) hoặc SMEWW 2130 B	A

QCVN 40:2011/BTNMT

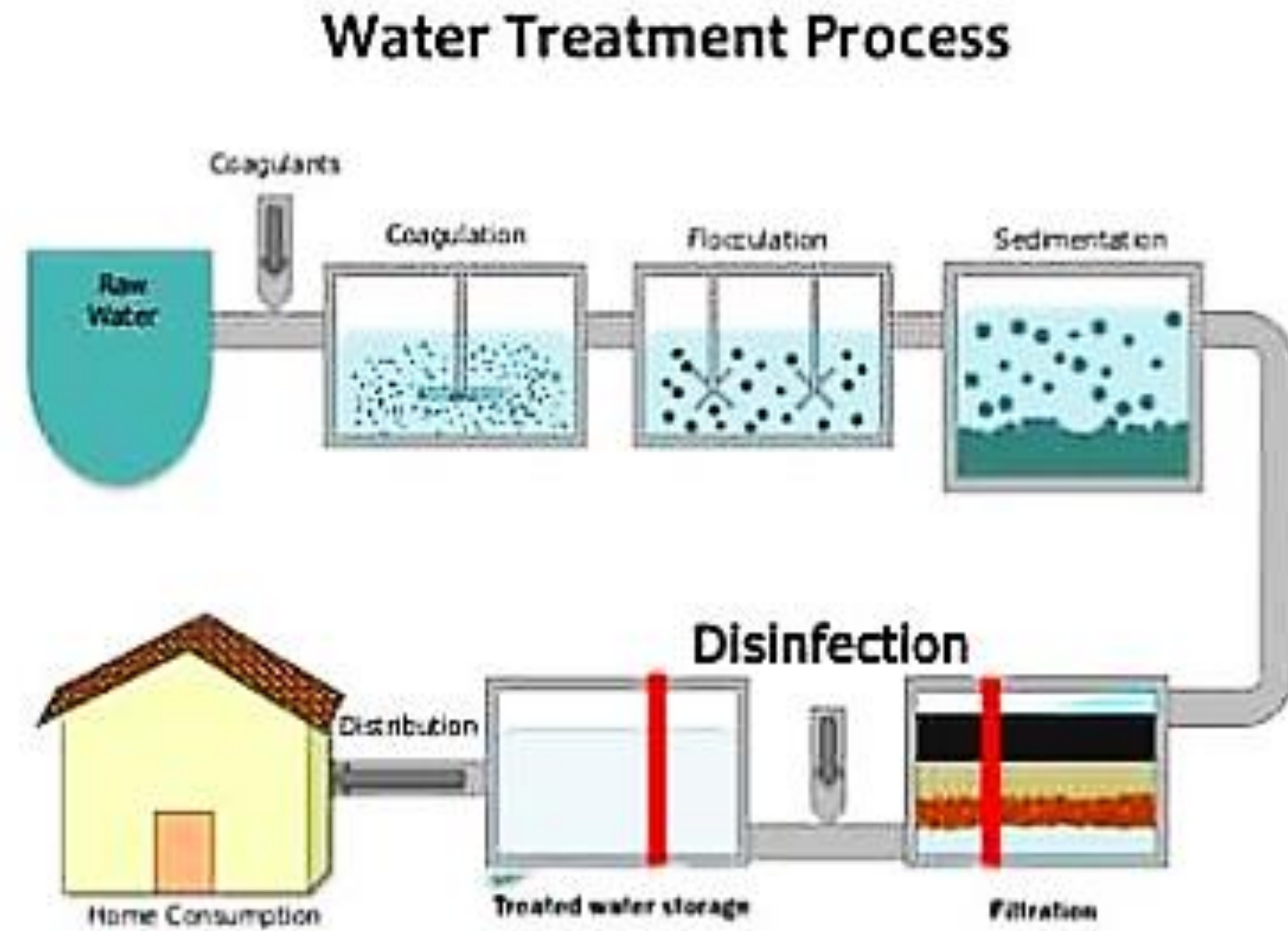
TT	Thông số	Đơn vị	Giá trị C	
			A	B
1	Nhiệt độ	oC	40	40
2	Màu	Pt/Co	50	150
3	pH	-	6 đến 9	5,5 đến 9
4	BOD5 (20oC)	mg/l	30	50
5	COD	mg/l	75	150
6	Chất rắn lơ lửng	mg/l	50	100

7.1 Introduction

The roles played by coagulation and flocculation

- ❑ Removal of colloidal substances from water
 - solid,
 - color,
 - turbidity
- ❑ Potable water requirements
 - health,
 - aesthetics,
 - economic

In wastewater treatment, sometimes, Coagulation and flocculation also remove COD, color, P, ...



7.1 Introduction

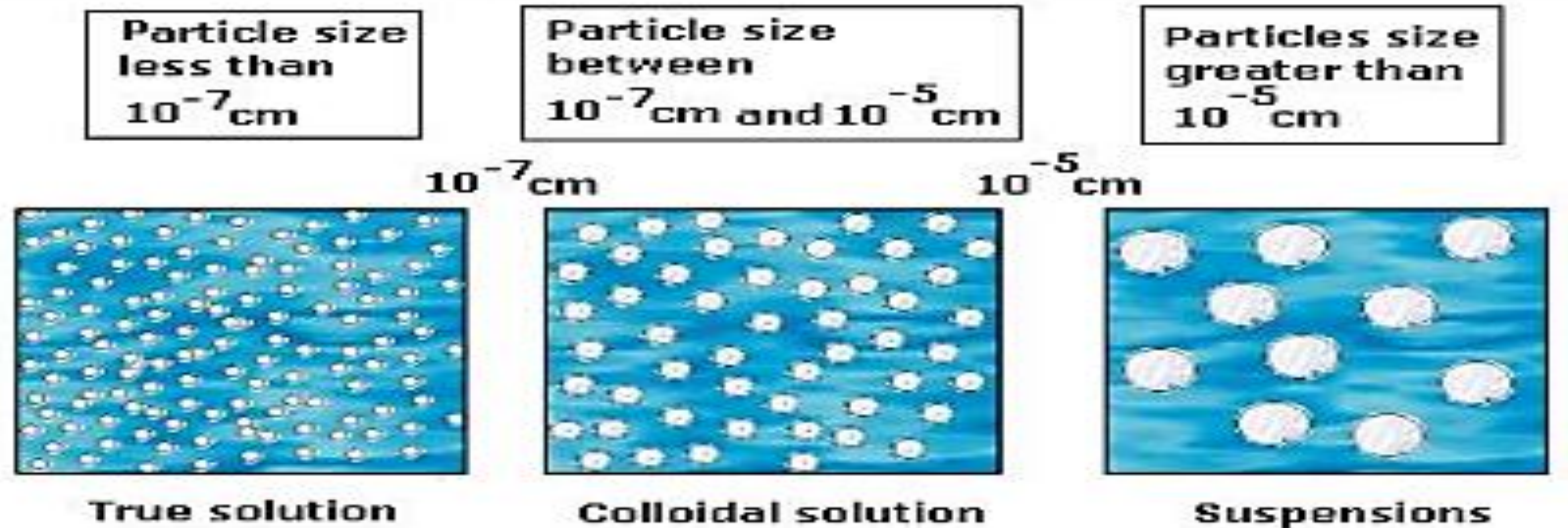
Why coagulation and flocculation?

Various sizes of particles in raw water

Particle diameter (mm)	Type	Settling velocity
10	Pebble	0.73 m/s
1	Course sand	0.23 m/s
0.1	Fine sand	0.6 m/min
0.01	Silt	8.6 m/d
0.0001 (10 micron)	Large colloids	0.3 m/y
0.000001 (1 nano)	Small colloids	3 m/million y

Gravity settling

7.2 Colloid



homogeneously distributed throughout the solutions, pass through ordinary filter paper as well as through animal membranes.
Sugar, NaCl, Ure, ...

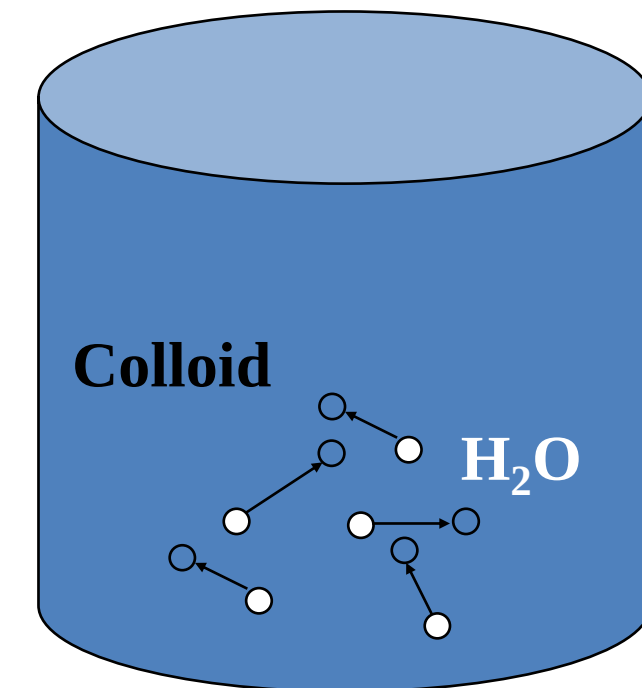
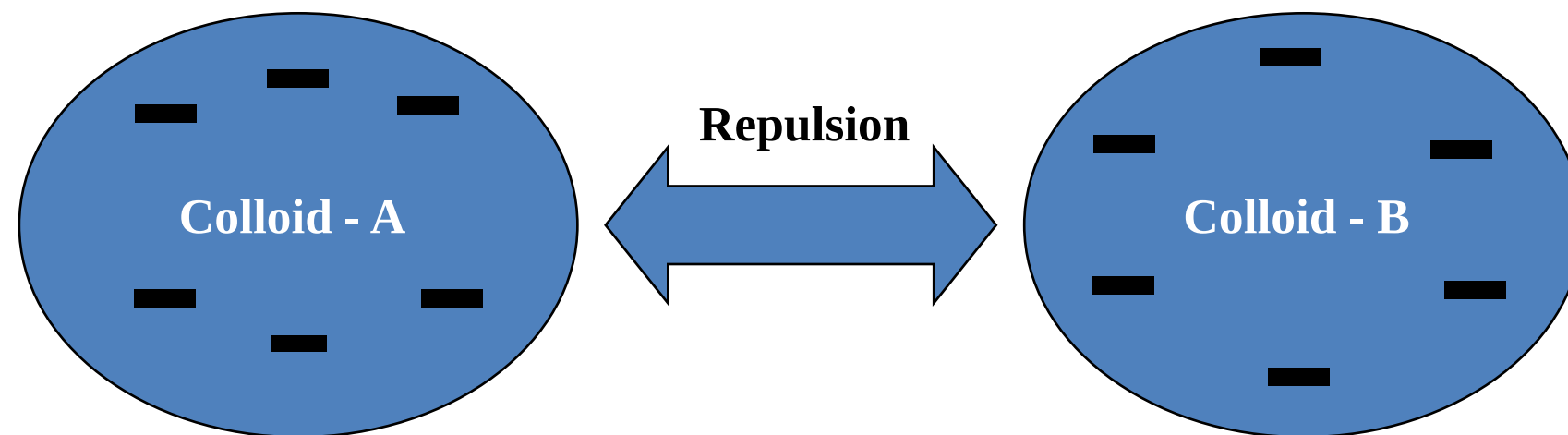
heterogeneous systems, larger size but not large enough to be seen with naked eye; can pass through ordinary filter paper but not through an animal membrane

heterogeneous system, visible to naked eye, neither pass through an animal membrane nor through an ordinary filter paper

7.2 Colloid

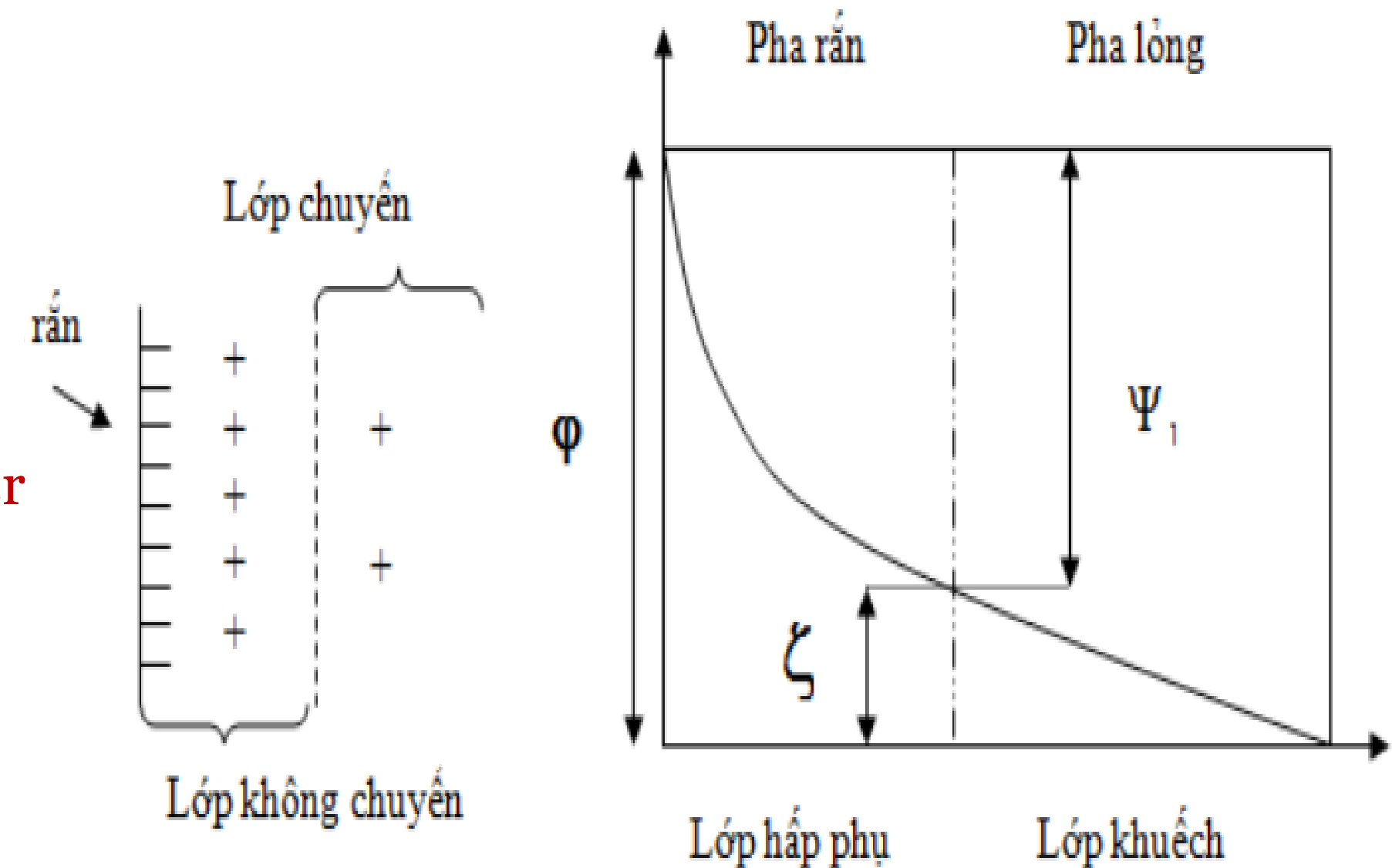
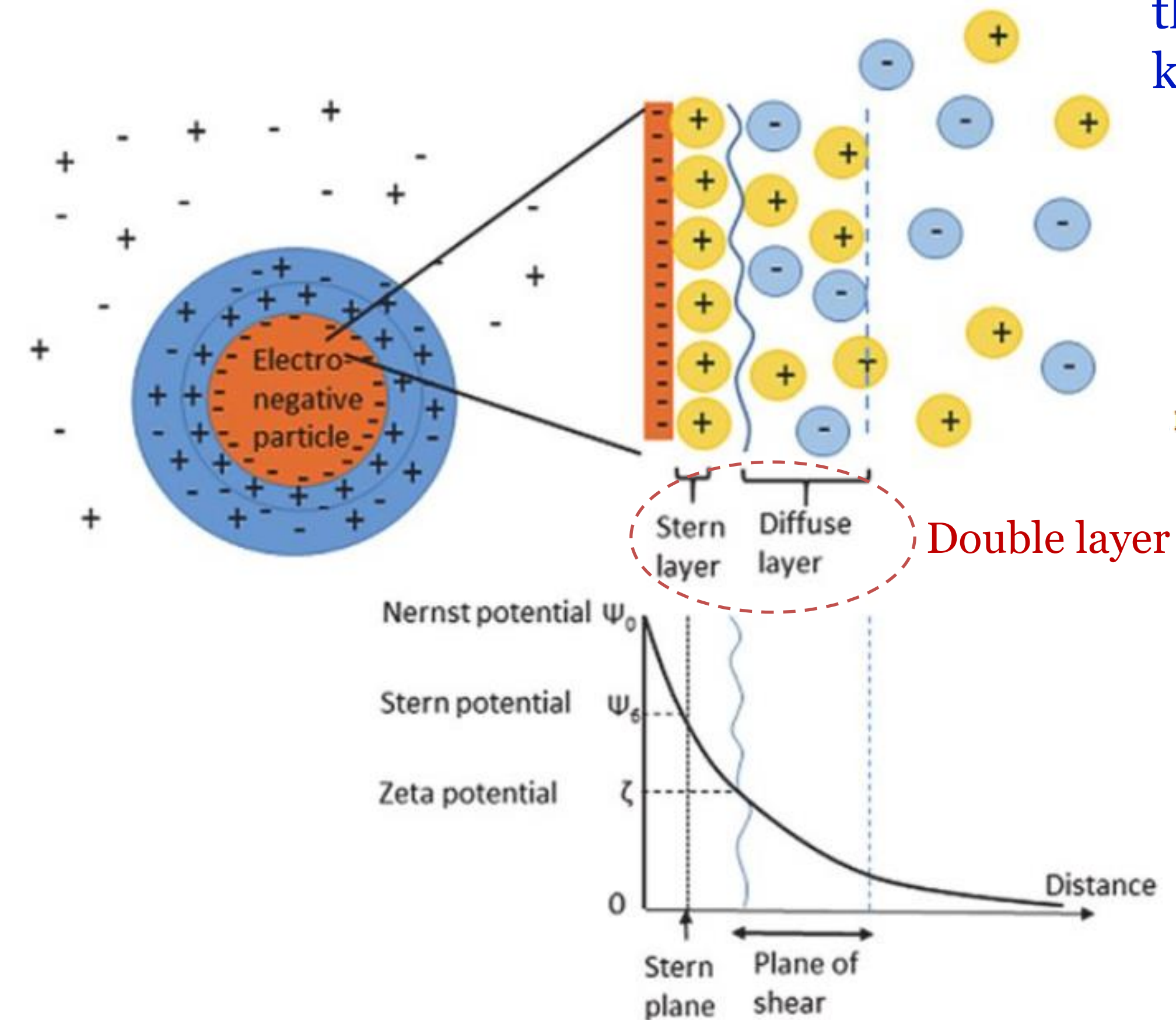
Colloid Stability

- ❑ Colloids have a net negative surface charge
- ❑ Electrostatic force prevents them from agglomeration (kết tụ)
- ❑ Brownian motion keeps the colloids in suspension (lơ lửng)
- ❑ Impossible to remove colloids by gravity settling



7.2 Colloid

Thế điện động (điện thế Zeta): Điện thế xuất hiện giữa lớp chuyển động và không chuyển động (lớp điện tích kép)



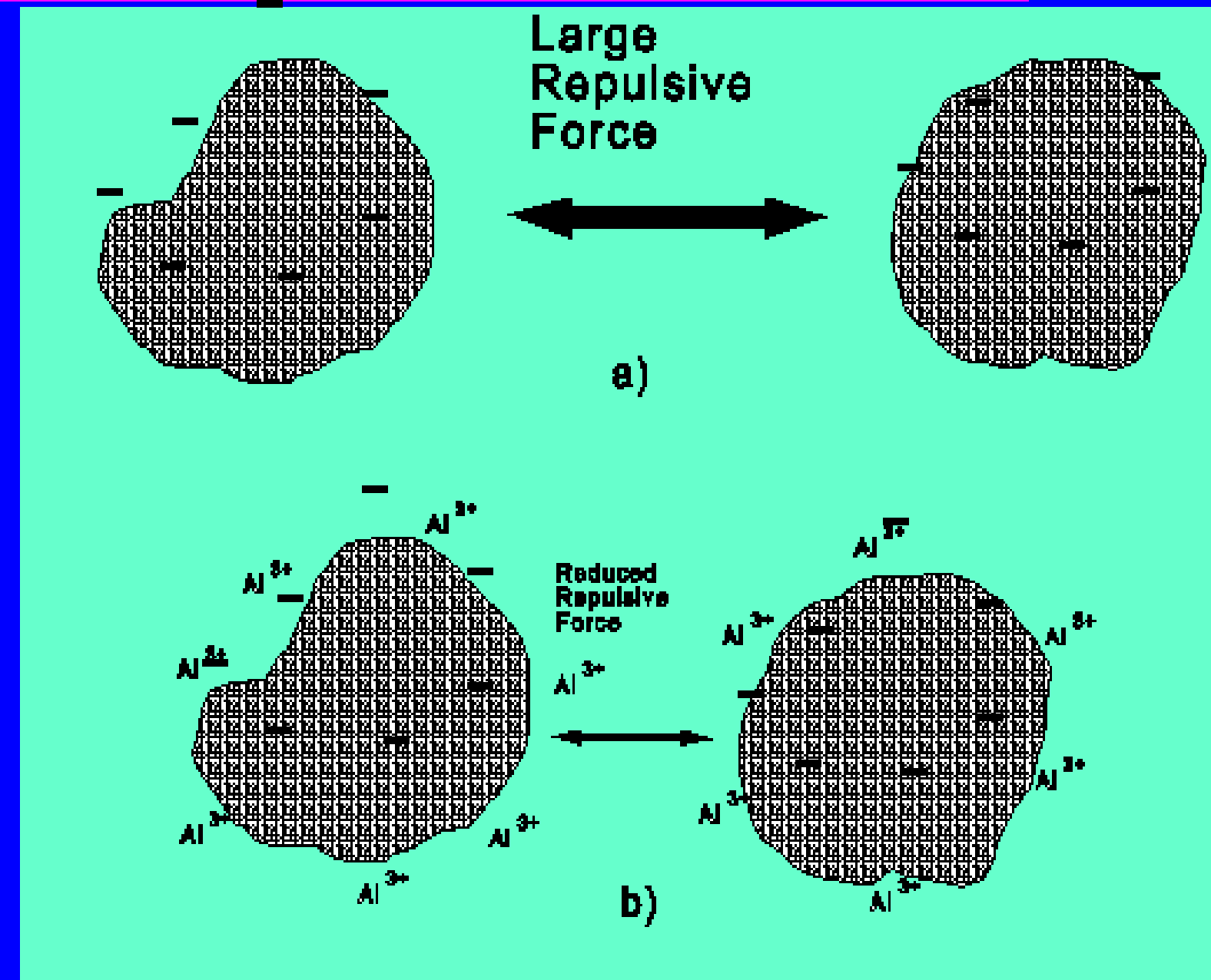
Thế điện động- điện tích kép

7.3 Definitions

- ❑ **Coagulation** is the destabilization of colloids by addition of chemicals that neutralize the negative charges
- ❑ The chemicals are known as coagulants, usually **higher valence cationic salts** (Al^{3+} , Fe^{3+} , etc.)
- ❑ Coagulation is essentially a chemical process

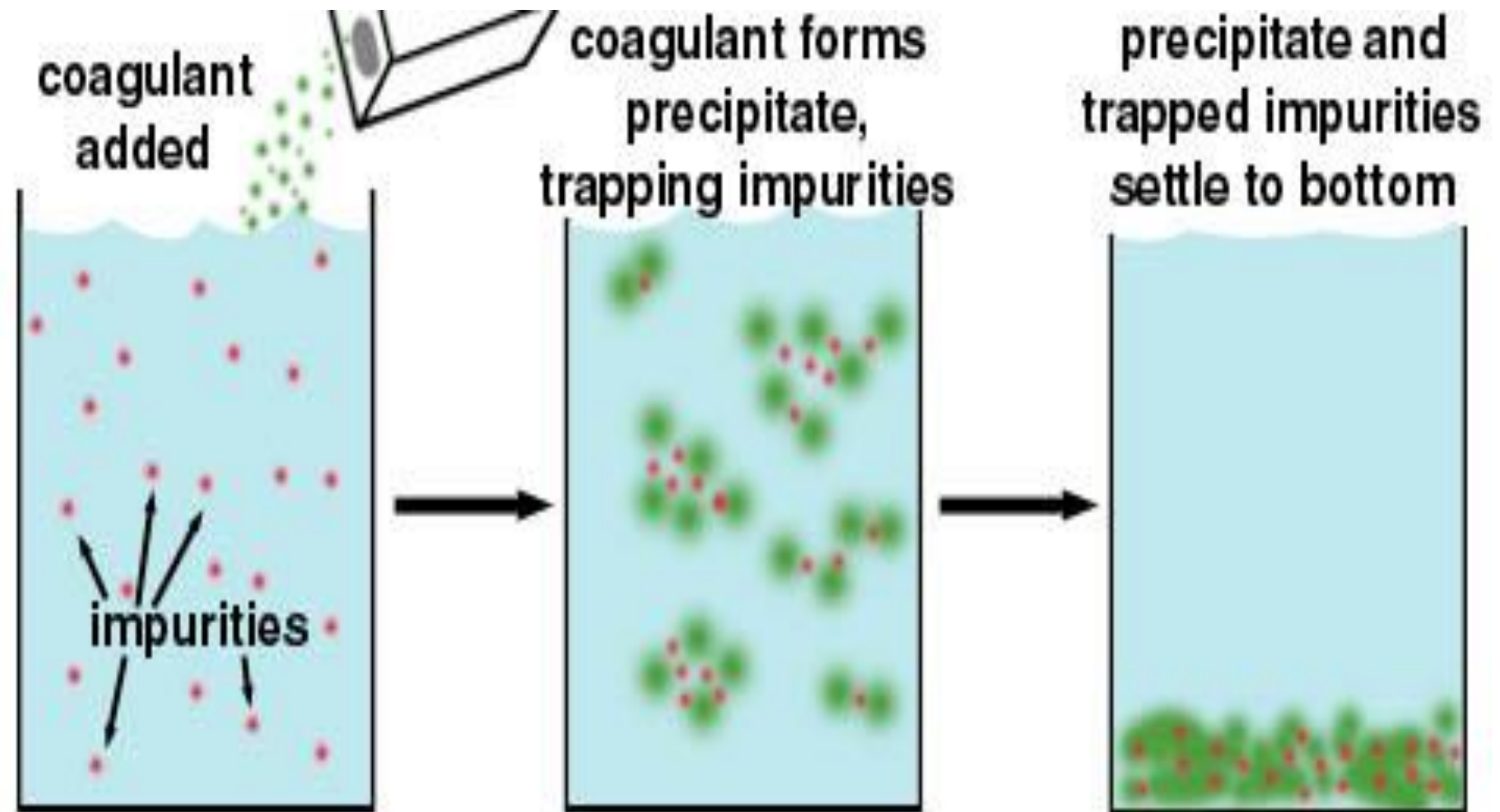
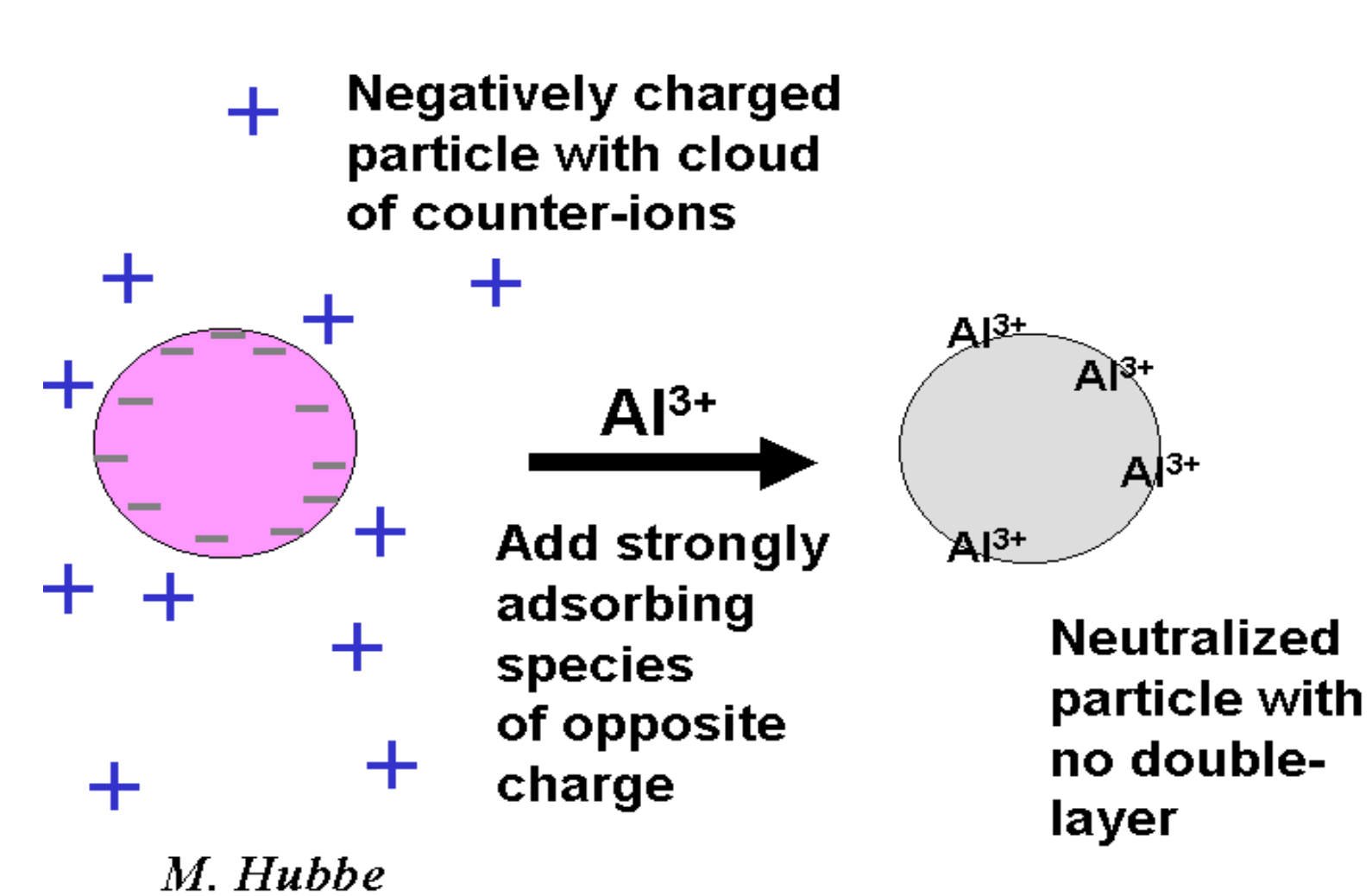
Destabilization: sự mất ổn định;
repulsive force: lực đẩy;
coagulant: chất keo tụ

Charge neutralization



7.3 Definitions

Coagulation aim



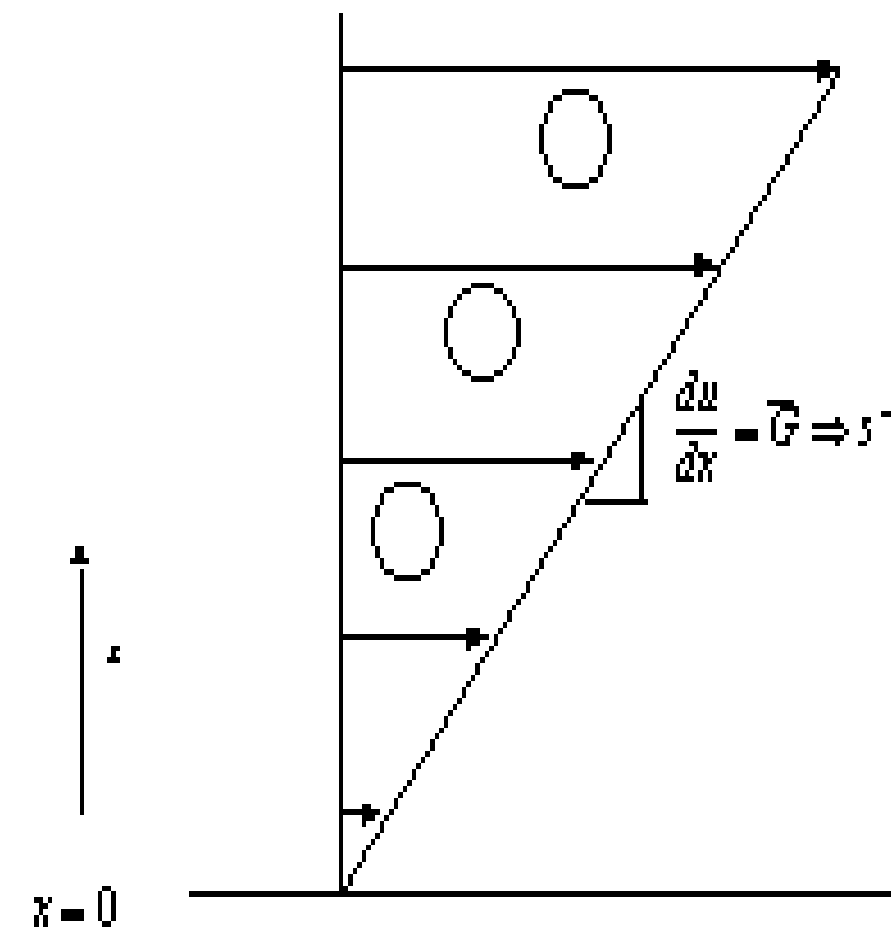
7.3 Definitions

Flocculation: Purpose

- ❑ Promote agglomeration of particles into larger floc
- ❑ Units often designed on the basis of mixing intensity as described by velocity gradient, G
 - some mixing is needed to keep particles in contact with other particles
 - too much mixing can cause floc break-up

Orthokinetic Flocculation

Fluid Shear

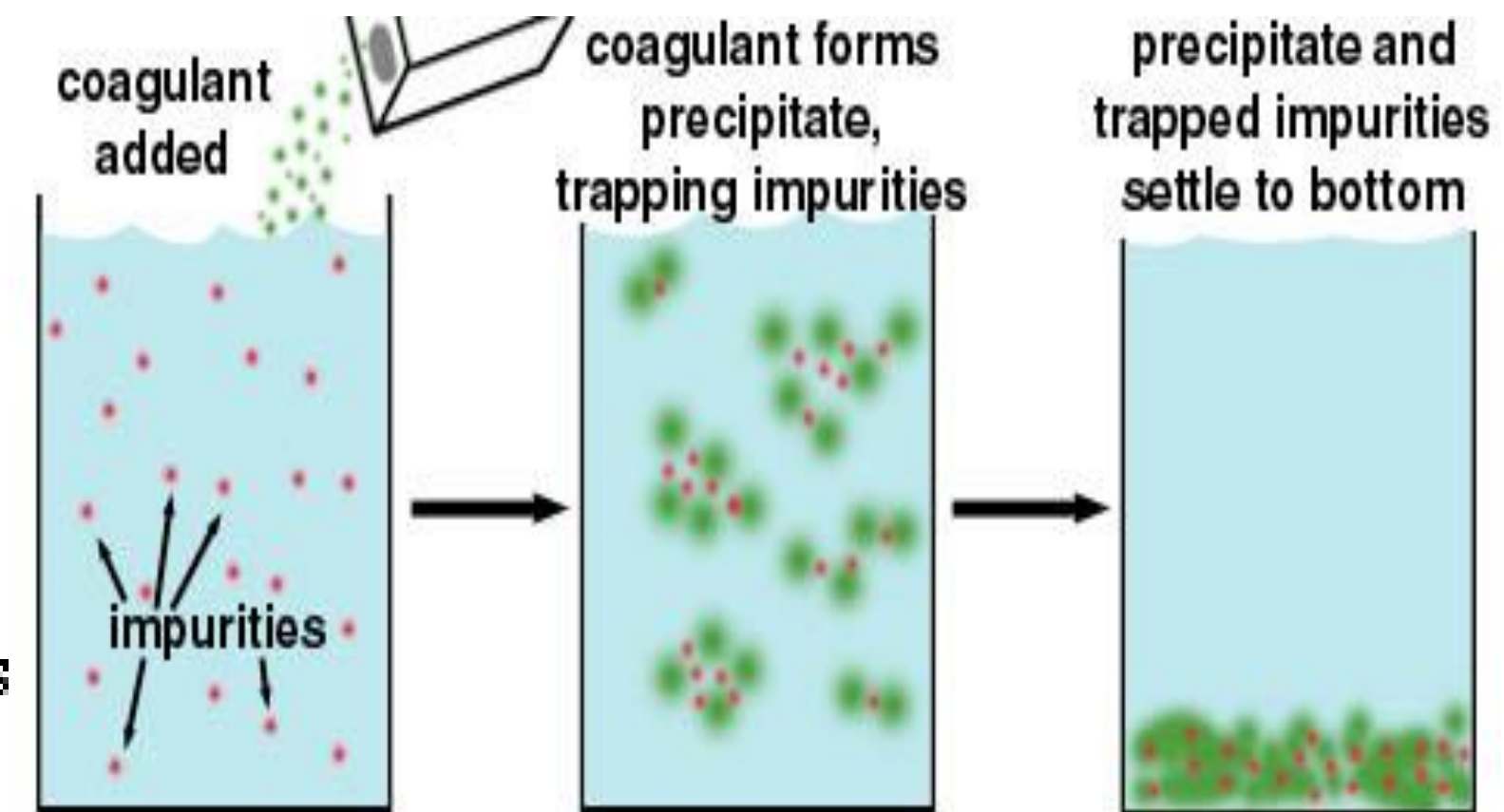
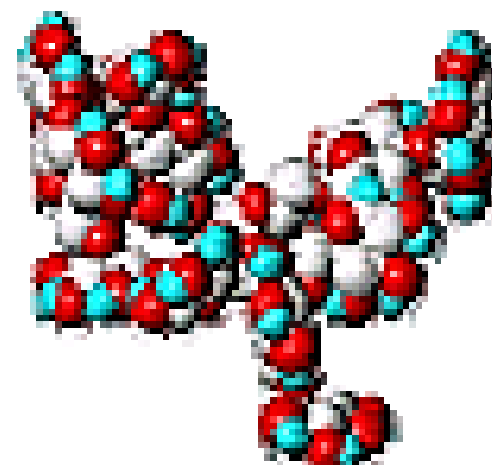


note: $u(x) = x \frac{\partial u}{\partial x}$

7.3 Definitions

Flocculation is the agglomeration of destabilized particles into a large size particles known as flocs which can be effectively removed by sedimentation or flotation.

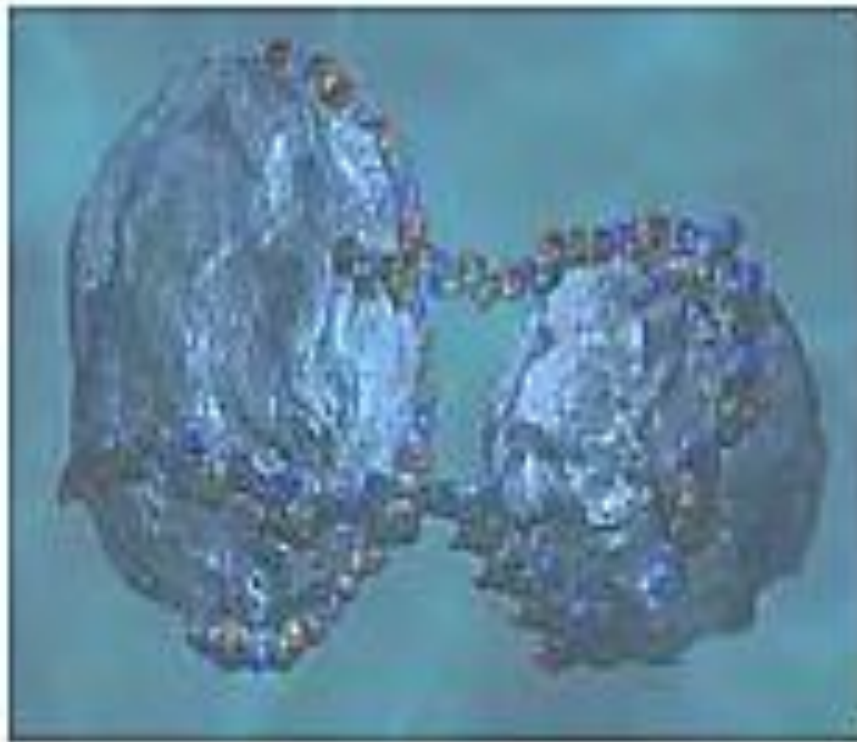
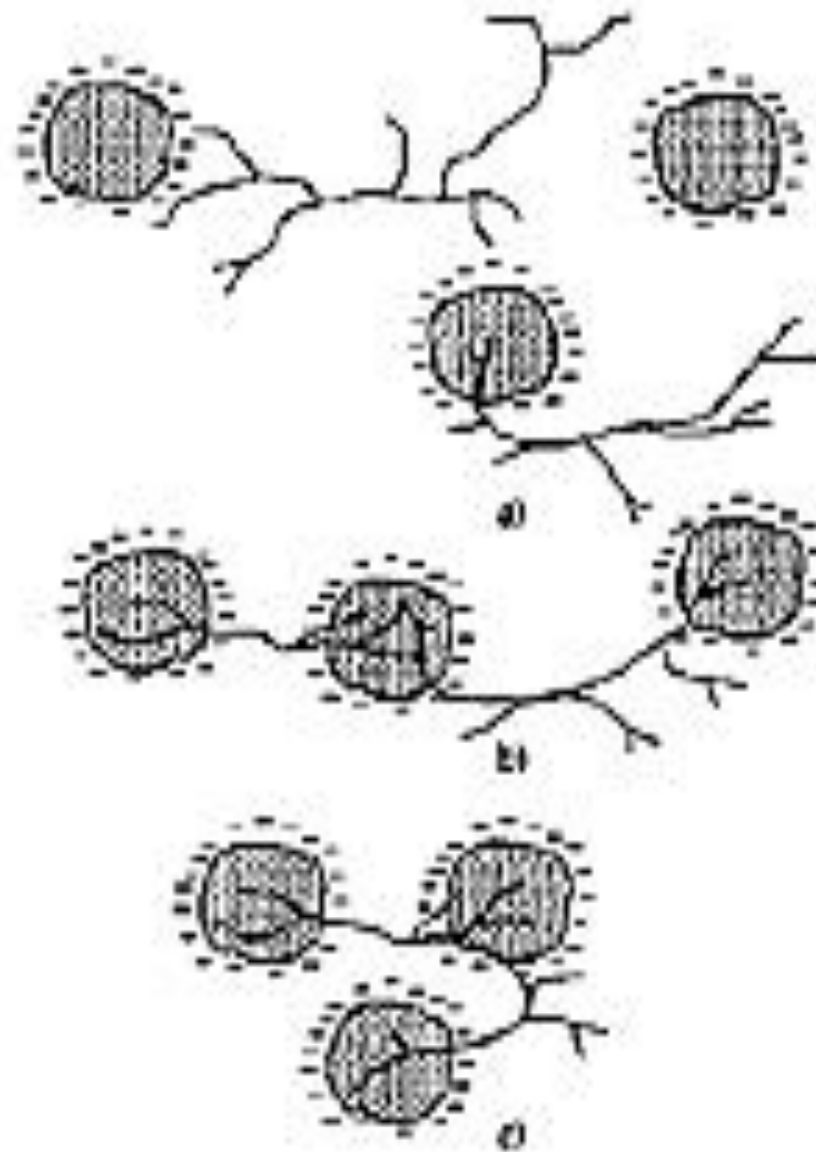
- Gentle mixing or *flocculation*, then causes the destabilized (reduced charge) colloids to cluster.
- Another method of enhancing agglomeration is to add organic polymers.
- These compounds consist of a long carbon chain with active groups such as amine, nitrogen, or sulfate groups along the chain.



7.3 Definitions

□ Flocculation aids

The chain is long enough to allow active groups to bond to multiple colloids



□ Floc formation with polymers

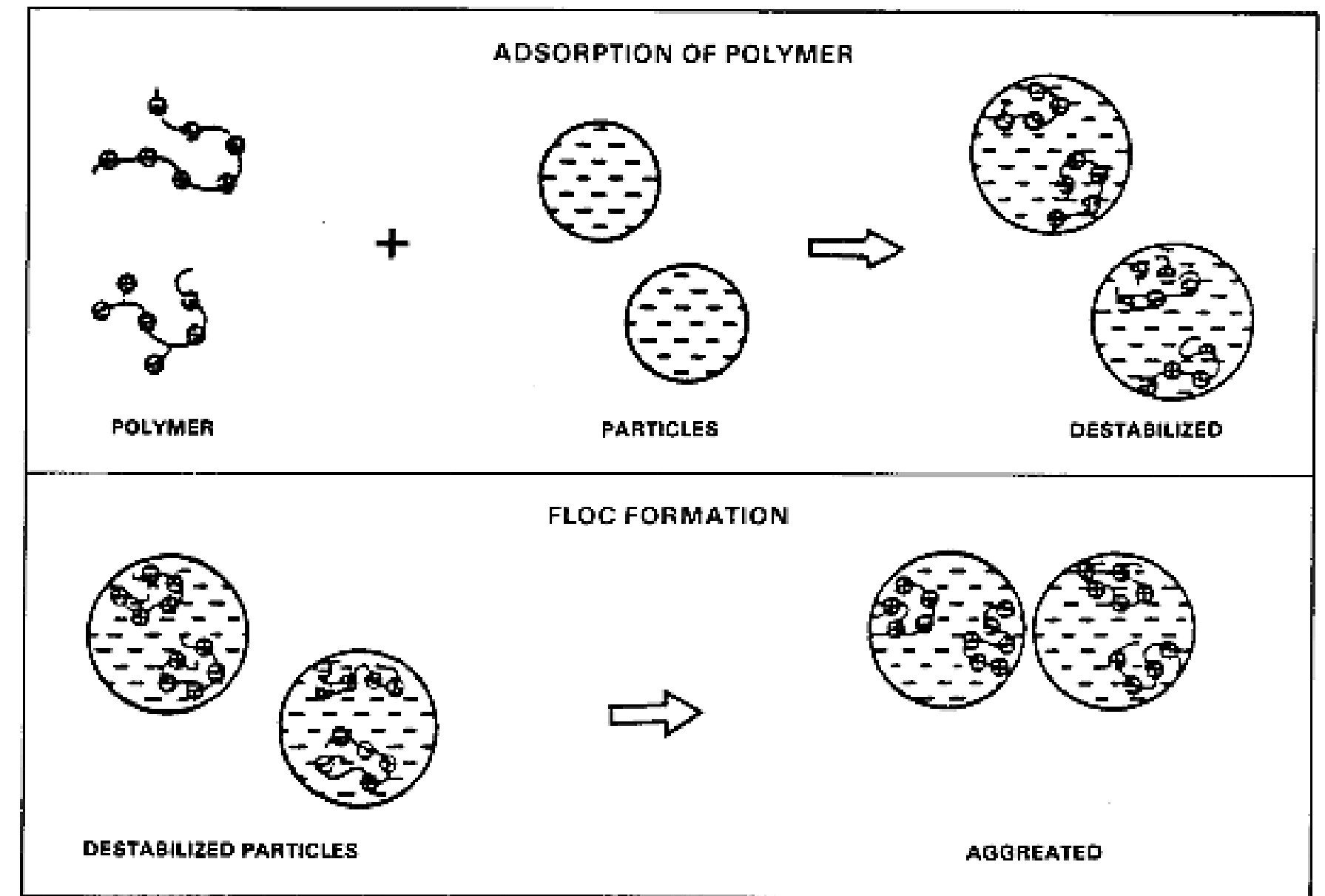


Figure 1-2. Floc formation process

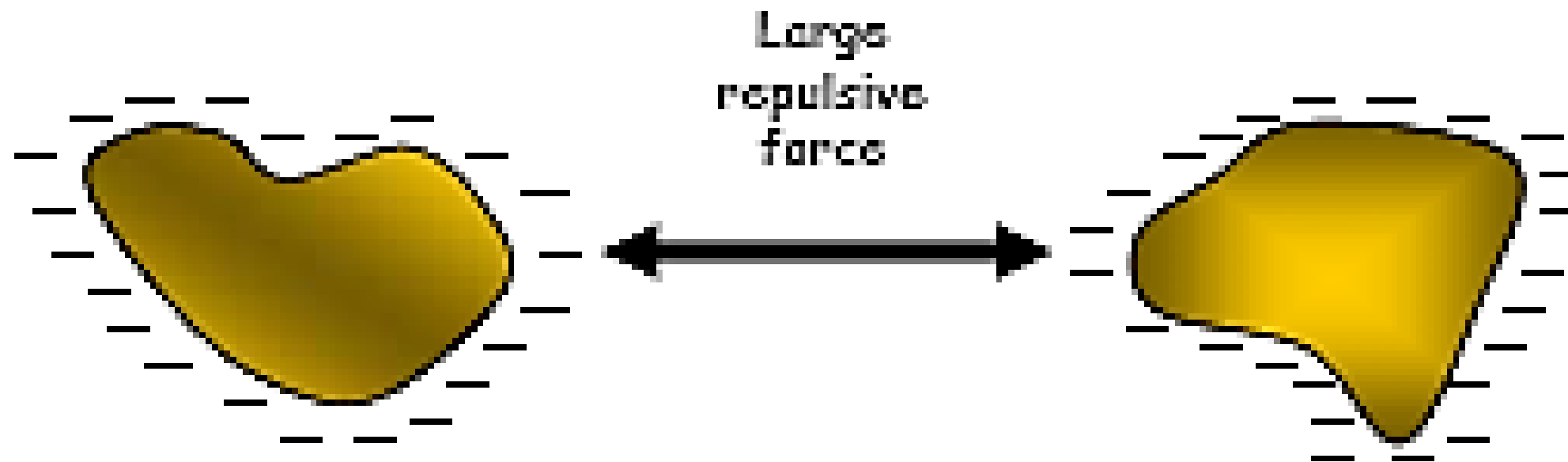
7.4 Colloid Destabilization

Colloidal interaction

- There are two major forces acting on colloids:

1) electrostatic repulsion

(simply, negative colloids repel other negatively charged colloids)

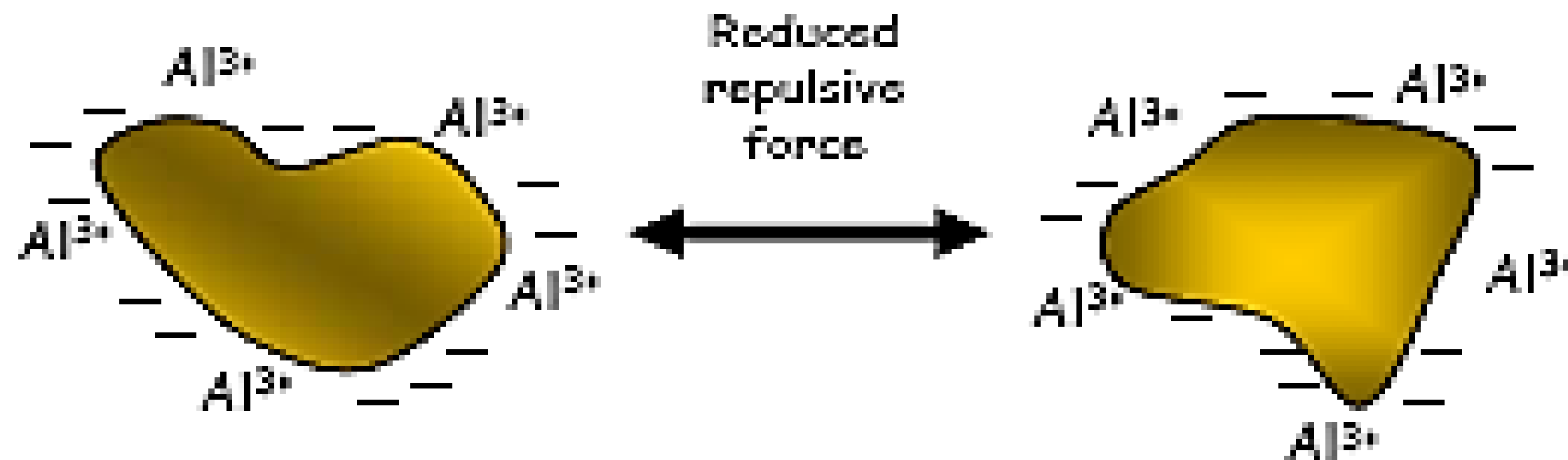


2) intermolecular, or van der Waals, attraction.

7.4 Colloid Destabilization

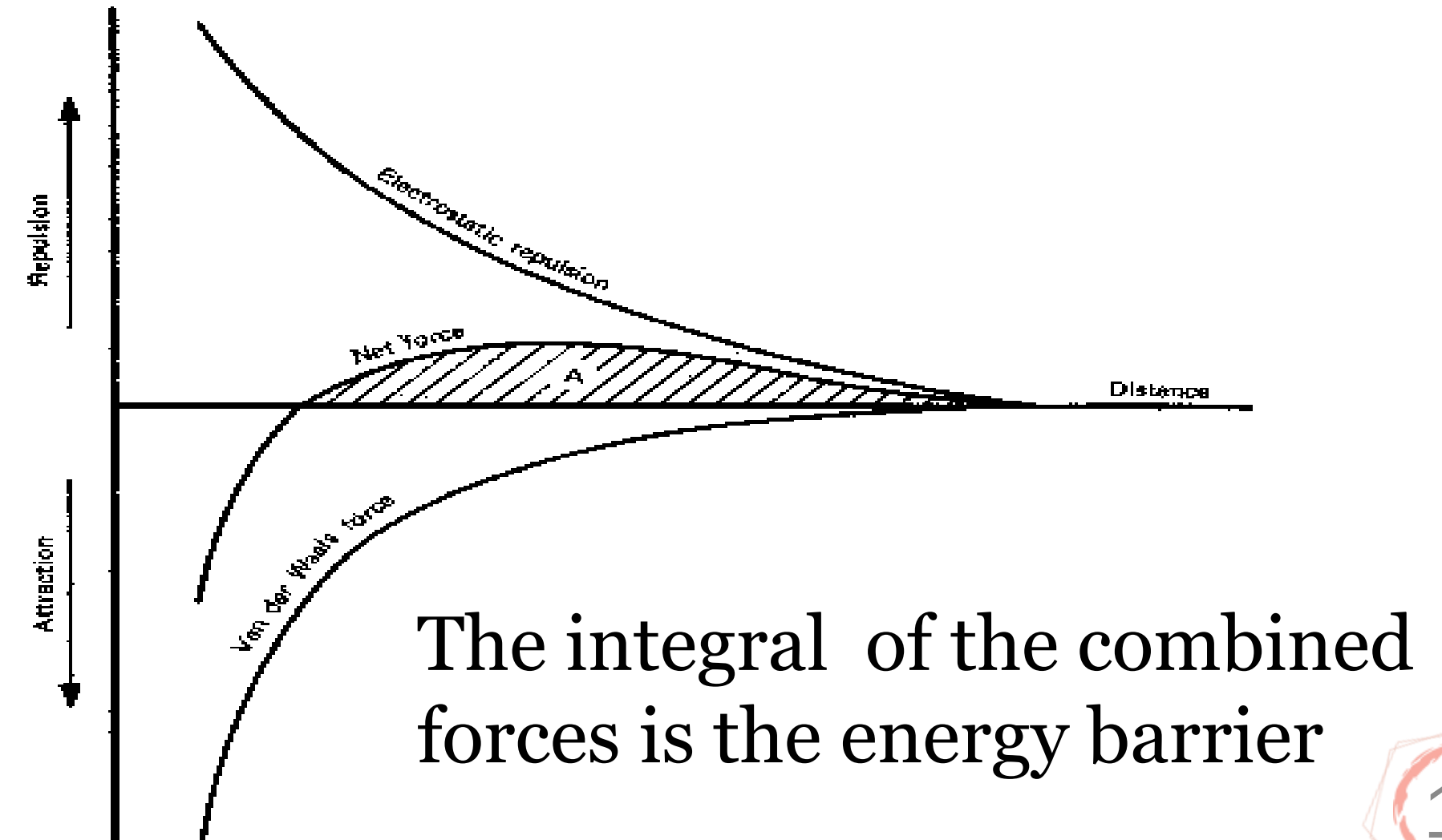
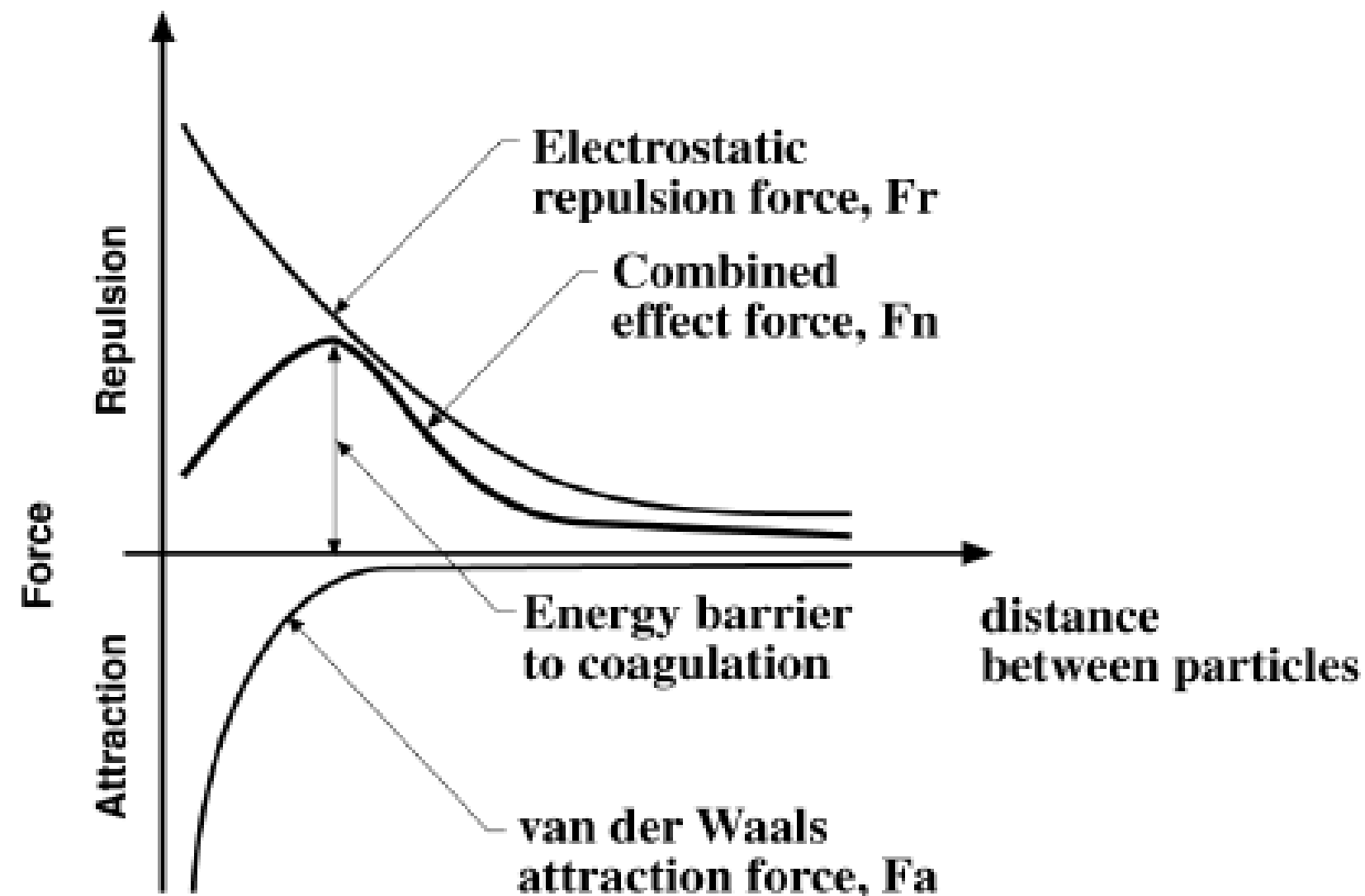
Charge reduction

- Coagulants can be used to reduce the electrostatic repulsive forces
- The electrostatic repulsion reduced by the addition of countercharged ions [Al^{3+}]



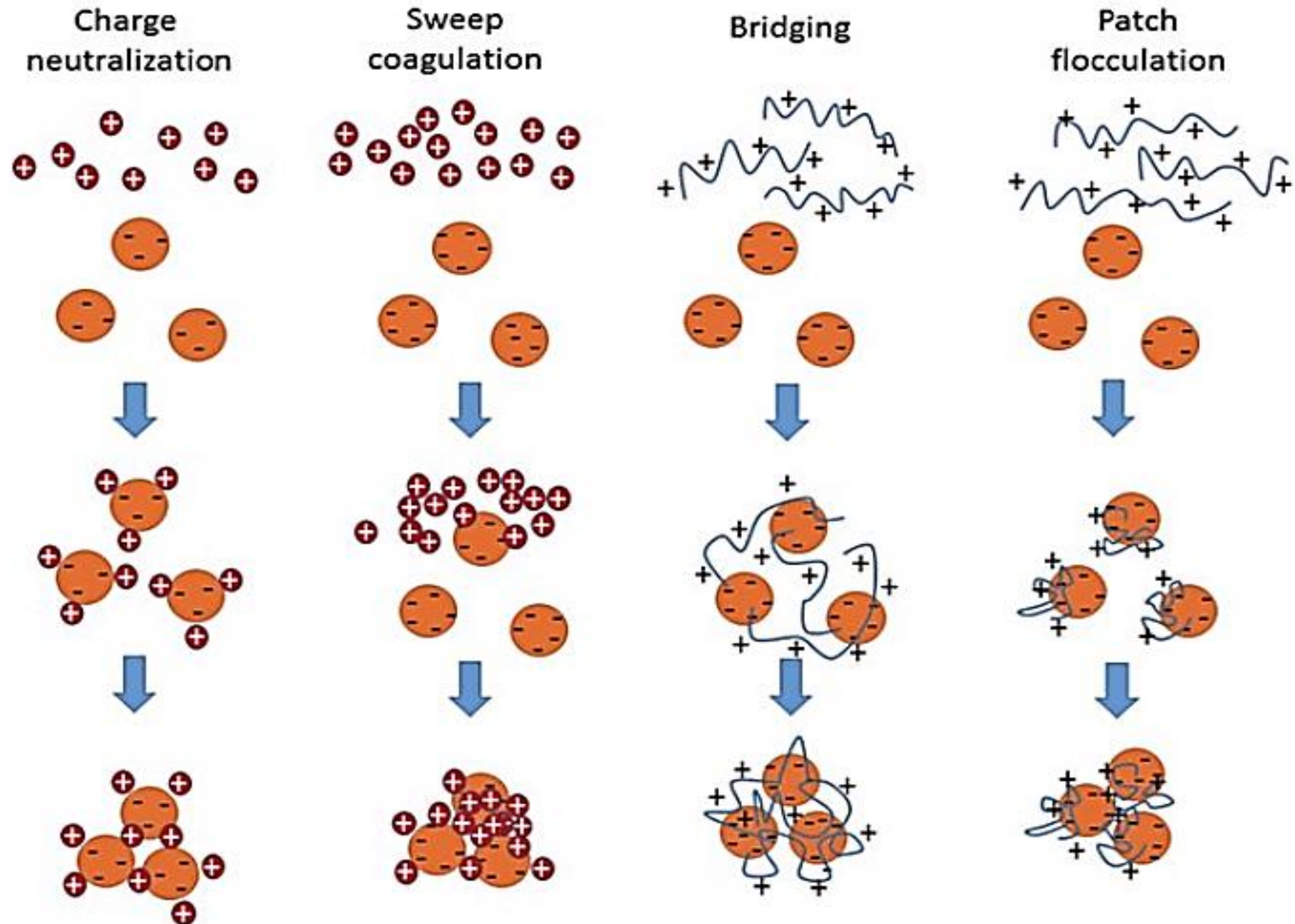
7.4 Colloid Destabilization

- ❑ Colloids can be destabilized by charge neutralization
- ❑ Positively charged ions (Na^+ , Mg^{2+} , Al^{3+} , Fe^{3+} etc.) neutralize the colloidal negative charges and thus destabilize them.
- ❑ With destabilization, colloids aggregate in size and start to settle



7.5 Mechanisms of coagulation and flocculation

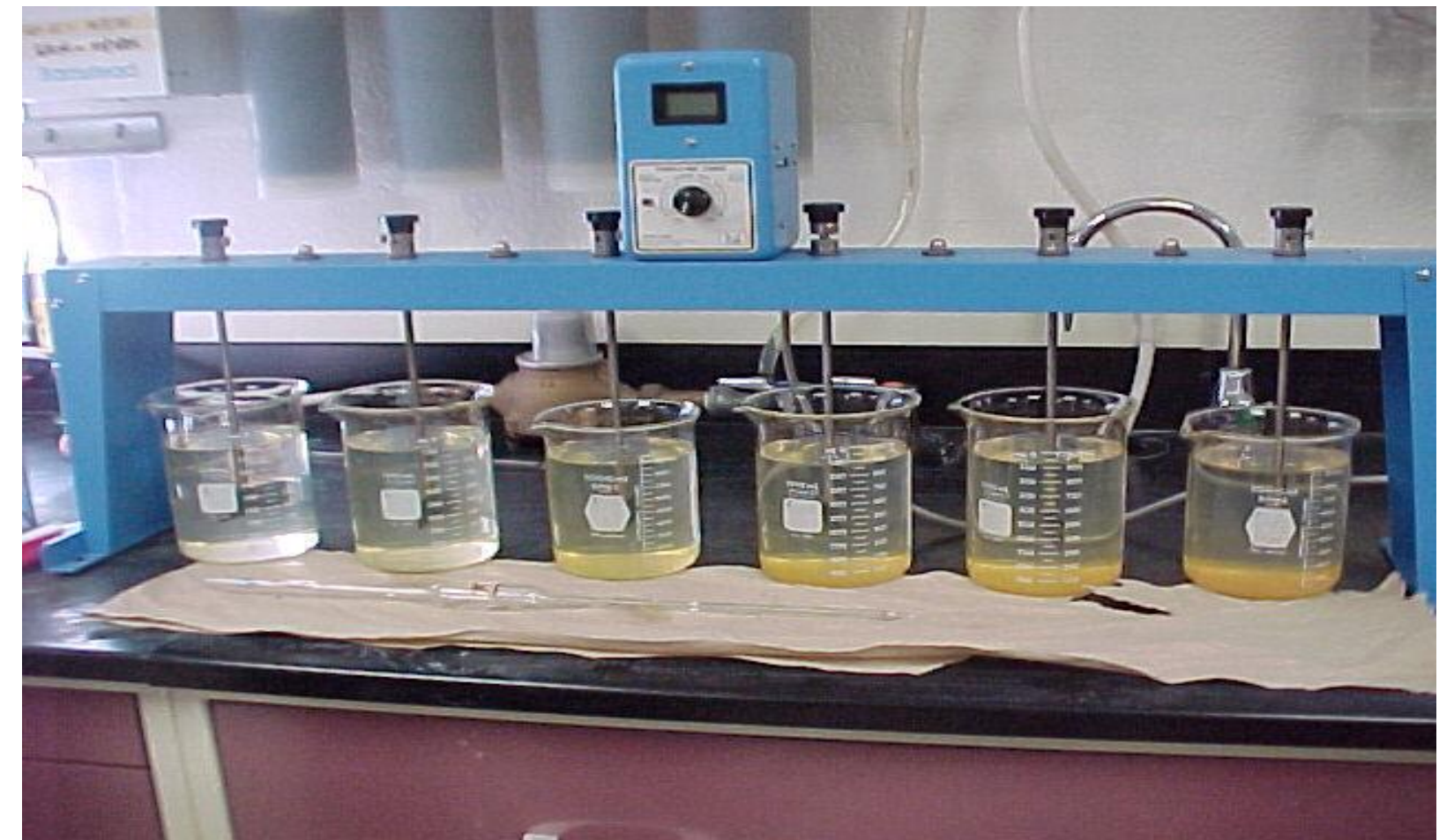
4 mechanisms



7.6 Factors influencing on coagulation and flocculation

- ❑ Properties of colloids in water
- ❑ pH
- ❑ Temperature
- ❑ Type of coagulants
- ❑ Dose of coagulants
- ❑ Dose of coagulant aids
- ❑ Mixing (slow or rapid)

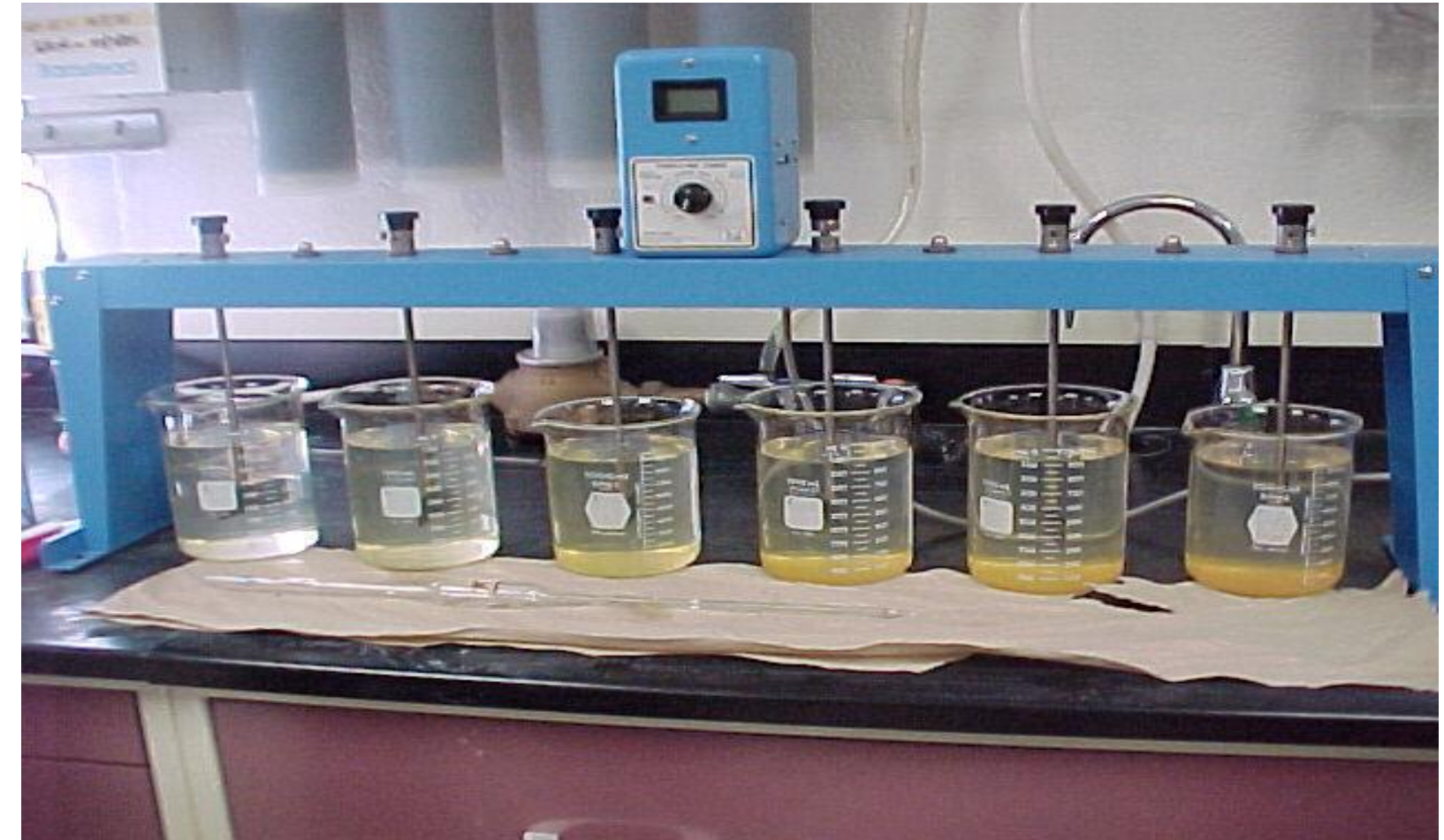
Jar Test Experiment



How to operate the coagulation and flocculation?

7. 6.1 Jar Test – for operation parameters

- ❑ The jar test – a laboratory procedure to determine the **optimum parameters** for coagulation and flocculation (pH and the optimum coagulant dose, ...)
- ❑ A jar test simulates the coagulation and flocculation processes



7. 6.1 Jar Test – determining optimum pH

Determination of optimum pH

- ❑ Fill the jars with raw water sample (500 or 1000 mL) – usually 6 jars
- ❑ Adjust pH of the jars while mixing using H_2SO_4 or NaOH /lime (pH: 5.0; 5.5; 6.0; 6.5; 7.0; 7.5)
- ❑ Add same dose of the selected coagulant (alum or iron) to each jar (Coagulant dose: 5 or 10 mg/L)

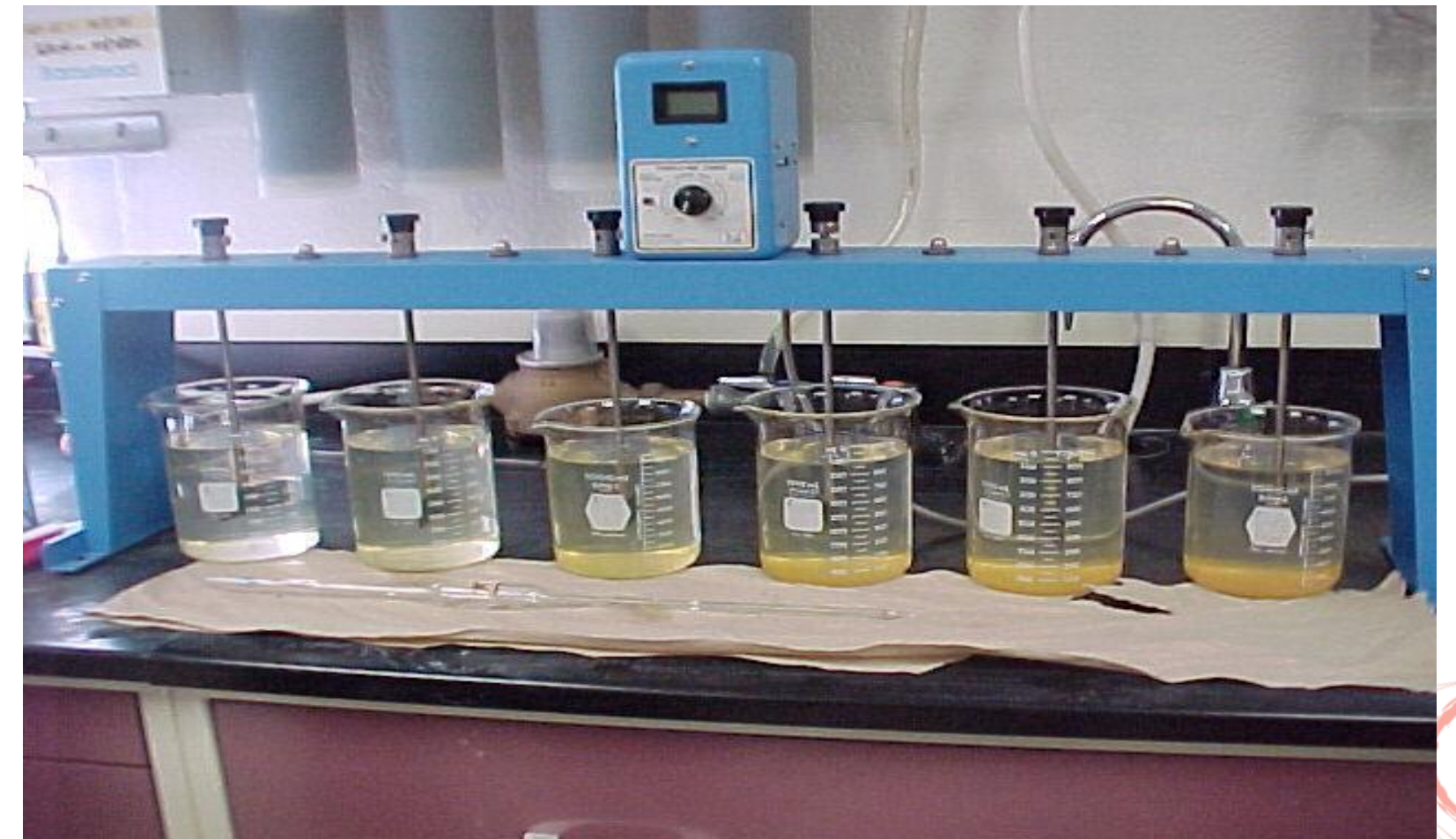


7.6.1 Jar Tests – determining optimum pH

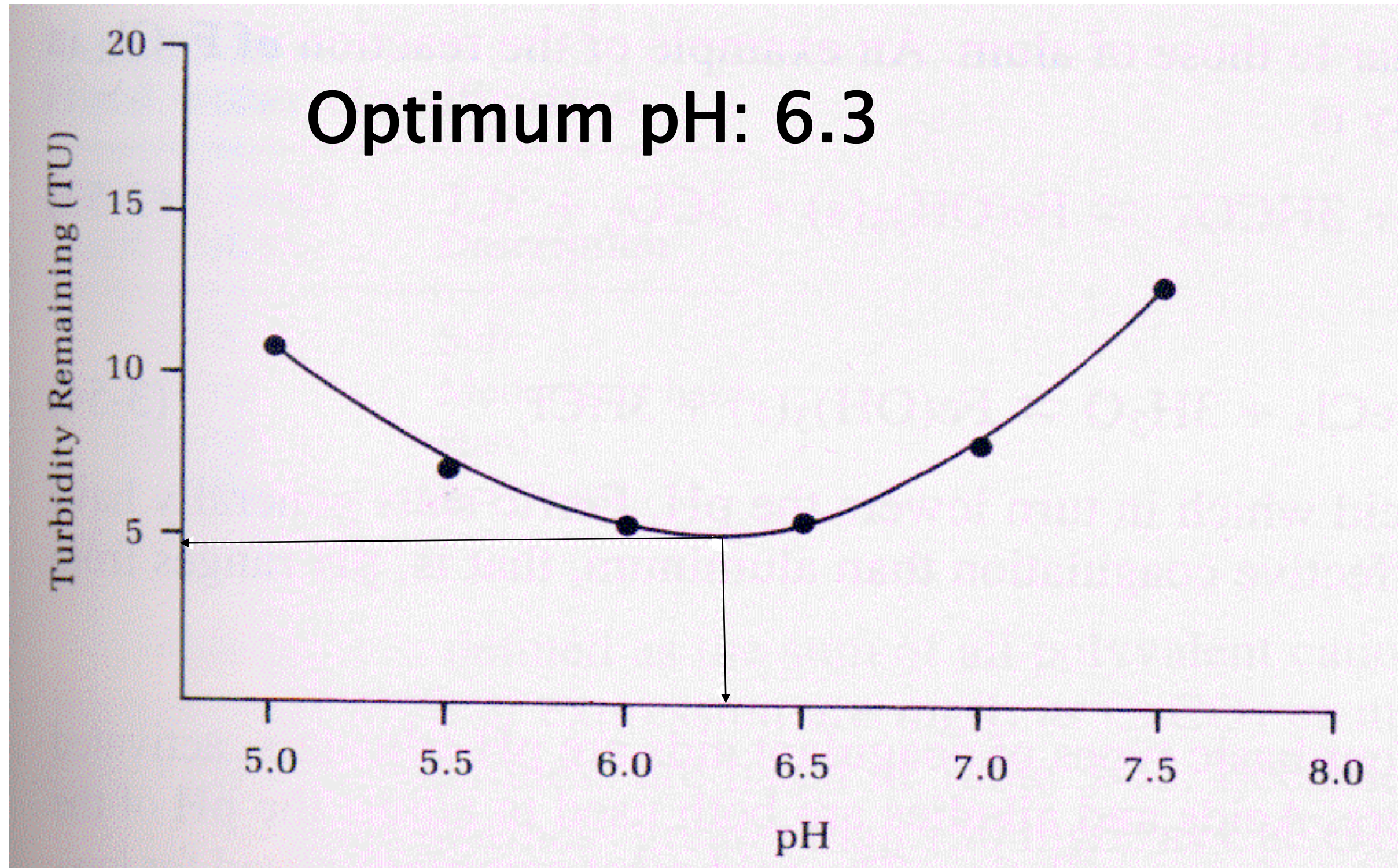
- ❑ Rapid mix each jar at 100 to 150 rpm for 1 minute. The rapid mix helps to disperse the coagulant throughout each container
- ❑ Reduce the stirring speed to 25 to 30 rpm and continue mixing for 15 to 20 mins

This slower mixing speed helps promote floc formation by enhancing particle collisions, which lead to larger flocs

- ❑ Turn off the mixers and allow flocs to settle for 30 to 45 mins
- ❑ Measure the final residual turbidity in each jar
- ❑ Plot residual turbidity against pH

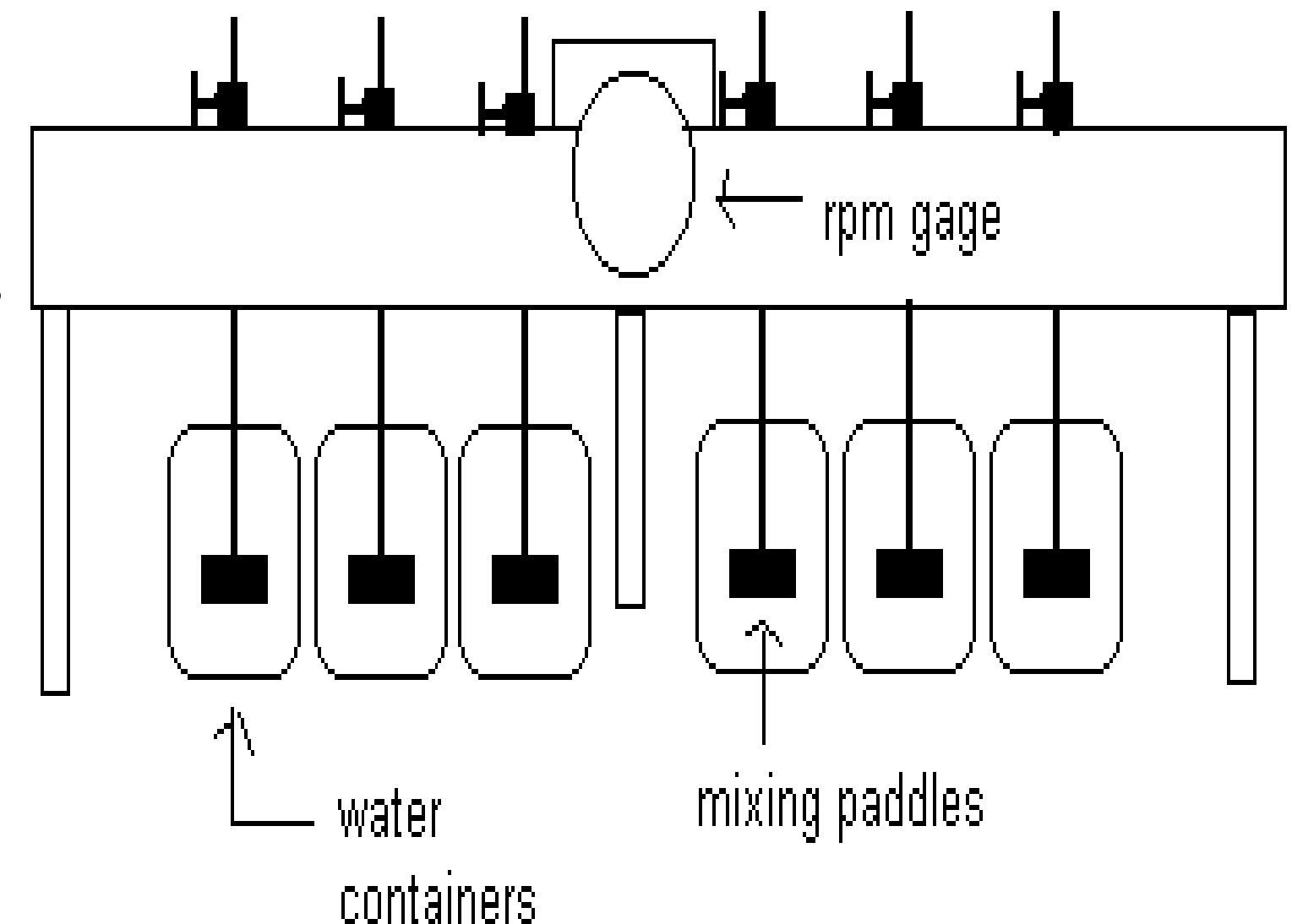


7.6.1 Jar Tests – optimum pH



7.6.2 Jar Tests – Optimum coagulant dose

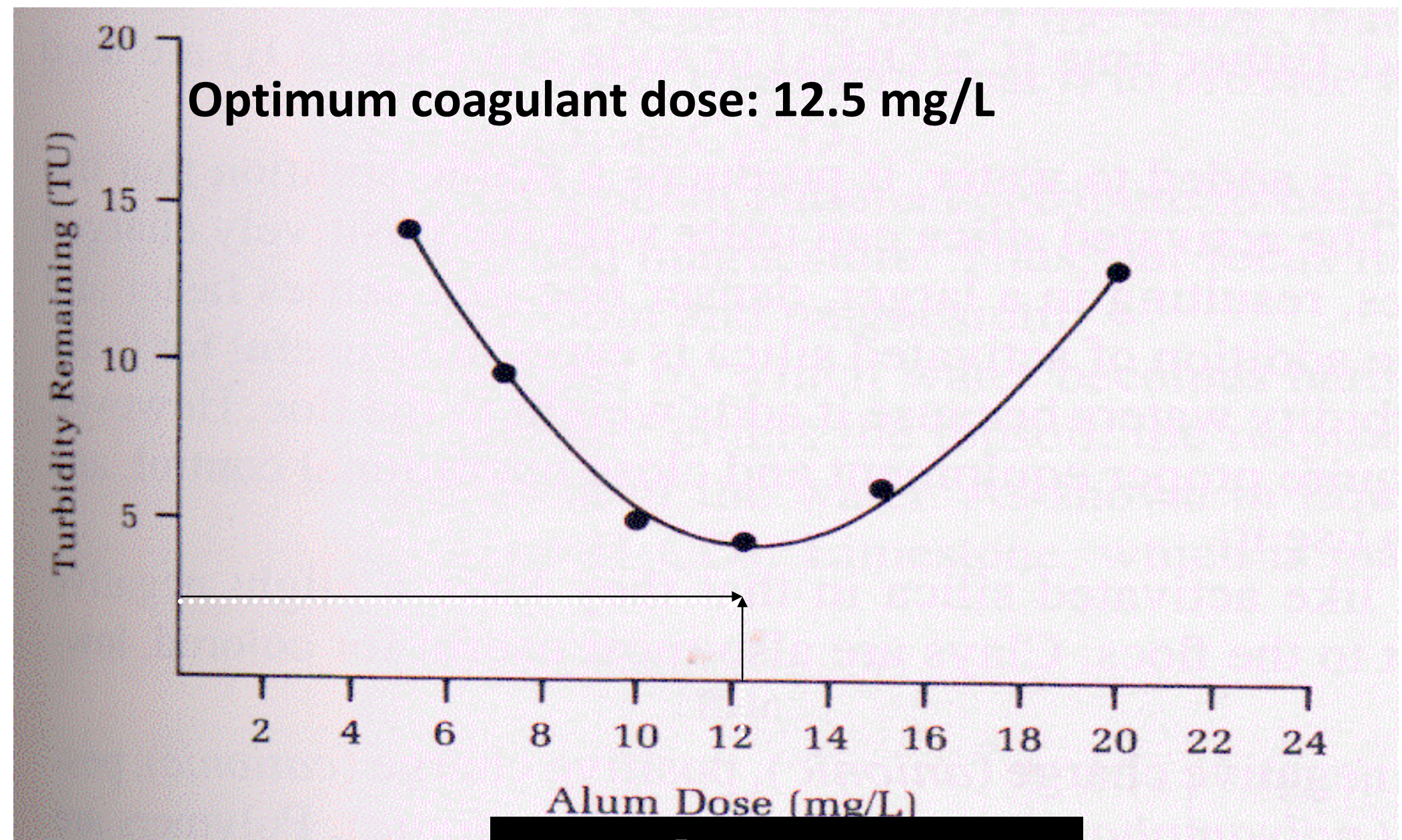
- ❑ Repeat all the previous steps
- ❑ This time adjust pH of all jars at optimum (6.3 found from first test) while mixing using H_2SO_4 or NaOH/lime
- ❑ Add different doses of the selected coagulant (alum or iron) to each jar
- ❑ Rapid mix each jar at 100 to 150 rpm for 1 minute. The rapid mix helps to disperse the coagulant throughout each container
- ❑ Reduce the stirring speed to 25 to 30 rpm for 15 to 20 mins



7.6.2 Jar Tests – Optimum coagulant dose

- ❑ Turn off the mixers and allow flocs to settle for 30 to 45 mins
- ❑ Then measure the final residual turbidity in each jar
- ❑ Plot residual turbidity against coagulant dose

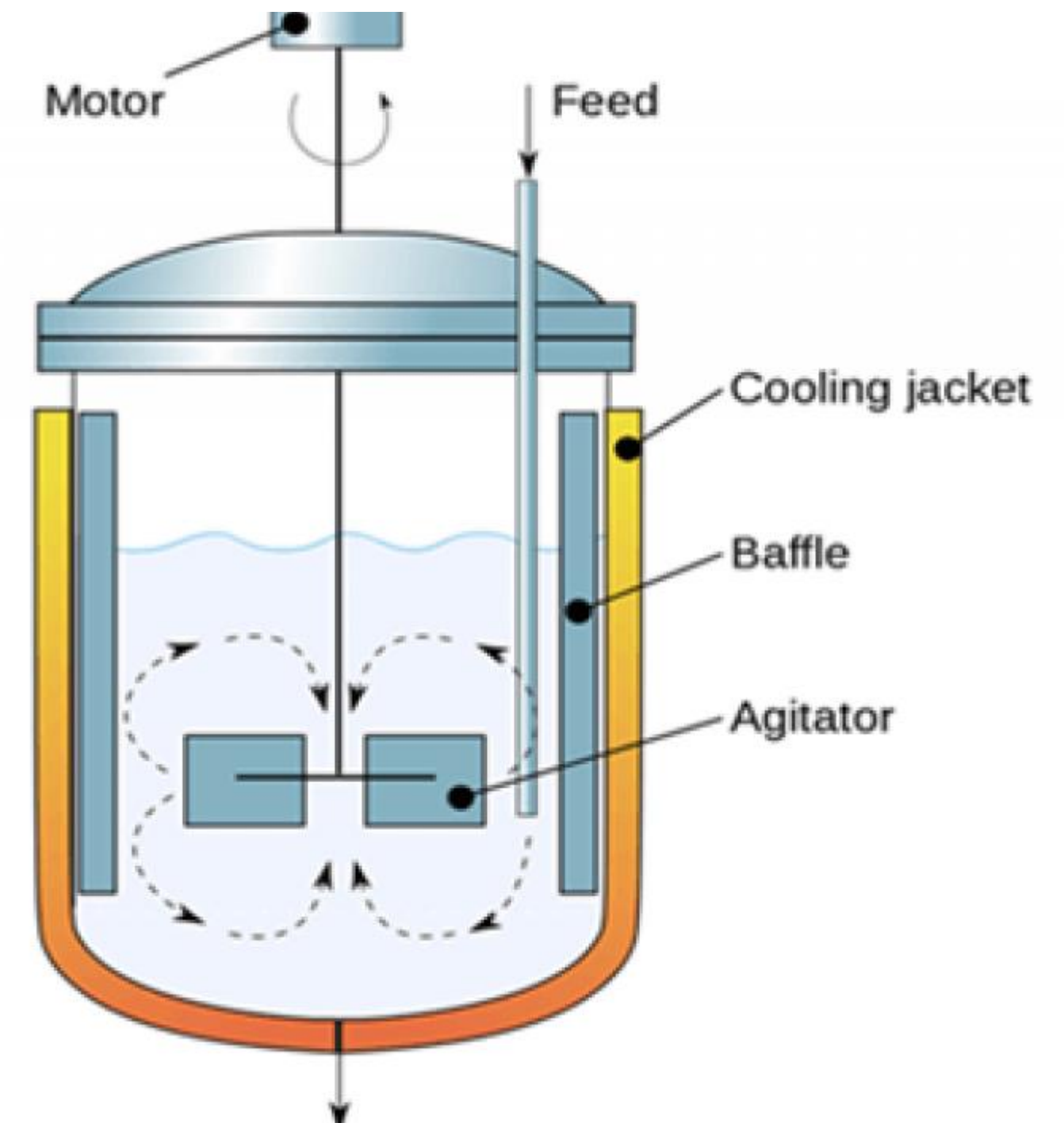
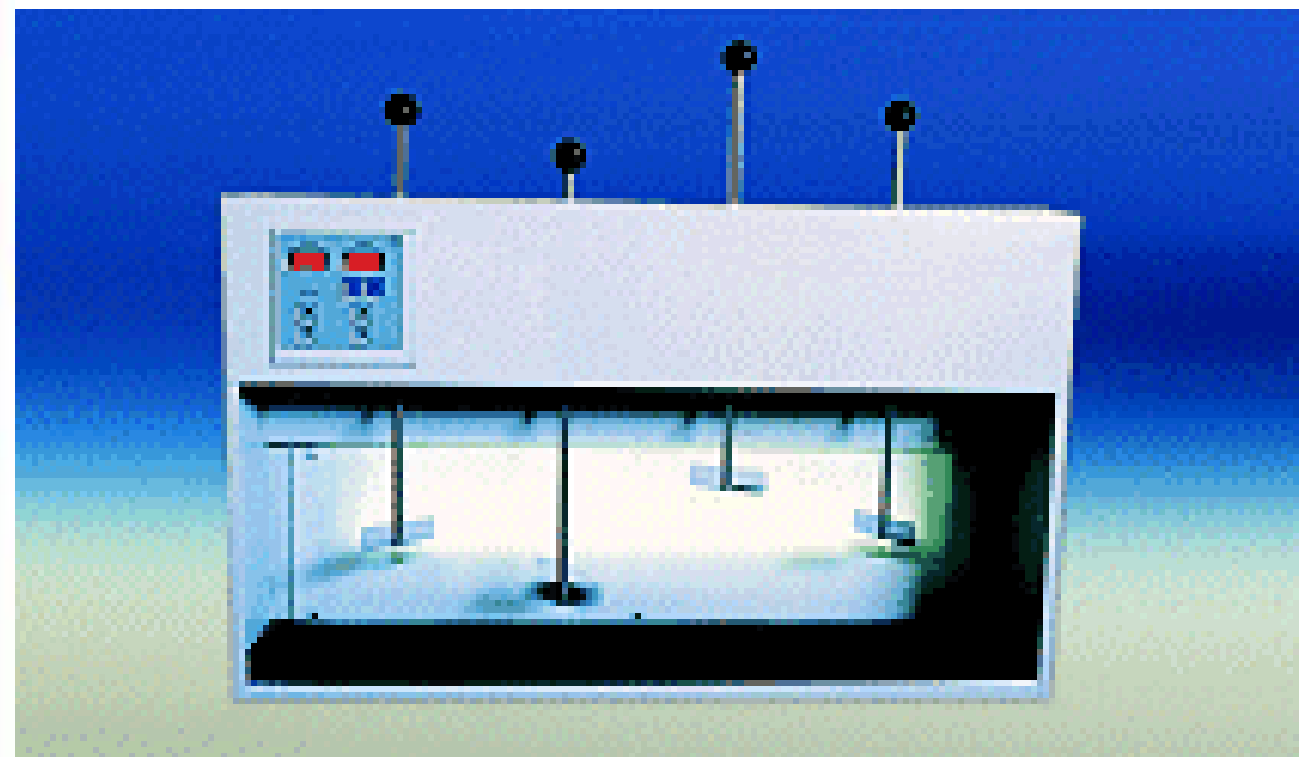
The coagulant dose with the lowest residual turbidity will be the optimum coagulant dose



Coagulant Dose mg/L

7.6.3 Coagulant Addition: Rapid mix

- ❑ Mixer: vertical shaft turbine impeller
- ❑ Tank: 3 to 10 ft diameter, flow through, top to bottom, 30 – 60s detention time



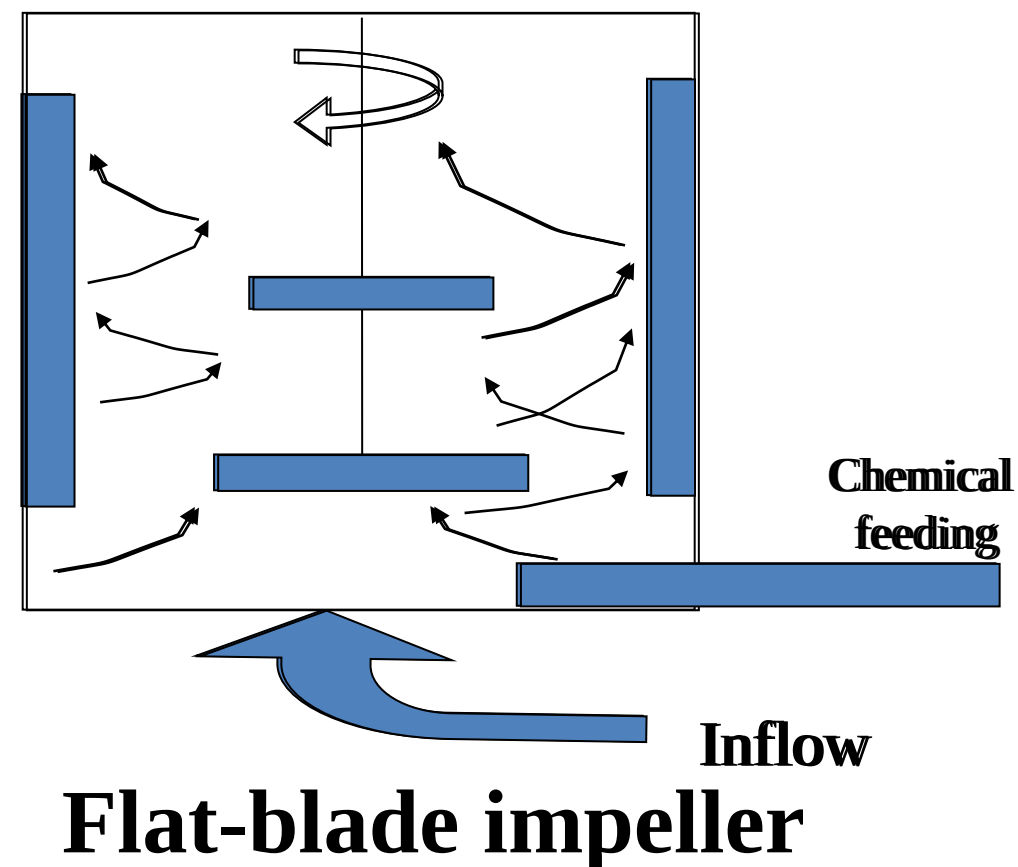
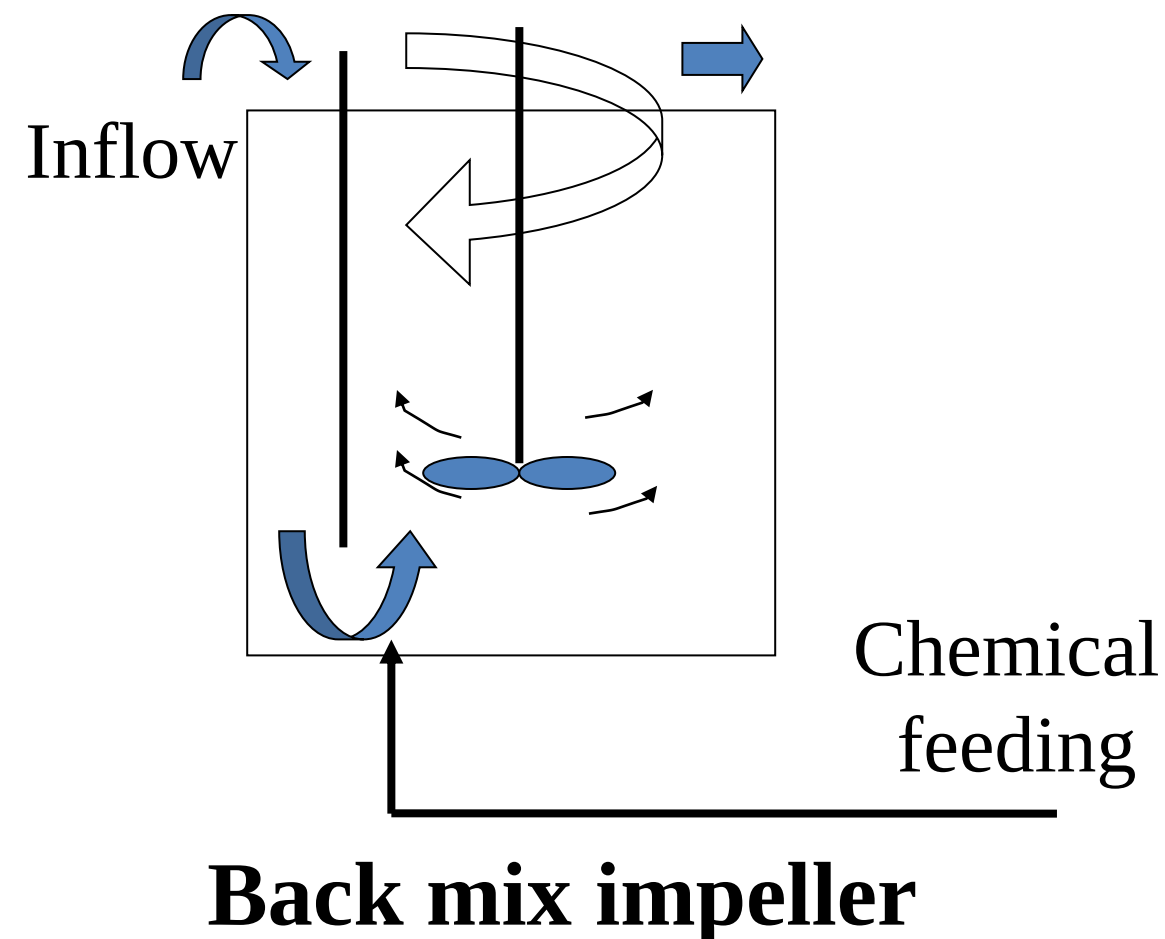
7.6.4 Mixing types for coagulation - Rapid mix

- ❑ Hydraulic Jump: Hydraulic Jump creates turbulence and thus help better mixing.
- ❑ In-line flash mixing



Hydraulic Jump

- ❑ Mechanical mixing

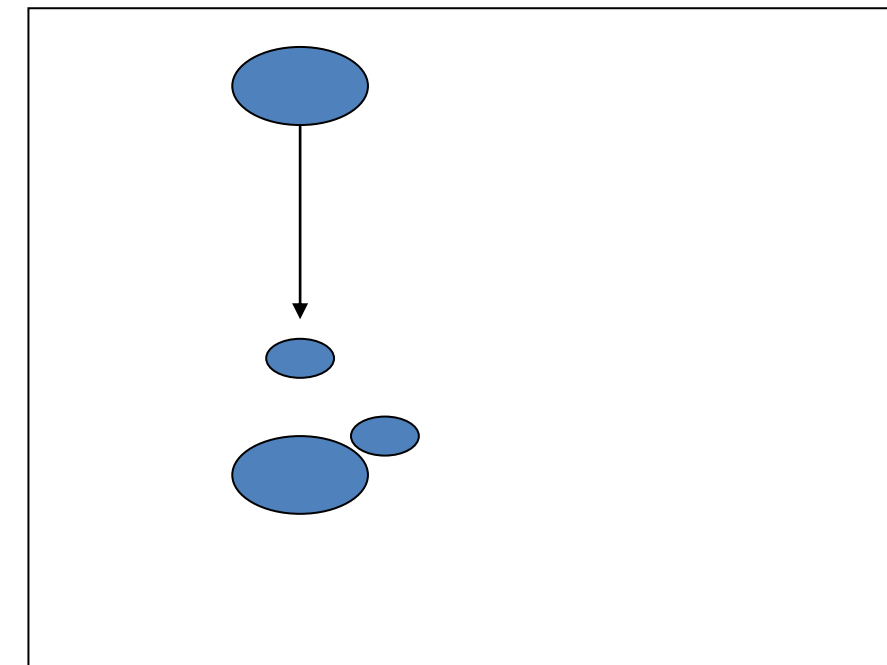
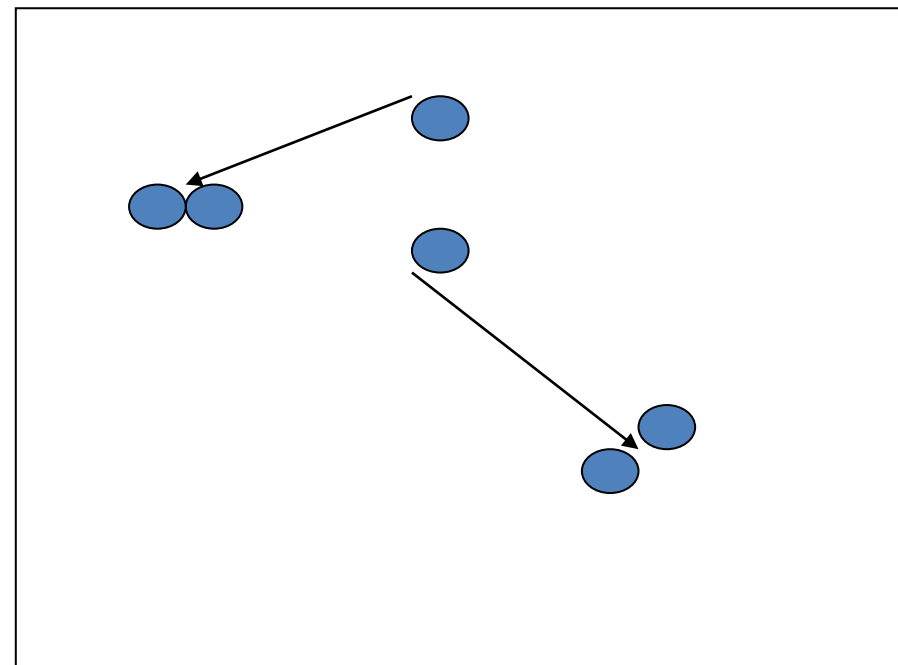


7.6.5 Mixing for Flocculator (Slow & Gentle mixing)

- ❑ Flocculators are designed mainly to provide enough interparticle contacts to achieve particles agglomeration so that they can be effectively removed by sedimentation or flotation

Transport Mechanisms

- ❑ Brownian motion: for relatively small particles which follow random motion and collide with other particles (perikinetic motion)
- ❑ Differential settling: Particles with different settling velocities in the vertical alignment collide when one overtakes the other (orthokinetic motion)



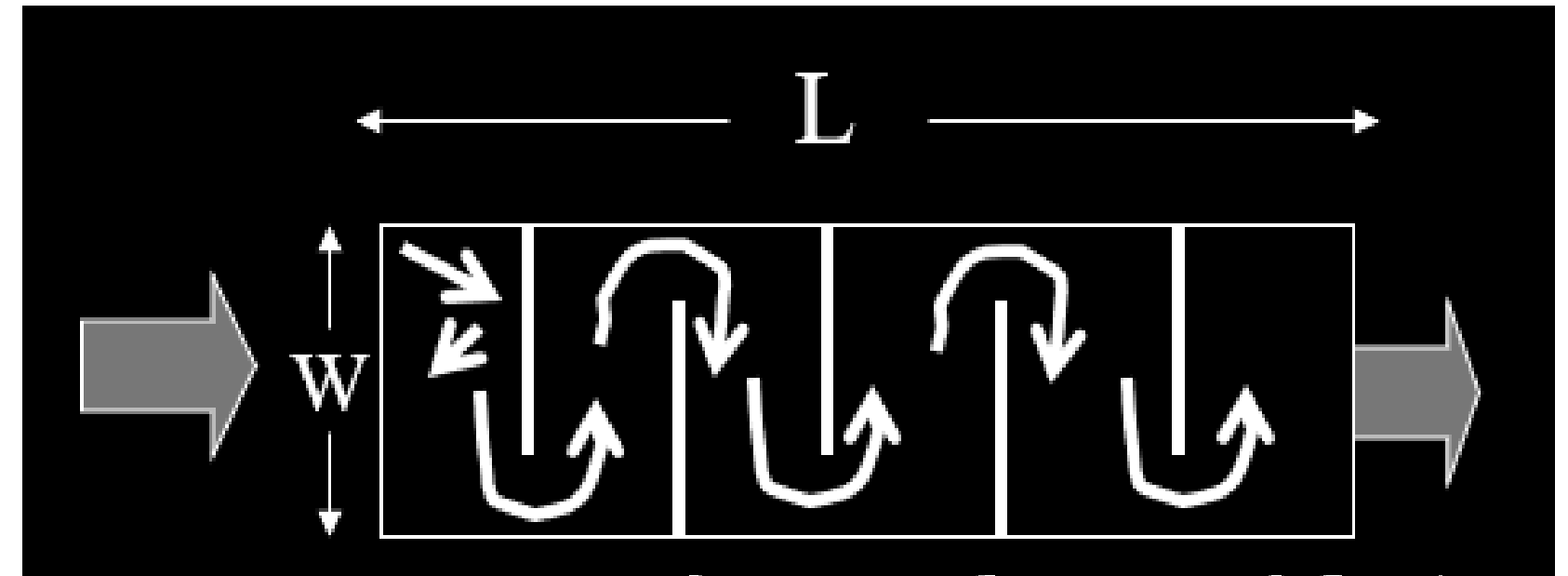
7.6.5 Mixing for Flocculator (Slow & Gentle mixing)

Hydraulic Flocculation

- Horizontally baffled tank

The water flows horizontally.

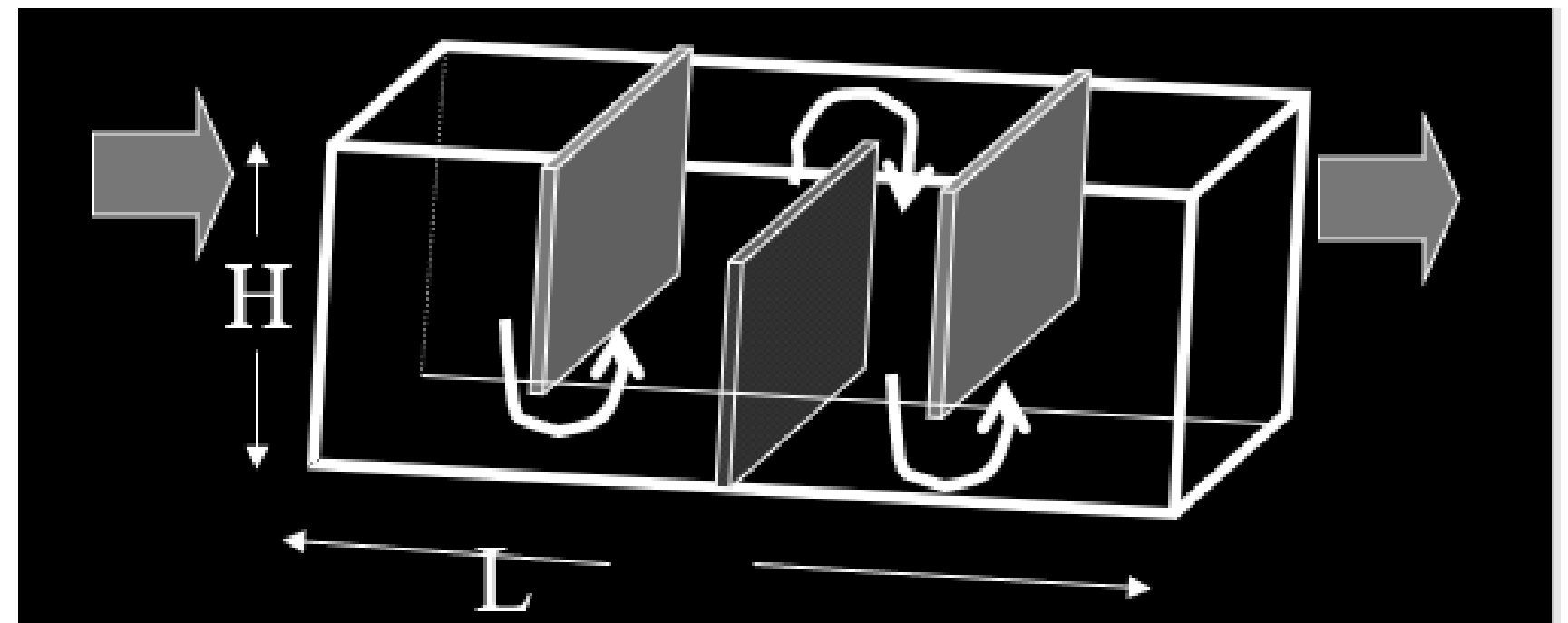
The baffle walls help to create turbulence and thus facilitate mixing



- Vertically baffled tank

The water flows vertically.

The baffle walls help to create turbulence and thus facilitate mixing



7.6.5 Mixing for Flocculator (Slow & Gentle mixing)

Hydraulic Flocculation

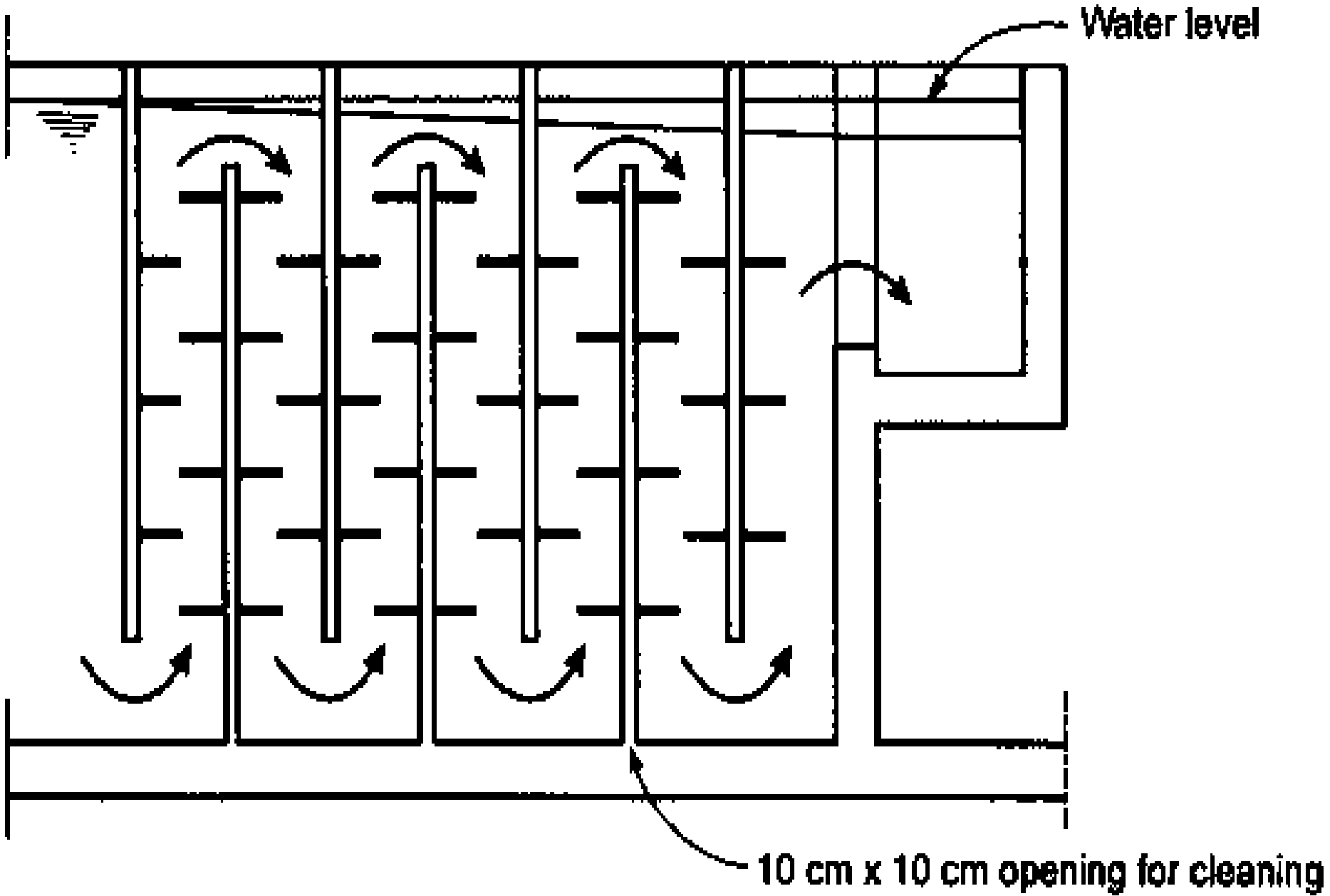
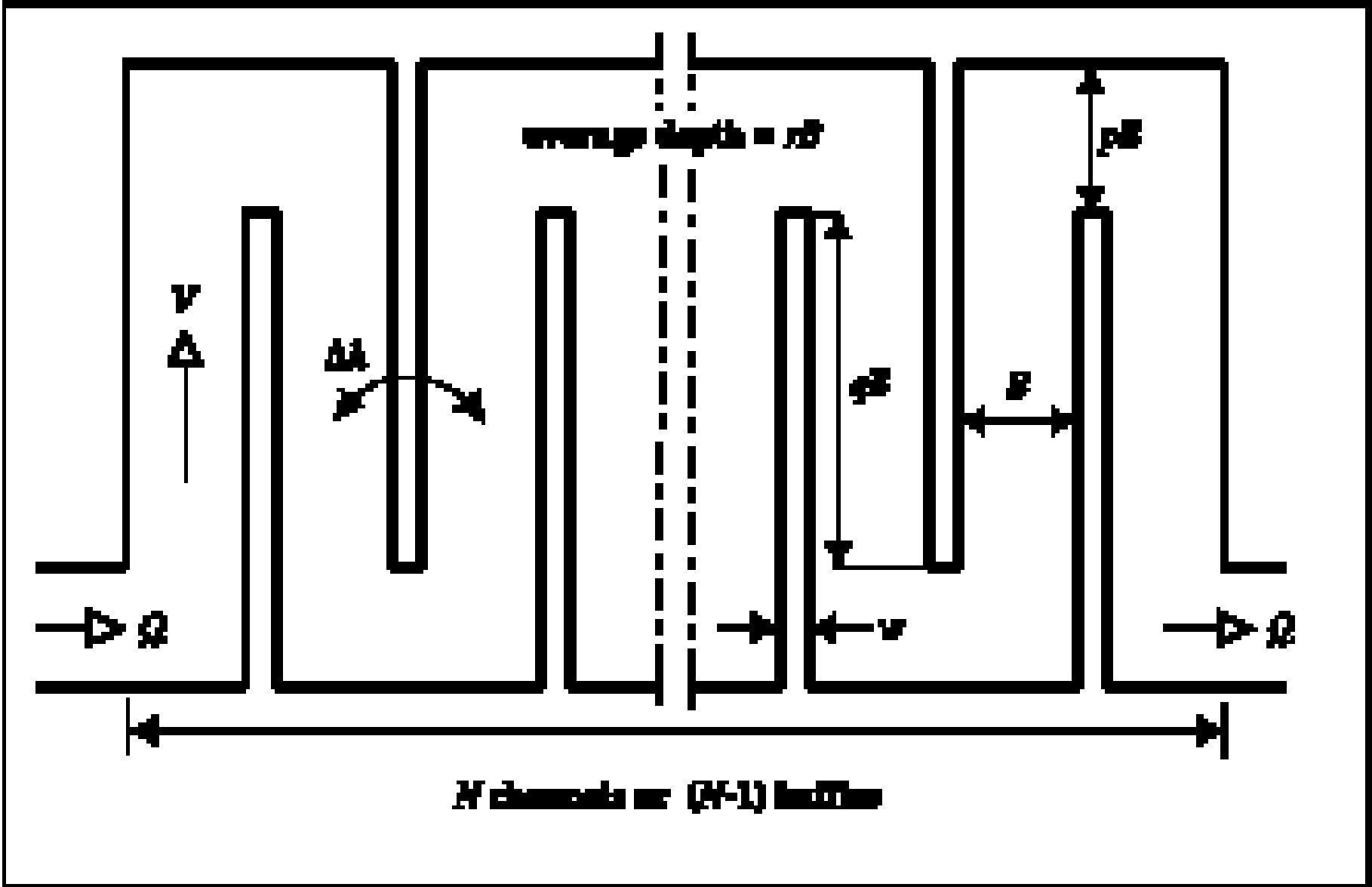
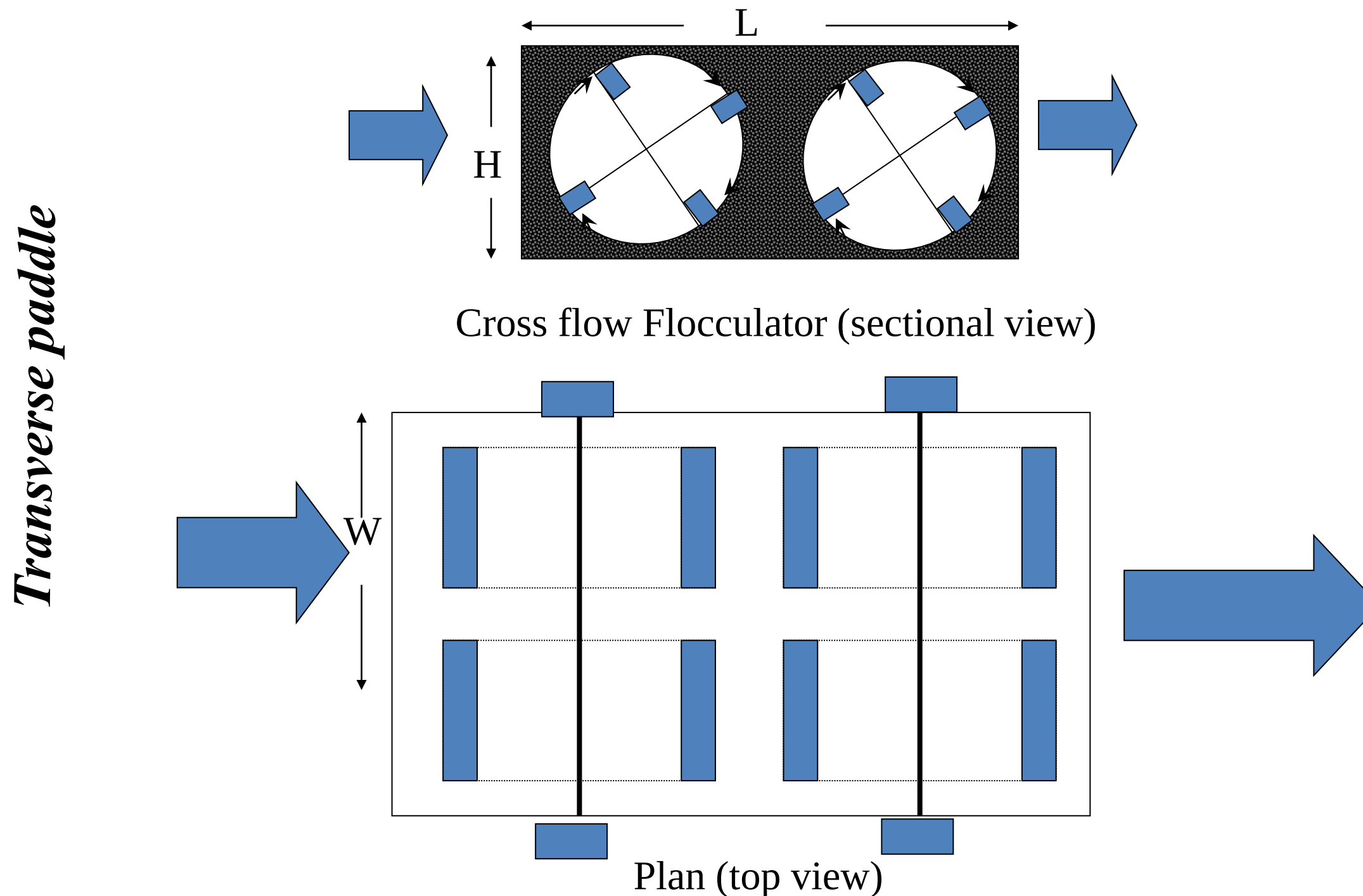


Figure 1 | Schematic layout of an around-the-end hydraulic flocculator, showing the notation used in this paper.

7.6.5 Mixing for Flocculator (Slow & Gentle mixing)

- ❑ **Mechanical Flocculator:** Flocculators are designed mainly to provide enough interparticle contacts to achieve particles agglomeration so that they can be effectively removed by sedimentation or flotation



7.7 Chemicals

- Chemically, coagulant chemicals are either metallic salts (such as alum) or polymers.
- **Polymers** are man-made organic compounds made up of a long chain of smaller molecules.
- Polymers can be either **cationic** (positively charged), **anionic** (negatively charged), or **nonionic** (neutrally charged.)

7.7.1 Coagulants

Primary coagulants

Aluminum sulfate (Alum)
 $\text{Al}_2(\text{SO}_4)_3$

Ferrous sulfate
 FeSO_4

Ferric sulfate
 $\text{Fe}_2(\text{SO}_4)_3$

Ferric chloride
 FeCl_3

- **Aluminum Sulfate (Alum)** – One of the most commonly used water treatment chemicals in the world. Alum is manufactured as a liquid, from which the crystalline form is dehydrated.
- **Aluminum Chloride** – A second choice to Alum as it is more expensive, hazardous and corrosive.
- **Polyaluminum Chloride (PAC) & Aluminum Chlorohydrate (ACH)**
- **Ferric Sulfate & Ferrous Sulfate** – Ferric sulfate is the more commonly used, but ferrous sulfate is typically used in applications where a reducing agent or excess soluble iron ions are required. Iron coagulants work similarly to aluminum coagulants but the cost may vary based on the local source of supply.
- **Ferric Chloride** – The least expensive inorganic coagulant, as it is generated as a waste material from steel making operations. However, it's by far the most corrosive and hazardous inorganic coagulant, and its use is limited to facilities equipped to handle it safely.

7.7.1 Coagulants

□ Alum

Advantages of alum are,

- It readily dissolves with water, and
- It does not cause the unsightly reddish brown staining of floors, walls and equipment like ferric sulphate.

Disadvantages of alum are,

- It is effective only at certain pH range, and
- good flocculation may not be possible with alum in some waters.

7.7.1 Coagulants

Advantages of Ferrous sulphate

- Ferric hydroxide is formed at low pH values, so that coagulation is possible with ferric sulphate at pH values as low as 4.0.
- Ferric hydroxide is insoluble over a wide range of pH values than aluminum hydroxide except for the zone of 7.0 to 8.5.
- The floc formed with ferric coagulants is heavier than alum floc.
- The ferric hydroxide floc does not redissolve at high pH values.
- Ferric coagulants may be used in color removal at the high pH values required for the removal of iron and manganese and in softening of water.

7.7.2 Coagulant aids

- Coagulant aid is an inorganic material, when used along with main coagulant, improves or accelerates the process of coagulation and flocculation by producing quick forming, dense and rapid-settling flocs.
- Coagulant aids when added increase the density to slow-settling flocs and toughness to the flocs so that they will not break up during the mixing and settling processes.

7.7.2 Coagulant aids

Common coagulant aids

- Bentonite
- Calcium carbonate
- Sodium silicate
- Anionic polymer
- Non ionic polymer



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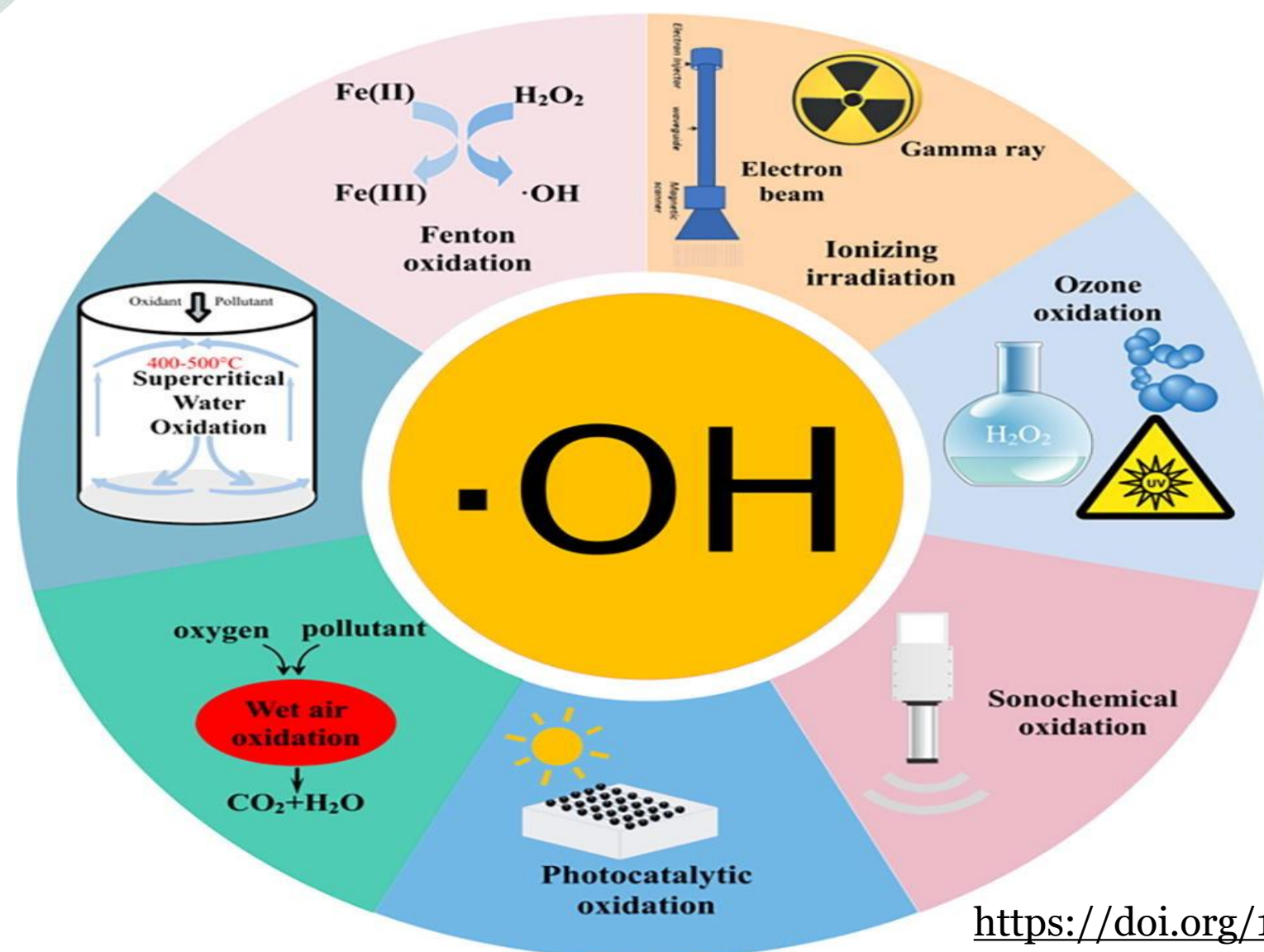
HCMC UNIVERSITY OF TECHNOLOGY AND EDUCATION

Faculty of Chemical and Food Technology



Chapter 8

ADVANCED OXIDATION PROCESSES - AOPs (CÁC QUÁ TRÌNH OXI HÓA BẬC CAO)



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9.1 Introduction

- ❑ **Advanced oxidation processes** (AOPs) are a set of chemical treatment procedures designed to **remove organic (and sometimes inorganic)** materials in water and wastewater by oxidation through reactions with hydroxyl radicals ($\bullet\text{OH}$).
- ❑ AOPs rely on in-situ production of **highly reactive hydroxyl radicals** ($\bullet\text{OH}$).
- ❑ These reactive species are **the strongest oxidants** that can be applied in water and **can oxidize virtually any compound** present in the water matrix, often at a diffusion-controlled reaction speed.
- ❑ Consequently, $\bullet\text{OH}$ reacts **unselectively** once formed and **contaminants** will be quickly and efficiently fragmented and converted into small inorganic molecules.

9.1 Introduction

- ❑ Hydroxyl radicals are produced with the help of one or more **primary oxidants**:
 - Ozone,
 - Hydrogen peroxide,
 - Oxygenand/or **energy sources** (e.g. ultraviolet light) or **catalysts** (e.g. titanium dioxide).
- ❑ **Precise, pre-programmed dosages, sequences and combinations** of these reagents are applied in order to obtain a maximum $\bullet\text{OH}$ yield.
- ❑ The AOP procedure is particularly useful for **removing biologically toxic or non-degradable pollutants**.
- ❑ The contaminant materials are converted into stable inorganic compounds such as water, CO_2 and salts.

9.1 Introduction

Chemistry in AOPs could be essentially divided into three parts:

1. Formation of $\cdot\text{OH}$;
2. Initial attacks on target molecules by $\cdot\text{OH}$ and their breakdown to fragments;
3. Subsequent attacks by $\cdot\text{OH}$ until ultimate mineralization.

Advanced chemical oxidation processes make use of (chemical) oxidants to reduce COD/BOD levels, and to remove both organic and oxidisable inorganic components.

9.1 Introduction

The mechanism of $\cdot\text{OH}$ production highly depends on the type of AOP technique that is used:

- ☐ Fenton oxidation
- ☐ Ozonation
- ☐ UV/H₂O₂
- ☐ Photocatalytic oxidation
- ☐ ...

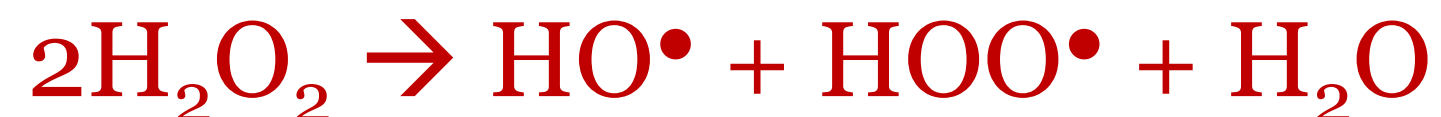
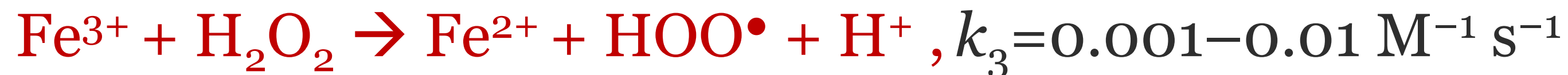
9.2 AOP processes

9.2.1 Fenton process

□ **Fenton processes** involve the reactions of **peroxides (mostly H₂O₂)** with **iron ions** to form **active oxygen moieties** that oxidize the target compounds.

□ Fenton's reagent is a solution of hydrogen peroxide (H₂O₂) with ferrous iron (typically iron(II) sulfate, FeSO₄) as a catalyst that is used to oxidize contaminants or waste waters. Fenton's reagent can be used to destroy organic compounds.

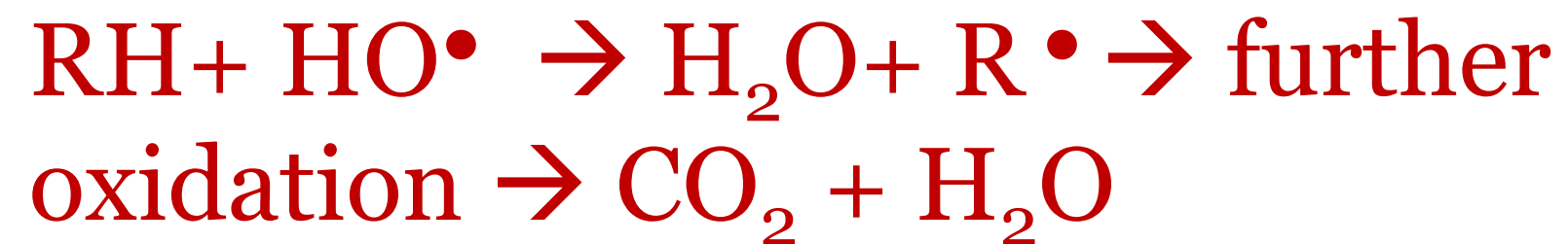
Other metals are also used as catalyst for Fenton reaction



9.2 AOP processes

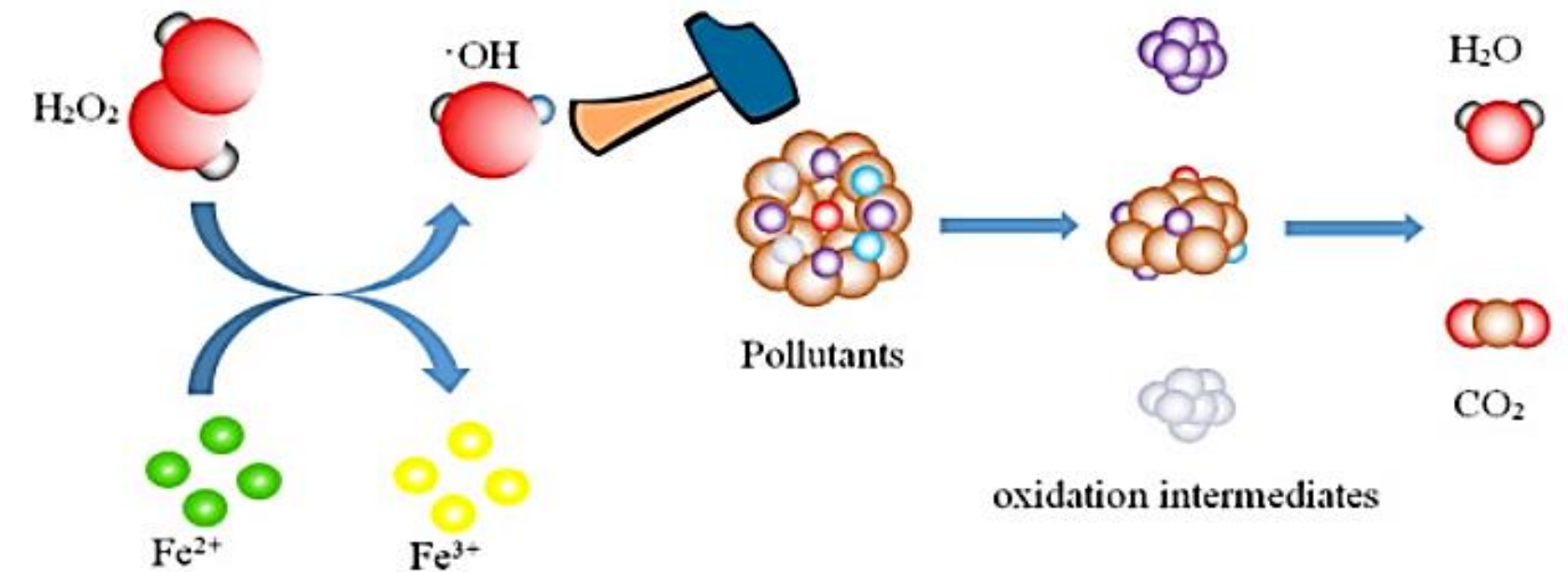
9.2.1 Fenton process

Hydroxyl radicals can oxidise organics (RH) by abstraction of protons producing organic radicals (R●), which are highly reactive and can be further oxidized.

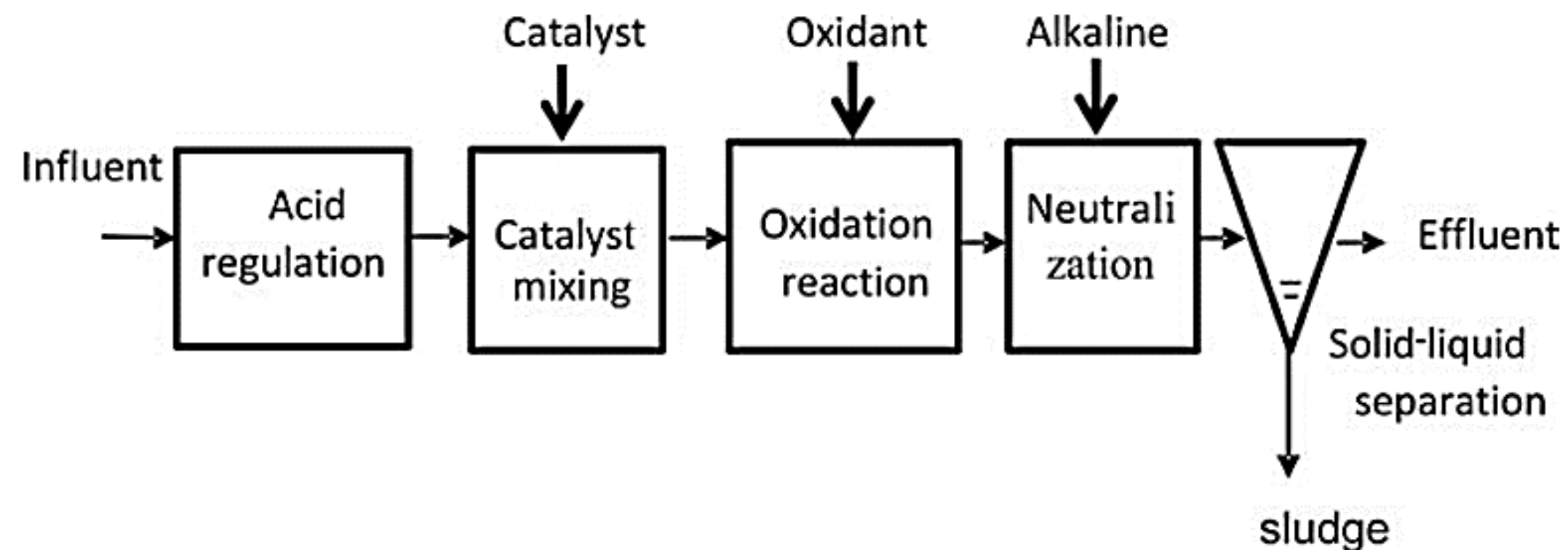


9.2 AOP processes

9.2.1 Fenton process



- ❑ The **free radicals** generated by this process then engage in secondary reactions.
- ❑ Oxidation of an organic compound by Fenton's reagent is **rapid** and **exothermic** and results in the oxidation of contaminants to primarily **carbon dioxide and water**.



9.2 AOP processes

9.2.1 Fenton process

Advantages and disadvantages

Fenton process is the most popular because of its some exciting advantages such as wide application range, strong anti-interference ability, simple operation and rapid degradation and mineralization.

Homogeneous Fenton's Process

- | Advantages | Disadvantages |
|---|---|
| <ul style="list-style-type: none">- Easy operation- Chemical oxygen demand (COD) removal- Biodegradability improvement- Established technology | <ul style="list-style-type: none">- Sludge formation- Operation in acidic conditions (pH ~3)- Requires neutralization of pH |

Removal of non-biodegradable
contaminants

High cost
(chemicals)

9.2 AOP processes

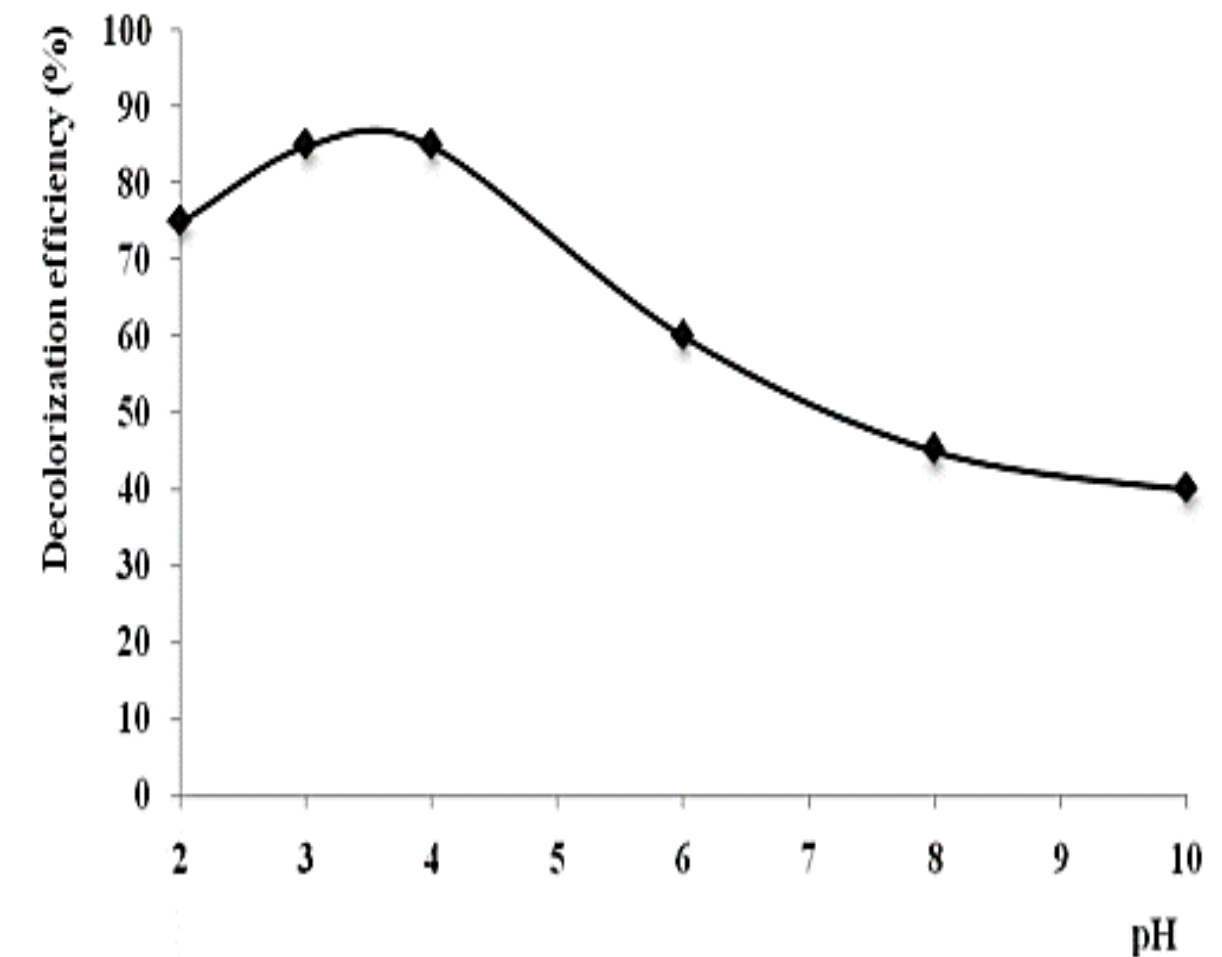
9.2.1 Fenton process

Effect of pH on formation of free radicals

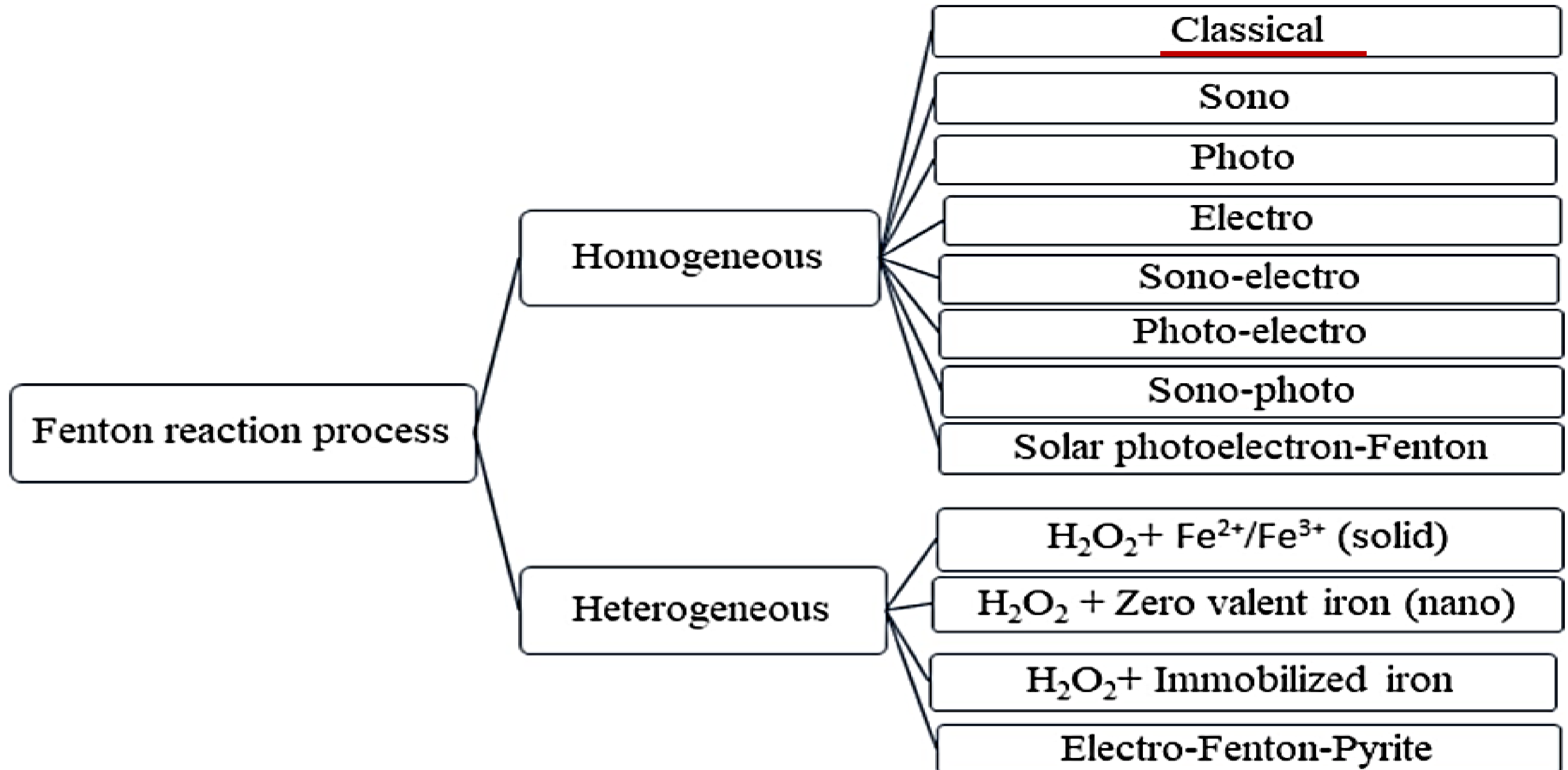
□ pH affects the reaction rate due to a variety of different reasons.

Impacts of operation pH on reaction rate

Low pH	Formation of $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ complex, hence reducing Fe^{2+} for radical generation
	Scavenging of $\cdot\text{OH}$ by excess H^+
High pH	Lower redox potential of $\cdot\text{OH}$
	Self-decomposition of H_2O_2 due to decreased stability at high pH
	Precipitation of $\text{Fe}(\text{OH})_3$ species in solution



9.2 AOP processes

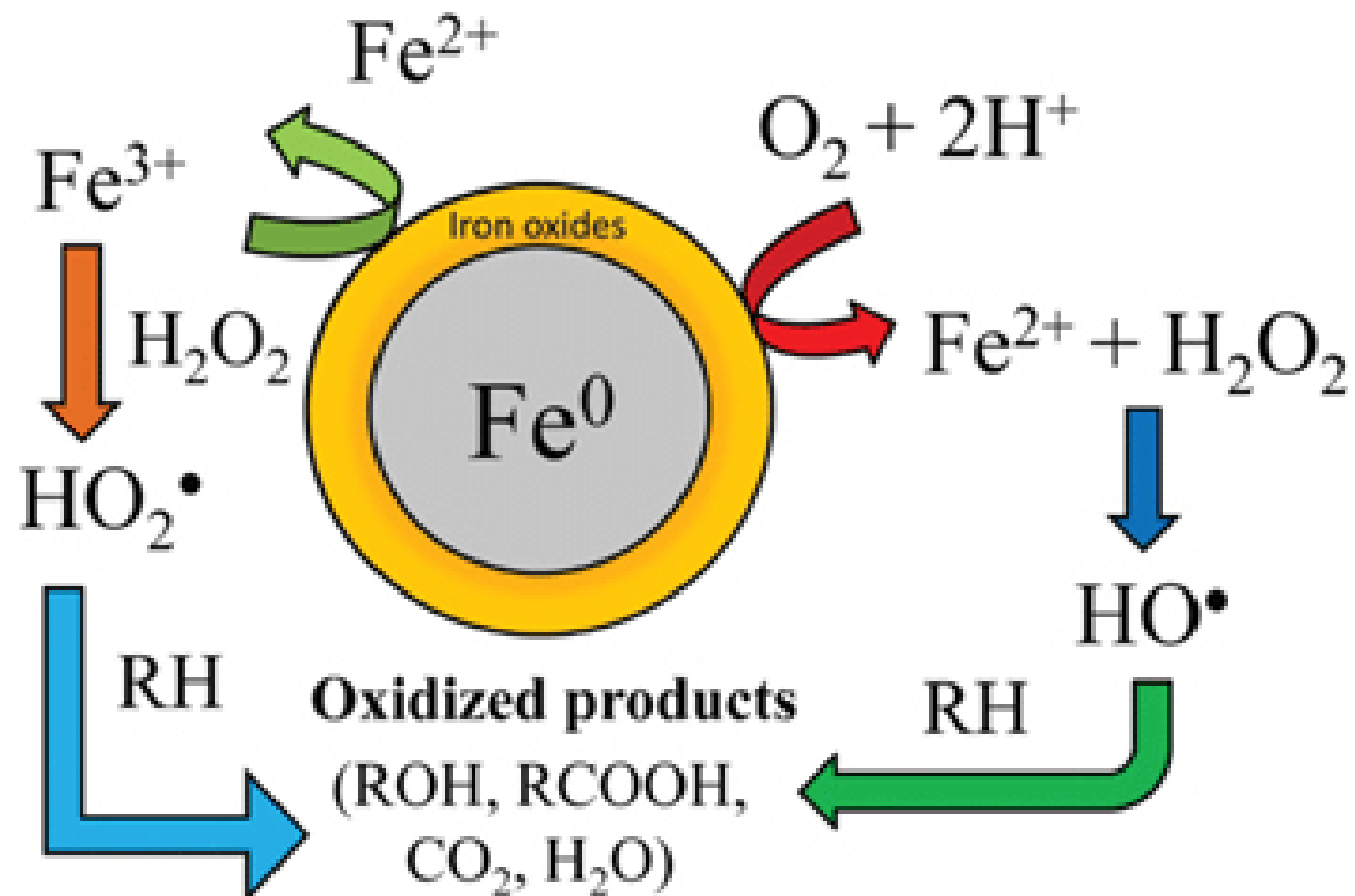


9.2 AOP processes

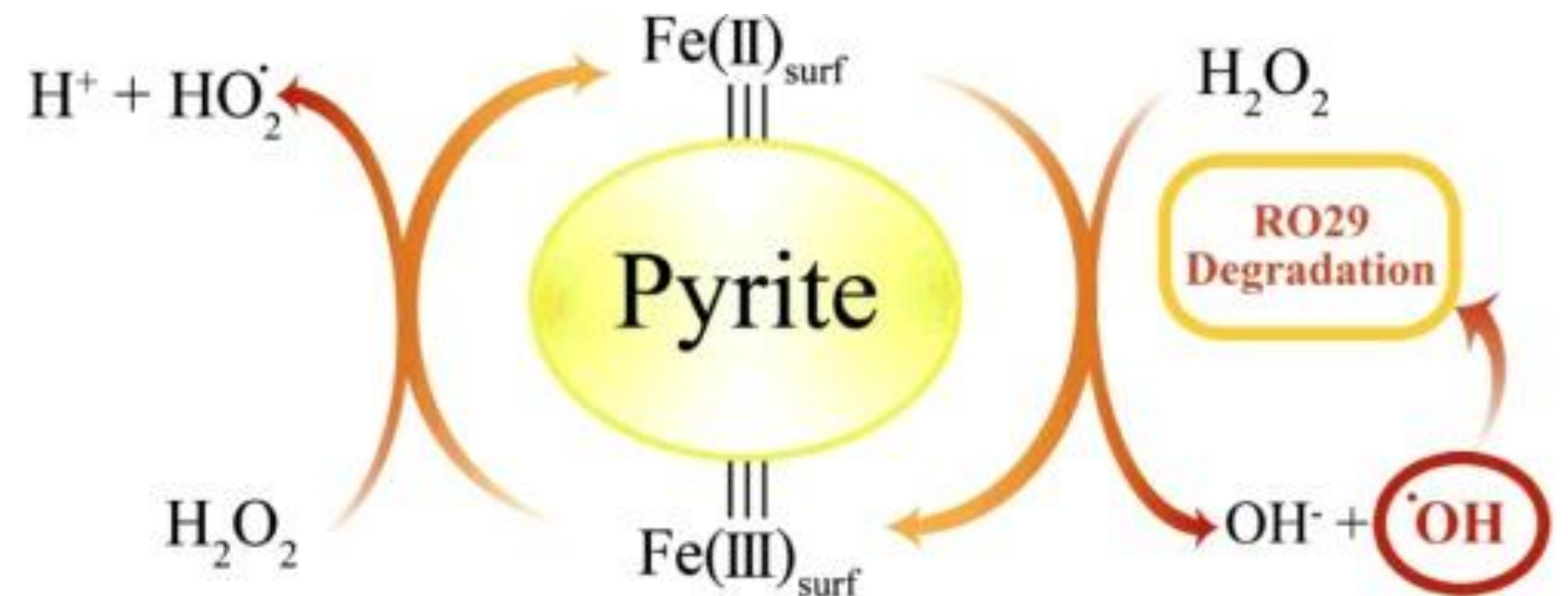
9.2.1 Fenton process

Heterogeneous Fenton process

In heterogeneous Fenton process, the radical $\bullet\text{OH}$ generation takes place from the catalytic decomposition of H_2O_2 mainly in acidic medium (pH 3), using as catalyst iron supported on a porous material. However, this reaction usually takes place in wide range of pH.



Less iron loss,
less pH effect,
low cost



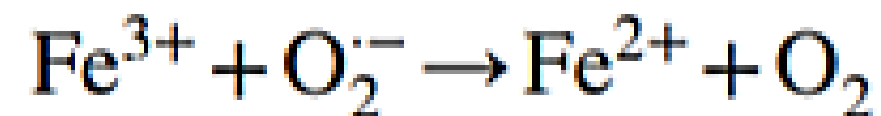
9.2 AOP processes

9.2.1 Fenton process

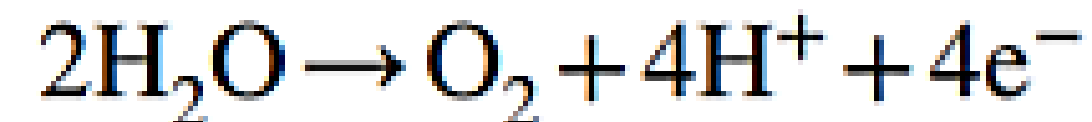
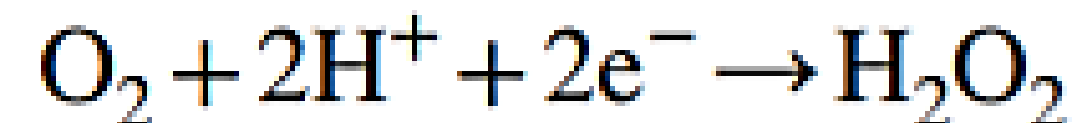
Photo-Fenton processes



Ultrasound Fenton processes



Electro-Fenton processes



Zero-valent-induced Fenton processes

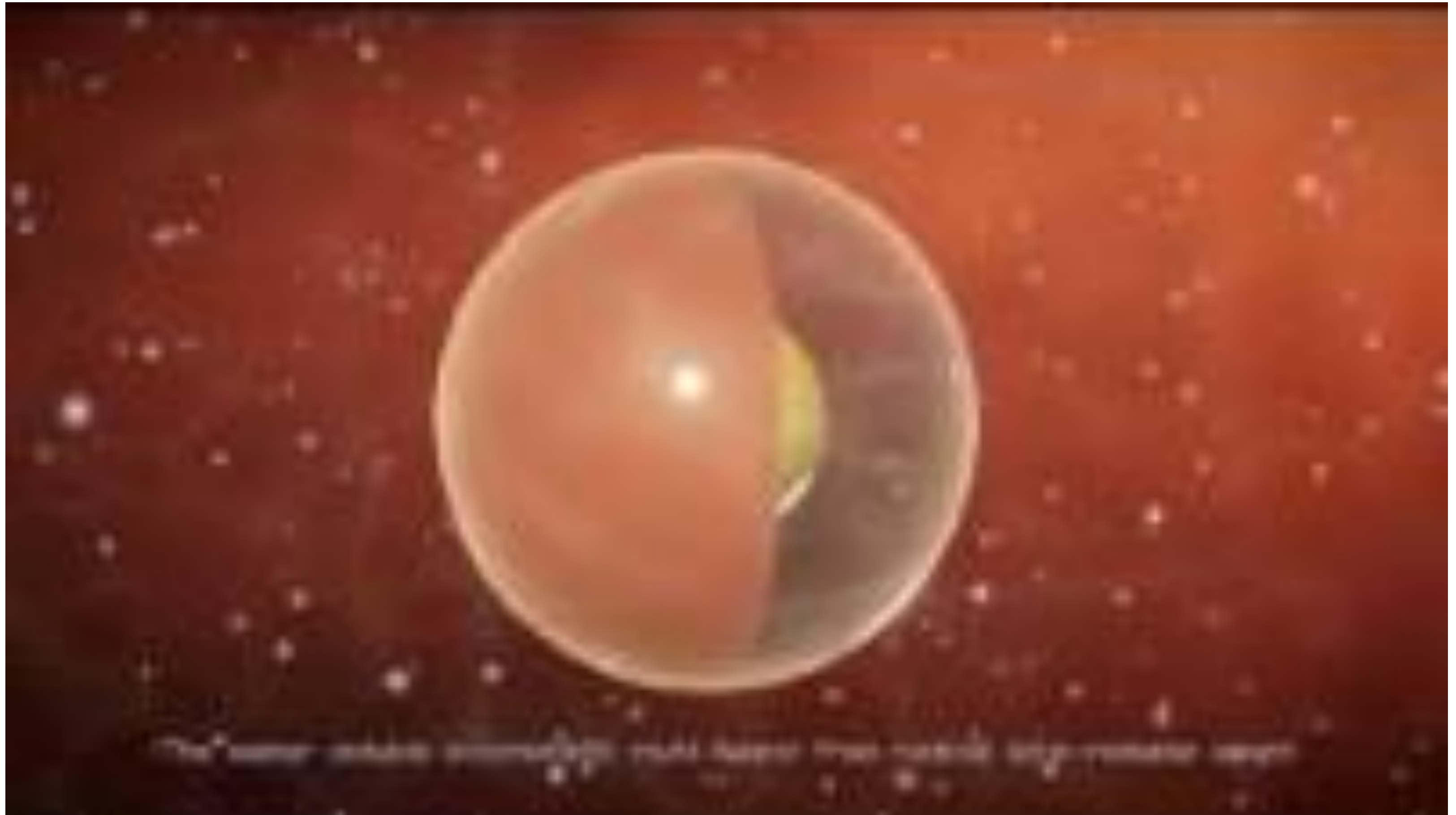


9.2 AOP processes

9.2.1 Fenton process

TABLE 1: Varieties of Fenton reactions.

	Reagents	pH	Iron loss	Light
Classic Fenton	H ₂ O ₂ , Fe ²⁺	2–4	Yes	No
Fenton-like	H ₂ O ₂ , Fe ³⁺	2–4	Yes	No
Photo-Fenton	H ₂ O ₂ , iron complex, free iron ions	Acidic to neutral	Yes	Yes
Heterogeneous Fenton	H ₂ O ₂ , solid iron oxide	Wide range	No	No
Electro-Fenton	Electrogenerated H ₂ O ₂ , free iron ions; H ₂ O ₂ , electrogenerated Fe ²⁺	2–4	Yes	
Heterogeneous photo-Fenton	H ₂ O ₂ , solid iron oxide	Wide range	No	Yes
Heterogeneous photoelectro-Fenton	Electrogenerated H ₂ O ₂ , solid iron oxide	Wide range	No	Yes



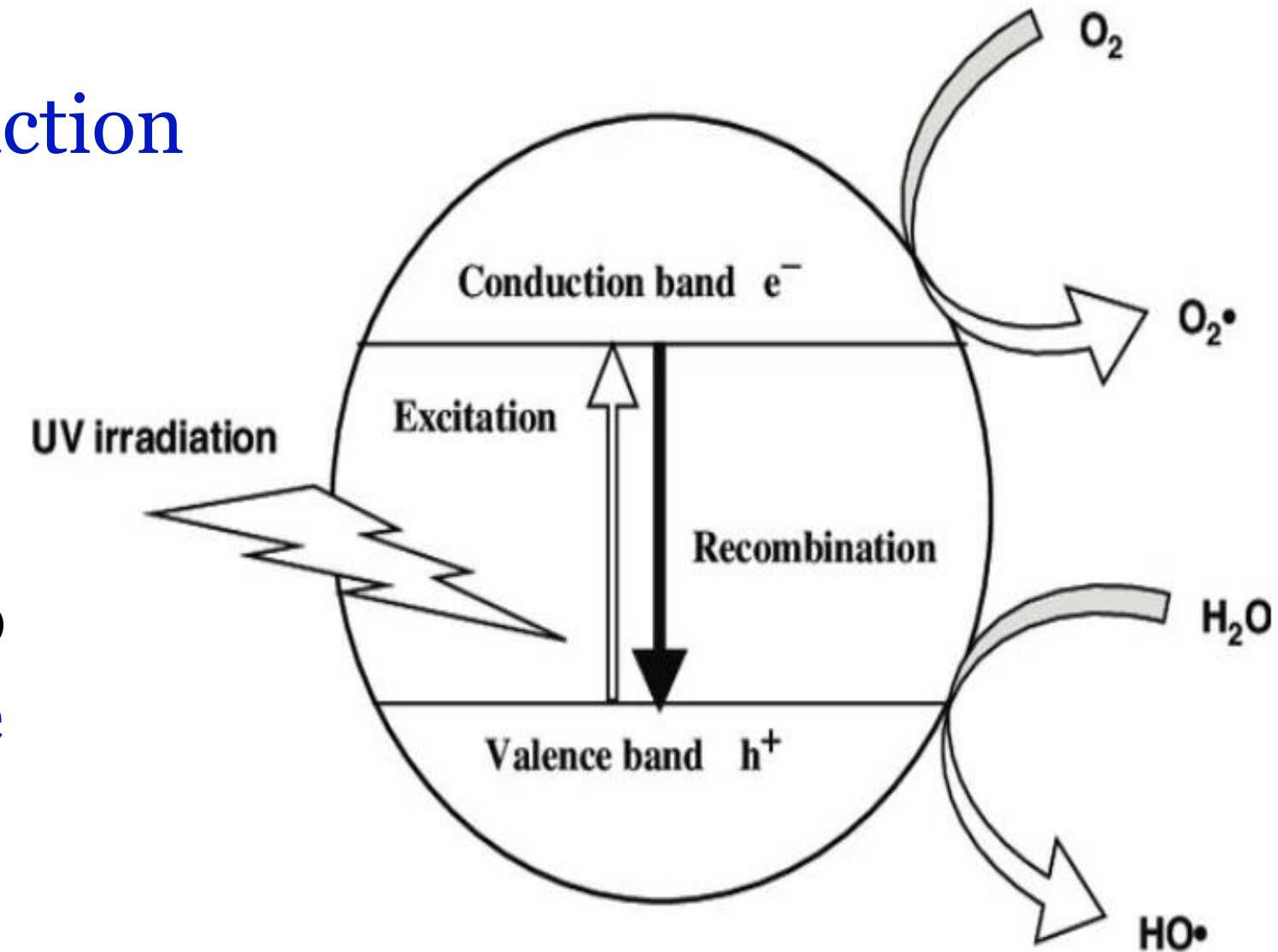
<https://www.youtube.com/watch?v=G1FwD1kja58>

9.2 AOP processes

Photocatalysis $\neq$ catalyst

9.2.2 Photocatalysis

- ❑ Photocatalysis is the **acceleration** of a **photoreaction** in **the presence** of a **catalyst**.
- ❑ In catalysed photolysis, light is absorbed by an adsorbed substrate.
- ❑ In photogenerated catalysis, the photocatalytic activity depends on the ability of the catalyst to create **electron–hole pairs**, which **generate free radicals** (e.g. hydroxyl radicals: $\bullet\text{OH}$) able to **undergo secondary reactions**.
- ❑ Its practical application was made possible by the discovery of water electrolysis by means of titanium dioxide (TiO_2).



9.2 AOP processes

9.2.2 Photocatalysis

Usually five steps are distinguished during photocatalysis:

- ❑ transfer of pollutants to the photocatalyst's surface;
- ❑ adsorption of pollutants on the surface;
- ❑ photonic activation and decomposition of adsorbed molecules;
- ❑ reaction product's desorption;
- ❑ removal of reaction products from the photocatalyst's surface.

9.2 AOP processes

9.2.2 Photocatalysis

